On Hybrid Inverse Problems for Partial Differential Equations

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Preface

This thesis was prepared as a partial fulfilment of the requirements for acquiring a joint-PhD Degree at The Technical University of Denmark (DTU) and The Norwegian University of Technology and Science (NTNU). The work was carried out during the period 15.12.2015 - 15.08.2019, in the Section for Scientific Computing at The Department for Applied Mathematics and Computer Science at DTU, at the Department of Mathematical Sciences at NTNU, and at The Department of Statistics at University of Chicago. The main supervisor of the PhD project has been associate professor Kim Knudsen (DTU), and co-supervisors has been associate professor Mirza Karamehmedović (DTU) and professor Anton Evgrafov (NTNU/DTU).

The aim of this thesis is to present the scientific work I have done and some of the things I have studied during my PhD project. The work has revolved around analysis of hybrid inverse problems, and the scientific results appear in the scientific papers in the appendix. Chapters 1-6 of the thesis aspire to present some of the things I've come to know and appreciate while studying inverse problems, and to set the stage for the content of the scientific work in the appendix. The appendix consists of 4 scientific papers concerning various topics related to hybrid inverse problems:

1 G. Bal and A. Kirkeby, "Photo-acoustic tomography in the rotating setting." *Inverse Problems*, DOI: 10.1088/1361-6420/ab315c, 2019.

2 B. Jensen, A. Kirkeby and K. Knudsen, "Feasibility of Acousto-electric tomography." *Submitted manuscript.*

3 M. Karamehmedović, A. Kirkeby, and K. Knudsen. "Stable source reconstruction from a finite number of measurements in the multi-frequency in-

verse source problem." *Inverse Problems*, DOI: 10.1088/1361-6420/aaba83, 2018.

4 M. Karamehmedović, A. Kirkeby, M. T. R. Henriksen, "Stability of the inverse source problem for the Helmholtz equation in $\mathbf{R}^3$." *Submitted manuscript.*

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Summary (English)

This thesis is concerned with the mathematical analysis of inverse problems for partial differential equations, and with a specific type of such problems known as *hybrid inverse problems*. Chapters 1-4 present various aspects of the theory of inverse problems for partial differential equations. Chapters 5-6 introduce the topic of hybrid inverse problems and content of the scientific papers in the appendix.

The scientific papers are concerned with the analysis of hybrid inverse problems. Specifically, we consider photoacoustic tomography and acousto-electric tomography, as well as the multi-frequency inverse source problem for the Helmholtz equation.

For photoacoustic tomography, we analyze the stability and uniqueness of the inverse problem of reconstructing the optical coefficients of an object in a situation where measurement data is collected by a measurement device that can rotate around the object.

For acousto-electric tomography, we provide a method to reconstruct the electrical conductivity from boundary measurements of the voltage. Moreover, we investigate the feasibility of acousto-electric tomography for medical imaging by examining how reconstructions depend on the strength of the acousto-electric coupling.

In the case of multi-frequency measurements of solutions to the Helmholtz equation, we provide a detailed characterization of sources that can be stably reconstructed. The characterization depends on the measurement frequencies, and

we devise an effective and stable reconstruction method.

Summary (Danish)

Denne afhandling vedrører den matematiske analyse af inverse problemer for partielle differentialligninger og en bestemt type af sådanne problemer kendt som *hybride inverse problemer*. Kapitlerne 1 - 4 præsenterer forskellige aspekter af teorien om inverse problemer for partielle differentialligninger. Kapitlerne 5 - 6 introducerer emnet hybride inverse problemer og indholdet af de videnskabelige artikler i appendiks.

De videnskabelige artikler vedrører analyse af hybride inverse problemer. Specifikt overvejer vi fotoakustisk tomografi og akusto-elektrisk tomografi såvel som det inverse kilde-problemet for Helmholtz-ligningen med flere frekvenser.

For fotoakustisk tomografi analyserer vi stabilitet og entydighed for det inverse problem, hvor målet er at rekonstruere de optiske koefficienter i et objekt i situationen, hvor måledata indsamles af en måleenhed, der kan rotere rundt om objektet.

For akusto-elektrisk tomografi fremlægger vi en metode til at rekonstruere den elektriske ledningsevne fra grænsemålinger af spændingen. Desuden undersøger vi hvorvidt akusto-elektrisk tomografi kan bruges til medicinsk billeddannelse ved at undersøge, hvordan rekonstruktioner afhænger af styrken af den akustisk-elektriske kobling.

I tilfælde af multifrekvensmålinger af løsninger til Helmholtz-ligningen giver vi en detaljeret karakterisering af kilder, der kan rekonstrueres stabilt. Karakteriseringen afhænger af målefrekvenserne, og vi foreslår en effektiv og stabil metode til rekonstruktioner.

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Chapter 1

Introduction

We consider partial differential equations equations of the form

$$L(p)u = f.$$

Here, $L(p)$ is a partial differential that depends on parameters p, and f is some function. Solving a partial differential equation (PDE) means finding a function u that, in some sense, satisfies the equation. A remarkable thing about solutions to PDEs is that they accurately describe physical phenomena, a fact that has been coined as "The unreasonable effectiveness of mathematics in the natural sciences". A more prosaic remark is that PDEs allow one to write down relationships between different physical quantities, and if one can solve these equations, they will reveal precise information about the interaction and behavior of the quantities involved. The study of PDEs is a central part of mathematics and includes a vast range of techniques and tools. Sometimes techniques are tailored for a specific PDE; other times one can study a large class of PDEs using the same approach. There are, however, still three fundamental questions one usually asks when confronted with a PDE:

- Existence: Is there a solution to the equation?
- Uniqueness: Is the solution unique?
- Stability: Does the solution stably depend on the data?

These three criteria were formulated by Jacques Hadamard, and an equation for which the answer to all three questions is positive is said to be *well-posed*. If this is not the case, i.e., if one or more criteria has a negative answer, the PDE is said to be *ill-posed*. According to Hadamard, only well-posed problems were worth studying. In this thesis we will be concerned with both ill-posed and well-posed problems. The PDEs we study will be well-posed, and their corresponding inverse problems will be ill-posed. The viewpoint we take is the following: Assume that we have a PDE $L(p)u = f$ that models a physical phenomenon. The quantity u depends on parameters p and f, called the data, and the specific form of the PDE. u could be a propagating acoustic pressure wave resulting from a sound source f within a material with sound speed p, or the gravitational force from an object with mass density f. Assume now that we have analyzed the PDE, and that it is found to be well-posed. Then a solution exists, it is unique, and it depends stably on the data (p, f). It then makes sense to write the equation in *operator form*:

$$u = K(p, f)$$

Here K is a solution operator, an operator that maps the data into the solution of the PDE. In general, the solution operator is not given to us explicitly, e.g., in terms of an integral operator or solution formula, but the well-posedness of the PDE means that it exists. By studying the properties of the PDE, we study the properties of the solution operator.

Let us now consider an opposite problem connected to PDEs, the so-called *inverse problem*, and explain why solution operators for PDEs play an important role in analyzing and solving such problems.

Inverse problems concern the task of finding a cause to an observation. This undertaking makes sense when the cause cannot be observed directly, but only through its action on an observable quantity. We call the observation a measurement, and the cause that leads the measurement is our (unknown) quantity of interest. For example, if we want to know the temperature on our balcony, we might know that the temperature is the cause of the volume expansion of mercury. If we have an old-fashioned thermometer, we can then observe the effect of the temperature on the mercury(the measurement), and with enough knowledge of our equipment, we can infer what the temperature (the cause) must be. The causality is important; it is not the volume expansion that determines the temperature. For many things that are not directly observable, their interaction with the surroundings is governed by the laws of physics and can be described by PDEs. If the PDE(s) governing this interaction is known, one can make a model connecting the unknown quantity with an observable quantity in terms of the PDE. This is often the starting point for the mathematical analysis of inverse problems, and the goal is to invert the operator that maps the unknown

quantity to the measurement, i.e., to obtain the unknown quantity. In the context of inverse problems, this operator is often called the forward operator (in contrast to the inverse operator) or the measurement operator. Inverting the measurement operator is closely connected to inverting the solution operator of the PDE.

We consider a concrete example of an inverse problem involving a PDE. Consider an unknown heat source f that is placed within some thermally homogeneous object occupying a region Ω in $\mathbb{R}^3$ with boundary $\partial\Omega$. We assume that the temperature u on the boundary of the object is 0 and that the system is in an equilibrium state, so that it does not change with time. The dependence of the temperature is then given by the *Poisson equation*,

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{1.1}$$

Our quantity of interest is the unknown and not directly observable heat source f within the object. We assume that we have a measurement g of the heat flux on the boundary of the object, i.e., $g = \frac{\partial u(x)}{\partial \nu}, x \in \partial\Omega$, where ν is the outward normal vector on $\partial\Omega$. The inverse problem is then:

> Find the heat source f such that $g = \frac{\partial u(x)}{\partial \nu}, x \in \partial\Omega$, where u satisfies the Poisson equation (1.1).

As we will see, the Poisson equation is indeed well-posed, and so it makes sense to talk about a solution operator K such that $K(f) = u$. Assuming the solution is smooth enough, we use our solution operator to construct a measurement operator:

$$Mf = \frac{\partial}{\partial \nu} K(f)\big|_{\partial\Omega}. \tag{1.2}$$

The measurement operator M relates the quantity of interest and the measurement. If M is invertible, the solution to our inverse problem is given by $f = M^{-1}g$, where M^{-1} is the inverse of M. However, in many problems of interest, and most of the problems studied in this thesis, inverting the measurement operator is a very delicate procedure. The inverse might not yield a unique solution, or it might be unbounded, leading to an unstable dependence on the measurement; in other words, the inverse problem is *ill-posed*. Indeed, inverse problems are often called ill-posed problems.

The ability to solve inverse problems is of immense importance science, engineering, medical care, navigation, communication and many other important areas of modern society. It requires the interdisciplinary strength of mathematics, physics, engineering and computer science; physics to explain and model

the world, engineering to measure what is going on, computer science to compute, and mathematics to make sense of it all. This thesis aims to present some essential aspects of the mathematical theory for inverse problems where the underlying physics is modelled by PDEs. Moreover, it presents a particular type of inverse problems that appear in context of medical imaging, known as "hybrid inverse problems". These inverse problems have in common that they rely on the coupling of different physical processes to reconstruct the unknown quantities, and they require delicate mathematical analysis. Hybrid inverse problems is also the unifying topic of the scientific papers in the appendix, and we hope that the text will put the reader in a position to comprehend and enjoy the contents of these works.

On the organization and content of this thesis

The ambition of Chapters 2-4 is communicate some of the most important elements of the mathematical theory of inverse problems that I have come to know and appreciate during my PhD, as well as to prepare the reader for Chapters 5 and 6, and the scientific papers in the appendix. Although most of the material in these chapters is more or less well-known to people working in inverse problems, it is accompanied by my personal take on the matters and hence may offer some new perspectives.

Chapter 5 contains an introduction to hybrid inverse problems, and Chapter 6 is concerned with explaining the content and contributions of the scientific papers in the appendix.

The intended audience is anyone interested in the topic, but a good mathematical background is certainly useful.

The chapters are organized as follows:

In Chapter 2 we introduce the mathematical framework for inverse problems for PDEs. Section 2.1 introduces some essential tools and concepts from functional analysis needed for such study. Sections 2.2-2.3 is concerned with compactness and compact operators, some of the key concepts needed to understand the difficulties encountered in inverse problems. In Section 2.4 we briefly consider non-linear problems and linearization.

Chapter 3 is concerned with a qualitative understanding of inverse problems for PDEs. Here, we study how information is transmitted by PDEs and get a better understanding of the underlying difficulties involved in solving inverse problems. In Sections 3.3-3.4 we introduce the tools of pseudo-differential operators and

microlocal analysis and demonstrate how they can be used to explain why some inverse problems are severely ill-posed, while others can be considered only moderately ill-posed or even well-posed.

In Chapter 4 we return to the material introduced in Chapter 2. We consider spectral decomposition of compact operators in Section 4.1. This is an essential tool for qualitative and quantitative aspects of all inverse problems, not only those arising from PDEs. In Section 4.2 we introduce the concept of regularization, a type of methods used to find approximate solutions to ill-posed problems, and we show how these methods is connected to spectral decomposition.

Since the material in Chapters 2-4 is fairly standard, we have chosen to collect the references in a separate section at the end of each of the chapters instead of referencing on the go.

In Chapter 5 we introduce the aforementioned hybrid inverse problems. We make the case for why these problem are of particular interest and survey several more and less recent results on their mathematical properties.

In Chapter 6 we describe the results, novelties and insights of the scientific papers in the appendix, as well as how they contribute to the understanding and advancement of hybrid inverse problems.

Last but not least, the appendix contains the previously mentioned four scientific papers, and it is my hope that these will be fully appreciated after reading through Chapters 1-6.

Chapter 2

Functional analysis and PDE

Inverse problems are concerned with the study of the properties of the inverse of the measurement operator. As we could see from equation (1.2), this means studying the properties of the solution operator for the particular PDE. In this chapter we formalize the considerations from the introduction and examine the connection between PDEs and inverse problems using the tools of functional analysis. We introduce concepts that allow us to make the introductory considerations precise. The natural function spaces for solutions to PDEs are Sobolev spaces, and when we consider solution and measurement operators as mappings between such spaces and take into consideration the natural topology for a measurement setting, we will see that many solution operators are compact operators, and that this causes problems when we want to invert them.

2.1 Preliminaries: Distributions, Sobolev Spaces and Linear Operators

Let Ω be an open domain in $\mathbb{R}^n$. We denote by $\mathcal{D}(\Omega)$ the space of C^∞-functions with compact support in Ω. $\mathcal{D}(\Omega)$ is often called the space of test-functions. A distribution Λ on Ω is a linear functional on $\mathcal{D}(\Omega)$ that is bounded in the

following way: For any compact set $\mathcal{K} \subseteq \Omega$ there is some $m \in \mathbb{N}_0$ and a constant C such that

$$|\Lambda(\varphi)| \leq C\|\varphi\|_{C^m(K)}, \quad \forall \varphi \in \mathcal{D}(\Omega), \mathrm{supp}(\varphi) \subseteq K.$$

Recall that $\|u\|_{C^m(K)} = \sum_{|\alpha| \leq m} \sup_{x \in K} |\partial^\alpha u(x)|$. The space of distributions on $\mathcal{D}(\Omega)$ is denoted by $\mathcal{D}'(\Omega)$. We will be especially interested in distributions that can be represented by integrals of certain functions against the test-functions. For $p \geq 1$, let $f \in L^p_{loc}(\Omega)$, i.e., the space of Lesbegue measurable functions such that $\int_K |f|^p \mathrm{d}x$ is finite for every compact $K \subset \Omega$. We define Λ_f by the formula

$$\Lambda_f(\varphi) = \int_\Omega f\varphi \mathrm{d}x, \quad \forall \varphi \in \mathcal{D}(\Omega).$$

Assuming supp $\varphi \subset K$, and recalling that $L^p_{loc}(\Omega) \subset L^1_{loc}(\Omega)$ for $p \geq 1$, we have that

$$\Lambda_f(\varphi) \leq \int_K |f\varphi| \mathrm{d}x \leq \int_K |f| \mathrm{d}x \|\varphi\|_{C(K)} = C\|\varphi\|_{C(K)},$$

and so Λ_f is a distribution on $\mathcal{D}(\Omega)$. We say that the distribution Λ_f is identified with the function f.

For a distribution Λ and a multi-index α, derivatives of are defined by

$$\partial^\alpha \Lambda(\varphi) = (-1)^{|\alpha|} \Lambda(\partial^\alpha \varphi). \tag{2.1}$$

Since $\partial^\alpha \varphi \in \mathcal{D}(\Omega)$, it follows that any derivative of a distribution is also a distribution. Again, we are interested in distributions for which the derivative can be identified with a function $f \in L^p_{loc}(\Omega)$. This leads to the concept of weak derivatives: Let Λ_f be the distribution identified with the function $f \in L^p_{loc}(\Omega)$, and let α be some multi-index. If there is a $g \in L^p_{loc}(\Omega)$ such that

$$\partial^\alpha \Lambda_f(\varphi) = (-1)^{|\alpha|} \int_\Omega f \partial^\alpha \varphi \mathrm{d}x = \int_\Omega g\varphi \mathrm{d}x, \quad \forall \varphi \in \mathcal{D}(\Omega),$$

then g is called the α'th weak derivative of f, and we make the identification $\partial^\alpha \Lambda_f = \Lambda_g$. Note that a function f has continuous derivatives of order $|\alpha|$, the formula

$$(-1)^{|\alpha|} \int_\Omega f \partial^\alpha \varphi \mathrm{d}x = \int_\Omega \partial^\alpha f \varphi \mathrm{d}x$$

is the result of integration by parts. But the integral on the left-hand side makes sense even if f is less regular than that, i.e., if $\partial^\alpha f$ does not exist in the classical sense. Hence the concept of weak derivatives extends the concept of derivatives.

Definition 2.1 Sobolev spaces: Let Ω be an open domain in $\mathbb{R}^n$, and $m \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}$. Then

$$H^m(\Omega) = \{u \in \mathcal{D}(\Omega)' : D^\alpha u \in L^2(\Omega), \forall |\alpha| \leq m\},$$

i.e., $H^m(\Omega)$ is the space of all $u \in L^2_{loc}(\Omega)$ such that their weak derivatives $D^\alpha u, |\alpha| \leq m$ belong to $L^2(\Omega)$.

When $H^m(\Omega)$ is equipped with its usual inner product and norm

$$(u,v)_{H^m(\Omega)} = \sum_{|\alpha|\leq m} (D^\alpha u, D^\alpha v)_{L^2(\Omega)}, \quad \|u\|_{H^m(\Omega)} = (u,u)^{1/2}_{H^m(\Omega)},$$

it is Hilbert space, i.e., a Banach space with an inner product. Later we will also encounter spaces H^s where $s \in \mathbb{R}$. The space $H^m_0(\Omega)$ is the completion of $\mathcal{D}(\Omega)$ in $H^m(\Omega)$, and is also a Hilbert space. For $m = 1$ it consists of all $u \in H^1(\Omega)$ such that the trace of u is zero on $\partial\Omega$.

When working with the Fourier transform, we use the Schwartz space of rapidly decaying functions on $\mathbb{R}^n$, denoted $\mathcal{S}$. $\mathcal{S}$ is the space consisting of functions $\varphi \in C^\infty(\mathbb{R}^n)$ such that

$$|\varphi|_{\alpha,\beta} = \sup_{x\in\mathbb{R}^n} |x^\alpha D^\beta \varphi(x)| < \infty,$$

for all multi-indices α, β. $|\cdot|_{\alpha,\beta}$ is a system of semi-norms, and this system makes $\mathcal{S}$ a Fréchet space, but this will not be of direct importance. The space of distributions on $\mathcal{S}$, called the tempered distributions, is denoted by $\mathcal{S}'$, are the linear functionals on $\mathcal{S}$ that for some multi-indices α, β satisfy

$$|\Lambda(\phi)| \leq C|\varphi|_{\alpha,\beta}, \quad \forall \varphi \in \mathcal{S}.$$

For $p \geq 1$ one has that $\mathcal{S} \subset L^p(\mathbb{R}^n)$, and as for $\mathcal{D}'$, any function $f \in H^m(\mathbb{R}^n)$ gives rise to a distribution $\Lambda_f \in \mathcal{S}'$ through the integral

$$\Lambda_f(\varphi) = \int_{\mathbb{R}^n} f(x)\varphi \mathrm{d}x,$$

and we say that Λ_f is identified with f. Derivatives of tempered distributions is defined as in (2.1). Moreover, since $\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}$ we have that $\mathcal{S}' \subset \mathcal{D}'(\mathbb{R}^n)$. For $u \in \mathcal{D}'(\Omega)$, we say that $u = 0$ on the open set $\mathcal{O} \subset \Omega$ if $u(\varphi) = 0$ for all $\varphi \in C^\infty_0(\mathcal{O})$. We can then define the support of a distribution:

$$\text{For } u \in \mathcal{D}'(\Omega), \quad \operatorname{supp}(u) = \Omega \setminus \bigcup \{\mathcal{O} : \mathcal{O} \text{ is open in } \Omega, u = 0 \text{ on } \mathcal{O}\}.$$

We denote by $\mathcal{E}'(\Omega)$ the space of distributions with compact support in Ω, and note that $\mathcal{E}'(\Omega) \subset \mathcal{S}'$.

For two distributions u, v in $\mathcal{D}'(\Omega)$ (or any of the other spaces), we say that $u = v$ if

$$u(\varphi) = v(\varphi), \quad \forall \varphi \in \mathcal{D}(\Omega).$$

If u, v belong to some Sobolev space $H^m(\Omega)$ and $u = v$ as distributions, we have that

$$\int_\Omega (u - v)\varphi \mathrm{d}x = 0 \quad \forall \varphi \in \mathcal{D}(\Omega),$$

and so $u = v$, except possibly for a set of Lesbegue measure zero. In the following, we will almost exclusively work with distributions in Sobolev spaces. In this setting, we treat (and refer to) distributions as functions with the properties that comes from being a member of a certain space.

The Sobolev spaces are suitable spaces to look for solutions to PDEs. To see an example of this, we introduce the representation theorem and the Lax-Milgram theorem, and apply it to solve the introductory problem (1.1).

For a Hilbert space H, the space of bounded linear operators $l : H \to \mathbb{K}$, is called the dual space of H, denoted H^*, and its members are called linear functionals. $\mathbb{K}$ is called the field of scalars and will always be either $\mathbb{R}$ or $\mathbb{C}$. The norm of $l \in H^*$ is defined as

$$\|l\|_{H^*} = \sup_{\|x\|_H = 1} |l(x)|.$$

Any $f \in H$ generates a linear functional l_f through the inner product:

$$l_f(x) = (f, x)_H, \tag{2.2}$$

and by Cauchy-Schwarz inequality, we have $|l_f(x)| \leq \|f\|_H \|x\|_H$, and consequently $\|l_f\|_{H^*} \leq \|f\|_H$. In fact, every linear functional $l \in H^*$ can be written as l_f in (2.2). This is the content of the Riesz representation theorem:

Theorem 2.2 *Let H be a Hilbert space, and H^* its dual space. Then, for any linear functional $l \in H^*$ there exists a unique $f \in H$ such that*

$$l(x) = (f, x)_H, \quad \forall x \in H.$$

Moreover, $\|l\|_{H^*} = \|f\|_H$.

Due to the Riesz theorem, one often says that a Hilbert space is identified with its dual.

A closely related concept is that of continuous bilinear forms. A continuous bilinear form on a Hilbert space H is a function $B : H \times H \to \mathbb{R}$ that is linear in each argument, and continuous in the sense that

$$|B(u, v)| \leq C\|u\|_H \|v\|_H, \quad \forall u, v \in H.$$

B is called coercive if, in addition, $B(u, u) \geq b\|u\|_H^2, \forall u \in H$ for some constant $b > 0$. The next result is known as the Lax-Milgram theorem.

Theorem 2.3 *Let B be a positive definite, continuous bilinear form on a Hilbert space H, and let $l \in H^*$ be a linear functional on H. Then there exists a unique $u \in H$ such that*

$$B(u, v) = l(v), \quad \forall v \in H.$$

In addition, $\|u\|_H \leq b^{-1}\|l\|_{H^*}$.

Returning to (1.1), we apply theorem 2.3 to show that the equation is well-posed. We proceed by multiplying both sides of the equation by a function $v \in H_0^1(\Omega)$. Integrating both sides, we have (formally)

$$-\int_\Omega \Delta u v \mathrm{d}x = \int_\Omega \nabla u \cdot \nabla v \mathrm{d}x = \int_\Omega f v \mathrm{d}x. \tag{2.3}$$

The integral on the right-hand side will be our functional $l \in (H_0^1(\Omega)^* = H^{-1}(\Omega)$. Assuming that $f \in L^2(\Omega)$ and applying the Cauchy-Schwarz inequality we have that $l(v) \leq C\|f\|_{L^2}\|v\|_{H_0^1}$. It is uncomplicated to verify that

$$B(u, v) = \int_\Omega \nabla u \cdot \nabla v \mathrm{d}x$$

is indeed a continuous bilinear form on $H_0^1(\Omega)$. Due to the Poincaré inequality, we have that $\|u\|_{H_0^1} \leq C\|\nabla u\|_{L^2}$, and it follows that $\|u\|_{H_0^1} \leq C\|\nabla u\|_{L^2} = CB(u, u)$ and that B is positive definite. Hence, we can use the Lax-Milgram Theorem to conclude that there exists a unique $u \in H_0^1(\Omega)$ that solves (1.1) in the sense that

$$\int_\Omega \nabla u \cdot \nabla v \mathrm{d}x = \int_\Omega f v \mathrm{d}x, \quad \forall v \in H_0^1(\Omega).$$

In addition, the solution u is bounded in terms of f. We have that $\|u\|_{H_0^1} \leq C\|l\|_{H^{-1}}$, and from the estimate

$$\sup_{\|v\|_{H_0^1}=1} |l(v)| \leq \sup_{\|v\|_{H_0^1}=1} C\|f\|_{L^2}\|v\|_{H_0^1} = C\|f\|_{L^2},$$

we have that $\|u\|_{H_0^1} \leq C\|f\|_{L^2}$. We are now in position to determine if our introductory problem is well-posed. First off, the Lax-Milgram theorem guarantees existence and uniqueness of a solution, and due to the linearity of the PDE, the bound we just found implies that for two different $f_1, f_2 \in L^2(\Omega)$ with corresponding solutions u_1, u_2, we have that $\|u_1 - u_2\|_{H_0^1} \leq C\|f_1 - f_2\|_{L^2}$. This inequality shows that if the data f_1, f_2 is close (in L^2), then the corresponding solutions u_1, u_2 are close (in H_0^1). As a consequence, a small change of the data will only lead to a small change in the solution, and hence one says that the solution depends stably on the data. In fact, such dependence is what is required for a solution to "stably depend on the data" in the sense of Hadamard's well-posedness criteria. Hence we conclude that problem (1.1) is well-posed.

Returning to our introductory considerations about solution operators, it thus makes sense to write the solution u to (1.1) as

$$u = K(f).$$

In fact, K is a linear operator, since the solution u depends linearly on f, i.e., $K(f_1 + f_2) = Kf_1 + Kf_2 = u_1 + u_2$. In addition, it is a bounded operator between $L^2(\Omega)$ and $H_0^1(\Omega)$, since $\|Kf\|_{H_0^1} = \|u\|_{H_0^1} \leq C\|f\|_{L^2}$.

Though elementary, the above analysis of the Poisson equation allows us to determine interesting properties of the solution operator. A major quality of functional analysis is that similar analysis can be applied to a large classes of PDEs. For example, the Poisson equation belongs to the class of second-order elliptic PDEs, i.e., PDEs for which $L(p)$ is of the form

$$L(p)u = \sum_{i,j=1}^{n} (a^{ij}(x)u_{x_i})_{x_j} + \sum_{i=1}^{n} b^i(x)u_{x_i} + c(x)u,$$

and the coefficients a^{ij} satisfies

$$\sum_{i,j=1}^{n} a^{ij}(x)\xi_i\xi_j \geq C|\xi|^2, \qquad \text{for all } x \in \Omega, \xi \in \mathbb{R}^n \text{ and for some } C > 0$$

It turns out that for second-order elliptic PDE $L(p)u = f$, a similar treatment as for the Poisson equation can be carried out to show existence and regularity properties of the solution.

Knowing that the solution operator K is a bounded linear operator acting between certain functions spaces, we now introduce some properties of linear operators before we explore the concept of compactness and compact operators.

Linear operators

In the following, X and Y will be Banach spaces. Recall that an operator $T : X \to Y$ is linear if $T(ax_1 + bx_2) = \alpha Tx_1 + \beta Tx_2$ for all $x_1, x_2 \in X$ and scalars $\alpha, \beta \in \mathbb{K}$. For a linear operator $T : X \to Y$ we define the (operator) norm

$$\|T\| = \sup_{\|x\| \leq 1} \|Tx\|_Y,$$

The T is said to be a bounded operator if $\|T\| < \infty$. The space of all linear(but not necessarily bounded) operators from X to Y is denoted by $\mathcal{L}(X, Y)$. We now define continuity for a linear operator; this is the type of continuity that that will be required when we consider stable dependence on data for inverse problems.

Definition 2.4 An operator $T \in \mathcal{L}(X, Y)$ is continuous if for any $\varepsilon > 0$ we can find $\delta > 0$ such that if

$$\|x_1 - x_2\|_X < \delta \quad \text{then} \quad \|Tx_1 - Tx_2\|_Y < \varepsilon, \quad \forall x_1, x_2 \in X.$$

An important property of linear operators is the equivalence between continuity and boundedness, as stated in the next theorem.

Theorem 2.5 *A linear operator $T \in \mathcal{L}(X, Y)$ is bounded if and only if it is continuous.*

For $T \in \mathcal{L}(X, Y)$, the kernel (or null-space) of T is the set

$$\text{Ker } T = \{x \in X : Tx = 0\}.$$

The image (or range) of T is the set

$$\text{Im } T = \{y \in Y : \exists x \in X, y = Tx\}.$$

A linear operator in $\mathcal{L}(X, Y)$ is said to be *injective* if $T(u) = T(v)$ implies $u = v, \forall u, v \in X$. Equivalently, T is injective when Ker $T = \{0\}$. If Im $T = Y$ we say that T is *surjective* and if T is both injective and surjective, it is said to be *bijective*. We will see that lack of injectivity and surjectivity of the measurement operator are two recurring problems that make inverse problems hard to solve.

Denote by I_X and I_Y the identity operators on X and Y, i.e., $I_X x = x$ and $I_Y y = y$ for all $x \in X$ and $y \in Y$, respectively. We define the inverse of a bounded linear operator.

Definition 2.6 Let $T \in \mathcal{L}(X, Y)$ be bounded. T is said to be invertible if there exists a linear operator $A : Y \to X$ such that $AT = I_X$ and $TA = I_Y$. The operator A is called the inverse of T and denoted by T^{-1}.

If T is injective, we have $T(T^{-1}(ay_1 + by_2)) = ax_1 + bx_2 = T(aT^{-1}y_1 + bT^{-1}y_2)$, and it follows that $T^{-1}(ay_1 + by_2) = aT^{-1}y_1 + bT^{-1}y_2$, so the inverse of an injective, bounded linear operator is again a linear operator.

The next theorem gives conditions on T that guarantee the boundedness (and hence continuity) of the inverse T^{-1}. It requires that T is bijective.

Theorem 2.7 *Let $T \in \mathcal{L}(X, Y)$ be bounded is bijective. Then T has a linear bounded inverse $T^{-1} \in \mathcal{L}(Y, X)$.*

Let us return again to our solution operator K for the Poisson equation (1.1). We now identify K as a bounded linear operator from $L^2(\Omega)$ to $H_0^1(\Omega)$, and ask if we can apply theorem 2.7 to conclude that K has a continuous inverse. From our analysis of the well-posedness of (1.1), we know that K is injective. Whether K is surjective is a more delicate question, and the answer is negative. Using regularity theory, i.e., the detailed study of the regularity of solutions to PDEs, one can show that when the source term f is in $L^2(\Omega)$, the corresponding solution u is in the space $H^2(\Omega) \cap H_0^1(\Omega)$, i.e., of higher regularity than shown by the initial investigation. Since $H^2(\Omega) \cap H_0^1(\Omega) \neq H_0^1(\Omega)$, this shows that $K : L^2(\Omega) \to H_0^1(\Omega)$ is not surjective, and so we cannot use theorem 2.7 to conclude that it has a bounded inverse. The natural question then becomes: why not define K as a mapping from $L^2(\Omega)$ to $H^2(\Omega) \cap H_0^1(\Omega)$ and consider the problem solved? The answer to this question is that it is not a good idea to define oneself away from a problem, and the reason for this will become clear in a few pages.

2.2 Compactness

A basic question one can ask about a sequence of vectors $\{x_n\}_{n\in\mathbb{N}}$ in a set (or space) $U \subset X$ is whether the sequence converges to some element x that also belongs to U (or X). It turns out that asking for the whole sequence to converge is often too restrictive, and that a more appropriate question is if the sequence has a convergent subsequence with a limit $\tilde{x}$ that is in U(or X). More precisely, can we find some $\tilde{x} \in U$ and a subsequence $\{x_{n_k}\}_{k\in\mathbb{N}} \subset \{x_n\}_{n\in\mathbb{N}}$ such that for any $\varepsilon > 0$ there is a $N \in \mathbb{N}$ such that $\|x_{n_k} - \tilde{x}\|_X < \varepsilon$ for all $k \geq N$?

In $\mathbb{R}^n$, this question is settled by the Bolzano-Weierstrass theorem, which states that if U (the set containing the elements x_n of the sequence) is closed and bounded, then $\{x_n\}_{n\in\mathbb{N}}$ has a subsequence $\{x_{n_k}\}_{k\in\mathbb{N}}$ converging to some element $\tilde{x} \in U$.

In the case that X is an infinite dimensional space the Bolzano-Weierstrass theorem does not hold, as illustrated by the following example: Let X be a Hilbert space with an orthonormal basis $\{e_n\}_{n\in\mathbb{N}}$, and let B_1 be the closed unit ball in X, i.e., $B_1 = \{x \in X : \|x\|_X \leq 1\}$. Then B_1 is closed and bounded in X, and $\{e_n\}_{n\in\mathbb{N}} \subset B_1$. However, for any $i \neq j$, we have that $\|e_i - e_j\|_X^2 = 2$, and so there cannot be a convergent subsequence. Hence, for a set $U \subset X$ in an infinite dimensional space, being closed and bounded is not enough to contain a convergent subsequence. The above example also gives some insights into why this happens: Since the sequence have infinitely many "directions" (represented by the vectors in the orthonormal basis) for the sequence to wander off, it can

always take some new direction and "avoid" converging in the classical sense. In $\mathbb{R}^n$, this is not an option.

We now define compactness.

Definition 2.8 A subset U of a metric space X is compact if every sequence in U has a convergent subsequence. If the set U has a compact closure in X it is said to be relatively compact.

We see that compact sets are sets that have the feature of the closed and bounded sets in $\mathbb{R}^n$ with respect to convergence of subsequences. One reason for the interest in compact sets is that they allow for the generalization of important results about functions. A good example of this is the extreme value theorem: If $f : X \to \mathbb{R}$ is a continuous function, then for any compact subset $U \subset X$, there are $x_{min}, x_{max} \in U$ such that $f(x_{min}) \leq f(x) \leq f(x_{max}), \forall x \in U$. However, we will be mainly interested in using compactness to characterize solution operators to PDE. To this end, we also need the notion of compact embedding: This concerns the convergence of subsequences in subspaces.

Definition 2.9 A subspace $Y \subset X$ is said to be compactly embedded in X if

- Boundedness: $\|u\|_X \leq C\|u\|_Y$ for $u \in Y$ and some constant C.
- Any bounded sequence in Y has a subsequence that converges to some element in X, i.e., for $\{y_n\}_{n\in\mathbb{N}} \in Y$ with $\sup_n \|y_n\|_Y < \infty$, there is some $y \in X$ and a subsequence $\{y_{n_k}\}_{k\in\mathbb{N}}$ such that

$$\lim_{k\to\infty} \|y_{n_k} - y\|_X = 0.$$

If the above criteria hold, we write $Y \subset\subset X$.

Equivalently, one can say that $Y \subset\subset$ if the unit ball $B_1 \subset Y$ is relatively compact in X.

Maintaining the analogy of the infinite number of directions, we can think of a space Y being compactly embedded in another space X if the bounded sets in Y are not "too infinite" when viewed as subsets of X.

Bounded, finite-dimensional subspaces are examples of relatively compact subsets. This follows from the fact that any finite-dimensional space is isomorphic to $\mathbb{R}^n$, and an application of the Bolzano-Weierstrass theorem. The Arzelà-Ascoli theorem gives conditions for subsets of the continuous functions to be

compact. It asserts that for $K \subset \mathbb{R}^n$ compact, a subset $U \subset C(K, \mathbb{R}^m)$ is compact if and only if it is closed, bounded and equicontinuous. We get another, more illuminating example of compactness by considering the sequence space

$$\ell^2 = \{x = (x_1, x_2, ...) : \|x\|_2^2 = \sum_{n=1}^{\infty} |x_i|^2 < \infty\}.$$

Our candidate for a compact subset will be the set

$$U = \{x \in \ell^2 : \|x\|_2 \leq C, |x_n| \to 0 \text{ as } n \to \infty\}.$$

Let $\{x^k\}_{k\in\mathbb{N}}$ be a sequence in U. We wish to show that we can find a convergent subsequence. For $m \in \mathbb{N}$, let $x^{k,m}$ be a truncated version of x^k, i.e., $x^{k,m} = (x_1^k, x_2^k, ..., x_m^k, 0, 0, ...)$. Then, for fixed m, U is equivalent to a closed and bounded subset of $\mathbb{R}^m$, and it follows from the Bolzano-Weierstrass theorem that we can find a convergent subsequence in the sequence of truncated sequences $\{x^{k,m}\}$. We now use a diagonalization technique to extract a subsequence that is convergent for any $m \in \mathbb{N}$: let $\{x^{k_j,1}\}$ be the convergent subsequence for $m = 1$, and $\{x^{k_j^1}\}$ the corresponding subsequence of $\{x^k\}$. For $m = 2$, let $\{x^{k,2}\}$ be a convergent subsequence extracted from $\{x^{k_j^1}\}$, and $\{x^{k_j^2}\}$ the corresponding subsequence of $\{x^{k_j^1}\}$. Iterating this argument in m, we denote $\{x^{k_j}\}$ the subsequence of $\{x^k\}$ such that $\{x^{k_j,m}\}$ is convergent for any $m \in \mathbb{N}$. Next, we note that since $\|x^{k,m} - x^k\|_2 \to 0$ as $m \to \infty$, there is some n such that $\|x^{k,n} - x^k\|_2 < \varepsilon/3$ for all k, since $\|x^k\|_2 \leq C$. For such n, we can find N such that $\|x^{k_j,n} - x^{k_i,n}\|_2 < \varepsilon/3$ for all $i, j > N$. Hence, for all $i, j > N$,

$$\|x^{k_j} - x^{k_i}\|_2 \leq \|x^{k_j} - x^{k_j,n}\|_2 + \|x^{k_j,n} - x^{k_i,n}\|_2 + \|x^{k_i,n} - x^{k_i}\|_2 \leq \varepsilon,$$

and since ℓ^2 is complete and $\varepsilon > 0$ was arbitrary, we get that $\{x^{k_j}\}$ is a convergent subsequence of $\{x^k\}$, and thus that U is compact in ℓ^2. What we see here is that the notion of decay at infinity is essential. Comparing with the example of the orthonormal basis, it is clear that the we do not face this problem with elements in U since its members have a decay at infinity.

Of special importance to us is compactness in Sobolev spaces. By looking at the case of a periodic domain, we make the connection between regularity and decay of Fourier coefficients, and the same "decay at infinity" argument as above allows us to show compact embedding of Sobolev spaces.

Let $\Omega = (0, 2\pi)$. For f in $L^2(\Omega)$, we have

$$f(x) = \sum_{n\in\mathbb{Z}} \hat{f}_n e^{inx}, \quad \text{where } \hat{f}_n = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{inx} \mathrm{d}x.$$

By Definition 2.1,

$$H^m(\Omega) = \{u \in \mathcal{D}'(\Omega) : u^{(n)} \in L^2(\Omega), 0 \leq n \leq m\}.$$

If $f \in H^1(\Omega)$, we have

$$\frac{1}{2\pi}\int_\Omega f^{(1)}(x)e^{inx}\mathrm{d}x = (-in)\hat{f}_n.$$

Parseval's identity says that $\|f\|^2_{L^2(\Omega)} = 2\pi\sum_{n\in\mathbb{Z}}|f_n|^2$, and so we have that

$$\|f\|^2_{H^1(\Omega)} = \|f\|^2_{L^2(\Omega)} + \|f^{(1)}\|^2_{L^2(\Omega)} = 2\pi\sum_{n=-\infty}^{\infty}(1+n^2)|f_n|^2$$

Extending this observation, one can show that an equivalent definition of $H^m(\Omega)$ is

$$H^m(\Omega) = \{f \in L^2(\Omega) : \|f\|^2_{H^m(\Omega)} = \sum_{n\in\mathbb{Z}}(1+n^2)^m|\hat{f}_n|^2 < \infty\}. \tag{2.4}$$

In fact, using this formulation one can extend the range of Sobolev exponents m from $\mathbb{N}_0$ to $\mathbb{R}$. We see that the requirement for f to be in $H^m(\Omega)$ becomes a requirement on the decay of its Fourier coefficients: if $\{f_n\}$ are the Fourier coefficients of any $f \in L^2(\Omega)$, then a function $\tilde{f}$ having the coefficients $\{f_n(1+n^2)^{-m}\}$ will belong to $H^m(\Omega)$.

We can now show that $H^1(\Omega)$ is compactly embedded in $L^2(\Omega)$. Due to Parseval's identity, we know that $L^2(\Omega)$ and ℓ^2 are isomorphic. We denote by U the space of Fourier coefficients of a bounded set of functions $U_{H^1} = \{u \in H^1 : \|u\|_{H^1(\Omega)} \leq C\}$. From (2.4) we infer that for any $f \in U_{H^1}$ $\hat{f}_n \to 0$ as $|n| \to \infty$. We now take a sequence $\{f^k\}_{k\in\mathbb{Z}} \subset U_{H^1}$, and let $\{\hat{f}^k\}_{k\in\mathbb{Z}} \subset U$ be the corresponding sequence of Fourier coefficients. By the same diagonalization argument we used in the ℓ^2-example, it follows that $\{\hat{f}_k\}$ has a subsequence converging to some $\hat{f} \in \ell^2$, and by the isomorphism, $\{f^k\}_{k\in\mathbb{Z}}$ has a subsequence converging to f, the unique function identified by $\hat{f}$. Since the sequence was arbitrary, we can conclude that $H^1(\Omega)$ is compactly embedded in $L^2(\Omega)$.

The connection between regularity, decay of the Fourier transform and compactness holds in great generality. Roughly speaking, the relationship between the three seemingly different concepts goes like this: Functions in higher order Sobolev spaces, i.e., functions of more regularity, have faster decaying Fourier transforms. And spaces of functions with faster decaying Fourier transforms are compactly embedded in spaces of functions with slower decaying Fourier transforms, i.e., spaces with less regular functions.

The decay characterization of Sobolev spaces on $(0, 2\pi)$ extends to $\mathbb{R}^n$, and the norm is given by $\|u\|^2_{H^s(\mathbb{R}^n)} = \|(1+|\xi|^2)^{s/2}\hat{u}\|_{L^2(\mathbb{R}^n)}$, for any $s \in \mathbb{R}$. In fact, for smooth, bounded domains $\Omega \subset \mathbb{R}^n$, we can define $H^s(\Omega)$ as functions in $L^2(\Omega)$ such that the norm

$$\|u\|_{H^s(\Omega)} = \inf\left\{\|(1+|\xi|^2)^{s/2}\hat{v}\|_{L^2(\mathbb{R}^n)} : v \in H^s(\mathbb{R}^n), v\big|_\Omega = u\right\}$$

is finite, and for $s \in \mathbb{N}_0$ this definition is equivalent to Definition 2.1.

We now present two theorems that demonstrate the connections sketched out above.

Theorem 2.10 *Sobolev embedding: Let $\Omega \subset \mathbb{R}^n$ be open. For $k \geq 0$ and $s > k + n/2$, $H^s(\Omega)$ is embedded in $C^k(\Omega)$, i.e., for any $u \in H^s(\Omega)$ we have that*

$$\|u\|_{C^k(\Omega)} \leq C\|u\|_{H^s(\Omega)}$$

for some constant $C > 0$.

This theorem relates Sobolev regularity with classical continuity and differentiability, and it implies that if u is in H^s with s large enough, we can find a $\tilde{u} \in C^k_0$ such that $u = \tilde{u}$ almost everywhere.

Interpreting the next result in light of the decay Fourier transform, we see that a slight increase of decay of the Fourier transform, and hence of smoothness, is enough to ensure compactness:

Theorem 2.11 *Compact embedding: Let $\Omega \subset \mathbb{R}^n$ be a smooth and bounded domain. For any $s \in \mathbb{R}$ and $\varepsilon > 0$, $H^{s+\varepsilon}(\Omega)$ is compactly embedded in $H^s(\Omega)$.*

We have now seen some important aspects of compactness. As we go on to introduce compact operators, i.e., operators that map bounded sets into (relatively) compact sets, these aspects will serve as a background to understand their properties.

2.3 Compact Operators

It turns out that compactness is essential in explaining why solving inverse problems is a complicated affair. We will see that many of the operators that we are interested in inverting are *compact operators*, and that such operators do not have a bounded inverse.

Definition 2.12 An operator $T \in \mathcal{L}(X, Y)$ is called compact if for each bounded set $U \subset X$, the set $T(U)$ is relatively compact in Y.

A simple example of a compact operator is given be what we might call the "sampling operator": Let $\Omega \subset \mathbb{R}^n$ be a compact set, and let $X_s = \{x_i\}_{i=1}^N$ be a finite set of points in Ω. For functions $f \in C(\Omega)$, define the sampling operator S as the values of the function f at the points X_s,

$$Sf = (f(x_1), f(x_2), ..., f(x_N)).$$

Since S has finite dimensional range, the closure of image $S(U)$ of any bounded set $U \in C(\Omega)$ is a closed and bounded set in $\mathbb{R}^N$, and by the Bolzano-Weierstrass theorem it is compact and so $S \in \mathcal{L}(C(\Omega), \mathbb{R}^N)$ is a compact operator.

We give another example: For the continuous function

$$K(x, y) = \begin{cases} \frac{1}{\pi}(\pi - x)y, & 0 \leq y \leq x \leq \pi, \\ \frac{1}{\pi}(\pi - y)x, & 0 \leq x \leq y \leq \pi, \end{cases}$$

define the operator $A : L^2([0, \pi]) \to L^2([0, \pi])$ by

$$Af(x) = \int_0^\pi K(x, y)f(y)\mathrm{d}y, \quad x \in [0, \pi],$$

A is a solution operator to the 1-D Laplace equation, i.e., $u(x) = Af$ solves

$$u''(x) = f(x), \quad u(0) = u(\pi) = 0.$$

We have shown that $u \in H_0^1(0, \pi)$, and we know from Theorem 2.11 that $H_0^1(0, \pi)$ is compactly embedded in $L^2(0, \pi)$, and hence for every bounded set $U \subset L^2(0, \pi)$ the closure of the set $A(U)$ has a convergent subsequence.

We now present some important properties of compact operators.

Theorem 2.13 *Let X, Y, Z be Banach spaces.*

1. *If $K \in \mathcal{L}(X, Y)$ is compact it is also bounded.*
2. *If $T \in \mathcal{L}(X, Y)$ and $K \in \mathcal{L}(Y, Z)$ and K, T are bounded, then KT is a compact operator if either T or K is compact.*
3. *Let $K_n \in \mathcal{L}(X, Y)$ be a sequence of compact operators converging to K in the operator norm. Then K is compact.*
4. *If $\dim(Im\ K)$ is finite, $K \in \mathcal{L}(X, Y)$ is compact.*

5. *If $K \in \mathcal{L}(X, Y)$ is a compact operator and $\dim X = \infty$, then K does not have a bounded inverse.*

We sketch the proof of assertion 5: Assume, to the contrary, that the compact operator $K \in \mathcal{L}(X, Y)$ has a bounded inverse K^{-1}. Then $K^{-1} \in \mathcal{L}(Y, X)$, and by assertion 1 and 2, $K^{-1}K$ is a compact operator. But $K^{-1}K = I$, and I cannot be compact, since that would mean that the unit ball in X is compact.

Intermission on topology in metric spaces

Given a vector space X, we are free to specify a metric on this space (although the choice might exclude a lot of members). It is common to say that the metric induces a topology on the space. The topology, amongst other things, specifies what it takes for vectors to be close to each other, i.e., convergence and continuity. For example, if $X = L^2(\Omega)$, the metric is given by

$$d_{L^2}(f, g) = \left(\int_\Omega |f(x) - g(x)|^2 \mathrm{d}x \right)^{1/2},$$

but if $X = C^k(\Omega)$, then

$$d_{C^k}(f, g) = \sum_{n \leq k} \sup_{x \in \Omega} |f^{(n)}(x) - g^{(n)}(x)|.$$

Now, given f, g that are members of both $C^k(\Omega)$ and $L^2(\Omega)$, we want to check if the two functions are close, e.g., if $d_{L^2}(f, g) < \varepsilon$ and $d_{C^k}(f, g) < \varepsilon$. It is clear that the requirements for being close are more comprehensive in terms of in the topology of $C^k(\Omega)$ compared to that of $L^2(\Omega)$: Closeness in $C^k(\Omega)$ requires $|f - g|$ and derivatives $|f^{(n)} - g^{(n)}|$ for n up to order k to be small in the whole domain. In contrast, $L^2(\Omega)$ requires only that the Lebesgue integral of $|f - g|^2$ over Ω is small, and is oblivious to derivatives. The difference is made clear by considering the sequence $\{n^{-1} \sin(nx)\}_{n\mathbb{N}}$ on $\Omega = (0, \pi)$. A quick calculation shows that

$$\|n^{-1} \sin(nx)\|_{L^2(\Omega)} \leq C_1/n \quad \text{while } \|n^{-1} \sin(nx)\|_{C^k(\Omega)} \leq C_2 n^{k-1}, \tag{2.5}$$

for constants $C_1, C_2 > 0$, showing that the sequence converges to 0 in $L^2(\Omega)$, but diverges in $C^k(\Omega)$ for $k > 0$, even though the function values gets arbitrarily close to 0.

This leads us to the next observation, a more practical consideration: Whichever topology we choose for our spaces should reflect what we can measure. For example, if one measures the concentration of CO_2 in the atmosphere, it is unrealistic to think than one can measure the derivative of the concentration, yet alone the k'th derivative. Another equally important point is that real measurements will inevitably be contaminated by noise. If the true CO_2-concentration is given by some function $c(x,t)$, a more truthful model of a measurement $m(x,t)$ would be $m(x,t) = c(x,t) + \varepsilon(x,t)$, where $\varepsilon(x,t)$ is some unknown, hopefully small, function representing the noise. However, in most cases we have no idea about the regularity properties of the noise, so trying to compute the derivative will be unstable, i.e., the noise is amplified. This what happens in (2.5).

One can also illustrate this in terms of linear operators. Assume $A \in \mathcal{L}(X,Y)$ is some operator mapping the concentration $c \in X$ of CO_2 in the atmosphere to temperature $t \in Y$. Let's suppose the measurement space X is equipped with the $C^k(\Omega)$-topology, the temperature space is $Y = L^2(\Omega)$ and we have a noisy measurement $m = c(x,t) + \varepsilon(x,t)$. We would like to know how much the temperature t_m deviates from the actual temperature t. We then have the estimate

$$\|t_m - t\|_{L^2(\Omega)} = \|Ac - Am\|_{L^2(\Omega)} = \|A\varepsilon\|_{L^2(\Omega)} \leq C\|\varepsilon\|_{C^k(\Omega)}.$$

This is not useful, since taking the C^k-norm of ε would make the estimate useless; from (2.5) we observed that even the smallest oscillations, typical of random nature of noise, results in large values of the C^k-norm. The conclusion is that we cannot really say much about the temperature from our measurements. In contrast, if the role of the spaces X and Y were interchanged, we would have

$$\|t_m - t\|_{C^k(\Omega)} \leq C(\|\varepsilon\|_{L^2(\Omega)}).$$

In this case, if $|\varepsilon|$ is small throughout Ω it would imply that the estimated temperature t_m would not differ much from the true temperature t. This latter scenario is clearly preferable in applications. For that reason (and also due to its Hilbert space structure) it is common to choose L^2 as the topology on the measurement space. Other popular choices are H^s-spaces with moderate values of s, L^p spaces, or the maximum norm $\|u\|_C$.

In light of the topological considerations, let us return to our example of the inverse problem for the Poisson equation (1.1): Given the normal derivative $g = \frac{\partial u}{\partial \nu}|_{\partial\Omega}$, find f such that $\Delta u = f$, $u|_{\partial\Omega} = 0$.

Recall that we found that for $f \in L^2(\Omega)$, we have $u \in H^2(\Omega) \cap H_0^1(\Omega)$. In terms of the solution operator K, we have that $K \in \mathcal{L}(L^2(\Omega), H^2(\Omega) \cap H_0^1(\Omega))$. Now

we introduce the Neumann trace operator γ_1: This is a linear operator taking a function defined on the interior of Ω to its normal derivative on $\partial\Omega$. One can show that $\gamma_1 : H^2(\Omega) \to H^{1/2}(\partial\Omega)$, i.e., $\frac{\partial u}{\partial \nu} \in H^{1/2}(\Omega)$. By applying γ_1 to K we now obtain our measurement operator M from (1.2): $Mf = \gamma_1 K f$. It follows from our analysis that $M \in \mathcal{L}(L^2(\Omega), H^{1/2}(\Omega))$.

Considering a practical measurement setting, we now define the measurement operator as a map from $L^2(\Omega) \to L^2(\partial\Omega)$. Since $H^{1/2}(\partial\Omega)$ is compactly embedded in $L^2(\partial\Omega)$, this implies that our measurement operator $M = \gamma_1 K \in \mathcal{L}(L^2(\Omega), L^2(\partial\Omega))$ is a compact operator.

And now for the bad news: Assertion 5 of Theorem 2.13 implies that M *does not have a bounded inverse*. Recalling Theorem 2.5, this implies that the inverse M^{-1} cannot be continuous, i.e., we cannot make a stability estimate of the form

$$\|f_1 - f_2\|_{L^2(\Omega)} = \|M^{-1}g_1 - M^{-1}g_2\|_{L^2(\Omega)} \leq C\|g_1 - g_2\|_{L^2(\partial\Omega)}.$$

In addition, Ker M = Ker $\gamma_1 = H_0^2(\Omega)$, i.e., the trace operator associates a kernel to the measurement operator. In conclusion, our inverse problem does not depend stably on the data, lacks uniqueness and, according to Hadamard, is ill-posed.

To emphasize that this ill-posedness is not something we can define our way out of, we consider what happens if we assume that $K \in \mathcal{L}(L^2(\Omega), H^{1/2}(\partial\Omega))$. For some heat source f, let $m = u + \varepsilon$ be the measurement, where $\varepsilon \in L^2(\partial\Omega)$ is the noise and $u = Kf \in H^{1/2}(\partial\Omega)$ is the true solution. We consider the error of the reconstruction f_ε and get that

$$\|f - f_\varepsilon\|_{L^2(\Omega)} = \|K^{-1}u - K^{-1}m\|_{L^2(\Omega)} \leq \|K^{-1}\|\|\varepsilon\|_{H^{1/2}(\partial\Omega)}$$

and since $\|\varepsilon\|_{H^{1/2}(\partial\Omega)}$ might be arbitrarily large, we have not gained anything.

This specific result is, unfortunately, not the consequence of some pathological example that we have cooked up. Quite the opposite, it is an intrinsic feature of many inverse problems, and the characteristic that makes them difficult to solve. In many cases, solutions to PDEs have more regularity and belong to spaces that are compactly embedded in the natural space for measurements. Gaining some understanding of why and when this is so, or why and when it is not, is the goal of the next chapter.

2.4 Non-linearity and Linearization

So far we have considered only linear operators, and when these operators are solution operators to PDEs, we are restricted to consider linear PDEs that depend linearly on the unknown quantity. However, there are many non-linear inverse problems, i.e., problems where the measured or observable quantity depend non-linearly on the unknown quantity. For such problems, the analysis becomes more complicated and, as is the case with the analysis of non-linear PDEs, there is not a general theory. Rather, each problem must be tackled with special techniques and ingenuity. Nevertheless, in many cases one can seek an approximate solution by the method of linearization. Linearization is interesting both from the analytic and practical framework, and allows for analysis and solution methods using the standard methods.

An interesting example of a non-linear inverse problem that can be tackled by linearization is the so-called inverse medium problem in acoustics. In a medium with an unknown, inhomogeneous sound speed $c(x)$, one sends plane waves $u^i = \mathrm{e}^{ik\xi\cdot x}$ with wave number $k > 0$ and in the direction $\xi \in S^2$ into the medium. Here S^2 denotes the unit sphere in $\mathbb{R}^3$. The incoming wave is scattered by the inhomogeneous sound speed, and the total wave field satisfies

$$\begin{aligned} \Delta u + k^2 n(x) u &= 0, \\ u &= u^s + u^i, \\ \lim_{r\to\infty} r\left(\partial_r u^s - iku^s\right) &= 0. \end{aligned} \tag{2.6}$$

Here, u^s is the scattered field; u is the total field; $n(x) = (c_0/c(x))^2$ is the index of refraction; $k = \omega/c_0$ is the wave number and c_0 is the background wave speed. The Sommerfeld radiation condition ensures that the scattered wave u^s is outgoing and that the equation has a unique solution. Assuming $m(x) = 1-n(x)$ is piecewise continuous and has compact support in some sphere B_0, one can show that the above equation admits a unique solution $u \in H^1_{loc}(\mathbb{R}^3)$. The scattered field u^s has the asymptotic expansion

$$u^s(x) = \frac{\mathrm{e}^{ik|x|}}{|x|} U(\hat{x}, \xi) + \mathcal{O}(|x|^{-2}), \quad \hat{x} = x/|x|, \quad \text{as } |x| \to \infty, \tag{2.7}$$

known as the far field expansion. The function $U(\hat{x}, \xi)$ is the far field pattern, and is given by the formula

$$U(\hat{x}, \xi) = -\frac{k^2}{4\pi} \int_{\mathbb{R}^3} \mathrm{e}^{ik\hat{x}\cdot y} m(y) u(y) \mathrm{d}y. \tag{2.8}$$

Assuming we can measure the scattered field $u^s(x)$ on the boundary ∂B of some large sphere containing the support of m and from incoming plane waves

from all direction $\xi \in S^2$, we define the measurement operator $M : L^2(B_0) \to L^2(S^2 \times S^2)$ by $M : n \mapsto U(\hat{x}, \xi)$. The inverse medium scattering problem is the following:

Given the far field pattern $U(\hat{x}, \xi)$ for all $\hat{x}, \xi \in S^2$, reconstruct n.

Since u depends on n through (2.6), it follows from (2.8) that the measurement operator M is non-linear in n.

We now briefly demonstrate how linearization can be used to simplify and gain understanding of the inverse problem. To this end, we introduce the Fréchet derivative. An operator $K : X \to Y$, where X and Y are Banach spaces, is Fréchet differentiable at $x_0 \in U$, U open in X, if there exists a bounded linear operator $F'(x_0) \in \mathcal{L}(X, Y)$ such that

$$\lim_{\|h\|_X \to 0} \frac{\|F(x_0) - F(x_0 + h) + F'(x_0)h\|_Y}{\|h\|_X} = 0$$

Let v be the solution to

$$\Delta v_\xi + k^2 n(x) v_\xi = -k^2 q_\delta u_\xi, \tag{2.9}$$

where $q_\delta \in L^2(B_0)$ and u is the solution to (2.6) with $u^i = \mathrm{e}^{ik\xi\cdot x}$, and where v satisfies the Sommerfeld radiation condition. It can be shown that the Fréchet derivative $M'(n)$ of M at n is given by the far field expansion of v, i.e.,

$$M'(n)q_\delta(\hat{x}, \xi) = -\frac{k^2}{4\pi} \int_{\mathbb{R}^3} \mathrm{e}^{ik\hat{x}\cdot y}(m(y)v_\xi(y) + q_\delta(y)u_\xi(y))\mathrm{d}y, \tag{2.10}$$

and that $M'(n)$ is a injective, compact operator in $\mathcal{L}(L^2(B_0), L^2(S^2 \times S^2))$. Hence, if the unknown index of refraction is assumed to be some perturbation q_δ of a known $n(x)$, we have uniqueness of the linearized inverse problem. Moreover, equation (2.10) is linear in the unknown q_δ which means that the inverse problem of reconstructing q_δ can be treated by classical methods.

Notice how the Fréchet derivative depends (through v_ξ) on the solution to the PDE (2.9) that is linear in the unknown q_δ. This is typically the case when one consider linearized solution operators to PDEs, and it affirms the importance of understanding solution operators to linear PDEs. However, in general, one needs the Fréchet derivative to be bijective to guarantee local uniqueness of a non-linear inverse problem, and some problems are so non-linear that analysis and inversion based on linearized problem is not really useful.

References

The introductory material in Section 2 on distributions and Sobolev spaces can be found in the the monographs [21, 22, 27, 31, 59]. The material on linear operators is taken mainly from the monographs [21, 45, 46, 51]. The material in Sections 2.2-2.3 is a collection of results from the monographs [21, 28, 45, 46, 51] and some of the examples and results are inspired by the monographs on inverse problems [25, 40, 55]. The example in the Section 2.4 is taken from [25] and the discussion is motivated by [24, 25, 38, 55]. These monographs are good resources on the topic. For some interesting recent results on the stability of non-linear problems and their linearization, we also recommend the paper [62].

Chapter 3

Qualitative Analysis

In the previous chapter, we saw that that when the measurement is given by a compact operator, the inverse problem will be ill-posed. This is, however, a rather general observation, and in itself not very informative. In this chapter, we will explore the connections between the Fourier transformation and PDEs, and show how this can provide us with insight on the difficulties encountered in many inverse problems. Moreover, it will allow us to understand why some inverse problems are hopelessly ill-posed, while others are well-posed.

3.1 Wave Equation vs. Heat Equation

A simple and illuminating example of two inverse problems with truly different properties comes from the wave equation and the heat equation. Respectively, these equations are the canonical examples of hyperbolic and parabolic PDEs, and some of the simplest mathematical models to describe the behavior of heat and waves.

For some interval $I = (-L, L)$, let $f(x) \in C_0(I)$ be a source function. Denote

by $v(x,t)$ the solution to the wave equation,

$$\begin{aligned} &\left(\frac{\partial^2}{\partial t^2} - c\frac{\partial^2}{\partial x^2}\right) v(x,t) = 0, \quad (x,t) \in \mathbb{R} \times \mathbf{R}^+ \\ &v(x,0) = f(x), \quad v_t(x,0) = 0, \quad x \in \mathbb{R}, \end{aligned} \tag{3.1}$$

where $c > 0$ is the constant wave speed. Further, let $h(x,t)$ be the solution to the heat equation,

$$\begin{aligned} &\left(\frac{\partial}{\partial t} - k\frac{\partial^2}{\partial x^2}\right) h(x,t) = 0, \quad (x,t) \in \mathbb{R} \times \mathbb{R}^+ \\ &v(x,0) = f(x), \quad x \in \mathbb{R}, \end{aligned} \tag{3.2}$$

where $k > 0$ is the constant diffusion coefficient.

For some point $x_m > L$, i.e., outside the support of the source f, we measure v and h in during the time interval $\mathcal{T} = (0, T]$. We define the measurement operators

$$\begin{aligned} M_v f &= v(x_m, t)\big|_{t\in\mathcal{T}}, \\ M_h f &= h(x_m, t)\big|_{t\in\mathcal{T}}. \end{aligned}$$

We now pose the two inverse problems:

1. Given a measurement g_v, find $f \in C_0(I)$ such that $M_v f = g_v$.
2. Given a measurement g_h, find $f \in C_0(I)$ such that $M_h f = g_h$.

By D'Alembert's principle and the fundamental solution for the heat equation, respectively, the solutions to (3.1) and (3.2) are given by

$$v(x,t) = \frac{1}{2}(f(x-ct) + f(x+ct)) \quad \text{and} \quad h(x,t) = \frac{1}{\sqrt{4\pi kt}} \int_{\mathbb{R}} \mathrm{e}^{-\frac{(x-y)^2}{4kt}} f(y)\mathrm{d}y.$$

Hence the measurement operators take the form

$$M_v f(t) = \frac{1}{2} f(x_m - ct), \quad t \in \mathcal{T}, \tag{3.3}$$

$$M_h f(t) = \frac{1}{\sqrt{4\pi kt}} \int_{\mathbb{R}} \mathrm{e}^{-\frac{(x_m-y)^2}{4kt}} f(y)\mathrm{d}y, \quad t \in \mathcal{T}. \tag{3.4}$$

We assume the measurement time T is sufficiently long for all of the wave to propagate through the point x_m during the acquisition time. It is now easy

to see that $M_v : C_0(I) \to C_0(T)$. Further, the inverse is given by the formula $(M_v^{-1} g_v)(x) = 2g_v(\frac{x_m - x}{c})$, since

$$(M_v^{-1} M_v f)(x) = 2\frac{1}{2} f\left(x_m - c\frac{x_m - x}{c}\right) = f(x).$$

M_v^{-1} is clearly bounded, with $\|M_v^{-1}\| = 2$, and existence and uniqueness follows from the formula. Hence, we conclude that inverse problem 1 is well-posed.

For inverse problem 2 the situation is quite different. For $t > 0$, $u(x,t)$ is infinitely differentiable in both variables, as can be seen by direct calculation. Hence $M_h f : C_0(I) \mapsto C^\infty(\mathcal{T})$. Since $C^\infty(\mathcal{T}) \subset H^s(\mathcal{T}), \forall s \in \mathbb{R}$, we can show that $C^\infty(\mathcal{T})$ is compactly embedded in $C^k(\mathcal{T})$ for any $k \in \mathbb{N}_0$. The argument goes as follows: For $s > k + 1/2$, we know from theorem 2.10 that $H^s(\mathcal{T})$ is continuously embedded in $C^k(\mathcal{T})$, i.e., $\|u\|_{C^k(\mathcal{T})} \leq C\|u\|_{H^s(\mathcal{T})}$ for every $u \in H^s(\mathcal{T})$. Recalling Theorem 2.11 we have that $H^{s+1}(\mathcal{T}) \subset\subset H^s(\mathcal{T})$. Due to the continuous embedding, it follows that convergence in $H^s(\mathcal{T})$ implies convergence in $C^k(\mathcal{T})$. Since k and s were arbitrary, we can conclude that that $C^\infty(\mathcal{T}) \subset\subset C^k(\mathcal{T})$.

As a consequence, M_h is a compact operator from $C_0(I)$ to $C^k(\mathcal{T})$ *for any* k. Recalling our discussion on the topology of the measurement space, this is bad news; even if we had perfect knowledge of the 1729'th derivative, we could not hope to stably invert our measurement.

We can thus conclude that inverse problem 1 for the wave equation is well-posed, while inverse problem 2 for the heat equation is ill-posed. The question now is if we can understand why this is so. What are the characteristics of the PDEs, and hence with the underlying physics, that yields these differences?

3.2 Information

In a more abstract form, an inverse problem can be considered as process of determining some unknown information from some known and causally related information. If we call the unknown piece of information X, and the known information Y, the goal is to figure out as much as possible about X from Y. A natural question to ask when solving such problems is whether all the information about X is present in Y? Next, one could ask the more practical question whether one can extract that information, and what then happens if there are errors in the known information?

In the particular case of inverse problems for PDEs, the unknown and the known

information are functions, and their relation is given by PDEs. What is then the information contained in a function? This question is, in fact, very deep, and it is far beyond the scope and intent of this thesis to discuss this. Nonetheless, we make the following suggestion for which properties we want the information to have:

Given the information, one should be able to identify the corresponding function through a general procedure. We also want the information to reflect some of the important properties of the function in order able to gain some qualitative understanding of the function from its information. Further, it should be possible to extract the same type of information from a large class of functions.

As the reader might have already guessed, our choice for describing the information of a function is the Fourier transform or, in some cases, the Fourier series. For a function $f \in L^1(\mathbb{R}^n)$, the Fourier transform is defined as

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi} f(x)\mathrm{d}x. \tag{3.5}$$

We denote the operator taking $f \mapsto \hat{f}$ by $\mathcal{F}$. By the Riemann-Lebesgue theorem, $\mathcal{F} \in \mathcal{L}(L^1(\mathbb{R}^n), C_0(\mathbb{R}^n))$. By the Plancherel theorem $\mathcal{F}$ extends continuously to $\mathcal{L}(L^2(\mathbb{R}^n), L^2(\mathbb{R}^n))$, and we have the relation

$$(f, g)_{L^2(\mathbb{R}^n)} = (2\pi)^{-n}(\hat{f}, \hat{g})_{L^2(\mathbb{R}^n)}, \quad \forall f, g \in L^2(\mathbb{R}^n).$$

As a consequence, $(f, g - (2\pi)^{-n}\mathcal{F}^*\mathcal{F}g)_{L^2(\mathbb{R}^n)} = 0$, and hence $(2\pi)^n\mathcal{F}^*\mathcal{F} = I$, where I is the identity. It follows that $\mathcal{F}^{-1} = (2\pi)^{-n}\mathcal{F}^*$, and so $\mathcal{F} \in \mathcal{L}(L^2(\mathbb{R}^n), L^2(\mathbb{R}^n))$ is invertible and the inverse is given by

$$\left(\mathcal{F}^{-1}\hat{f}\right)(x) = (2\pi)^{-n} \int_{\mathbb{R}^n} \mathrm{e}^{ix\cdot\xi} \hat{f}(\xi)\mathrm{d}\xi.$$

In addition, the Fourier transform maps the space of rapidly decaying functions $\mathcal{S}$ to itself, i.e., $\mathcal{F} : \mathcal{S} \to \mathcal{S}$. By duality, one can thus extend $\mathcal{F}$ to a continuous, invertible mapping on the space of tempered distributions $\mathcal{S}'$, and

$$\langle \hat{u}, v\rangle = \langle u, \hat{v}\rangle, \quad u \in \mathcal{S}', v \in \mathcal{S}.$$

From the information perspective, we notice that Fourier transform immediately meets two of our criteria: If we say that the information of a function $f \in L^2(\mathbb{R}^n)$ is its Fourier transform $\hat{f} = \mathcal{F}f$, then $\hat{f}$ identifies f by $\mathcal{F}^{-1}$, and we can extract the same type of information from a large class of functions, e.g., for all f in $L^2(\mathbb{R}^n)$. What type of information does the Fourier transform hold? The Fourier transform of a function can be understood as a decomposition into plane waves

in the form $e^{ix\cdot\xi}$, with direction $\xi/|\xi| \in S^{n-1}$ and frequency $|\xi|$. We say that $\hat{f}(\xi)$ is in the frequency domain, and it is often referred to as the spectrum of $f(x)$. One can understand $\hat{f}(\xi)$ as the information needed to construct $f(x)$ from plane waves, and $\mathcal{F}^{-1}$ is the construction procedure.

Recalling Section 2.2, we saw that there is a connection between the decay of the Fourier transform and the regularity of the function. The requirement that $\|(1+|\xi|^2)^{s/2}\hat{u}\|_{L^2(\mathbb{R}^n)}$ is finite is met if

$$|\hat{u}(\xi)| \leq C|\xi|^{-((s+n)/2+\varepsilon)} \quad \text{for } \varepsilon > 0 \quad \text{as } |\xi| \to \infty,$$

and we see that the required decay increases with both regularity and the dimension. The Paley-Wiener theorem give a precise relation between the smoothness and support of a function or distribution and the behavior of its Fourier transform.

Theorem 3.1 *For $f \in C_0^\infty(\mathbb{R}^n)$, and $\chi(x)$ the characteristic function for supp f, the Fourier transform of f is an entire analytic function that is rapidly decaying, i.e.,*

$$|\hat{f}(\xi)| \leq C_N(1+|\xi|)^{-N}e^{\chi(im(\xi))}, \quad \textit{for any } N \in \mathbb{N}, \xi \in \mathbb{C}^n.$$

Vice versa, the inverse Fourier transform of an entire analytic function satisfying the above estimate belongs to $C_0^\infty(\mathbb{R}^n)$.

The decay-regularity-support relations will serve as our way of obtaining qualitative understanding of functions from their Fourier transform.

We can now begin to analyze how information is transmitted in PDEs. As a first example, assume that L is a constant coefficient partial differential operator of order m,

$$L(D) = \sum_{|\alpha|\leq m} c_\alpha D^\alpha,$$

where α is a multi-index, $c_\alpha \in \mathbb{C}$ and $D^\alpha = (-i)^{|\alpha|}\partial_{x_1}^{\alpha_1}\cdots\partial_{x_n}^{\alpha_n}$. For u with sufficient regularity, we have that $\widehat{D^\alpha u} = \xi^\alpha \hat{u}$. The polynomial $l(\xi)$, defined as

$$l(\xi) = \sum_{|\alpha|\leq m} c_\alpha \xi^\alpha,$$

is called the symbol of L. For $f \in L^2(\mathbb{R}^n)$, we consider the PDE

$$Lu = f.$$

Assuming that a solution u exists, we take the Fourier transform on both sides and get that

$$l(\xi)\hat{u} = \hat{f}. \tag{3.6}$$

Proceeding formally, ignoring possible zeros of $l(\xi)$, we now get

$$\hat{u}(\xi) = \frac{\hat{f}(\xi)}{l(\xi)}, \tag{3.7}$$

and have an equation relating the unknown information and the measured information, i.e., the Fourier transform of the source and the solution to the PDE. We see that $\hat{u}$ is given by a manipulation of $\hat{f}$ by the symbol $l(\xi)$. To appreciate why such a description is useful, we consider a concrete inverse problem. Let $L = -\Delta + 1$, and $f \in L^2(\mathbb{R}^n)$, and let u be the solution to

$$Lu = f. \tag{3.8}$$

Given u, we want to reconstruct f. Since $l(\xi) = |\xi|^2 + 1$, we have

$$\hat{u}(\xi) = \frac{\hat{f}(\xi)}{|\xi|^2 + 1}.$$

Notice how $\frac{1}{l(\xi)}$ acts a weight function, controlling for each ξ how $\hat{f}(\xi)$ is mapped to $\hat{u}(\xi)$. We see that it reduces the magnitude of the frequency information by a factor of $|\xi|^{-2}$. Recalling our initial questions about information, we can conclude that all the information in f is present in u, but that the high frequency information is increasingly suppressed by $l(\xi)^{-1}$. As a consequence, inversion is unstable: Assume we have a noisy measurement $u_\varepsilon = u + \varepsilon \in L^2(\mathbb{R}^n)$ of u, and we want to find an approximation f_ε to f. We have that

$$\begin{aligned} \|f - f_\varepsilon\|_{L^2(\mathbb{R}^n)} &= (2\pi)^{-n}\|\hat{f} - \hat{f}_\varepsilon\|_{L^2(\mathbb{R}^n)} \\ &= (2\pi)^{-n}\|\hat{f} - l(\xi)\hat{u}_\varepsilon\|_{L^2(\mathbb{R}^n)} \\ &= (2\pi)^{-n}\|l(\xi)\hat{\varepsilon}\|_{L^2(\mathbb{R}^n)}. \end{aligned} \tag{3.9}$$

What we see here is the manifestation of the unbounded operator L in the frequency domain. It is expressed by the magnification of the Fourier transform of the noise by the symbol $l(\xi)$. Since we cannot expect the noise to have a rapidly decaying Fourier transform, the error can be arbitrarily large, and our approximation might be arbitrarily bad.

We now introduce some notions from the theory of pseudo-differential operators and show that the same insights generalizes to a large class of operators known as elliptic operators.

3.3 Pseudo-differential Operators

For differential operators with spatially varying coefficient, attempts to write down an equation in the form (3.6) becomes quite intractable. For an operator $L(x, D)$ on the form

$$L(x, D) = \sum_{|\alpha| \leq m} c_\alpha(x) D^\alpha, \tag{3.10}$$

one writes instead

$$L(x, D)u = (2\pi)^{-n} \int_{\mathbb{R}^n} \mathrm{e}^{i\xi \cdot x} l(x, \xi) \hat{u}(\xi) \mathrm{d}\xi, \tag{3.11}$$

where

$$l(x, \xi) = \sum_{|\alpha| \leq m} c_\alpha(x) \xi^\alpha. \tag{3.12}$$

$l(x, \xi)$ is still called the symbol of L, and $L(x, D)$ is called a Pseudo-differential operator (ψDO). The theory of ψDOs is concerned with the analysis of mapping properties of such operators from knowledge of their symbols, and will allow us to generalize the insights about how differential operators modify the Fourier transform to a large class of problems. We now give a brief description of the main tools from the classical theory of pseudo-differential operators.

ψDOs are defined by their symbols; we thus start by introducing a suitable class of symbols.

Definition 3.2 For $m \in \mathbb{R}$, the space of symbols of order m on $\mathbb{R}^n$, denoted S^m, is the space of all $a \in C^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ such that the estimate

$$|\partial_\xi^\alpha \partial_x^\beta a(x, \xi)| \leq C_{\alpha,\beta}(1 + |\xi|)^{m-|\alpha|}$$

holds for all $(x, \xi) \in \mathbb{R}^n \times \mathbb{R}^n$ and all multi-indices $(\alpha, \beta) \in \mathbb{N}_0^n \times \mathbb{N}_0^n$.

For a symbol $a \in S^m$, the corresponding pseudo-differential operator $A(x, D)$ is defined as

$$A(x, D)u = (2\pi)^{-n} \int_{\mathbb{R}^n} \mathrm{e}^{i\xi \cdot x} a(x, \xi) \hat{u}(\xi) \mathrm{d}\xi, \quad \forall u \in \mathcal{S}. \tag{3.13}$$

We say that $A(x, D)$ is an operator of order m.

One can verify that S^m includes the symbols of linear partial differential operators of order $m' \leq m$ with smooth and bounded coefficients. An important property of symbols is that for $a \in S^m, b \in S^{m'}$, we have that $ab \in S^{m+m'}$.

Often, the highest order term in a symbol determines the properties of the corresponding pseudo-differential operator. One can show that for a sequence of symbols $\{a_{m_j}\}_{j\in\mathbb{N}_0}$ in which each $a_{m_j} \in S^{m_j}$, and $m_j \to -\infty$ as $j \to \infty$, there exist a symbol $a \in S^{m_0}$ such that for each N

$$a - \sum_{j=0}^{N} a_{m_j} \in S^{m_N}.$$

The sum $\sum_j a_{m_j}$ is called an asymptotic expansion for a, written

$$a \sim \sum_{j\in\mathbb{N}_0} a_{m_j}.$$

The highest order symbol a_{m_0} is called the principal symbol of a. For linear partial differential operators, the principal symbol is given by highest order term D^α. Note that the asymptotic expansion is neither unique nor convergent.

ψDOs are defined on the Schwarz space $\mathcal{S}$ of rapidly decreasing functions, i.e., $A(x,D) \in \mathcal{L}(\mathcal{S},\mathcal{S})$, and the mapping $a(x,\xi) \mapsto A(x,D)$ is continuous. To extend the domain of definition to $\mathcal{S}'$, we need the adjoint. The adjoint of a ψDO A is an operator A^* such that

$$(Au,v)_{L^2(\mathbb{R}^n)} = (u,A^*v)_{L^2(\mathbb{R}^n)}, \quad \forall u,v \in \mathcal{S}.$$

Its existence and form are given by the following theorem:

Theorem 3.3 *For a symbol $a \in S^m$ and its corresponding ψDO $A(x,D)$, there exists a unique adjoint $A^*(x,D)$, and its symbol a^* belongs to S^m and has the asymptotic expansion*

$$a^*(x,\xi) \sim \sum_{\alpha\in\mathbb{N}_0^n} \frac{1}{\alpha!}\partial_\xi^\alpha D_x^\alpha \bar{a}(x,\xi).$$

Due to the existence of the adjoint A^*, one can extend the domain of definition to the tempered distributions. For $u \in \mathcal{S}'$, the action of A on u is given by the action of the adjoint on $\mathcal{S}$:

$$\langle Au,v\rangle = \langle u,\overline{A^*v}\rangle, \quad u \in \mathcal{S}', v \in \mathcal{S}.$$

A key feature of ψDOs is that they form an algebra. For operators $A(x,D)$ and $B(x,D)$ with symbols $a \in S^m$ and $b \in S^{m'}$ we have that the compositions AB

and BA are again ψDO's with symbols $a \circ b, b \circ a \in S^{m+m'}$. The symbol $a \circ b$ has the asymptotic expansion

$$a \circ b(x, \xi) \sim \sum_{\alpha \in \mathbb{N}_0^n} \frac{1}{\alpha!} \partial_\xi^\alpha a(x, \xi) D_x^\alpha b(x, \xi). \tag{3.14}$$

Let us now see what happens to the regularity of a function in a Sobolev space when it is acted on by a pseudo-differential operator.

Theorem 3.4 *Let $A(x, D)$ be a ψDO with symbol $a(x, \xi) \in S^m$. Then $A(x, D)$ maps $H^s(\mathbb{R}^n)$ to $H^{s-m}(\mathbb{R}^n)$.*

Compare this result with our initial considerations on how the symbol manipulates the spectrum and we see that the observation carries over to ψDO's, but in a local (x-dependent) way. For each x the symbol $a(x, \xi)$ manipulates $\hat{u}$ as in (3.6), and we have that a differential operator $L(x, D)$ of order m with symbol $l(x, \xi) \in S^m$ maps a function $u \in H^s(\mathbb{R}^n)$ to $L(x, D)u \in H^{s-m}(\mathbb{R}^n)$.

In the constant coefficient situation, we formally obtained a solution operator by dividing the source term by the symbol. This can be made precise for the class of ψDO's that are elliptic. Elliptic operators have so-called parametrices, also known as approximate inverses, and the allow us to assess the regularity of the solutions to PDEs.

Definition 3.5 The ψDO $A(x, D)$ of order m is said to be elliptic if there exists positive constants C and R such that its symbol $a(x, \xi)$ satisfies

$$|a(x, \xi)| \geq C|\xi|^m \text{ when } |\xi| \geq R, \forall x \in \mathbb{R}^n.$$

The idea behind the construction of the parametrix is this: if we take a cut-off function $\psi(\xi) \in C_0^\infty(\mathbb{R}^n)$ such that $\psi(\xi) = 1$ for $|\xi| \leq R$, and $\psi(\xi) = 0$ for $|\xi| \geq 2R$, we can define the symbol

$$q(x, \xi) = \frac{(1 - \psi(\xi))}{a(x, \xi)} \in S^{-m}. \tag{3.15}$$

Taking $Q(x, d)$ to be the corresponding ψDO, the symbol of the composition $QA(x, D)$ then has a principal symbol

$$(s \circ a)_0(x, \xi) = 1 \quad \text{for } |\xi| > 2R.$$

One can then show that the parametrix is an inverse of A modulo a smooth term.

Theorem 3.6 *For an elliptic ψDO $A(x, D)$ with symbol $a \in S^m$, there exists a parametrix $Q(x, D)$ such that*

$$QA(x, D) = I + K,$$

where I is the identity operator and $K : \mathcal{S}' \to C^\infty(\mathbb{R}^n)$ is a smoothing operator. Furthermore, the operator Q is a ψDO of order $-m$ with symbol $q \in S^{-m}$.

Consider now the PDE

$$L(x, D)u = f, \tag{3.16}$$

where $f \in L^2(\mathbb{R}^n)$. If $L(x, D)$ is an elliptic partial differential operator with symbol $l(x, \xi) \in S^m$, we can apply theorem 3.6 to get

$$u = Qf + Ku.$$

The remainder term Ku is in C^∞, and since the paramatrix Q is of order $-m$ it follows from Theorem 3.4 that $Qf \in H^m(\mathbb{R}^n)$. If the PDE is supplemented with appropriate additional conditions that ensures uniqueness of the solution, the remainder can be dropped and we have that

$$\|u\|_{H^m(\mathbb{R}^n)} \leq C\|f\|_{L^2(\mathbb{R}^n)}.$$

Moreover, we now have a precise generalization of our initial observation that holds for a large class of operators. For equations of the form (3.16), where L is an elliptic ψDOs of order $m > 0$, the solution is the inverse Fourier transform of $\hat{f}$ suppressed by a symbol that is proportional to $|\xi|^{-m}$. Hence, the observation about transfer of information from source to solution in a PDE made in the beginning holds true for a large class of differential operators. However, the problems studied here are somehow artificial, since they are posed on unbounded domains and since we consider the unrealistic situation of having knowledge of the full solution. Still, many of the techniques introduced can be extended to tackle bounded domains, boundary conditions etc., and the main takeaway remains: the solution to many PDEs is related to the suppressed Fourier transform of the source term, and this results in ill-posedness for the inverse problem.

We continue now to refine this analysis through the use of microlocal analysis, and apply the results to hyperbolic PDEs to explain some underlying differences leading to inverse problem with very different characteristics.

3.4 Microlocal Analysis

When studying PDEs, it is interesting to locate where the solutions are non-smooth. The non-smooth points are called singular and the set of all such points

of a given distribution u is called the singular support of u, denoted sing supp(u). The singular support is defined as follows: let U be the open set of all points x where u is smooth, i.e., those x where we can find a neighborhood U_x such that $u|_{U_x} = v|_{U_x}$ for some $v \in C^\infty$. Then

$$\text{sing supp}(u) = \mathbb{R}^n \setminus U.$$

In microlocal analysis, one augments the concept of singular support by adding extra information about the behavior of the localized Fourier transformation of the function. For some function u, we find a localized Fourier transformation at a point x_0 by multiplying u with $\varphi \in C_0^\infty(U)$, where U is small, compact set containing x_0 and $\varphi(x_0) = 1$, and computing $\mathcal{F}(u\varphi)(\xi)$. The wave front set of u, denoted WF(u), is defined as the complement of the points $x \in \mathbb{R}^n$ and directions $\xi \in \mathbb{R}^n$ for which the localized Fourier transform is rapidly decaying.

Definition 3.7 For a distribution $u \in \mathcal{S}'$, the pair $(x_0, \xi_0) \in \mathbb{R}^n \times \mathbb{R}^n \setminus \{0\}$ *does not* belong to WF(u) if there is a $\varphi \in C_0^\infty(\mathbb{R}^n)$ with $\varphi(x_0) = 1$ such that

$$\mathcal{F}(u\varphi)(\xi) \leq C_N(1 + |\xi|)^{-N}, \quad \forall N \in \mathbb{N},$$

for $\xi \neq 0$ such that $\big|\xi/|\xi| - \xi_0/|\xi_0|\big| < \varepsilon$, for some $\varepsilon > 0$.

An elementary property wave front set is that the projection of the wave front set on to its first coordinate is the singular support, i.e.,

$$P_x(\text{WF}(u)) = \text{sing supp}(u).$$

A distribution with a particularly simple wave front set is the Dirac delta distribution. For any $\varphi \in C_0^\infty(\mathbb{R}^n)$ we have that

$$\int_{\mathbb{R}^n} \mathrm{e}^{-ix\cdot\xi}\varphi(x)\delta(x)\mathrm{d}x = \varphi(0).$$

Hence, the localized Fourier transform in not decaying in any direction $\xi \in \mathbb{R}^n \setminus \{0\}$, and

$$\text{WF}(\delta) = \big\{(0, \xi), \xi \in \mathbb{R}^n \setminus \{0\}\big\}.$$

Another example is when $u(x) = H(x_1)$, where $H(x)$ is the Heaviside step function,

$$H(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

By a similar calculation one can show that

$$\text{WF}(u) = \big\{(x, \xi) : x = (0, x_2, ..., x_n), \xi = (\xi_1, 0, ..., 0), \xi_1 \neq 0\big\}.$$

Hence, the wave front set contains the points where u is discontinuous, and the directions in frequency space normal to the discontinuity.

We are interested in understanding what happens to the wave front set of a function when it is mapped to the solution of a PDE. Since the wave front set contains the non-smooth, high frequency parts of a function, i.e., "the finer details", it is of interest to understand how this information is transferred to the solution. Another important aspect is that in practice, one is more likely to only know parts of the solution, e.g., the trace on some hypersurface or its restriction to some smaller set, and we want to obtain insights about the relationship of the Fourier transform of the unknown and the measurement in such situations.

The following theorem, describing what is known as the *pseudo-local property* is useful.

Theorem 3.8 *Let $A(x, D)$ be a ψDO and $u \in \mathcal{S}'$. Then $WF(Au) \subset WF(u)$.*

This means that when a pseudo-differential operator acts on a distribution, it cannot increase its wave front set. The theorem immediately implies the following result.

Theorem 3.9 *For an elliptic differential operator*

$$L(x, D) = \sum_{|\alpha| \leq m} c_\alpha(x) D^\alpha, \quad c_\alpha(x) \in C^\infty(\mathbb{R}^n),$$

let u be the solution to

$$Lu = f$$

for some $f \in \mathcal{S}'$. Then $WF(u) \subset WF(f)$.

The result follows from the fact that L has a parametrix Q (that is also a ψDO) and that $u = Qf + Kf$, where $Kf \in C^\infty(\mathbb{R}^n)$ does not affect the wave front set, and Theorem 3.8.

We now give an example of how this property can be used to analyze an inverse problem. The example is inspired by the recent hit TV-series "Chernobyl". Assume we want to find the strength and precise distribution of a source of the radioactive isotope Iodine-131 located somewhere in Ukraine. The Iodine is transported through the air by means of convection and diffusion processes, and there is some decay due to its relatively short half-life. Assuming the wind velocity field $v(x)$ and the Iodine source is stationary and that the diffusion and

decay is homogeneous in space, a model for the concentration $u(x)$ of Iodine-131 is

$$-a\Delta u + \nabla \cdot (vu) + cu = s, \quad u(x) \to 0 \quad \text{as } x \to \infty.$$

where $a > 0$ is the diffusion coefficient, $c \geq 0$ is the decay rate, and $s(x)$ is the Iodine-131 source. It is natural to assume that $s(x)$ is compactly supported in some domain Ω. Since Iodine-131 is highly dangerous, we cannot measure the concentration anywhere near the source, and we must rely on having knowledge of u in some compact domain $\Omega_M \subset \mathbb{R}^2 \setminus \Omega$, assuming here that the surface of the earth resembles the plane globally[1]. From such measurements, we want to estimate the Iodine source.

The symbol of the governing differential operator is

$$l(x, \xi) = a|\xi|^2 - i\xi \cdot v + \nabla \cdot v + c.$$

We assume $v \in C^\infty(\mathbb{R}^2)$ is bounded, and note that the term $a|\xi|^2$ is positive and grows faster than any of the other terms. We can thus find some $R > 0$, $0 < C < a$, such that when $|\xi| \geq R$, we have $(a - C)|\xi|^2 - |\xi \cdot v| + \nabla \cdot v + c > 0$; hence the operator is elliptic of order $m = 2$. Applying Theorem 3.6, we find that if the radiation source $s \in L^2(\mathbb{R}^2)$, the solution belongs to $H^2(\mathbb{R}^2)$. By taking $\varphi \in C_0^\infty(\mathbb{R}^2)$ to be a smooth characteristic function for our measurement domain Ω_M we can model our measurement as $m = \varphi u$. With the aid of Theorem 3.9, we have that $\mathrm{WF}(u) \subset \mathrm{WF}(s) \subset \Omega$, and since $\Omega_M \cap \Omega = \emptyset$ we can conclude that $m \in C_0^\infty(\mathbb{R}^n)$, and that the Fourier transform $\mathcal{F}(m)(\xi)$ is rapidly decaying, i.e., $|\mathcal{F}(m)(\xi)| \leq C_N(1 + |\xi|)^{-N}$ for any N. Comparing with the example (3.8) where the decay of the Fourier transform was proportional to $|\xi|^{-2}$, our situation now looks worse, and the high frequency information of the source is highly suppressed. In addition is the problem of possibly introducing a kernel in the measurement operator when one only has access to parts of the solution. To tackle such problems one can sometimes rely on unique continuation results, but these have limited practical use, as they rely on knowledge of derivatives.

Fortunately, the situation is not all bad. As we saw in the introductory example, some inverse problem are actually fairly well-posed. We now turn our attention towards some problems in this category; These are the hyperbolic PDEs. Hyperbolic PDEs often describe wave phenomena and the behavior of quantities and systems that respect certain conservation laws (momentum, energy, mass, etc.). The most classical example of a hyperbolic PDE is perhaps the Cauchy problem for the linear wave equation:

$$(\partial_t^2 - c\Delta)u(x, t) = 0, \quad u(x, 0) = f(x), \quad \partial_t u(x, 0) = g. \tag{3.17}$$

[1] This is an increasingly popular view.

The solution operators of hyperbolic PDEs have the tendency to displace the wave front set in space, a phenomenon often referred to as propagation of singularities. This allows for high frequency information to be captured in the measurements and more favorable circumstances for inverse problems.

We start by looking at the transport equation. Let

$$L(x, D) = \partial_t + \sum_{j=1}^{n} c_j(x)\partial_x.$$

We consider the initial value problem

$$L(x, D)u(t, x) = 0, \quad u(0, x) = f(x), \quad (t, x) \in (0, \infty) \times \mathbb{R}^n, f \in C_0(\mathbb{R}^n). \tag{3.18}$$

The solution is obtained by the method of characteristics.

Let $\Phi(t, x) = (\phi_1(t, x), ..., \phi_n(t, x)) \in \mathbb{R}^n$ be the solution to the system of differential equations

$$\frac{\mathrm{d}\phi_j(t, x)}{\mathrm{d}t} = c_j(\Phi(x, t)), \quad \phi_j(0, x) = x_j, \quad 1 \leq j \leq n, \tag{3.19}$$

Assuming the coefficients $c(x)_j$ are $C^\infty(\mathbb{R}^n)$, one can show that for some time interval $[0, T]$ there exists a unique solution $\Phi(t, X)$ for $t \in [0, T]$ that is invertible, and in fact a diffeomorphism, i.e., both $\Phi, \Phi^{-1} \in C^\infty(\mathbb{R}^n)$ for $t \in [0, T]$. Φ is called a flow of the vector field $(c_1, ..., c_n)$. Then, for $v(x, t) = u(\Phi(t, x), t)$ we have $\partial_t v(x, t) = 0$ and $v(x, 0) = f(x)$, and so the solution to (3.18) is given by

$$u(x, t) = f(\Phi^{-1}(t, x)), \quad (t, x) \in [0, T] \times \mathbb{R}^n. \tag{3.20}$$

The initial condition propagates along the flow. This implies that the singular support of f is also propagated accordingly, i.e.,

$$\text{sing supp}(u(x, t)) = \Phi(t, \text{sing supp}(f(x)), \quad t \in (0, T],$$

since if $x_0 \in \text{sing supp}(f(x))$, then $u(\Phi(t, x_0), t) = f(x_0)$ and so $\Phi(t, x_0) \in \text{sing supp}(u(x, t))$, while if $u(\Phi(t, x_0), t) \in C^\infty$, then $x_0 \notin \text{sing supp}(f)$.

The situation therefore differs from what we have seen earlier, as the flow can displace and distort the singular support. One consequence of this is that the solution is not more regular than the initial data. In fact, one can show the stability estimate

$$\sup_{t \in [0,T]} \mathrm{e}^{-\lambda t} \|u(t, \cdot)\|_{H^s(\mathbb{R}^n)} \leq \|f\|_{H^s(\mathbb{R}^n)}, \quad \lambda > 0, s \in \mathbb{R}.$$

Considering again the information perspective, this is good news; it means that the solution operator does not inescapably suppress the information from the source term.

As an example of this, we consider a version of (3.18) in $\mathbb{R}^2$, with constant coefficients v_1, v_2. This could, for example, be a model of the non-diffusive flow of a substance on the sea surface, where $v = (v_1, v_2)$ is the vector describing the sea current. Assume the initial concentration of the substance is given $s(x) \in C_0(\mathbb{R}^n)$, with compact support in some domain Ω. Next, assume we can measure $u(x,t)$ along some straight line segment Γ located upstream of the source, such that $\Gamma \cdot v = 0$, and $\Gamma \cap \Omega = \emptyset$.

Solving (3.19), we find that the flow is

$$\Phi(t,x) = \begin{bmatrix} x_1 + v_1 t \\ x_2 + v_2 t \end{bmatrix}.$$

Writing $v = |v|(\cos(\alpha), \sin(\alpha))$, we can parameterize Γ as

$$\Gamma(\tau) = (L\cos(\alpha) - \tau\sin(\alpha), L\sin(\alpha) + \tau\cos(\alpha)), \quad \tau \in [-M, M],$$

for L sufficiently large. Using formula (3.20), we find that the measurement is given by

$$\begin{aligned} m(\tau,t) =& u(x,t)\big|_{x=\Gamma(\tau)} = s(\Phi^{-1}(t,x))\big|_{x=\Gamma(\tau)} \\ =& s(L\cos(\alpha) - \tau\sin(\alpha) - v_1 t, L\sin(\alpha) + \tau\cos(\alpha) - v_2 t). \end{aligned} \tag{3.21}$$

Taking the Fourier transform of $m(\tau,t)$, we get

$$\begin{aligned} \hat{m}(\xi) &= \int_{-\infty}^{\infty}\int_0^{\infty} s(L\cos(\alpha) - \tau\sin(\alpha) - v_1 t, L\sin(\alpha) + \tau\cos(\alpha) - v_2 t)\mathrm{e}^{-i\xi\cdot(\tau,t)}\mathrm{d}t\mathrm{d}\tau \\ &= \int_{-\infty}^{\infty}\int_0^{\infty} s(\cos(\alpha)(L - |v|t) - \tau\sin(\alpha), \sin(\alpha)(L - |v|t) + \tau\cos(\alpha))\mathrm{e}^{-i\xi\cdot(\tau,t)}\mathrm{d}t\mathrm{d}\tau \\ &= \frac{\mathrm{e}^{i\xi_2 L/|v|}}{|v|}\int_{-\infty}^{\infty}\int_{-\infty}^{L} s(\cos(\alpha)u - \tau\sin(\alpha), \sin(\alpha)u + \tau\cos(\alpha))\mathrm{e}^{-i\xi\cdot(\tau,u/|v|)}\mathrm{d}t\mathrm{d}\tau \\ &= \frac{\mathrm{e}^{i\xi_2 L/|v|}}{|v|}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} s(x_1,x_2)\mathrm{e}^{-i(\xi_1,\xi_2/|v|)\cdot(x_1,x_2)}\mathrm{d}x_1\mathrm{d}x_2 \\ &= \frac{\mathrm{e}^{i\xi_2 L/|v|}}{|v|}\hat{s}(\xi_1, \xi_2/|v|). \end{aligned}$$

We have assumed here that L, M and the measurement time is sufficiently large for the integration to be carried out over $\mathrm{supp}(s)$.

The result is perhaps not surprising, but compared with the Iodine-131 example, we see that information is transferred to the solutions and measurements without the spectrum being suppressed. In the expression

$$\hat{m}(\xi_1, \xi_2) = \frac{\mathrm{e}^{i\xi_2 L/|v|}}{|v|}\hat{s}(\xi_1, \xi_2/|v|)$$

there is no additional decay in $\hat{m}(\xi)$, and since $\left|\mathrm{e}^{i\xi_2 L/|v|}/|v|\right| = 1/|v|$, it is clear that all the information present in s is present in the measurement. Consequently, we can conclude that the inverse problem is well-posed.

In many cases, one can construct solution operators (or parametrices) to hyperbolic PDEs that belong to a class of operators called Fourier integral operators (FIOs), for which the pseudo-differential operators are a subset. In general, FIOs are allowed to have a more complicated phase function than the ΨDOs, and take the form

$$Au(x) = \int \mathrm{e}^{i\varphi(x,\xi)} a(x,\xi)\hat{u}(\xi)\mathrm{d}\xi.$$

Fourier integral operators can propagate the singularities of their input function, and the propagation relies on the spectral directions ξ of the singularities. Therefore, the wave front set is necessary to understand the behavior of such operators.

Let us first give a proper definition of FIOs. We require that the phase function $\varphi(x,\xi)$ is real-valued and homogeneous of order 1 in ξ, and that it is smooth, i.e., $\varphi(x,\xi) \in C^\infty(\mathbb{R}^m, \mathbb{R}^n \setminus \{0\})$. Note that the dimension of x and ξ is not required to agree. In addition, we require that $(\partial_x\varphi, \partial_\xi\varphi) \neq 0, \forall(x,\xi)$, where ∂_x denotes the gradient with respect to x etc.. Last, φ is called non-degenerate if the set of vectors

$$\{(\partial_x\partial_{\xi_j}\varphi, \partial_\xi\partial_{\xi_j}\varphi), 1 \leq j \leq n : \partial_\xi\varphi = 0\}$$

is linearly independent.

Definition 3.10 Let $a(x,\xi) \in S^m$ be a symbol of order m and φ a phase function. For $u \in C_0^\infty(\mathbb{R}^n)$, the Fourier integral operator A is defined by

$$Au(x) = \int_{\mathbb{R}^n} \mathrm{e}^{i\varphi(x,\xi)} a(x,\xi)\hat{u}(\xi)\mathrm{d}\xi. \tag{3.22}$$

As with pseudo-differential operators, Fourier integral operators have an extension from $C_0^\infty(\mathbb{R}^n)$ to continuous linear maps in $\mathcal{L}(\mathcal{E}'(\mathbb{R}^n), \mathcal{D}'(\mathbb{R}^m))$. We now give a version of Hörmanders "propagation of singularities" theorem and apply that to the transport equation and Cauchy problem for the wave equation.

Theorem 3.11 *For $u \in \mathcal{E}'(\mathbb{R}^n)$ and a Fourier integral operator A, assume that the phase function φ is non-degenerate and that $\partial_x\varphi(x,\xi) = 0$ implies that $(\partial_\xi\varphi(x,\xi), 0) \notin WF(u)$. Then*

$$WF(Au) \subset \big\{(x, \partial_x\varphi(x,\xi)) : (\partial_\xi\varphi(x,\xi), \xi) \in WF(u)\big\}.$$

We now put this rather mystifying result to work. Returning to the constant coefficient transport equation, we recall that the solution was given by $u(x,t) = s(\Phi^{-1}(t,x))$. Written as

$$u(x,t) = (2\pi)^{-2} \int_{\mathbb{R}^2} \mathrm{e}^{i\xi\cdot\Phi^{-1}(t,x)} \hat{s}(\xi)\mathrm{d}\xi$$

we have the phase function

$$\varphi(x,\xi) = \xi \cdot \Phi^{-1}(t,x) = \xi \cdot \begin{bmatrix} x_1 - v_1 t \\ x_2 - v_2 t \end{bmatrix}.$$

Here we consider the phase function to depend on the time t as a parameter. It is quick to check that it fulfils the required criteria and that it is non-degenerate, and hence the solution is given by the action of FIO on the source. We have that $\partial_x\varphi = \xi$, and so $\partial_\xi\varphi = 0 \implies \xi = 0$, and since $(\partial_\xi\varphi(x,0), 0) \notin \mathrm{WF}(s)$ (since $\xi = 0$ is never in the wave front set), we can apply Theorem 3.11 and get that

$$\mathrm{WF}(u(x,t)) \subset \left\{(x,\xi) : (x_1 - v_1 t, x_2 - v_2 t, \xi_1, \xi_2) \in \mathrm{WF}(s), \quad \right\}.$$

Again, the result is not surprising: the singular support of the source propagates along the flow, and since the characteristic curves are straight lines, the spectral directions originally in $\mathrm{WF}(s)$ remain the same in $\mathrm{WF}(u)$.

Next, we consider the wave equation (3.17). We assume $f \in \mathcal{E}'(\mathbb{R}^n)$ and $g = 0$. Taking the Fourier transform in x, we have

$$\partial_t^2 \hat{u}(\xi,t) + c\xi^2 \hat{u}(\xi,t) = 0, \quad \hat{u}(\xi,0) = \hat{f}(\xi), \quad \partial_t\hat{u}(\xi,0) = 0. \tag{3.23}$$

The above equation is an ordinary differential equation for each $\xi \in \mathbb{R}^n$, and has the solution

$$\hat{u}(\xi,t) = A(\xi)\mathrm{e}^{itc|\xi|} + B(\xi)\mathrm{e}^{-itc|\xi|}.$$

Imposing the initial condition, we find that

$$\hat{u}(\xi,t) = \frac{1}{2}\hat{f}(\xi)\left(\mathrm{e}^{itc|\xi|} + \mathrm{e}^{-itc|\xi|}\right).$$

Taking the inverse Fourier transform yields

$$u(x,t) = \frac{1}{2}(2\pi)^{-n} \int_{\mathbb{R}^n} \mathrm{e}^{ix\cdot\xi} \left(\mathrm{e}^{itc|\xi|} + \mathrm{e}^{-itc|\xi|}\right) \hat{f}(\xi)\mathrm{d}\xi. \tag{3.24}$$

Writing

$$A_\pm f = \frac{1}{2}(2\pi)^{-n} \int_{\mathbb{R}^n} \mathrm{e}^{ix\cdot\xi \pm itc|\xi|} \hat{f}(\xi)\mathrm{d}\xi,$$

we have that $u(x,t) = A_- f + A_+ f$. We claim that $\varphi_\pm(x,\xi) = \xi \cdot (x \pm ct\xi/|\xi|)$ is a phase function, and hence that $A_\pm$ are FIOs with symbol $a(x,\xi) = 1 \in S^0$. Indeed, $\varphi_\pm$ is real-valued and homogeneous of order 1 in ξ, and has the desired smoothness properties. Also, since $\partial_x \partial_{\xi_j} \varphi_\pm = e_j$, where e_j is the j'th unit vector in $\mathbb{R}^n$, $\varphi_\pm$ is non-degenerate. Considering again t as a parameter, we have for each $t > 0$

$$\partial_x \varphi_\pm(x,\xi) = \xi, \quad \partial_\xi \varphi_\pm(x,\xi) = x \pm ct\xi/|\xi|,$$

and we can apply Theorem 3.11. Hence, we get that

$$\mathrm{WF}(u(x,t)) \subset \big\{(x,\xi) : (x \pm ct\xi/|\xi|, \xi) \in \mathrm{WF}(f)\big\}.$$

We see above that the spectral direction ξ plays a role. If for some x_0, we have $(x_0,\xi_1), (x_0,\xi_2) \in \mathrm{WF}(f)$, then the singularity at x_0 propagates along the curves $x_{1,\pm}(t) = x_0 \pm ct\xi_1/|\xi_1|$ and $x_{2,\pm}(t) = x_0 \pm ct\xi_2/|\xi_2|$. In that sense, the wave front set of the wave equation tend to be more "spread out" than for the transport equation, like the waves propagating away from the splash where the rock hit the quiet pond.

We consider now the inverse problem of reconstructing the initial condition f in (3.17) from measurements of the solution. We assume $f \in L^2(\Omega)$ has compact support in some domain $\Omega \subset \mathbb{R}^3$, and that $g = 0$. Further, we assume that we can measure the wave $u(x,t)$ on the smooth surface Γ of a convex domain containing Ω for a sufficiently long time interval $[0,T]$. We denote the measurement $m(x,t) = Mf = u(x,t)\big|_{x\in\Gamma, t\in[0,T]}$, where M is the measurement operator, a restriction of the FIO in (3.24) to Γ. The inverse problem is to reconstruct f from m. Due to the constant sound speed c, the above analysis of the wave front set guarantees that no waves are trapped within B, i.e., all the singularities of f pass through some part of Γ. As a consequence, one can show that $\mathrm{Ker} M = \{0\}$ and that

$$\|f\|_{L^2(\Omega)} \le C\|Mf\|_{H^1([0,T]\times\Gamma)} = C\left(\int_0^T \|m(t,\cdot)\|_{L^2(\Gamma)} + \|m'(t,\cdot)\|_{L^2(\Gamma)}\mathrm{d}t\right)^{1/2}.$$

Hence M is an invertible mapping in the space $\mathcal{L}(L^2(\Omega), H^1([0,T]\times\Gamma))$, and the inverse problem is considered to be moderately ill-posed. The result above also

holds in the case of variable wave speed and less optimal measurement surfaces; then the analysis of the wave front set depend on the wave speed, and one has to assume that the wave speed is non-trapping, i.e., that all singularities can propagate to the measurement.

We end this section by returning to our two initial inverse problems for the wave equation (3.1) and the heat equation (3.2). Taking the Fourier transform in space of (3.2), we get that

$$\partial_t \hat{h}(\xi,t) + \xi^2 \hat{h}(\xi,t) = 0, \quad t > 0, \xi \in \mathbb{R}, \hat{h}(\xi,0) = \hat{f}(\xi),$$

and hence $\hat{h}(\xi,t) = \mathrm{e}^{-t\xi^2}\hat{f}(\xi)$. The measurement operator is then given by

$$M_h f = (2\pi)^{-1} \int_{\mathbb{R}} \mathrm{e}^{ix_m\xi} \mathrm{e}^{-t\xi^2} \hat{f}(\xi) \mathrm{d}\xi. \tag{3.25}$$

For the wave equation, we have that

$$M_v f = \frac{1}{2} f(x_m - ct) = \frac{1}{2}(2\pi)^{-1} \int_{\mathbb{R}} \mathrm{e}^{i(x_m - ct)\xi} \hat{f}(\xi) \mathrm{d}\xi. \tag{3.26}$$

In light of what we have seen in this chapter, the difference could not be more pronounced. The measurement operator for the heat equation is a pseudo-differential operator with a symbol $\mathrm{e}^{-t\xi^2}$ that belongs to $S^{-\infty}$, and hence the Fourier transform of the source is suppressed by an exponentially decaying weight function. On the other hand, the measurement operator of the wave equation is a Fourier integral operator with symbol $1/2$ and phase function $\phi(x,\xi) = (x_m - ct)\xi$, and the singularities are propagated to the measurement.

There is an interesting physical explanation for the different characteristics of the equations: parabolic and elliptic PDEs describe physical processes that are dominated by diffusion and describe the aggregate behavior of the random, erratic motions of a large number of particles. Diffusion has a tendency to smooth out details, making it hard to recover the cause of a specific state. For hyperbolic equations the situation is quite different. For example, in the case of acoustic waves, the wave propagates by compression and rarefaction of the medium; the particles do not travel, they only move in a periodic fashion around their equilibrium state while the wave energy passes through them. As a consequence, waves tend to maintain their shape while they propagate through space, and hence they are good transporters of information. One can perhaps summarize the difference between the two regimes by noting that it is no wonder that the preferred way of communication for humans is by sound, and not by gas or heat.

The different regimes of hyperbolic and parabolic or elliptic PDEs are often encountered in practice, and the insights and methods emphasized in this chapter serves as useful tools for analyzing qualitative aspects of inverse problems

related to such equations. For example, and understanding of how the singularities of a certain equation propagate can help us to understand how to make good measurements. Or the realization that the information in the measurement is to suppressed to allow for a reconstruction can lead us to conclude that we need additional measurements to succeed. However, the applicability of these methods to the quantitative aspects of inverse problems, i.e., the construction of methods to solve inverse problems, is limited. Better suited for this is the spectral theory of compact operators and regularization methods that we introduce in the next chapter.

References

The material on the Fourier transform in Section 3.2 can be found in the monographs [31, 38, 58]. A nice presentation of the material on pseudo-differential operators in Section 3.3 is given in [5]; see also [27, 31]. The material on microlocal analysis in Section 3.4 is found in [5, 27, 38], and the results on hyperbolic PDEs appear in [5, 28]. The material on Fourier integral operators is mainly taken from [5, 26], and the material on the inverse problem for the wave equation is from the paper [63].

Chapter 4

Spectral Theory and Regularization

In this chapter we shall again consider compact operators $K \in \mathcal{L}(X, Y)$, where X and Y will be Hilbert spaces. We will introduce some elegant results on their spectral decomposition. These results are both of qualitative interest and immense practical value. The latter will be clear when we make the connection to regularization methods, i.e., methods that can be used to find stable, approximate solutions to inverse problems of the form $Kx = y$.

Spectral decomposition is a remarkable display of how compactness allows for results in finite-dimensional spaces to be generalized to the infinite-dimensional setting. In linear algebra, a matrix $A \in \mathbb{R}^{n\times n}$ that is symmetric has an eigendecomposition. That A has an eigendecomposition means we can find a set of vectors $\{v_i\}_{i=1}^n$ such that $(v_i, v_j)_{\mathbb{C}^n} = \delta_{i,j}$, called the eigenvectors of A, and a set of corresponding real numbers $\{\lambda_i\}_{i=1}^n$ such that $Av_i = \lambda_i v_i$ for each $1 \le i \le n$. As a consequence, we can write

$$A = V^*\Lambda V, \quad \text{where } V = \begin{bmatrix} v_1^T \\ \vdots \\ v_n^T \end{bmatrix} \quad \text{and } \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Having an eigendecomposition is very useful; say we want to solve the inverse

problem

$$\text{Given } b \in \mathbb{R}^n, \quad \text{find } x \in \mathbb{R}^n \quad \text{such that } Ax = b.$$

The eigenvectors form an orthonormal basis for $\mathbb{R}^n$ and we can decompose the vector b as $b = \sum_{i=1}^n b_i v_i$, where $b_i = b \cdot v_i$. If A is invertible we then have that

$$x = A^{-1}b = \sum_{i=1}^n \frac{b_i}{\lambda_i} v_i.$$

In addition to being a simple expression for the inverse, this equation allows us to see in detail how A^{-1} acts on each component of the decomposition of b. This becomes very useful when we use spectral theory to analyze the ill-posedness of inverse problems and suggest regularization strategies.

4.1 Spectral Theorems

Consider now the infinite-dimensional setting. First, we recall the adjoint operator and some of its properties. For any compact operator $K \in \mathcal{L}(X, Y)$, there exists a unique operator $K^* \in \mathcal{L}(Y, X)$, called the adjoint of K, such that

$$(Ku, v)_Y = (u, K^* v)_X, \quad \forall u \in X, v \in Y.$$

For K^*, we have

- $\|K^*\| = \|K\|$
- $\text{Im } K^\perp = \text{Ker } K^*$ and $\text{Ker } (K^*)^\perp = \overline{\text{Im } K}$.

An operator $K \in \mathcal{L}(X, X)$ is said to be self-adjoint if $K = K^*$. We first introduce the spectral theorem for self-adjoint, compact operators.

Theorem 4.1 *Let $K \in \mathcal{L}(X, X)$ be a self-adjoint compact operator. Then there exists a sequence of real numbers $\{\lambda_n\}_{n\in\mathbb{N}}$, called the eigenvalues of K, and corresponding set of eigenvectors $\{v_n\}_{n\in\mathbb{N}}$ such that*

$$Kv_n = \lambda_n v_n, \quad \forall n \in \mathbb{N}.$$

The sequence of eigenvalues have 0 as its only accumulation point and at least one $\lambda_n \neq 0$. In addition, all the eigenvectors are orthonormal and form a basis for Im K, and the operator has the decomposition

$$K = \sum_{n\in\mathbb{N}} \lambda_n (\cdot, v_n)_X v_n.$$

We thus get an analogous result as for the finite-dimensional setting, but now X is an arbitrary Hilbert space, and K is any self-adjoint, compact operator. The compactness of the operator is seen in the decay of the eigenvalue sequence, and since the eigenvectors tend to become increasingly oscillating as the index increases, the spectral decomposition reflects the observations from Chapter 3 on how compact solution operators to PDEs tend to dampen the high-frequency information of the source term.

A classic example of such a spectral decomposition is found for the Poisson equation on the disk $D_R \subset \mathbb{R}^2$ of radius R. As we saw in Chapter 2, the solution operator $K : L^2(D_R) \to L^2(D_R)$ taking a source f in the PDE

$$-\Delta u = f \quad \text{in } D_R, \quad u = 0 \quad \text{on } \partial D_R \tag{4.1}$$

to the solution u, is compact in $\mathcal{L}(L^2(D_R), L^2(D_R))$. It is also self-adjoint, since Δ is, and so K has a spectral decomposition

$$Kf = \sum_{m\in\mathbb{Z}} \sum_{n\in\mathbb{N}} \lambda_{m,n} (f, v_{m,n})_{L^2(D_R)} v_{m,n}, \tag{4.2}$$

where

$$\lambda_{m,n} = \left(\frac{j_{m,n}}{R}\right)^{-2}, \quad v_{m,n} = (R^2 \pi J_{m+1}^2(j_{m,n}))^{-1/2} J_m\left(\frac{j_{m,n}}{R} r\right) e^{im\theta}.$$

Here, (r, θ) are polar coordinates; $J_m(z)$ is the Bessel function of order m and $j_{m,n}$ is the $n'th$ root of J_m. Hence, for any $f \in L^2(D_R)$, the corresponding solution u to (4.1) is given by

$$u = \sum_{m\in\mathbb{Z}} \sum_{n\in\mathbb{N}} \lambda_{m,n} (f, v_{m,n})_{L^2(D_R)} v_{m,n}.$$

Moreover, since Ker $K = \{0\}$, the eigenvectors form an orthonormal basis for $L^2(D_R)$. By expanding f in this basis, we see how the compact operator scales the generalized Fourier coefficients $f_{m,n} = (f, v_{m,n})_{L^2(D_R)}$ of f by multiplication with the decreasing eigenvalues $\lambda_{m,n} \sim (m + \pi n)^{-2}$.

We now introduce the singular value decomposition (SVD), a spectral decomposition for operators that are not self-adjoint. The SVD is of special interest, since it allows us to study operators that act between different spaces, e.g., the mapping of a source or boundary condition in a PDE to the trace of the solution on the boundary.

Note that if $K \in \mathcal{L}(X, Y)$ and $T = K^*K \in \mathcal{L}(X, X)$, then $(Tu, v)_X = (K^*Ku, v)_X = (Ku, Kv)_Y = (u, K^*Kv)_X = (u, Tv)_X$, thus T is self-adjoint. This observation that grants us the next result.

Theorem 4.2 *Let $K \in \mathcal{L}(X, Y)$ be compact operator with adjoint K^*. Then there exists a singular system $\{\sigma_n, \varphi_n, \phi_n\}_{n\in\mathbb{N}}$ with the following properties:*

- *The sequence $\{\sigma_n\}$, called the singular values, is real-valued and tends to 0 as $n \to \infty$.*
- *$K\varphi_n = \sigma_n\phi_n$ and $K^*\phi_n = \sigma_n\varphi_n$ for all $n \in \mathbb{N}$.*
- *The sets $\{\varphi_n\}$ and $\{\phi_n\}$, called the singular vectors, form orthonormal bases for Ker $(K)^\perp \subset X$ and Im $K \subset Y$, respectively.*

The singular system of K admits the decomposition

$$K = \sum_{n\in\mathbb{N}} \sigma_n(\cdot, \varphi_n)\phi_n, \tag{4.3}$$

known as the singular value decomposition.

Unless otherwise stated, we will assume the singular system are indexed such that the singular values are decreasing with n, i.e., $\sigma_n \geq \sigma_{n+1}$.

Returning again to our introductory example of the Poisson equation, we recall that we found the measurement operator M to be a compact operator $M : L^2(\Omega) \to L^2(\partial\Omega)$. Using a quadrature method, we discretize the solution operator to equation (4.1) and compute a numerical approximation M^h to M. By computing the discrete SVD of M^h, we get an approximation $\{\sigma_n^h, \varphi_n^h, \phi_n^h\}$ to the singular system $\{\sigma_n, \varphi_n, \phi_n\}$ associated with M. Figure 4.1 shows a plot of some of the singular values and singular vectors.

When a compact operator K is discretized and represented as a (for simplicity, square) matrix $K^h \in \mathbb{C}^{n\times n}$ the inherited properties of K may result in K^h being singular. If that is not the case and K^h is invertible, the unboundedness of K^{-1} usually manifests itself by a large condition number $\kappa(K^h) = \frac{\sigma_{max}(K^h)}{\sigma_{min}(K^h)}$, and the discrete inverse problem inherits the instability of its continuous version.

We now consider the inverse problem of solving $Kx = y$ for x, with (a possibly noisy) y given, and $K \in \mathcal{L}(X, Y)$ compact. Formally, we get that

$$x = \sum_{n\in\mathbb{N}} \sigma_n^{-1}(y, \phi_n)_Y \varphi_n. \tag{4.4}$$

However, the sequence $\{\sigma_n^{-1}\}$ diverges as $n \to \infty$, and unless $\sigma_n^{-1}(y, \phi_n)_Y \to 0$ sufficiently fast, the right hand side of (4.4) is divergent. This is the unboundedness of K^{-1} manifesting itself through the SVD.

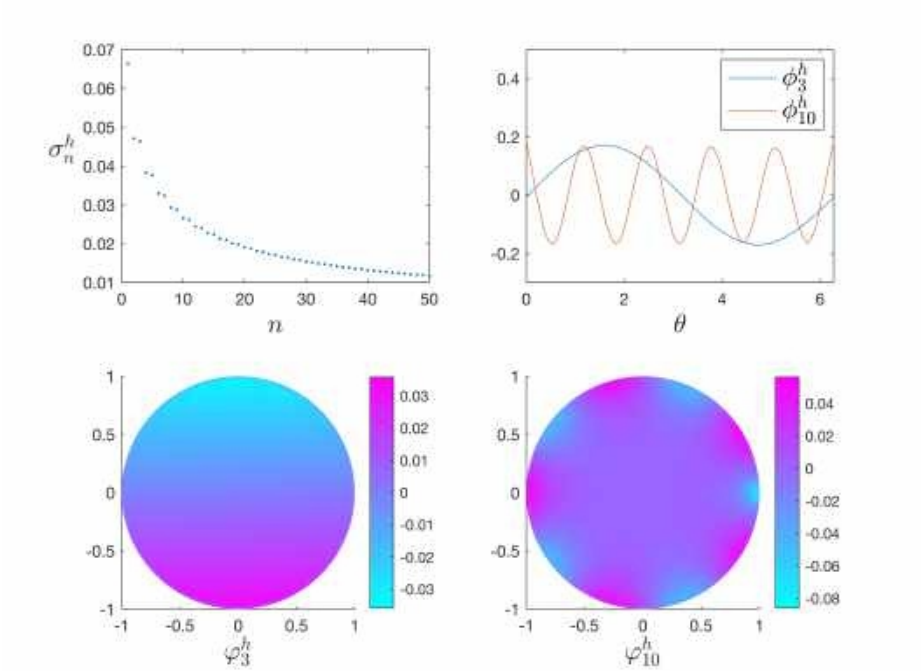

Figure 4.1: The top-left plot shows the first 50 singular values of the matrix K^h. The fast decay is evident. The top-right plot shows two singular vectors ϕ_n^h. Notice the resemblance with trigonometric basis functions and the increasing frequency. The two bottom plots show the corresponding singular vectors φ_n^h. The bottom-right plot displays an interesting feature of singular vectors: the main variation of φ_{10} happens near the measurement boundary, and this reveals the fact that one can mainly reconstruct features close to the boundary.

Conditions for when (4.4) is valid are given by the Picard-criterion:

Theorem 4.3 *For a compact operator $K \in \mathcal{L}(X, Y)$ and $y \in Y$, the equation*

$$x = K^{-1}y$$

has a solution if and only if $y \in Ker\ (K^*)^\perp$ and

$$\sum_{n\in\mathbb{N}} \frac{1}{\sigma_n^2} |(y, \phi_n)_Y|^2 < \infty.$$

Then the solution is given by (4.4).

The decay rate of the singular values is sometimes used to describe the ill-posedness of an inverse problem [55], and the following categorization is devised: Let $\{\sigma_n\}_{n\in\mathbb{N}}$ be the singular values of a compact operator K. The inverse problem of solving the equation $Kx = y$ for x is said to be:

- Mildly ill-posed if $\sigma_n = \mathcal{O}(n^{-\beta}), \quad 0 < \beta \leq 1.$
- Moderately ill-posed if $\sigma_n = \mathcal{O}(n^{-\beta}), \quad \beta > 1.$
- Severely ill-posed if $\sigma_n = \mathcal{O}(\mathrm{e}^{-n\beta}), \quad \beta > 0.$

For example, we see that the solution operator to the Poisson equation (4.2) gives rise to a moderately ill-posed inverse problem. This categorization is sometimes useful, but does not take into account the presence of a kernel in K.

The conditions of the Picard criterion are often not met. However, we will see that equation (4.4) is important in the development of approximate, stable solution methods, so-called regularization methods.

4.2 Regularization

When the measurement operator is compact the inverse is an unbounded operator. This means that we cannot hope to solve the inverse problem exactly; Naturally, the question arises if we can solve the problem approximately and in a stable way. The theory of regularization is developed to find such approximate, stable solutions and is made rigorous by the concept of a regularization scheme:

Definition 4.4 Let $K \in \mathcal{L}(X, Y)$ be an injective compact operator[1]. Let $\{R_\alpha\}_{\alpha\in I}$ be a family of bounded operators from Y to X, where $\alpha \in I \subset \mathbb{R}^+$. We require that $R_\alpha \in \mathcal{L}(Y, X)$ for each $\alpha > 0$, and that

$$R_\alpha Kx \to x \quad \text{as } \alpha \to 0, \quad \forall x \in X.$$

The family of operators $\{R_\alpha\}_{\alpha\in I}$ is called a regularization scheme for K, and the parameter α is called the regularization parameter.

The operators R_α are bounded approximations of K^{-1}. The idea of the regularization scheme is this: Let $y \in \operatorname{Im} K$, and denote by $y_\varepsilon \in Y$ a noisy measurement of y, i.e., $y_\varepsilon = y + \varepsilon$ for some $\varepsilon \in Y$. We denote by x the true unknown, and

[1] If K is not injective, we can work with the restriction $\tilde{K} : \operatorname{Ker}(K)^\perp \to Y$

by x_ε^α the approximate unknown coming from inverting y_ε, i.e., $x_\varepsilon^\alpha = R_\alpha y_\varepsilon$. We then have

$$\begin{aligned}\|x_\varepsilon^\alpha - x\|_X &= \|R_\alpha y_\varepsilon - K^{-1}y\|_Y = \|R_\alpha(y_\varepsilon - y) + (R_\alpha - K^{-1})y\|_Y \\ &\le \|R_\alpha(y_\varepsilon - y)\|_Y + \|(R_\alpha - K^{-1})y\|_Y \\ &\le \|R_\alpha\|\|\varepsilon\|_Y + \|(R_\alpha - K^{-1})y\|_X.\end{aligned} \tag{4.5}$$

The aim is, of course, to have $\|x_\varepsilon^\alpha - x\|_X$ as small as possible. We see that it is bounded by two terms; on one hand, the term $\|R_\alpha\|\|\varepsilon\|_Y$ involves the noise and the operator norm of R_α. Since $R_\alpha y \to K^{-1}y$ as $\alpha \to 0$, one can show that $\|R_\alpha y_\varepsilon\|_Y \to \infty$ as $\alpha \to 0$. On the other hand, the second term $\|(R_\alpha - K^{-1})y\|_X$ must approach 0 as $\alpha \to 0$, and accounts for the approximation error. Hence, the quality of the approximated solution x_ε^α depends on the choice of regularization parameter α, and this comes down to finding a balance between approximation of the operator and amplification of noise. Finding such a balance is often referred to as a regularization strategy.

Given a regularization scheme $\{R_\alpha\}_{\alpha\in I}$ for a compact operator $K \in \mathcal{L}(X,Y)$, we say that $x^\alpha = R_\alpha y$ is a regularized solution to $Kx = y$.

Spectral decomposition and the SVD is very useful for understanding and constructing regularization schemes. Consider a compact operator $K \in \mathcal{L}(X,Y)$ with a SVD in the form (4.3). To construct a regularization scheme, we introduce the weight function $r(\alpha,\beta) : \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}$ with the properties that

$$|r(\alpha,\beta)| \le C\beta, \quad \lim_{\alpha\to 0} r(\alpha,\beta) = 1 \text{ for } 0 < \beta \le \|K\|,$$

for some $C = C(\alpha) > 0$. For $\alpha > 0$ we define the operator R_α as

$$R_\alpha = \sum_{n=1}^{\infty} \frac{r(\alpha,\sigma_n)}{\sigma_n}(\cdot,\phi_n)_Y \varphi_n. \tag{4.6}$$

Since

$$\|R_\alpha y\|_X^2 \le C^2 \sum_{n=1}^{\infty} |(y,\phi_n)_Y|^2 \le C^2\|y\|_Y^2, \quad \alpha > 0,$$

we see R_α is bounded. Next, we check if R_α is an approximate inverse for K as $\alpha \to 0$. By the orthonormality of the singular vectors we have that $(R_\alpha Kx, \varphi_n)_X = r(\alpha,\sigma_n)(x,\varphi_n)_X$, and so

$$\|R_\alpha Kx - x\|_X^2 = \sum_{n=1}^{\infty} |(R_\alpha Kx - x, \varphi_n)_X|^2 = \sum_{n=1}^{\infty} (r(\alpha,\sigma_n) - 1)^2 |(x,\varphi_n)_X|^2.$$

Since we have required K to be injective, $\{\varphi_n\}$ is an orthonormal basis for X. Thus for any $\varepsilon > 0$ there is some N such that

$$\sum_{n=N+1}^{\infty} |(x, \varphi_n)_X|^2 \leq \frac{\varepsilon}{2(\tilde{C}+1)^2},$$

where $\tilde{C}$ is the constant bounding $r(\alpha, \beta)$. For all $n \leq N$ we have $\sigma_n \geq \sigma_{N+1} > 0$, and by the requirement on $r(\alpha, \beta)$, we can choose $\alpha > 0$ such that $(r(\alpha, \sigma_n) - 1)^2 \leq \varepsilon/(2\|x\|_X^2)$ for all $n \leq N$. Hence

$$\|R_\alpha Kx - x\|_X \leq \sum_{n=1}^{N} \frac{\varepsilon}{2\|x\|_X^2} |(x, \varphi_n)_X|^2 + \frac{\varepsilon}{2} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since ε was arbitrary, it follows that R_α is an approximate inverse of K as $\alpha \to 0$ and hence $\{R_\alpha\}_{\alpha>0}$ is a regularization scheme.

The idea behind the spectral regularization scheme comes from the observation following (4.4), i.e., that the divergence of reciprocal sequence $\{\sigma_n^{-1}\}$ of singular values is the cause[2] of the unboundedness. By modifying the coefficients in the formal expansion of K^{-1}, we avoid the amplification and obtain a bounded operator that is an approximation to the inverse. Many regularization methods use this principle, for example the classical Tikhonov method. The Tikhonov method is often formulated as an optimization problem: Given $y \in Y$, a regularized solution x^α to $Kx = y$ is found by

$$x^\alpha = \operatorname{argmin}_{x \in X} \|Kx - y\|_Y^2 + \alpha \|x\|_X^2. \tag{4.7}$$

The above formulation demonstrate the balance between approximation and amplification of noise. The first term controls the approximation error, while the second term controls the norm of x^α and hence the amplification of noise. α acts as a tuning parameter that must be properly adjusted to find the appropriate balance between the two. One can show that the solution to (4.7) is given by

$$x^\alpha = \sum_{n=1}^{\infty} \frac{\sigma_n}{\sigma_n^2 + \alpha} (y, \phi_n) \varphi_n.$$

Comparing now with (4.6), we see that the Tikhonov method gives rise to a spectral regularization scheme$\{R_\alpha\}_{\alpha>0}$ with

$$R_\alpha = \sum_{n=1}^{\infty} \frac{r(\alpha, \sigma_n)}{\sigma_n} (\cdot, \phi_n) \varphi_n, \quad \text{where } r(\alpha, \beta) = \frac{\beta^2}{\beta^2 + \alpha}.$$

[2] or the symptom?

Another spectral regularization method is the so-called truncated singular value decomposition (TSVD). In TSVD, the operator R_α is defined as

$$R_\alpha = \sum_{\sigma_n \leq \alpha} \frac{1}{\sigma_n} (\cdot, \phi_n)_Y \varphi_n. \tag{4.8}$$

It is not complicated to show that $\{R_\alpha\}_{\alpha>}$ is a regularization scheme. Successful use of a TSVD-regularization requires precise knowledge of the behavior of the singular values of the measurement operator.

To briefly illustrate a regularization method, we apply TSVD-regularization to the discretized inverse problem for the Poisson equation, i.e., the reconstruction of the source in (4.1) from boundary measurements of the trace of the derivative. We form the discrete approximation M^h and take a source $f^h \in \text{Ker}\ (M^h)^\perp$, a linear combination of 10 singular vectors, and produce the discrete measurement[3] $m^h = M^h f^h$. We then add random noise ε^h with a noise level of 10%, i.e., $\|\varepsilon^h\|_2 \leq 0.1\|m^h\|_2$, and do a TSVD-reconstruction of f^h_α from $m^h_\varepsilon = m^h + \varepsilon^h$. The top-right plot shows the relative error $\|f^h - f^h_\alpha\|_2/\|f^h\|_2$ for different values of the regularization parameter α. Here one can observe the balancing of the approximation error and noise amplification. The top-right plot shows f^h and the bottom-left plot shows the reconstruction f^h_α for the optimal choice of α. The bottom-right plot shows their pointwise absolute difference.

Spectral analysis and regularization methods are indispensable tools for inverse problems. However, there are some problems that are so ill-posed that even with a noise free measurement we cannot hope to reconstruct the unknown. In fact, our initial inverse problem of reconstructing the source in the Poisson equation from boundary measurements belongs to this category of problems. The kernel of the measurement operator is considerable, and little information about the source is present in the measurement. This is not suprising, given the loss of dimension(source is in $\mathbb{R}^2$, measurement is in $\mathbb{R}$) and the rapid decay of the singular values. As a consequence, we can hardly reconstruct anything. Figure 4.3 shows the result of an application of the TSVD-regularization method with the same regularization parameter as in the reconstruction in Figure 4.2. Here, the source f^h is a characteristic function for a disk centered at $x = (0.2, 0.1)$ with radius 0.3, and f^h_α is the reconstruction from a noise free measurement. We can hardly see anything of the source in the reconstruction, and this can be anticipated from inspecting the singular vectors in Figure 4.1. It is common to say that the reconstruction suffers from low resolution in situations like this, i.e., when the finer details of the unknown cannot be seen in the reconstruction.

To obtain useful reconstructions for inverse problems of this kind, one needs

[3] To avoid a so-called *inverse crime*, we should produce the measurements using, e.g., a finer mesh, but this example is for illustration purposes.

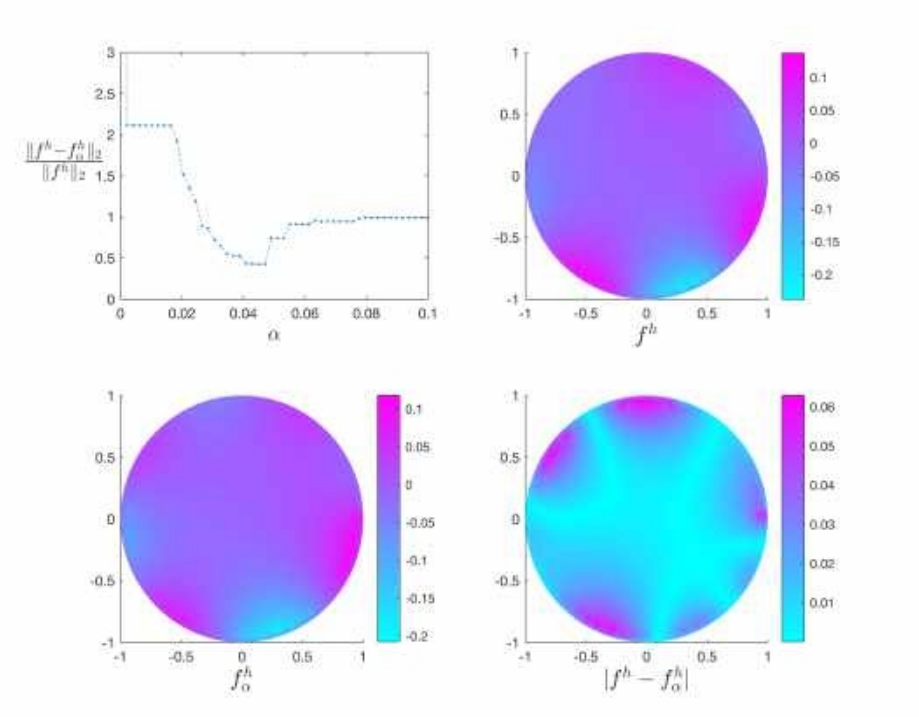

Figure 4.2: Relative error as a function of regularization parameter, and plots of source, reconstructed source from noisy measurements and pointwise absolute error.

more information than what one can get from the boundary measurements alone. In the next chapter, we look at some inverse problems in this category, where one utilizes the coupling of different physical phenomenon to acquire extra information, often in term of some internal information about the unknown.

References

For the spectral theory of compact operators there are many good resources, e.g., the monographs [21, 24, 45, 46]. For singular value decomposition the presentation in [45] is great, and for more advanced material, the monograph [30] is recommend. For the discretization and numerical treatment of inverse problems for compact operators, see [33, 34, 55]. The material on regularization in Section 4.2 can be found in [25, 32, 34, 45].

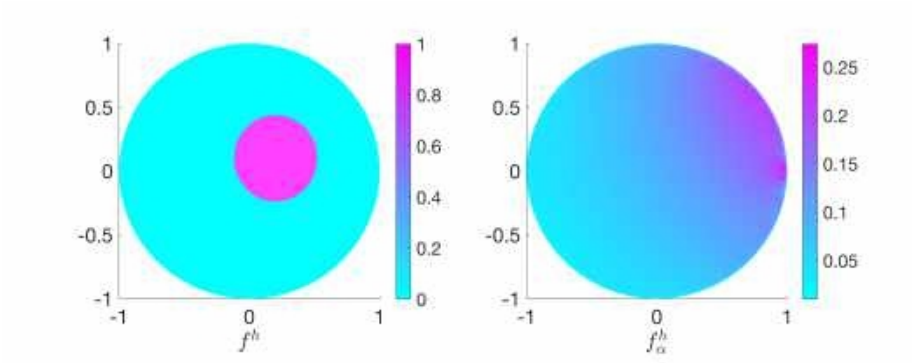

Figure 4.3: Left: source term f^h in equation (4.1). Right: reconstruction from noise free boundary measurement of f^h.

Chapter 5

Hybrid Inverse Problems

The inverse problems that are successfully solved in real-world applications are often not too ill-posed. Ultrasound imaging (UI), magnetic resonance imaging (MRI) and X-ray/CT imaging are examples of medical imaging modalities for which the underlying physics are described by PDEs that gives rise to moderately ill-posed inverse problems [60, 64]. The physical process that connects the measurement with the unknown quantities is often of wave- or transport type, and the amount of information in the measurements allows for good reconstructions. Each of these modalities makes it possible to obtain images in a non-destructive way of certain parameters in the interior of the object of interest. In ultrasound imaging, transducers send high frequency acoustic waves into the object and measure the reflected waves. By applying techniques such as beamforming and time-travel tomography, one obtains a high resolution image of the acoustic impedance in the interior of the object. In magnetic resonance imaging, the application of strong magnetic fields and radio waves allow one to detect the concentration of hydrogen atoms with high accuracy. In X-ray imaging/CT, one reconstructs the X-ray absorption coefficient, and the inverse problem is solved by inverting a Radon transform. The reconstructed "parameters" from these modalities contain information about various features of the human physiology, and such information is used by medical practitioners and researchers to obtain trustworthy and valuable diagnostic information, and to understand the governing mechanisms of the human body. As a consequence, they are ubiquitous in modern science and healthcare.

However, some highly interesting parameters are invisible to these modalities. A famous example of such a parameter is the electrical conductivity. The electrical conductivity is a parameter that describes a material's ability to conduct electrical currents, and is known to vary significantly between various types of materials [55, 60]. This variation makes it a so-called "high-contrast" parameter, and knowledge of the electrical conductivity is known to provide important information about, for example, the occurrence of cancerous tissue in humans and of pulmonary function of medical patients; or in the industrial setting, leaks in pipes and impurities in metals; cf. [39, 43, 55].

Unlike variations in wave speed or mass density, variations in the electrical conductivity do not cause scattering of acoustic waves, and hence it is not visible in ultrasound images. Neither does it interfere directly with the magnetic fields in MRI or X-ray beams in CT, making it "invisible" to these modalities [6]. It does, however, influence the flow of electrical current in the material:

Assume we are interested in knowing the electrical conductivity $\sigma(x)$ within some object occupying a region $\Omega \subset \mathbb{R}^d, d = 2, 3$. By connecting a set of electrodes to the boundary $\partial\Omega$ of the object, one applies an electric (voltage) potential $f(x)$ to the boundary of the object; this causes an electrical current pattern to form within the object. A widely accepted model for the electrical potential distribution $u(x)$ in Ω is given by the PDE [3, 55, 60],

$$\begin{cases} -\nabla \cdot (\sigma(x)\nabla u(x)) = 0, & x \in \Omega, \\ u(x) = f(x), & x \in \partial\Omega. \end{cases} \tag{5.1}$$

One then measures the electrical current $g_i = \sigma(x)\partial_\nu u(x)\big|_{\partial\Omega}$ on the boundary of the object. This is done for several different potential distributions f_i applied to the boundary, the inverse problem is to reconstruct the electrical conductivity σ inside Ω from the pairs $\{(g_i, f_i)\}$. This imaging method is known as electrical impedance tomography (EIT) and was first tested in the mid 1980s [36]. Since then, EIT has been subject to a lot of attention from both experimenters, medical practitioners and mathematicians [36, 55, 66]. Although commercial EIT-systems exist and are in use, there is a problem: The inverse problem of EIT is severly ill-posed, and the reconstructed images suffer from low spatial resolution and are highly unstable towards measurement noise and modelling errors [3, 36, 55]. To see why this is the case, we will briefly review the mathematical properties of EIT.

We assume the domain of interest Ω to be a smooth, bounded and connected in $\mathbb{R}^d$, $d = 2, 3$. The electrical conductivity σ is considered to belong to $L^\infty(\Omega)$ and to be real-valued, positive and bounded in the sense that there is some constant $C_\sigma > 0$ such that $C_\sigma^{-1} \leq \sigma(x) \leq C_\sigma$ almost everywhere. The Dirichlet-to-Neumann operator is defined as the operator that for a given σ takes the bound-

ary potential f as argument and returns the electrical current $\sigma(x)\partial_\nu u(x)\big|_{\partial\Omega}$ on the boundary $\partial\Omega$, where u is the solution to (5.1). We denote the Dirichlet-to-Neumann operator by Λ_σ. Using the tools from Chapter 2, it can be shown that Λ_σ is a bounded operator in $\mathcal{L}(H^{1/2}(\partial\Omega), H^{-1/2}(\partial\Omega))$ [3]. Moreover, we have the identity

$$(\Lambda_\sigma f, g)_{L^2(\partial\Omega)} = \int_\Omega \sigma \nabla u \cdot \nabla v \mathrm{d}x.$$

Here $u \in H^1(\Omega)$ is the solution to (5.1) with the boundary condition $f \in H^{1/2}(\partial\Omega)$, and $g = v\big|_{\partial\Omega}$ for $v \in H^1(\Omega)$.

Can we know Λ_σ? Knowing Λ_σ is equivalent to knowing the electrical current on the boundary *for every* voltage potential, i.e., knowing all possible pairs $\{f, \Lambda_\sigma f\}$. Assuming this rather demanding requirement to be fulfilled, we arrive at the so-called Calderón problem, named after Alberto Calderón who was the first to investigate this problem the 1980-paper "On an inverse boundary value problem" [23]:

Given Λ_σ, determine and recover σ.

This is a non-linear inverse problem, since $u = u(\sigma)$ clearly depends on σ and $\Lambda_\sigma f = \sigma(x)\partial_\nu u(\sigma)\big|_{\partial\Omega}$, and Calderón showed in his paper that the linearized problem is injective. Under the assumption that $\sigma \in C^2(\Omega)$ it has been shown that the mapping $\sigma \to \Lambda_\sigma$ is injective, i.e., that knowing Λ_σ uniquely determines σ. However, as previously mentioned, the inverse problem of EIT has very poor stability properties making the recovery problematic. In fact, the best possible stability estimate is

$$\|\sigma_1 - \sigma_2\|_{L^\infty(\Omega)} \leq C \left(\log(1 + \|\Lambda_{\sigma_1} - \Lambda_{\sigma_2}\|^{-1})\right)^{-\delta},$$

for a dimension dependent $\delta \in (0,1)$ [4, 54]. Here $\|\Lambda_\sigma\|$ is the operator norm on $\mathcal{L}(H^{1/2}(\partial\Omega), H^{-1/2}(\partial\Omega))$. This estimate seem perhaps mystifying at first, but a closer look reveals that even very precise knowledge of the Dirichlet-to-Neumann operator is not enough to obtain a good reconstruction of σ.

5.1 Hybrid Imaging Methods

The ill-posedness of the EIT-problem is a fact of nature, and cannot be circumvented by clever algorithms and increasingly improved measurement equipment. Therefore, the practical potential of EIT remains limited [3, 55]. EIT is not the only imaging method that suffers from such instability issues. Other examples are optical tomography, where one aims to reconstruct the optical parameters

of an object from boundary measurements of light [12], and microwave imaging, where one aims to reconstruct certain electrical properties of the object from measurements of electromagnetic waves [61]. In all these methods one attempts to reconstruct interior parameters from boundary measurements of a diffusive physical process.

As a way to overcome the limitations of these methods, a set of novel imaging techniques known as *hybrid imaging methods*[1] has been proposed. The unifying idea behind these methods is to utilize certain coupling mechanisms between different physical phenomena in a way that allows one to measure additional information, and that this information can make a severely ill-posed inverse problem less so [3, 6, 10, 13, 47, 50, 60].

By understanding the coupling mechanism and the governing physics of the coupled phenomena, one devises a way to combine different imaging modalities and equipment such that one can measure the additional information resulting from the coupling. Typically, one of the involved physical processes gives rise to an an inverse problem which is severely ill-posed(like EIT). The other, coupled physical process must be of a different type; its role is to either precisely measure or influence the other process though the coupling mechanism. Hence it has to be of a more controllable, precise nature and act as a good carrier of information. Recalling the development in Chapter 3, this suggests that the coupled process should be of wave type, i.e., a physical phenomenon described by a hyperbolic PDE.

Coupled physical phenomena are everywhere. A "striking" example of such a coupling comes from lightning and thunder; when a flash of lightning appears, it instantaneously heats up the air in its immediate surroundings. The heat causes the air to expand rapidly, and this creates an acoustic shock wave, the thunder, that can be observed far away from the lightning even if it is not visible to the eye. Though overly ferocious, this example is typical for the kind coupling mechanism that is present in hybrid inverse problems. Moreover, it is a good example what we saw in Chapter 3, i.e, of the fundamentally different ways the heat equation and wave equation transfer information. Had the heat not been turned into an acoustic wave, we would have a hard time noticing the lightning from far away, being forced to detect the resulting temperature changes.

[1] Also called "multi-wave imaging" or "coupled-physics imaging".

5.2 Acousto-electric Tomography

We now make matters concrete and give an example of a hybrid inverse problem involving EIT. In Acousto-electric tomography (AET), the idea is to combine the electro-static process involved in EIT with the acoustic waves generated by ultrasonic transducers. The coupling mechanism is a phenomenon known as the *acousto-electric effect*, where acoustic pressure causes a small, localized perturbation in the electrical conductivity of the medium it propagates through [44, 57]. The use of this effect for medical imaging was first proposed in [69].

We consider a setup for the EIT problem in which we apply a current f to the boundary $\partial\Omega$ and measure the resulting potential $u\big|_{\partial\Omega}$. In addition, we assume we can control the acoustic pressure $p(x,t)$ within the domain Ω, for example by the use of an ultrasonic transducer. A first-order approximation of the acousto-electric effect is

$$\sigma_p(x,t) = \sigma(x)(1 + \eta p(x,t)),$$

where $\eta \ll 1$ is the material dependent coupling parameter that determines the strength of the acousto-electric effect [3, 48]. Figure 5.1 illustrates a phantom conductivity and how it changes when it is perturbed by the acoustic wave penetrating the material. The time-dependent measurement $g_p(x,t) = u_p(x,t)\big|_{\partial\Omega}$ is now governed by the PDE

$$\begin{cases} -\nabla \cdot (\sigma_p(x,t)\nabla u_p(x,t)) = 0, & x \in \Omega, \\ \sigma_p(x,t)\partial_\nu u_p(x,t) = f(x), & x \in \partial\Omega. \end{cases} \tag{5.2}$$

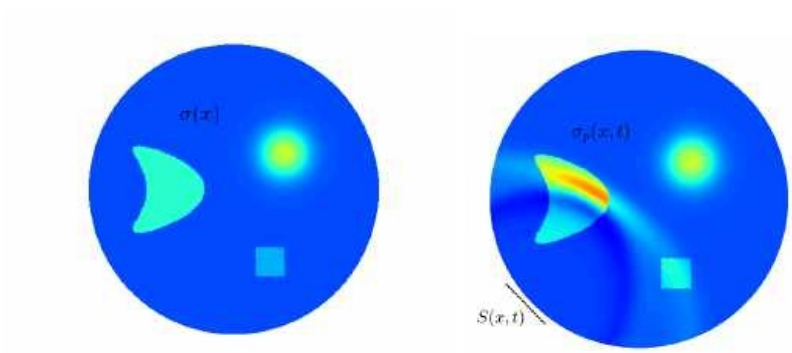

Figure 5.1: Right: the phantom conductivity σ. Left: The perturbed conductivity σ_p. $S(x,t)$ is the source of the acoustic wave.

Does this new equation add anything to the table? It does. To see this, denote by the unperturbed solution to (5.2) with $p = 0$ by u, and multiply (5.2) by u and vice versa with u_p. Integrating by parts and taking the difference we obtain

$$(f, g_p - g)_{L^2(\partial\Omega)} = -\eta \int_\Omega p(x,t)\sigma(x)\nabla u(x) \cdot \nabla u_p(x,t)\mathrm{d}x.$$

Under the (reasonable) assumption that the perturbation σ_p is small, we linearize u_p around $p = 0$, i.e., $u_p \approx u$. We then arrive at the equation

$$I(t) \approx -\eta \int_\Omega p(x,t)\sigma(x)|\nabla u|^2 \mathrm{d}x.$$

We now have a linear integral equation relating the known time-dependent measurement functional $I(t) = (f(x), g_p(x,t) - g(x))_{L^2(\partial\Omega)}$ and the so-called power density $H(x) = \sigma(x)|\nabla u(x)|^2$, with a integral kernel given by the acoustic pressure $p(x,t)$:

$$I(t) = -\eta \int_\Omega p(x,t)H(x)\mathrm{d}x, \quad t \in [0,T]. \tag{5.3}$$

If $p(x,t) \in L^2(\Omega \times [0,T])$ and we consider the integral equation as a mapping $L^2(\Omega) \to L^2([0,T])$, then it is a compact operator and equation 5.3 is known as a Fredholm equation of the first kind [45]. Still, if the applied pressure field is chosen with care, the inverse problem of reconstructing the power density $H(x)$ from $I(t)$ is only moderately ill-posed, and one can find approximate solutions using regularization methods. For example, for a highly focused pressure field, some authors have used the approximation $p(x,t) \approx \delta(x(t)-x)$, where $\delta(x(t)-x)$ is the Dirac delta distribution centered at a known position $x(t) \in \Omega$ [6]. If one can focus the pressure at all points $x(t) \in \Omega$, the problems becomes simple. On the other hand, if $p(x,t)$ is assumed to be plane waves of the form $\mathrm{e}^{i(\xi\cdot x - ct)}$ for sufficiently many different directions and frequencies ξ, then (5.3) resembles a Fourier transformation. In reality, the pressure is likely to be neither a plane wave or a fully focused wave, but somewhere in between, as in Figure 5.1.

If $H(x)$ can be reconstructed, it is apparent that the "hybridization" of the EIT problem is beneficial. We have augmented the boundary measurements with internal information in terms of the power density, and we now have to reconstruct $\sigma(x)$ from the knowledge of $\{f, g, H\}$, subject to

$$\begin{cases} -\nabla \cdot (\sigma(x)\nabla u(x)) = 0, & x \in \Omega, \\ \sigma\partial_\nu u(x) = f(x), & x \in \partial\Omega. \end{cases} \tag{5.4}$$

This second step, often called the "quantitative step" is considered to be only moderately ill-posed in the situation when one has data from several well-chosen imposed current distributions, i.e., $\{f_i, g_i, H_i\}_{i=1}^N$ [3, 8, 13, 42].

5.3 Photo- and Thermoacoustic Tomography

In recent decades, multiple different hybrid imaging methods has been proposed; cf the monographs [3, 6, 10, 60]. Some of the more well-known hybrid imaging methods that have been studied both theoretically and in practice are Thermoacoustic tomography (TAT), Photoacoustic tomography (PAT) and Magnetic resonance electrical impedance tomography (MREIT). These methods all share the feature that the additional information is some internal data depending on the parameter(s) one wants to image.

In MREIT, the patient or object of interest is equipped with the usual EIT-equipment and put inside an MRI-scanner. The MRI scanner can then reconstruct one or more of the components of the magnetic field $B = (B_x, B_y, B_z)$ within the object. Ampère's law states that

$$\sigma \nabla u = \eta_m^{-1} \nabla \times B,$$

where η_m is the magnetic permeability of the material and $\sigma \nabla u$ is the current density appearing due to the boundary potential from the EIT-equipment. If all components of the magnetic field are available, a differentiation yields the internal data $\sigma \nabla u$. If only one or two components are available, rotating the patient or object will provide extra information, typically of the form $\sigma |\nabla u|$. Hence, similarly to AET the MRI-reconstruction augments the EIT-problem with interior information and the result is in an inverse problem with much better stability and resolution properties [3, 6].

As with AET, Photo- and Thermoacoustic tomography are examples of hybrid imaging methods that also rely on the coupling between sound and electricity, but in the opposite direction. In AET the acoustic wave is used to perturb the electrical conductivity and hence the electric potential. In PAT and TAT the absorption of applied electromagnetic radiation causes a fast expansion of the material that gives rise to an acoustic wave. By measuring this wave with ultrasonic transducers on the boundary, one can reconstruct internal data, and this data is used to stabilize an initially severly ill-posed problem, similar to both in MREIT and AET.

In PAT, the goal is to reconstruct the optical parameters of biological tissue. Similarly to the electrical conductivity, the optical parameters are high contrast parameters and are of importance in both diagnostic and therapeutic applications [12, 41, 68]. Biological tissue is highly scattering and absorbing of electromagnetic radiation (light) in the infra-red frequency range. A simple model often used to describe the radiation intensity $u(x)$ within an object occupying

some region Ω is

$$\begin{cases} -\nabla \cdot (D(x)\nabla u(x)) + \mu(x)u(x) = 0, & x \in \Omega, \\ u(x) = f(x), & x \in \partial\Omega. \end{cases} \tag{5.5}$$

This PDE is known as the diffusion approximation; more accurate models such as the radiative transport equation or Maxwell's equations are also used [53]. In this model $D(x)$ is the scattering parameter, and it appears as a diffusion coefficient due to the highly scattering nature of the material. $\mu(x)$ is the absorption coefficient and accounts for the spatial absorption of light [3, 12, 15]. The Dirichlet boundary condition $f(x)$ represents the boundary value of the applied light source, often a pulsed laser. In diffuse optical tomography, one aims to reconstruct the optical parameters in (5.5) by measuring the outgoing light around the boundary, but the scattering and absorption is known to decrease the resolution of reconstructions exponentially with depth [56, 65, 67, 68]. By contrast, the absorption is the property that makes PAT work. When the light is absorbed, there is quick but minute heating of the tissue. The heating causes a rapid expansion of the tissue, and this gives rise to an acoustic wave. The phenomenon is known as the *photo-acoustic effect* and was first reported by Alexander G. Bell in 1880 in the report "Upon the production and reproduction of sound by light" [19]. The effect is also known as thermoelastic expansion, and the essential mechanism is the conversion of optical energy in to acoustic energy. A model regularly used for the light-to-acoustic wave coupling is

$$\begin{cases} (\partial_t^2 - c^2(x)\Delta)p(x,t) = 0, & (x,t) \in (\mathbb{R}^d \times (0,\infty)), \\ p(x,0) = H(x), \quad \partial_t p(x,0) = 0, & x \in \mathbb{R}^d. \end{cases} \tag{5.6}$$

Above $p(x,t)$ is the acoustic pressure, $c(x)$ is the sound speed and $H(x)$ is the expansion caused by the photo-acoustic effect,

$$H(x) = \Gamma(x)\mu(x)u(x),$$

where $u(x)$ is the radiation intensity (solution to (5.5)), $\mu(x)$ is the absorption parameter and $\Gamma(x)$ is the so-called Grüneisen parameter, a temperature and material dependent parameter determining the strength of the expansion [3, 10, 60]. $H(x)$ is assumed to be supported in $\Omega \subset \mathbb{R}^d, d = 2, 3$. Equation (5.6) is known as the "free-space model", since it does not incorporate the presence of a boundary in the acoustic domain; other models that include boundary conditions and acoustic absorption can be found in, e.g., [3, 37]. The acoustic pressure wave is measured on a subset (or the whole) Σ of boundary $\partial\Omega$ of the object for some time interval $[0, T]$, i.e.,

$$m(x,t) = p(x,t)\big|_{x\in\Sigma, t\in[0,T]},$$

and the first step of PAT is to reconstruct $H(x)$ from $m(x,t)$. Recalling Section 3.4, we saw that (5.6) gives rise to a fairly well-posed inverse problem, due to

the fact that the wave front set propagates to the measurement. The problem has achieved a lot of attention from researchers; cf [49] for a review. In the situation that Σ fulfils geometric constraints ensuring that all singularities of $H(x)$ are present in the measurement, and that the sound speed is known and non-trapping (see [3, 49, 63]), $H(x)$ can be uniquely reconstructed from $m(x,t)$ and the inversion is stable in the sense that

$$\|H\|_{L^2(\Omega)}\| \leq C\|m\|_{H^1(\Sigma\times[0,T])}. \tag{5.7}$$

Assuming $H(x)$ is reconstructed, the next step, known as *qualitative photoacoustic tomography* (qPAT), is to reconstruct the optical parameters $D(x)$ and $\mu(x)$ from (5.5) and knowledge of $H(x) = \Gamma(x)\mu(x)u(x)$. In practice, the Grüneisen parameter $\Gamma(x)$ is also an unknown. In [15] it was shown that all three unknowns (Γ, D, μ) cannot be reconstructed simultaneously, even in the case of multiple boundary illuminations f_i and corresponding H_i's.

In the same paper, it was also shown that once any one of the three parameters are known, the other two are uniquely determined by a pair of reconstructed acoustic perturbations (H_1, H_2) arising from well-chosen boundary illuminations (f_1, f_2). Therefore, one of the parameters (often Γ) are considered known. At least for theoretical results, the choice of boundary illuminations is of importance. They should be chosen such that $|H_1(x)\nabla H_2(x) - H_2(x)\nabla H_1(x)| > 0$ in Ω [3, 15, 17]. In [17], existence of such "well-chosen" boundary illuminations is shown by the use of complex geometrical optics solution. The drawback of this result is that the illuminations rely on the unknown coefficients and hence they are not constructive. In fact, in [2] it is shown that in $\mathbb{R}^3$, for any illumination there is a set of coefficients such that $\nabla u = 0$ somewhere in Ω. Nonetheless, under the assumption that well-chosen boundary illuminations has been imposed, the following stability estimate is given in [17]:

In $\mathbb{R}^d$, consider Γ to be known and let (D, μ) and $(\tilde{D}, \tilde{\mu})$ two pairs of optical coefficients. Assume that all coefficients are bounded from below and above, that $D|_{\partial\Omega} = \tilde{D}|_{\partial\Omega}$ and that $\sqrt{D}, \sqrt{\tilde{D}} \in C^{k+2}(\Omega)$ and $\mu, \tilde{\mu} \in C^{k+1}(\Omega)$, $k \geq 3$. Then there exists boundary conditions $\{g_i\}_{i=1}^{2d} \in C^{k+1}(\partial\Omega)$ such that the following stability estimate holds,

$$\|D - \tilde{D}\|_{C^{k-1}(\Omega)} + \|\mu - \tilde{\mu}\|_{C^{k-1}(\Omega)} \leq C \sum_{i=1}^{2d} \|H_i - \tilde{H}_i\|_{C^k(\Omega)},$$

where $\{H_i\}_{i=1}^{2d}$, $\{\tilde{H}_i\}_{i=1}^{2d}$ are the acoustic expansions associated with (D, μ) and $(\tilde{D}, \tilde{\mu})$, resulting from the illuminations $\{g_i\}_{i=1}^{2d}$, respectively. The requirement that $k \geq 3$ is not optimal, and can be lowered if data from additional illuminations are available [15, 17].

Thermoacoustic tomography is another, closely related hybrid imaging method. In TAT, the patient or object of interest is illuminated with a short pulse of electromagnetic radiation in the microwave frequency range. As in PAT, and essentially due to the same principle [3, 49], the absorption of the electromagnetic radiation causes a rapid increase in temperature that gives rise to a thermoelastic expansion. The expansion is the physical coupling mechanism and gives rise to an acoustic wave. The acoustic wave satisfies (5.6), but the initial condition $H(x)$ is on the form $H(x) = \sigma(x)|u(x)|^2$. Here $\sigma(x)$ is the electrical conductivity of the material and the electric field $u(x)$ is assumed to satisfy a scalar approximation of the Maxwells equations [3, 9, 16]:

$$\begin{cases} -\Delta u(x) + (k^2 + ik\sigma(x))u(x) = 0, & x \in \Omega, \\ u(x) = f(x), & x \in \partial\Omega. \end{cases} \tag{5.8}$$

Above k is the wave number of the imposed radiation $f(x)$. The first step in TAT-reconstructions is the same as in PAT, and consists of reconstructing the acoustic expansion $H(x)$ from the acoustic measurements. The second, qualitiative step (qTAT) is to reconstruct the conductivity $\sigma(x)$ from knowledge of (possibly multiple) $\{H_i(x)\}$ and subject to (5.8). As in PAT, the stability of qTAT relies on well-chosen boundary illuminations [3]. In [16], the stability estimate

$$\|\sigma - \tilde{\sigma}\|_{H^s(\Omega)} \leq C\|H - \widetilde{H}\|_{H^s(\Omega)}, \quad s \geq 0,$$

is given under the assumption that σ and $\tilde{\sigma}$ is bounded in $H^s(\Omega)$ by some constant M, and that the illuminations is of the complex geometrical optics type. We therefore conclude that if these criteria are met, TAT successfully accomplishes its goal of stablizing the initial inverse problem.

We have now seen some examples of hybrid inverse problems, and motivated their importance in medical imaging. Moreover, we have seen how they utilize the favorable properties of wave propagation to increase the stability and quality of reconstructions in inverse problems that are otherwise too ill-posed. There are, however, still many unresolved problems in the area of hybrid inverse problems. In the next Chapter we examine the contributions of the scientific papers in the appendix to some of these problems.

Chapter 6

Contributions

All the papers in the appendix are to a various degree concerned with different aspects of hybrid inverse problems. In the papers "Photo-acoustic tomography in the rotating setting" and "Feasibility of Acousto-electric tomography" we investigate two prototypical hybrid inverse problems. In the former, we consider uniqueness and stability properties; in the latter, modelling, feasibility and reconstruction algorithms are investigated. In the papers "Stable source reconstruction from a finite number of measurements in the multi-frequency inverse source problem" and "Stability of the inverse source problem for the Helmholtz equation in $\mathbf{R}^3$" we consider an inverse problem in which the boundary measurements of a wave equation is augmented with more information in terms of measurements at multiple frequencies. We now give a more detailed description of the contents of these papers in light of preceding chapters.

Paper 1: "Photo-acoustic tomography in the rotating setting"

In the paper "Photo-acoustic tomography in the rotating setting" we consider a measurement set-up that occurs in some applications of PAT. We consider the PAT-model presented previously, but where the acoustic waves caused by the absorbed illumination can only be measured on a small measurement sur-

face $\Sigma_i \subset \Sigma$, where Σ is a surface surrounding the object Ω. In this situation, the acoustic expansion $H(x)$ cannot be reconstructed from a single measurement of the acoustic wave, as no single acoustic measurement contains enough information [14, 63].

To remedy this, the light source(s) and acoustic measurement device is instead rotated around the object, collecting new measurements at each position $\{\Sigma_i\}_{i=1}^N$ [20, 52]. The main objective of the paper is to investigate whether the optical parameters of the object can be stably reconstructed from such rotating measurements. The difficulty that arises in this setting is that without direct access to the acoustic perturbation $H(x)$, the typical two-step procedure shared by many hybrid inverse problems does not apply. Instead, one has to consider a coupled measurement operator, i.e., an (non-linear) operator that for an illumination f_j maps the coefficients (D, μ) to the measurement $V_{i,j} = p(x,t)\big|_{\Sigma_i \times [0,T]}$. We construct an approximation to this operator, a composition of a pseudo-differential operator and a Fourier integral operator. The composed operator is a FIO, and we study the propagation of singularities from (D, μ) to the measurements. Through microlocal analysis of the wave propagation and by building on the results in [63], we give a criterion for the measurement surfaces Σ_i that ensures that all the singularities of (D, μ) are captured in the measurements. Careful analysis then shows that the measurement operator is invertible for a proper choice of illuminations, and we get the following stability result:

Theorem 6.1 *Assume the Grüneisen coefficient is known. Let (D, μ) and $(\bar{D}, \bar{\mu})$ be two pairs of optical parameters, giving rise to measurements $V_{i,j}$ and $\bar{V}_{i,j}$, respectively. If supp$(\sigma - \bar{\sigma})$ and supp$(D - \bar{D})$ are contained in a sufficiently small ball $B_\varepsilon \subset \Omega$, there exists for each rotation $1 \leq i \leq M$ an open set of $2d$ illuminations $\{g_{i,j}\}_{j=1}^{2d}$ such that the following estimate holds for the corresponding measurements,*

$$\|D - \bar{D}\|_{L^2(\Omega)} + \|\mu - \bar{\mu}\|_{H^1(\Omega)} \leq C \left(\sum_{i=1}^{M} \sum_{j=1}^{2d} \|V_{i,j} - \bar{V}_{i,j}\|_{H^1([0,T] \times \partial\Omega_M)} \right). \quad (6.1)$$

The constant C depends on $D, \bar{D}, \sigma, \bar{\sigma}, c, \Omega$ and the illuminations, and d is the spatial dimension.

The result shows that the information contained in the rotating measurements is sufficient to obtain stable reconstructions of the optical parameters, and demonstrate the advantageous stability properties of PAT. The paper also demonstrates how the methods introduced in Chapter 3 can be used to analyze a highly non-trivial problem and obtain a qualitative understanding of its properties. More importantly, the analysis indicates that PAT (and similar hybrid

inverse problems) do not rely on the typical two-step inversion procedure to exhibit favorable properties.

Paper 2: "Feasibility of Acousto-electric tomography"

The paper "Feasibility of Acousto-electric tomography" is concerned with two aspects of acousto-electric tomography. The first aspect is to propose and implement a complete inversion framework for AET. Most of the current literature on AET occupied with the qualitative step of reconstructing the conductivity from the power density; cf. [3, 7, 35, 42]. We consider both steps, i.e., the reconstruction of $H(x)$ from boundary measurements of the perturbed potential and the subsequent qualitative step; this is what we mean by complete inversion framework.

The second aspect of the paper to investigate if AET is feasible with realistic parameter values. It is known that the acousto-electric effect is very weak, i.e., the perturbation of the electrical conductivity caused by the applied acoustic pressure is small [29]. Due to this, it is unknown if measurements of the perturbed potential allows for reconstructions of the power density. To investigate these matters, we create synthetic data from realistic parameters (pressure waves, conductivity, applied current) and apply the inversion method to measurements corresponding to values of the coupling parameter in a range relevant for imaging of biological tissue.

Recalling equation (5.3), the reconstruction of the power density from the boundary measurements is accomplished by solving

$$I(t) = -\eta \int_\Omega p(x,t)H(x)\mathrm{d}x, \quad t \in [0,T], \tag{6.2}$$

for $H(x)$. Here, $I(t)$ is computed from the boundary measurements and the pressure $p(x,t)$ is considered known. By discretizing $p(x,t)$ and $H(x)$ using a spectral basis, we turn (6.2) into a linear system $I^h = K^h H^h$, where K^h is a matrix of moderate size, H^h are the basis coefficients of H, and I^h is $I(t)$ sampled at discrete time steps. We solve the discretized inverse problem by a regularized least squares method,

$$H^h = \min_{\hat{H}\in\mathbb{R}^n} \|K^h\hat{H} - I\|_2^2 + \alpha\|\nabla\hat{H}\|_2^2. \tag{6.3}$$

From reconstructions of several H_i^h resulting from $N = 3$ different boundary

currents f_i, we reconstruct σ solving the minimization problem

$$\sigma = \min_{\hat{\sigma}} \sum_{i=1}^{N} \|H_i(\hat{\sigma}) - H_i^h\|_{L^1(\Omega)} + \beta|\hat{\sigma}|_{\mathrm{TV}}. \tag{6.4}$$

This problem is solved numerically by the algorithm developed in [42].

To generate realistic measurements we create a multi-physics model. We simulate pressure waves from an ultrasonic transducer with the acoustic simulation toolbox k-Wave, and then solve equation (5.2) using a finite element method, with $\sigma_p(x)$ perturbed by the generated acoustic waves. This is done for several applied boundary currents f_i. In addition, we add noise to the voltage measurements at noise levels in a range typically associated with EIT measurements before we compute the resulting measurement functional $I_i(t)$. All physical parameters involved in the simulations are chosen to have values relevant for the imaging of biological tissue, and we take values of the coupling constant to lie in a range between the highest and lowest values reported in the literature.

The results are quite uplifting: The reconstruction framework preforms well, and in Figure 6.1 we see a plot of a phanton conductivity distribution and the reconstruction from data with 0.01% noise. Here, the coupling parameter is mid-range and set to $10^{-8}\mathrm{Pa}^{-1}$. Some of the sharper details are lost, but the reconstruction captures most of the features of the phantom. The numerical results from experimentation with different noise levels and values of the coupling parameter indicate that coupling is just strong enough to allow for satisfactory reconstructions of the electrical conductivity at realistic noise levels, and that had the coupling parameter been a factor 10^{-1} smaller, reconstructions would not be feasible.

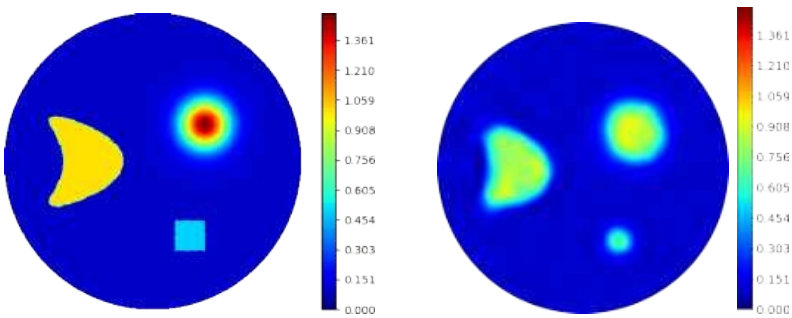

Figure 6.1: Left: The conductivity phantom. Right: The reconstruction of the same phantom from boundary measurements.

Paper 3-4: "Stable source reconstruction from a finite number of measurements in the multi-frequency inverse source problem" and "Stability of the inverse source problem for the Helmholtz equation in $\mathbf{R}^3$"

The papers "Stable source reconstruction from a finite number of measurements in the multi-frequency inverse source problem" and "Stability of the inverse source problem for the Helmholtz equation in $\mathbf{R}^3$" consider the inverse source problem for the Helmholtz equation in $\mathbb{R}^2$ and $\mathbb{R}^3$, respectively. This inverse problem has applications in both PAT and TAT, as well as other medical imaging methods; cf. [1, 11, 18]. It is a rewarding mathematical problem that allows one to study in depth the properties of inverse problems for wave propagation and utilize both spectral analysis and regularization methods.

We first introduce the probelm and describe the results from the papers. Next, we show how the inversion method from the papers can be applied to the reconstruction problem of the acoustic perturbation $H(x)$ in equation (5.6), i.e., to the first step of PAT and TAT.

The inverse source problem for the Helmholtz equation we consider is the following:

Let $R > R_0 > 0$ and $B_0, B \subset \mathbb{R}^d, d = 2, 3$, be concentric open balls centered at the origin, of radius R_0 and R, respectively. Assume the source $s \in L^2(B_0)$, and fix a positive frequency (wavenumber) k. Let u_k be the unique outgoing solution of the inhomogeneous Helmholtz equation in $\mathbb{R}^d$ with source term s,

$$\begin{cases} (\Delta + k^2)u_k &= s \quad \text{in } \mathbb{R}^d, \\ \lim\limits_{|x|\to\infty} |x|^{\frac{d-1}{2}}(\partial_{|x|} - ik)u_k &= 0 \quad \text{uniformly for } x/|x| \in S^{d-1}. \end{cases} \tag{6.5}$$

The solution u_k is given as a convolution of s with the fundamental solution,

$$u_k(x) = \int_{B_0} G_k(|x-y|)s(y)dy, \quad x \in \mathbb{R}^d, \tag{6.6}$$

where G_k is the outgoing fundamental solution of the Helmholtz equation in $\mathbb{R}^d$,

$$G_k(x,y) = \begin{cases} -\frac{i}{4}H_0^{(1)}(k|x-y|), & \text{in } \mathbb{R}^2, \\ \frac{e^{ik|x-y|}}{4\pi|x-y|}, & \text{in } \mathbb{R}^3. \end{cases} \tag{6.7}$$

Above, $H_0^{(1)}(z)$ is the Hankel function of the first kind and of order 0. The

measurement operator F_k at a frequency k maps the source to the measurement,

$$\begin{aligned} F_k \colon L^2(B_0) &\to L^2(\partial B), \\ s &\mapsto U_k = u_k\big|_{\partial B}, \end{aligned} \tag{6.8}$$

and F_k is a linear, compact operator given by the restriction of (6.6) to ∂B. We call U_k the measurement at frequency k. The *inverse source problem* associated with (6.5) is:

> Given a set of measurements $\{U_{k_j} = u_{k_j}|_{\partial B}\}$ performed over a frequency set $Q = \{k_j\}$, estimate the source term s.

Since we are only given the trace of the solution to (6.5) on ∂B, the inverse source problem with a single frequency measurement suffer from similar problems as the inverse problem for the Poisson equation that we saw earlier, i.e., it has a substantial kernel. The multi-frequency setting gives us more information; indeed, we show that with measurements at certain finite set of frequencies we can achieve stable, high resolution reconstructions. We present the background and main result in $\mathbb{R}^3$:

Let Λ_M be a finite set of eigenvalues for the Dirichlet-Laplacian on B_0, i.e., a set of positive real numbers $\{\lambda_n\}$ such that for each λ_n, there is one or more corresponding eigenfunctions φ_n satisfying the PDE

$$-\Delta\varphi_n(x) = \lambda_n\varphi_n(x), \quad x \in B_0, \quad \varphi_n(x) = 0, \quad x \in \partial B_0. \tag{6.9}$$

Next, we introduce the finite-dimensional source space associated with Λ_M,

$$S_{\Lambda_M}(B_0) = \operatorname{span}\{\varphi_n : \lambda_n \in \Lambda_M\}.$$

Since the set $\{\varphi_n\}_{n\in\mathbb{N}}$ is an orthonormal basis for $L^2(B_0)$, sources $s \in S_{\Lambda_M}(B_0)$ can be seen as projections of $s \in L^2(B_0)$ onto $S_{\Lambda_M}(B_0)$. The following theorem concerns the reconstructions of such sources from measurements $\{U_{k_j}\}_{k_j\in Q}$, where Q is the set of measurement frequencies.

Theorem 6.2 *Let $s \in S_{\Lambda_M}(B_0)$, for some finite set Λ_M of eigenvalues. There is a $\Delta k \in]0, 1/R_0]$ such that, if for each $\lambda_n \in \Lambda_M$ there is a measurement frequency $k_j \in Q$ satisfying $|k_j - \lambda_n^{1/2}| < \Delta k$, then the source s can be stably reconstructed from the set of measurements $\{U_{k_j}\}_{k_j\in Q}$.*

The theorem says that if we have measurements of the source at frequencies that are sufficiently close to the eigenvalues in Λ_M, then any source in $S_{\Lambda_M}(B_0)$ can be stably reconstructed from these measurements. Due to the increasing

density of the eigenvalues[1], we can find sets of measurement frequencies that are close to a much larger set of eigenvalues, and as a result one can reconstruct high-dimensional sources from measurements at a small number of frequencies. For general sources $s \in L^2(B_0)$ that don't necessarily belong to $S_{\Lambda_M}(B_0)$, we reconstruct a projection of s onto $S_{\Lambda_M}(B_0)$. In Figure 6.2 we show a reconstruction of a source in $\mathbb{R}^3$. The reconstruction uses measurements at 42 different frequencies, and the dimension of $S_{\Lambda_M}(B_0)$ is 3840. The measurements are contaminated with 10% noise, but still the reconstruction captures most features of the source well.

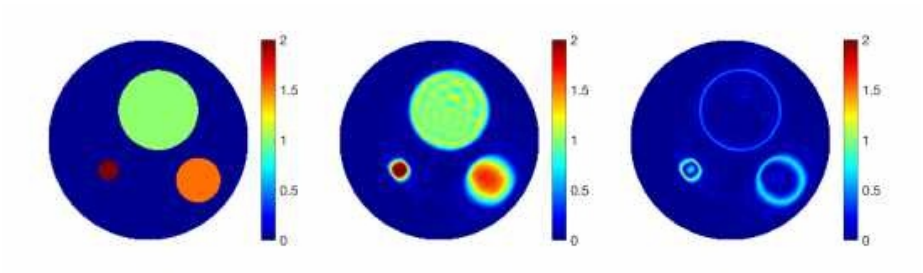

Figure 6.2: Plots are in the $(0, y, z)$-plane of B_0. Left: Source $s(x)$. Middle: Reconstruction $\hat{s}(x)$ of $s(x)$ from measurements $\{U_{k_j}\}_{k_j \in Q}$. Right: Pointwise absolute error $|\hat{s}(x) - s(x)|$.

In both papers we utilize a relationship between the singular vectors of the measurement operator F_k and the eigenvectors of the Dirichlet-Laplacian. By considering sources in $S_{\Lambda_M}(B_0)$, we explicitly construct a set of matrices that relate the multi-frequency measurements to the source. We analyze the invertibility of these matrices, and this results in a constructive understanding of the relationship between $S_{\Lambda_M}(B_0)$ and the measurement frequencies. By incorporating a characterization of the behavior of the singular values of the measurement operator, we devise a reconstruction method that resembles a TSVD-regularization method. The method proves to be remarkably stable with respect to noise in the measurements and has a low computational cost.

Relation to PAT and TAT

We now show that the inverse problem of reconstruction $H(x)$ in equation (5.6), i.e., the first step of PAT and TAT, can be posed as an inverse source problem for

[1] By increasing density we mean the increasing number of eigenvalues that lie in an interval of fixed length as the interval moves up the real line.

the Helmholtz equation. We denote by $\hat{p}(x,\omega)$ the temporal Fourier transform of the acoustic pressure,

$$\hat{p}(x,\omega) = \int_{\mathbb{R}} \mathrm{e}^{-i\omega t} p(x,t)\mathrm{d}t, \quad x \in \mathbb{R}^d, d = 2,3, \quad \omega \in \mathbb{R}. \tag{6.10}$$

We assume $p(x,t)$ satisfies equation (5.6), i.e.,

$$\begin{cases} (\partial_t^2 - c^2\Delta)p(x,t) = 0, & (x,t) \in (\mathbb{R}^d \times (0,\infty)), \\ p(x,0) = H(x), \quad \partial_t p(x,0) = 0, & x \in \mathbb{R}^d. \end{cases} \tag{6.11}$$

and that the wave speed c is constant. This is a reasonable assumption in, e.g., breast imaging where the tissue is assumed to be acoustically homogeneous. Taking the Fourier transform on both sides of (6.11), we find that $\hat{p}(x,\omega)$ satisfies the Helmholtz equation

$$(\Delta + k^2)\hat{p}(x,\omega) = ikH(x), \quad \text{for } k \in \mathbb{R}, \tag{6.12}$$

subject to the Sommerfeld radiation condition [25]. Here $k = \omega/c$ is the wave number and $H(x)$ is the acoustic perturbation. Let $m(x,t) = p(x,t)\big|_{\partial B \times (0,\infty)}$ be a boundary measurement of the pressure wave, where ∂B is the boundary of a sphere B surrounding the support B_0 of H. The temporal Fourier transform $\hat{m}(x,\omega)\big|_{\omega\in\mathbb{R}}$ of the measurement then satisfies (6.12) for each $\omega \in \mathbb{R}$, and the inverse problem of reconstructing H from the measurement $m(x,t)$ now appears in the form of a multi-frequency inverse source problem for the Helmholtz equation. The problem differs slightly from that considered in the two papers in that the source in (6.12) is multiplied by the wave number, but the following modification solves this:

Define $\hat{m}_{k_j} = \hat{m}(x,\omega_j)$, $\omega_j > 0$, and let U_{k_j} be the measurement of the solution to (6.5) with $H(x)$ as the source term and $k_j = \omega_j/c$. Due to the linearity of the measurement operator (6.6), we then have that

$$m_{k_j} = F_{k_j}(ik_jH) = ik_jF_{k_j}H = ik_jU_{k_j}.$$

Hence $U_{k_j} = m_{k_j}/ik_j$, and we can apply the reconstruction method for the multi-frequency inverse source problem to the data $\{m_{k_j}/ik_j\}_{k_j\in Q}$, where $Q = \{\omega_j/c\}$ and ω_j are the the frequencies for which we have computed the Fourier transform of $m(x,t)$.

Appendix

This appendix contain the following scientific papers.

1 G. Bal and A. Kirkeby, "Photo-acoustic tomography in the rotating setting." *Inverse Problems*, DOI: 10.1088/1361-6420/ab315c, 2019.

2 B. Jensen, A. Kirkeby and K. Knudsen, "Feasibility of Acousto-electric tomography." *Submitted manuscript.*

3 M. Karamehmedović, A. Kirkeby, and K. Knudsen. "Stable source reconstruction from a finite number of measurements in the multi-frequency inverse source problem." *Inverse Problems*, DOI: 10.1088/1361-6420/aaba83, 2018.

4 M. Karamehmedović, A. Kirkeby, M. T. R. Henriksen, "Stability of the inverse source problem for the Helmholtz equation in $\mathbf{R}^3$." *Submitted manuscript.*

Appendix A

Paper 1

Photo-acoustic tomography in the rotating setting

Guillaume Bal[1] and Adrian Kirkeby[2 3]

Abstract

Photo-acoustic tomography is a coupled-physics (hybrid) medical imaging modality that aims to reconstruct optical parameters in biological tissues from ultrasound measurements. As propagating light gets partially absorbed, the resulting thermal expansion generates minute ultrasonic signals (the photo-acoustic effect) that are measured at the boundary of a domain of interest. Standard inversion procedures first reconstruct the source of radiation by an inverse ultrasound (boundary) problem and second describe the optical parameters from internal information obtained in the first step.

This paper considers the rotating experimental setting. Light emission and ultrasound measurements are fixed on a rotating gantry, resulting in a rotation-dependent source of ultrasound. The two-step procedure we just mentioned does not apply. Instead, we propose an inversion that directly aims to reconstruct the optical parameters quantitatively. The mapping from the unknown (absorption and diffusion) coefficients to the ultrasound measurement via the unknown ultrasound source is modeled as a composition of a pseudo-differential operator and a Fourier integral operator. We show that for appropriate choices of optical illuminations, the above composition is an elliptic Fourier integral operator. Under the assumption that the coefficients are unknown on a sufficiently small domain, we derive from this a (global) injectivity result (measurements uniquely characterize our coefficients) combined with an optimal stability estimate. The latter is the same as that obtained in the standard (non-rotating experimental) setting.

1 Introduction

Photo-acoustic tomography (PAT) is a coupled-physics (also known as hybrid) imaging method that aims to reconstruct the optical parameters of biological tissues. The optical parameters are known to provide valuable, high-contrast information about for example cancerous tissue, and hence are of clinical interest [12, 13, 29, 35]. To image these parameters, PAT leverages the thermoelastic expansion generated by absorbed light. A domain of interest is illuminated by electromagnetic waves and the absorption of light within the

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sample causes a rapid expansion of the material proportional to the amount of absorbed photons; this conversion of light to ultrasound is what is known as the *photo-acoustic effect*. The expansion initiates a pressure wave, and this wave is recorded by acoustic detectors at the boundary of the sample. The objective of PAT is to reconstruct the optical parameters from such acoustic measurements.

A model approximating the propagation of light in the diffusive regime is the following [11, 17]:

$$\begin{aligned} -\nabla \cdot D\nabla u + \sigma u &= 0 \quad \text{in } \Omega, \\ u &= g \quad \text{on } \partial\Omega. \end{aligned} \tag{1}$$

Here $\Omega \in \mathbb{R}^d$ (typically $d = 2, 3$) is the object domain, $u(x)$ is the light/photon intensity, $D(x)$ is the diffusion coefficient and $\sigma(x)$ is the absorption parameter. The illumination is modelled as the boundary condition g. The thermo-elastic expansion is proportional to the amount of absorbed light, and is given by

$$H(x) = \mu(x)\sigma(x)u(x), \tag{2}$$

where $\mu(x)$ is a proportionality coefficient known as the Grüneisen coefficient. It is known that the reconstruction of the three parameters (σ, D, μ) is not possible without prior assumptions or multi-color measurements [8, 11]. In the rest of the paper, we assume $\mu(x)$ known.
To model the acoustic wave resulting from the thermoelastic expansion, we consider the linear wave equation

$$\begin{aligned} &(\partial_t^2 - c^2(x)\Delta)v(t,x) = 0, \quad \text{in } (0,\infty) \times \mathbb{R}^d, \\ &v(0,x) = H(x), \quad x \in \Omega, \\ &\partial_t v(0,x) = 0, \quad x \in \mathbb{R}^d, \end{aligned} \tag{3}$$

where $v(t,x)$ is the acoustic pressure and $c(x)$ is the sound speed, assumed to be known [32]. For some domain Ω_M such that $\Omega \subset \Omega_M$, one measures $v(t,x)$ for $(t,x) \in [0,T] \times \partial\Omega_M$ for some sufficiently large duration $T > 0$. We refer to this as the full data. The quantitative reconstruction of D and σ is then often considered in two successive steps:

1. The reconstruction of $H(x)$ from $v(t,x)\big|_{[0,T]\times\partial\Omega_M}$. In the case of full data, this is a well-posed problem and several inversion methods exist[20, 25, 32, 33].

2. The reconstruction of $D(x)$ and $\sigma(x)$ from $H(x)$ (assuming $\mu(x)$ known). This is quantitative PAT, which is known to be reasonably well-posed [11, 12, 17, 25].

The fact that both steps are well-posed is what makes PAT an attractive medical imaging modality. However, the requirements for the decoupling into two separate steps can be hard to meet in an experimental situation. Having access to a full measurement of the acoustic wave requires the illuminated object to be fully surrounded by acoustic measurement devices, a situation which is hard to achieve in most situations of interest. Since Step 1 cannot be stably completed without access to (sufficiently) full data, one instead relies on doing several partial measurements, changing the position and the illumination patterns for each measurement, for instance by rotating both the light source and ultrasound

detectors [13, 27]. In [10], such a situation, with the diffusion coefficient D constant, was considered. There it was shown that the absorption coefficient σ can be stably determined in situations where the object (or measurement device) is rotating. In the current paper we investigate the situation were both D and σ are unknown, and the measurements and illuminations are rotating. The presence of an additional, unknown parameter require in-depth mathematical analysis. We show that under certain assumptions on the parameters, this situation also allows for stable determination of both D and σ.

Let us conclude this introductory section by mentioning that PAT is one of the many hybrid (coupled-physics) medical imaging modalities that have emerged in recent years. For a brief list of mathematical description of such methods, we refer the reader to, e.g., [4, 5, 6, 23, 28, 30, 34] and their multiple references. The rest of the paper is organized as follows. The setting and main uniqueness and stability result are presented in section 2. The details of the derivation are collected in section 3 while some concluding remarks on the imaging modality and the mathematical tools are given in section 4.

2 Background and main results

2.1 Sound propagation

The linear acoustic wave equation is a reasonable description of sound propagation [10, 24, 32]. The sound speed $c(x)$ is considered known in this paper. Taking v to be the solution of

$$\begin{aligned} &(\partial_t^2 - c^2(x)\Delta)v(t,x) = 0, \quad \text{in } (0,\infty)\times\mathbb{R}^d, \\ &v(0,x) = f(x), \quad x\in\Omega, \\ &\partial_t v(0,x) = 0, \quad x\in\mathbb{R}^d, \end{aligned} \tag{4}$$

we define the acoustic measurement operator Λ as

$$\Lambda f = v(t,x)\big|_{[0,T]\times\partial\Omega_M}.$$

In the setting considered here, we do not have one, but rather a large number of ultrasound sources $f(x) = f_i(x)$ corresponding for instance to the rotating measurement setting, where a rotating set of light sources generates a different ultrasound source for each rotation. Since the ultrasound detectors also rotate in such a setting, $v(t,x) = \Lambda f$ is available only for x in the support of the rotating detector, which is a small fraction of the boundary $\partial\Omega_M$. As a consequence, we cannot reconstruct the whole $f(x)$ stably from such measurements [4]. All that we can reconstruct is the singularities of f that are visible from the available measurements. This mapping from the visible singularities of f to the available measurements is described by a standard microlocal procedure, which we now recall [32].

Consider the Hamiltonian system:

$$\begin{aligned} &\dot{X}(t) = \frac{c(X(t))}{2}\xi(t), \\ &\dot{\xi}(t) = -\frac{\nabla c(X(t))}{2}|\xi(t)|, \\ &X(0) = x_0, \quad \xi(0) = \xi_0. \end{aligned} \tag{5}$$

The curves $(X(t), \xi(t))$ are known as bi-characteristic curves, and each $X(t)$ is called a ray. We assume that $c(x) \in C^\infty(\mathbb{R}^d)$ is non-trapping, that is $|X(t)| \to \infty$ as $t \to \infty$, for all rays [24]. An important feature of the rays is that they are curves along which the propagation of the singularities of a wave occurs, a fact that motivates the preceding assumptions on the measurement geometry, see [10, 32]. To that end, we use the unit speed geodesics $\gamma_{x_0,\xi_0}(t)$, defined by the relation $\dot{\gamma}_{x_0,\xi_0}(t) = \dot{X}(t)/|\dot{X}(t)|$, where $X(t)$ is the solution to (5) with initial conditions (x_0, ξ_0).

We can then describe the measurement operator Λ has the following Fourier integral operator(FIO) [15, 32]:

$$v(t,x) = \Lambda f(t,x) = (2\pi)^{-d} \sum_{\tau=\pm} \int \mathrm{e}^{i\phi_\tau(t,x,\xi) - i\xi\cdot y} a_\tau(t,x,\xi) \hat{f}(\xi) \mathrm{d}\xi + Rf(t,x) \tag{6}$$

for $(t,x) \in [0,T] \times \partial\Omega_M$, where R is a linear operator with smooth Schwartz kernel. The phase functions $\phi_\pm$ are solutions to the eikonal equations

$$\mp\partial_t \phi_\pm = c(x)|\nabla_x \phi_\pm|, \quad \phi_\pm(0,x) = x \cdot \xi,$$

and homogeneous of order one in ξ. As usual,

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} \mathrm{e}^{-i\xi\cdot y} f(y) \mathrm{d}y.$$

Note that if $c = \text{const.}$, then $\phi_\pm = x \cdot \xi \pm c|\xi|t$, and the solution is exact. The function $a_\pm$ is called a classical amplitude of order 0, and satisfies a recursive transport equation; see ([15], page 128) for details.
We have assumed here that the FIO could be represented with a single (global) phase function ϕ_τ. This can always be done for sufficiently short times. For longer times, and to avoid the presence of caustics, the above operator should really be written as a composition of a finite number of such terms, or more generally as a globally defined FIO [15, 32]. To simplify notation, we represent our FIO, mapping all the singularities of f (that is f up to a smooth term) to the available measurements by (6).

2.2 Detector model

Assume a detector supported on Γ_d, a closed and bounded hypersurface in $\mathbb{R}^d$. The i'th measurement consists of acquiring $v(t,x)$ for $t \in [0,T]$ and $x \in \Gamma_i$, where Γ_i is a translation and rotation of Γ_d.
To obtain enough information on the optical coefficients, we make the following assumption on the measurement setting. We require overlapping measurements, i.e., that $\mu(\Gamma_i \cap (\cup_{j\neq i}\Gamma_j)) > 0$, for all i, where μ is the Lebesgue (surface) measure on Γ_d. For a collection of M measurements, we set the full measurement surface $\Gamma = \cup_{i=1}^M \Gamma_i$. We then assume we have a domain Ω_M such that $\Omega \subset \Omega_M$ and $\Gamma = \partial\Omega_M$. In the case of rotating measurements, we obtain that $\Gamma = \partial\Omega_M$ for some ball $\Omega_M = B_r$ of radius r containing Ω.

To describe our measurement setting, we follow the approach of [32] used to analyze the

inversion of the wave equation with incomplete data. Our requirements on Γ and T are the same as those required to get stable inversions with incomplete data. Define

$$\mathcal{G} = \{(t, x) : x \in \Gamma, 0 < t < s(x)\},$$

where $s(x)$ is a continuous function determining the temporal measurement interval. We introduce the function $\tau_\pm(x, \xi)$, defined as

$$\tau_\pm(x, \xi) = \max\{t \geq 0 : \gamma_{x,\xi}(\pm t) \in \Omega_M\},$$

where $\gamma_{x,\xi}(t)$ are the rays in (5). In order to make sure all of the singularities of the function $H(x)$ in (3) (and hence of the optical coefficients via (2)) are captured by the measurement, additional requirements on $\mathcal{G}$ (and hence Γ and the measurement interval length T) are

- $\forall x \in \Omega, \quad \exists z \in \Gamma$ such that $\mathrm{dist}(x, z) < s(z)$. Here $\mathrm{dist}(x, z)$ is the length of the geodesic connecting x and z, with respect to $\mathrm{d}s^2 = c^{-2}(x)\mathrm{d}x^2$. The conditions says that all of the wave should reach Γ during the measurement period.

- $\forall (x, \xi) \in T^*\Omega \backslash 0, \quad (\tau_k(x, \xi), \gamma_{x,\xi}(\tau_k(x, \xi)) \in \mathcal{G}$, for $k = +$ or $k = -$.
  Here $T^*\Omega = \Omega \times \mathbb{R}^d$ is the cotangent bundle of Ω, where we consider $\xi \neq 0$ only. The motivation for this condition is that every point in the wavefront set of H should reach the measurement surface.

The above conditions are satisfied if for example Γ is a sphere containing Ω and an interval such that $s(x) > \max_{x,z\in\bar{\Omega}_M} \mathrm{dist}(x, z)/2$ when the wave speed is constant. These conditions are also satisfied for non-trapping speeds for sufficiently long measurement times [32].

Last, for the measurement at Γ_i, let $\{\varphi_j\}_{j=1}^M$ be a partition of unity subordinate to $\{\Gamma_j\}_{j=1}^M$. Note that there is a $K_i \subset \Gamma_i \setminus (\cup_{j\neq i}\Gamma_j)$ such that $\varphi_i|_{K_i} = 1$, while $\varphi_j|_{K_i} = 0$ for $j \neq i$. Next, we take $\psi \in C_0^\infty([0, 2T])$ such that $\psi(t) = 1$ for $t \in [\varepsilon, T]$, with ε sufficiently small, and set $\chi_i(x, t) = \psi(t)\varphi_i(x)$. We now define our rotating measurement V_i at Γ_i by

$$V_i = \chi_i \Lambda f. \tag{7}$$

Note that $f \equiv H = \mu\sigma u$ above should really be read as $f = f_i = H_i = \mu\sigma u_i$, which is rotation dependent as the illumination $g = g_i$ in (1) rotates along with the detector Γ_i. There is therefore no hope to reconstruct all sources f_i from measurements of the form (7) (with f replaced by f_i) unless $\Gamma_i = \Gamma$. All the sources $f_i \equiv H_i$ have to be anchored to a rotation-independent object, namely the domain of interest modeled by the optical parameters (D, σ). This is the objective of the next section.

2.3 Light propagation and boundary conditions

The second order elliptic PDE (1) serves as a reasonable model for propagation of light in in highly scattering media such as biological tissues.
We now aim to understand how the optical parameters (D, σ) influence the ultrasound sources $H = \mu\sigma u$. Since $H \equiv H_i$ corresponds to rotating illuminations $g = g_i$ in (1), and rotating ultrasound measurements such as (7) do not allow full reconstructions of each H_i, we are also forced to understand such an influence locally, and in fact micro-locally. As in

the derivation of (6), the resulting pseudo-differential calculus requires enough smoothness for all the required Taylor-type expansions to make sense. We therefore assume that (D, σ) are smooth, as we did for the sound speed $c(x)$. A laborious, standard, tracking of all relevant calculations shows that only finitely many terms are necessary in each Taylor expansion. As a consequence, all results hold for (D, σ, c) of class C^k for k sufficiently large. We will present all results assuming $k = \infty$ both to simplify and stress that in practice, the difference between large k and yet larger k is somewhat immaterial.
Suppose that the object of interest occupies an open domain $\Omega \in \mathbb{R}^d, d = 2, 3$, where the boundary $\partial\Omega$ is C^∞. Assume $D \in C^\infty(\mathbb{R}^d)$, and such that there is a positive constant C_D and $C_D^{-1} \leq D \leq C_D$, and that $\text{supp}(D - 1) \subset \Omega$. For $0 \leq \sigma \in C_0^\infty(\mathbb{R}^d)$, assume there is some closed $\tilde{\Omega}$ such that $\Omega \subset \tilde{\Omega}$, and that $\text{supp}(\sigma) = \tilde{\Omega}$.

It is well-known in the non-rotating setting that stable reconstruction of both D and σ requires internal functionals $H(x)$ from multiple different illuminations, and that appropriate illuminations exist [11, 12]. Consider such a set of boundary conditions $\{g_j\}_{j=1}^N$. When the object is rotating and the detectors and illumination patterns are fixed, or equivalently, the detector and illumination patterns rotate, as for example in [14, 27], we get a different set of N illuminations for each rotation $i = 1, 2, ...M$, i.e., $\{g_{i,j}\}_{j=1}^N$. Now, for a fixed i, let $u_{i,j}$ be the solution to

$$\begin{aligned} -\nabla \cdot D\nabla u_{i,j} + \sigma u_{i,j} &= 0 \quad \text{in } \Omega, \\ u_{i,j} = g_{i,j} \quad \text{on } \partial\Omega, &\quad j = 1, 2, ..., N. \end{aligned} \tag{8}$$

If the $g_{i,j}$ are smooth, we have that $u_{i,j} \in C^\infty(\Omega)$ ([16], Theorem 3, Chapter 6.3).
We already mentioned that, regardless of the number of illuminations, one cannot reconstruct both D,σ and the Grüneisen coefficient μ [11]. To simplify notation, we assume μ known and set

$$H_{i,j}(x) = \sigma(x)u_{i,j}(x),$$

where $u_{i,j}$ is the solution to (8). Indexing the acoustic measurements accordingly, we have that

$$V_{i,j} = \chi_i \Lambda H_{i,j}, \quad \text{for } i = 1, 2, ..., M, \quad j = 1, 2, ..., N. \tag{9}$$

This provides a full description of our measurement setting. The parameter N provides the diversity in boundary illuminations that is necessary to stably reconstruct the optical parameters in the second step of standard quantitative PAT [11, 12]. The parameter M indicates the number of rotations necessary to obtain measurements over all of Γ, and so roughly corresponds to the ratio between the size (volume of the hypersurface) of Γ divided by that of the support of the rotating detector.

2.4 Main results

We can now state the main result of the paper. This is, under some restrictive assumptions, an injectivity and stability result. It does not provide a reconstruction algorithm, merely the reassurance that enough information has been collected to uniquely and stably characterize the unknown optical coefficients.
We therefore consider two pairs of admissible diffusion and absorption coefficients (D, σ) and $(\bar{D}, \bar{\sigma})$, satisfying the assumptions from Section 2.3. Since the PDO and FIO calculus

we use here is not meant to handle boundaries, we have to assume that the values of the coefficients agree in a neighborhood of $\partial\Omega$. For fixed i, j, let $u_{i,j}$ and $\bar{u}_{i,j}$ be solutions to (8) with illumination $g_{i,j}$ and optical parameters (D, σ), $(\bar{D}, \bar{\sigma})$, respectively. Denote $\delta H_{i,j} = \sigma u_{i,j} - \bar{\sigma}\bar{u}_{i,j}$, $\delta D = D - \bar{D}$ and $\delta\sigma = \sigma - \bar{\sigma}$. We collect the measurements for each illumination and write their difference as

$$\delta V_{i,j} = V_{i,j} - \overline{V}_{i,j} = \chi_i \Lambda \delta H_{i,j}, \quad \text{for } i = 1, 2, ..., M, \quad j = 1, 2, ..., N. \tag{10}$$

We use here the fact that the wave propagation step is linear: the measurements are linear in the source terms H_i. The full inverse problem, which maps (D, σ) to such measurements, is however nonlinear. In spite of this, we will show in the next section that the terms $\delta H_{i,j}$ can be written as a functional that is nonlinear in $(D, \sigma, \bar{D}, \bar{\sigma}, u_{i,j}, \bar{u}_{i,j})$ but linear in $(\delta D, \delta\sigma)$, essentially as a generalization of the fact that for any polynomial $p(x)$, we can find another polynomial $q(x, y)$ such that $p(x) - p(y) = q(x, y)(x - y)$.
Moreover, we will show in the next section that there exist open sets of illuminations $\{g_{i,j}\}$ such that the mapping from $(\delta D, \delta\sigma)$ to $\{\delta H_{i,j}\}$ may be described as an elliptic pseudo-differential operator (PDO) with a symbol that depends on $(D, \sigma, \bar{D}, \bar{\sigma}, u_{i,j}, \bar{u}_{i,j})$. The first result along these lines was obtained in the non-rotating setting (with $M \equiv 1$) in [25].
Combining the elliptic FIO in (6) with the above elliptic PDO shows that the mapping from $(\delta D, \delta\sigma)$ to δV_j is itself an elliptic FIO with symbol that depends on the $(D, \sigma, \bar{D}, \bar{\sigma}, u_{i,j}, \bar{u}_{i,j}, c(x))$, which must all be sufficiently smooth for the calculus to apply.
Such micro-local results provide optimal stability estimates as well as an injectivity result provided that a smoothing compact operator does not have eigenvalue one. This assumption is notoriously difficult to verify, unless we have recourse to a smallness assumption somewhere. In [10], where the PAT problem with constant D is considered, the smallness assumption was on the size of the absorption coefficient. Here, we make no assumption on the $O(1)$ size of the absorption and diffusion coefficients. Rather, we assume that the support of the domain where $\delta D = D - \bar{D}$ and $\delta\sigma = \sigma - \bar{\sigma}$ are unknown is itself sufficiently small. This will prove to be a sufficient assumption to obtain the following result.

Theorem 1. *If supp$(\sigma - \bar{\sigma})$ and supp$(D - \bar{D})$ are contained in sufficiently small ball $B_\varepsilon \subset \Omega$, there exist for each rotation $1 \leq i \leq M$ an open set of $2d$ illuminations $\{g_{i,j}\}_{j=1}^{2d}$ such that the following estimate holds for the corresponding measurements.*

$$\|\delta D\|_{L^2(\Omega)} + \|\delta\sigma\|_{H^1(\Omega)} \leq C \left(\sum_{i=1}^{M} \sum_{j=1}^{2d} \|\delta V_{i,j}\|_{H^1([0,T]\times\partial\Omega_M)} \right). \tag{11}$$

The constant C depends on $D, \bar{D}, \sigma, \bar{\sigma}, c, \Omega$ and the illuminations.

Additional remarks on the set of necessary illuminations will be provided in the next section. We note that in dimension $d \geq 3$, no boundary conditions g guarantee the necessary ellipticity assumptions to obtain the optimal stability estimates given in the above theorem independently of the coefficients (σ, D) [3, 4].

3 Construction of elliptic operator

In lines with [10, 25], we show that the problem mapping the unknown coefficients to the (also unknown) ultrasound sources $\delta H_{i,j}$ is described by an elliptic pseudo-differential

operator for an appropriate choice of illuminations and for smooth coefficients and solutions of (1).

3.1 Parametrix for $\delta H_{i,j}$

Let the situation be as described in Sections 2.3-2.4. Denote $\delta u_{i,j} = u_{i,j} - \bar{u}_{i,j}$. Subtracting the equations (8) for $u_{i,j}$ and $\bar{u}_{i,j}$, we obtain an equation for $\delta u_{i,j}$ in terms of δD and $\delta\sigma$:

$$\begin{aligned} -\nabla \cdot D\nabla\delta u_{i,j} + \sigma\delta u_{i,j} &= \nabla \cdot \delta D\nabla\bar{u}_{i,j} - \delta\sigma\bar{u}_{i,j} \quad \text{in } \Omega, \\ \delta u_{i,j} &= 0 \quad \text{on } \partial\Omega. \end{aligned} \tag{12}$$

We define the operators

$$\begin{aligned} P_l(x,\partial) &= -\nabla \cdot D(x)\nabla + \sigma(x), \\ P_r^{i,j}(x,\partial) &= \nabla\bar{u}_{i,j}(x) \cdot \nabla + \Delta\bar{u}_{i,j}(x), \end{aligned}$$

with symbols

$$\begin{aligned} p_l(x,\xi) &= D(x)\xi^2 - i\xi \cdot \nabla D(x) + \sigma(x) \in S^2(\Omega,\mathbb{R}^n), \\ p_r^{i,j}(x,\xi) &= -i\nabla\bar{u}_{i,j}(x) \cdot \xi + \Delta\bar{u}_{i,j} \in S^1(\Omega,\mathbb{R}^n). \end{aligned}$$

Recall that a function $p(x,\xi) \in C^\infty(\Omega \times \mathbb{R}^d)$ is said to be a symbol of class $S^m(\Omega \times \mathbb{R}^d)$, $m \in \mathbb{R}$, if for any multi-indices $\alpha, \beta \in \mathbb{N}^d$, there is a constant $C_{\alpha,\beta,\Omega,}$ such that

$$|D_x^\alpha D_\xi^\beta p(x,\xi)| \le C_{\alpha,\beta,\Omega}(1+|\xi|)^{m-|\beta|}, \quad \forall (x,\xi) \in \Omega \times \mathbb{R}^d.$$

In particular, the symbols p_l and $p_r^{i,j}$ are homogeneous in ξ, and their principal symbols are $p_{l,0}(x,\xi) = D(x)\xi^2$ and $p_{r,0}^{i,j}(x,\xi) = -i\nabla\bar{u}_{i,j} \cdot \xi$. Since $D(x)$ is positive on Ω, P_l is elliptic and $p_{l,0}(x,\xi) \sim \xi^2$. On the other hand, $p_{r,0}^{i,j}(x,\xi)$ is not elliptic at points (x_0,ξ_0) such that $\nabla\bar{u}_{i,j}(x_0) \cdot \xi_0 = 0$. Let Q be the parametrix (an inverse modulo smooth terms) of P_l, with symbol q and principal term q_0. Since $D(x), \sigma(x) > 0$ in Ω, we have that $q(x,\xi) = \frac{1}{\xi^2+1} \bmod S^{-3}(\Omega,\mathbb{R}^n)$, meaning that $q - \frac{1}{\xi^2+1} \in S^{-3}(\Omega,\mathbb{R}^d)$. Now we can solve (12) for $\delta u_{i,j}$, modulo smooth terms:

$$\delta u_{i,j} = QP_r^{i,j}\delta D - Q\bar{u}_{i,j}\delta\sigma.$$

The symbol $r^{i,j}$ of the composition $QP_r^{i,j}$ has the asymptotic expansion

$$r^{i,j} \sim \sum_{\alpha\in\mathbb{N}_0^d} \frac{1}{\alpha!}\partial_\xi^\alpha q D_x^\alpha p_r^{i,j}, \quad \text{where } D_x^\alpha = (-i)^{|\alpha|}\partial_x^\alpha,$$

and hence the principal part is

$$r_0^{i,j}(x,\xi) = \frac{-i\nabla\bar{u}_{i,j}(x) \cdot \xi}{\xi^2+1} \bmod S^{-2}(\Omega,\mathbb{R}^n).$$

Next, consider $\delta H_{i,j} = H_{i,j} - \bar{H}_{i,j} = \sigma u_{i,j} - \bar{\sigma}\bar{u}_{i,j}$. Rearranging, we have

$$\delta H_{i,j} = \sigma\delta u_{i,j} + \delta\sigma\bar{u}_{i,j},$$

and hence we can write $\delta H_{i,j}$ as the result of a pseudo-differential operator acting on δD and $\delta\sigma$, i.e,

$$\delta H_{i,j} = \sigma(QP_r^{i,j}\delta D - Q\bar{u}_{i,j}\delta\sigma) + \bar{u}_{i,j}\delta\sigma,$$

modulo smooth terms. We define

$$\mathcal{H}_D^{i,j} = \sigma Q P_r^{i,j}, \qquad \mathcal{H}_\sigma^{i,j} = \bar{u}_{i,j}(I - Q\sigma), \quad \text{and} \quad \mathcal{H}^{i,j} = [\mathcal{H}_D^{i,j}, \mathcal{H}_\sigma^{i,j}]. \tag{13}$$

In terms of (13) we have

$$\delta H_{i,j} = \mathcal{H}^{i,j}\begin{bmatrix}\delta D\\ \delta\sigma\end{bmatrix} = \mathcal{H}_D^{i,j}\delta D + \mathcal{H}_\sigma^{i,j}\delta\sigma + S_{i,j}\delta p,$$

where $S_{i,j}$ is a linear operator with smooth Schwartz kernel and $\delta p := (\delta D, \delta\sigma)$. Let us finally note that the principal symbols of $\mathcal{H}_D^{i,j}$ and $\mathcal{H}_\sigma^{i,j}$ are $h_{D,0}^{i,j}(x,\xi) = r_0^{i,j}(x,\xi)$ and $h_{\sigma,0}^{i,j}(x,\xi) = \bar{u}_{i,j}(x)$, respectively.

3.2 Coupling with wave propagation

From Section 2.1 and (9), the measurement $V_{i,j}$ is given by

$$V_{i,j} = \chi_i(2\pi)^{-d}\sum_{\tau=\pm}\int\int \mathrm{e}^{i\phi_\tau(t,x,\xi)-i\xi\cdot y}a_\tau(t,x,\xi)H_{i,j}(y)\mathrm{d}y\mathrm{d}\xi + \chi_i R H_{i,j}.$$

Letting $V_{i,j}$ and $\bar{V}_{i,j}$ be solutions to (4) with initial conditions $H_{i,j}$ and $\bar{H}_{i,j}$ respectively, we set $\delta V_{i,j} = V_{i,j} - \bar{V}_{i,j}$. Due to the linearity of (4), we have

$$\begin{aligned}\delta V_{i,j} &= \chi_i(2\pi)^{-d}\sum_{\tau=\pm}\int\int \mathrm{e}^{i\phi_\tau(t,x,\xi)-i\xi\cdot y}a_\tau(t,x,\xi)\delta H_{i,j}(y)\mathrm{d}y\mathrm{d}\xi + \chi_i R\delta H_{i,j}\\ &= \chi_i(2\pi)^{-d}\sum_{\tau=\pm}\int\int \mathrm{e}^{i\phi_\tau(t,x,\xi)-i\xi\cdot y}a_\tau(t,x,\xi)\left(\mathcal{H}_D^{i,j}\delta D + \mathcal{H}_\sigma^{i,j}\delta\sigma\right)\mathrm{d}y\mathrm{d}\xi + \chi_i R S_{i,j}\delta p.\end{aligned}$$

The composition of an FIO with a PDO is well defined: from ([15], Theorem 4.2), the resulting operator is again a FIO with the same phase function and with a amplitude function with asymptotic expansion given by

$$c_\tau^{i,j}(t,x,y,\xi) \sim \sum_{\alpha\in\mathbb{N}_0^d}\frac{i^{-|\alpha|}}{\alpha!}\partial_y^\alpha\left(a_\tau(t,x,\xi)\partial_\xi^\alpha(h_D^{i,j}(y,\xi) + h_\sigma^{i,j}(y,\xi))\right), \quad \tau = \pm. \tag{14}$$

From [32], we know that a_τ is a zeroth order amplitude with $a_{\tau,0} = \frac{1}{2}$. Hence the principal term of (14) is $\frac{1}{2}(h_{D,0}^{i,j}(y,\xi) + h_{\sigma,0}^{i,j}(y,\xi))$ mod $S^{-2}(\Omega\times\mathbb{R}^d), \tau = \pm$.
The forward map now takes the form

$$\delta V_{i,j} = \chi_i\Lambda\mathcal{H}^{i,j}\begin{bmatrix}\delta D\\ \delta\sigma\end{bmatrix} = \chi_i(2\pi)^{-d}\sum_{\tau=\pm}\int\int \mathrm{e}^{i\phi_\tau(t,x,\xi)-i\xi\cdot y}c_\tau^{i,j}(t,x,y,\xi)\begin{bmatrix}\delta D\\ \delta\sigma\end{bmatrix}\mathrm{d}y\mathrm{d}\xi, \tag{15}$$

modulo a smooth term $Q_{i,j}\delta p$ involving a linear operator $Q_{i,j}$ with smooth Schwartz kernel.

3.3 Time-reversal

To bring us back from boundary measurements to objects (such as δp) that are defined on the spatial domain Ω, we apply the time-reversal inversion to the composed operator in (15). The time reversal operator is an approximate inverse to Λ, which we denote by A [20, 32, 33]. For given data $h \in H^1([0,T] \times \partial\Omega_M)$, let w be the solution of

$$\begin{aligned}
&(\partial_t^2 - c(x)^2\Delta)w(t,x) = 0, \quad \text{in } (0,\infty) \times \mathbb{R}^n,\\
&w(t,x) = h, \quad (t,x) \in [0,T] \times \partial\Omega_M,\\
&w(T,x) = \phi,\\
&\partial_t w(T,x) = 0,
\end{aligned}$$

where ϕ solves $\Delta\phi = 0, \phi|_{\partial\Omega_M} = h(T,\cdot)$. Then $Ah = w\big|_{t=0,x\in\Omega}$. For $f \in C_0^\infty(\Omega)$, we have that $f - A\Lambda f = Kf$, where K is a compact operator with $\|K\| < 1$. See [32] for details. For a fixed measurement position i and illuminations $g_{i,j}$, we now consider the system

$$A \begin{bmatrix} \delta V_{i,1} \\ \vdots \\ \delta V_{i,N} \end{bmatrix} = A \begin{bmatrix} \chi_i \Lambda \mathcal{H}^{i,1} \\ \vdots \\ \chi_i \Lambda \mathcal{H}^{i,N} \end{bmatrix} \begin{bmatrix} \delta D \\ \delta\sigma \end{bmatrix} + AQ_i\delta p, \tag{16}$$

where we stacked the $\delta V_{i,j}$ and applied the time reversal operator to each side. We now define

$$\kappa_i = A \begin{bmatrix} \chi_i \Lambda \mathcal{H}^{i,1} \\ \vdots \\ \chi_i \Lambda \mathcal{H}^{i,N} \end{bmatrix}.$$

For a more compact notation in what follows, we write

$$\tilde{\chi}_i(x,\xi) = \frac{1}{2}\big(\chi_i(\tau_+(x,\xi), \gamma_{x,\xi}(\tau_+(x,\xi))) + \chi_i(\tau_-(x,\xi), \gamma_{x,\xi}(\tau_-(x,\xi)))\big).$$

We will need the following lemma.

Lemma 1. *The operator κ_i is a pseudo-differential operator with principal symbol κ_i^0 given by*

$$\kappa_i^0(x,\xi) = \begin{bmatrix} \tilde{\chi}_i(x,\xi)\frac{-i\nabla\bar{u}_{i,1}(x)\cdot\xi}{\xi^2+1} & \tilde{\chi}_i(x,\xi)\bar{u}_{i,1}(x) \\ \vdots & \vdots \\ \tilde{\chi}_i(x,\xi)\frac{-i\nabla\bar{u}_{i,N}(x)\cdot\xi}{\xi^2+1} & \tilde{\chi}_i(x,\xi)\bar{u}_{i,N}(x) \end{bmatrix}. \tag{17}$$

Proof. It follows the linearity of A and Theorem 3 in [32] that $A(\chi_i\Lambda)$ is a pseudo-differential operator of order zero with principal symbol $\tilde{\chi}_i(x,\xi)$. Since $A(\Lambda\mathcal{H}^{i,j}) = (A\Lambda)\mathcal{H}^{i,j}$, we have that $A\left(\chi_i\Lambda\mathcal{H}^{i,j}\right)$ is again a pseudo-differential operator with symbol $\tilde{\chi}_i(x,\xi)(h_D^{i,j}(x,\xi) + h_\sigma^{i,j}(x,\xi))$. Considering the expressions for $h_D^{i,j}(x,\xi)$ and $h_\sigma^{i,j}(x,\xi)$ found in Section 3.1, equation (17) follows. □

By applying the adjoint of κ_i, κ_i^* to (16) and summing over i we get the system

$$\delta\mathcal{V} = \sum_{i=1}^{M} \kappa_i^*\kappa_i \begin{bmatrix} \delta D \\ \delta\sigma \end{bmatrix} + P\delta p, \tag{18}$$

where

$$\delta\mathcal{V} = \sum_{i=1}^{M} \kappa_i^* A \begin{bmatrix} \delta V_{i,1} \\ \vdots \\ \delta V_{i,N} \end{bmatrix} \quad \text{and} \quad P = \sum_{i=1}^{M} \kappa_i^* A Q_i. \tag{19}$$

Now $\delta\mathcal{V}$ contains all the measured data, and the linear operator P has a smooth Schwartz kernel.

We thus have replaced our inversion problem by the analysis of equation (18).

3.4 Analysis of the system of pseudo-differential operators

We now investigate the system

$$\sum_{i=1}^{M} \kappa_i^* \kappa_i \begin{bmatrix} \delta D \\ \delta\sigma \end{bmatrix} + P\delta p = \delta\mathcal{V}. \tag{20}$$

Such a system is called elliptic when the principal symbol $\sum_{i=1}^{M} (\kappa_i^0)^* \kappa_i^0$ of the operator $\sum_{i=1}^{M} \kappa_i^* \kappa_i$ is full rank for all $(x,\xi) \in T^*\Omega$, [2, 19]. We show that there exist suitable sets of illuminations such that this is the case.

Since each of the terms in $\sum_{i=1}^{M} (\kappa_i^0)^* \kappa_i^0$ is positive semi-definite, the sum will also be positive semi-definite. Hence it suffices to show that for each $(x,\xi) \in T^*\Omega$, at least one term is positive definite, as this will then guarantee the positive definiteness and full rank property of the sum. Since $(\kappa_i^0)^* \kappa_i^0$ is positive when κ_i^0 is full rank, we aim to find illuminations such that for every $(x,\xi) \in T^*\Omega$, there is some κ_i^0 with full rank. Recall that

$$\kappa_i^0(x,\xi) = \begin{bmatrix} \tilde{\chi}_i(x,\xi) \frac{-i\nabla\bar{u}_{i,1}(x)\cdot\xi}{\xi^2+1} & \tilde{\chi}_i(x,\xi)\bar{u}_{i,1}(x) \\ \vdots & \vdots \\ \tilde{\chi}_i(x,\xi) \frac{-i\nabla\bar{u}_{i,N}(x)\cdot\xi}{\xi^2+1} & \tilde{\chi}_i(x,\xi)\bar{u}_{i,N}(x) \end{bmatrix}. \tag{21}$$

From the assumptions on the non-trapping wave speed and sufficiently long measurement time, we know that for each $(x,\xi) \in T^*\Omega$, at least one $\tilde{\chi}_i(x,\xi) > 0$. For $(x,\xi) \in T^*\Omega$ and such a rotation index $i = i(x,\xi)$, we need to show that $\kappa_i^0(x,\xi)$ is full rank.
We denote the determinant of 2×2 matrix consisting of rows m and n of (21) by $q_{m,n}(x,\xi)$. Then

$$q_{m,n}(x,\xi) = \tilde{\chi}_i^2(x,\xi) \frac{i}{1+\xi^2} \xi \cdot (\nabla\bar{u}_{i,m}\bar{u}_{i,n} - \nabla\bar{u}_{i,n}\bar{u}_{i,m}). \tag{22}$$

For $\kappa_i^0(x,\xi)$ to be full-rank, we thus require that there is at least one $q_{m,n}(x,\xi) \neq 0$ for every $(x,\xi) \in \Omega \times S^{d-1}$. Here S^{d-1} denotes the unit sphere in $\mathbb{R}^d$, and replaces $\mathbb{R}^d \setminus \{0\}$ since only the direction of ξ is of importance.
Hence we seek to find illuminations $\{g_{i,j}\}$ such that for some indices (m,n) it holds that

$$\tilde{q}_{m,n}(x,\xi) = \xi \cdot (\nabla\bar{u}_{i,m}\bar{u}_{i,n} - \nabla\bar{u}_{i,n}\bar{u}_{i,m}) \neq 0, \quad (x,\xi) \in \Omega \times S^{d-1}, \quad \forall 1 \leq i \leq M. \tag{23}$$

Expressions similar to (23) appears in the literature on PAT, e.g., in [4, 11, 12, 25].

For (23) to hold, we need that the vector fields

$$v_{m,n}(x) = (\nabla \bar{u}_{i,m}\bar{u}_{i,n} - \nabla \bar{u}_{i,n}\bar{u}_{i,m})(x)$$

constitute a basis for $\mathbb{R}^d$ for each $x \in \Omega$. In that way, a direction $\xi_0 \in S^{d-1}$ can never be orthogonal to all $v_{m,n}$. It is therefore clear that we must have at least $d+1$ different illuminations $\{g_{i,j}\}_{j=1}^{d+1}$. We will use complex geometric optics (CGO) solutions to show that there exists a set of $2d$ illuminations such that the vector fields $v_{m,n}$ does indeed form a basis for $\mathbb{R}^d$ for each $x \in \Omega$, and also show that under certain restrictions on the optical coefficients, the same boundary conditions will work for all rotations. We briefly introduce the CGO solutions first.

By the Liouville transformation $v = \sqrt{D}u$, equation (8) is written as

$$\begin{aligned} -\Delta v_{i,j} + q v_{i,j} &= 0, \quad \text{in } \Omega, \\ v_{i,j} &= \tilde{g}_{i,j}, \quad \text{on } \partial\Omega, \end{aligned} \tag{24}$$

where $q(x) = \Delta\sqrt{D}/\sqrt{D} + \sigma/D$. The CGO solutions are special solutions to (24) that are perturbations of complex plane waves $\mathrm{e}^{\rho\cdot x}$ of the form

$$v_\rho(x) = \mathrm{e}^{\rho\cdot x}(1 + \psi_\rho(x)),$$

where $\rho \in \mathbb{C}^d$ and $\rho \cdot \rho = 0$. For $|\rho|$ large enough, such solutions exist, and from ([12], Corollary 3.2), it is known that the perturbation term ψ_ρ satisfies the bound

$$\|\psi_\rho\|_{H^s(\Omega)} \leq C(\Omega)\frac{\|q\|_{H^s(\Omega)}}{|\rho|}, \tag{25}$$

where $s > d/2 + k$ for $k \geq 1$. By choosing the parameter ρ in a certain way, we can tailor these solutions to achieve solutions with the right properties. We present the result for the situation when $d = 3$.

Theorem 2. *For any rotation i there is a open set of illuminations $\{g_{i,j}\}_{j=1}^6$ such that for each $x \in \Omega$ the vector fields*

$$v_{n,m}(x) = (\nabla u_{i,m}u_{i,n} - \nabla u_{i,n}u_{i,m})(x), \quad 1 \leq n, m \leq 6,$$

form a basis for $\mathbb{R}^3$.

In addition, if D is constant in a neighborhood of the boundary, there exists a δ such that if $\|q\|_{H^{d/2+k+\varepsilon}} \leq \delta$, then the same set of illuminations work for all rotations.

By open set, we mean $\{\tilde{g}_{i,j}\}$ sufficiently close to $\{g_{i,j}\}$ in any topology of sufficiently smooth functions such as $C^2(\bar{\Omega})$. The proof combines ideas from [4] and [9].

Proof. We choose CGO-parameters $\rho_1 = t(\mathbf{e}_2 + i\mathbf{e}_1)$, $\rho_2 = t(\mathbf{e}_3 + i\mathbf{e}_1)$ and $\rho_3 = t(-\mathbf{e}_2 + i\mathbf{e}_1)$, and choose the corresponding (real-valued) solutions to be

$$\begin{aligned} u_{i,1} &= \mathrm{Re}\{D^{-1/2}v_{\rho_1}(x)\}, \quad u_{i,2} = \mathrm{Im}\{D^{-1/2}v_{\rho_1}(x)\}, \quad u_{i,3} = \mathrm{Re}\{D^{-1/2}v_{\rho_2}(x)\}, \\ u_{i,4} &= \mathrm{Im}\{D^{-1/2}v_{\rho_2}(x)\}, \quad u_{i,5} = \mathrm{Re}\{D^{-1/2}v_{\rho_3}(x)\}, \quad u_{i,6} = \mathrm{Im}\{D^{-1/2}v_{\rho_3}(x)\}. \end{aligned}$$

After some algebra, it follows that

$$
\begin{aligned}
|\det[\nabla u_{i,1} \quad \nabla u_{i,2} \quad \nabla u_{i,3}](x)| &= Ct^3 \mathrm{e}^{t(2\mathbf{e}_2+\mathbf{e}_3)\cdot x}|(1+O(t^{-1}))\cos(t\mathbf{e}_1\cdot x)|,\\
|\det[\nabla u_{i,1} \quad \nabla u_{i,2} \quad \nabla u_{i,4}](x)| &= Ct^3 \mathrm{e}^{t(2\mathbf{e}_2+\mathbf{e}_3)\cdot x}|(1+O(t^{-1}))\sin(t\mathbf{e}_1\cdot x)|,\\
u_{i,5}(x) &= \mathrm{e}^{-t\mathbf{e}_3\cdot x}\cos(t\mathbf{e}_1\cdot x)(1+O(t^{-1})),\\
u_{i,6}(x) &= \mathrm{e}^{-t\mathbf{e}_3\cdot x}\sin(t\mathbf{e}_1\cdot x)(1+O(t^{-1})),
\end{aligned}
$$

where $C > 0$ is independent of t. We take the smallest t so large that $|O(t^{-1})| \leq 1/2$ in all expressions above. Since the zero sets of the determinants are disjoint, it follows that there are complementary sets Ω_1, Ω_2 such that $\Omega = \Omega_1 \cup \Omega_2$ and that $\{\nabla u_{i,1}, \nabla u_{i,2}, \nabla u_{i,3}\}$ is a basis for $\mathbb{R}^3$ on Ω_1, and $\{\nabla u_{i,1}, \nabla u_{i,2}, \nabla u_{i,4}\}$ is a basis for $\mathbb{R}^3$ on Ω_2 (see [9] for details on the construction). Correspondingly, $u_{i,5}(x) \neq 0$ for $x \in \Omega_1$ and $u_{i,6}(x) \neq 0$ for $x \in \Omega_2$. As a consequence, the vector fields

$$
v_{5,1}, v_{5,2}, v_{5,3}, v_{6,1}, v_{6,2}, v_{6,4}
$$

form a basis for $\mathbb{R}^3$ for every $x \in \Omega$. Hence we can choose illuminations such that $g_{i,j} = u_{i,j}|_{\partial\Omega}$. Further, by continuity of the mapping $g \in C^2(\overline{\Omega}) \mapsto u \in C^1(\overline{\Omega})$, it suffices to take $g_{i,j}$ close to $u_{i,j}|_{\partial\Omega}$, i.e., $\|g_{i,j} - u_{i,j}\|_{C^2(\overline{\Omega})} \leq \varepsilon$, for some $\varepsilon > 0$ sufficiently small [4].

Now, let $R \in \mathbb{R}^{3\times 3}$ be a rotation matrix, and set $\tilde{q}(x) = q(Rx)$. Then equation (24) with q replaced by $\tilde{q}$ corresponds to a rotation i of the object or the illumination patterns. Let

$$
\tilde{v}_{\rho_1}(x) = \mathrm{e}^{\rho_i\cdot x}(1+\tilde{\psi}_{\rho_1}(x)), \quad \tilde{u}_{k,1} = \mathrm{Re}\{D^{-1/2}\tilde{v}_{\rho_1}(x)\}, \tag{26}
$$

where $\tilde{\psi}_{\rho_i}$ is the perturbation term with respect to $\tilde{q}$. We assume that $\|g_{i,1} - u_{i,j}|_{\partial\Omega}\|_{C^2(\overline{\Omega})} \leq \varepsilon/2$. Then

$$
\begin{aligned}
\|g_{i,1} - \tilde{u}_{k,1}\|_{C^2(\overline{\Omega})} &= \|g_{i,1} - u_{k,1} + u_{k,1} - \tilde{u}\|_{C^2(\overline{\Omega})}\\
&\leq \varepsilon/2 + \|u_{k,1} - \tilde{u}_{k,1}\|_{C^2(\overline{\Omega})}\\
&= \varepsilon/2 + \|\mathrm{Re}\{\mathrm{e}^{\rho_1\cdot x}(\psi_{\rho_1} - \tilde{\psi}_{\rho_1})\ \|_{C^2(\overline{\Omega})}\\
&\leq \varepsilon/2 + \|\mathrm{Re}\{\mathrm{e}^{\rho_1\cdot x}\}\|_{C^2(\overline{\Omega})}\frac{2}{|\rho_1|}\|q\|_{H^s},
\end{aligned}
$$

where the last inequality follows by Sobolev embedding and the bound in (25), setting $s > 3/2 + 2$, and the fact that $\|q\|_{H^s} = \|\tilde{q}\|_{H^s}$. The result now follows by requiring

$$
\|q\|_{H^s} \leq \delta = \frac{\varepsilon|\rho_1|}{4\|\mathrm{Re}\{\mathrm{e}^{\rho_1\cdot x}\}\|_{C^2(\overline{\Omega})}}.
$$

The same argument then works for all illuminations.

□

The drawback with using CGO-solutions is that the results are not constructive, since they rely on the unknown parameters. Numerical simulations [11] show that most illuminations will in fact work, but we know that in dimension $d \geq 3$, we cannot find illuminations that provide ellipticity conditions independently of the coefficients [3].

3.5 Stability and uniqueness

Written out, the principal part of measurement object $\delta\mathcal{V}$ in (19) is of the form

$$\delta\mathcal{V} = \begin{bmatrix} \delta\mathcal{V}_1 \\ \delta\mathcal{V}_2 \end{bmatrix}$$

with components

$$\delta\mathcal{V}_1 = \sum_{i=1}^{M}\sum_{j=1}^{N} \tilde{\chi}_i \frac{i\nabla\bar{u}_{i,j}(x)\cdot\xi}{\xi^2+1} A\delta V_{i,j} \quad \text{and} \quad \delta\mathcal{V}_2 = \sum_{i=1}^{M}\sum_{j=1}^{N} \tilde{\chi}_i \bar{u}_{i,j} A\delta V_{i,j}.$$

Note that for $\delta\mathcal{V}_1$ and $\delta\mathcal{V}_2$, we have by standard mapping properties of PDOs on Sobolev spaces that

$$\begin{aligned} \|\delta\mathcal{V}_1\|_{H^s} &= \left\| \sum_{i=1}^{M}\sum_{j=1}^{N} \tilde{\chi}_i \frac{i\nabla\bar{u}_{i,j}(x)\cdot\xi}{\xi^2+1} A\delta V_{i,j} \right\|_{H^s} \leq C_s \sum_{i=1}^{M}\sum_{j=1}^{N} \|A\delta V_{i,j}\|_{H^{s-1}}, \\ \|\delta\mathcal{V}_2\|_{H^s} &= \left\| \sum_{i=1}^{M}\sum_{j=1}^{N} \tilde{\chi}\bar{u}_{i,j} A\delta V_{i,j} \right\|_{H^s} \leq \tilde{C}_s \sum_{i=1}^{M}\sum_{j=1}^{N} \|A\delta V_{i,j}\|_{H^s}, \end{aligned} \tag{27}$$

for $s \in \mathbb{R}$ and constants $C_s, \tilde{C}_s > 0$.

We now associate to the system (18) two sets of integers $s = (s_1, s_2)$ and $t = (t_1, t_2)$ such that for each entry of the 2×2 symbol $\sum_{i=1}^{M}(\kappa_i)^*\kappa_i$ we have that $\left(\sum_{i=1}^{M}(\kappa_i)^*\kappa_i(\kappa_i^0)^*\kappa_i^0\right)_{(i,j)} \in S^{s_i-t_j}(\Omega,\mathbb{R}^n)$. Since for each i

$$(\kappa_i^0)^*\kappa_i^0(x,\xi) = \tilde{\chi}_i^2(x,\xi)\sum_{j=1}^{N} \begin{bmatrix} \frac{(\nabla\bar{u}_{i,j}(x)\cdot\xi)^2}{(\xi^2+1)^2} & \frac{i\nabla\bar{u}_{i,j}(x)\cdot\xi\bar{u}_{i,j}}{\xi^2+1} \\ \frac{-i\nabla\bar{u}_{i,j}(x)\cdot\xi\bar{u}_{i,j}}{\xi^2+1} & \bar{u}_{i,j}^2 \end{bmatrix},$$

we can choose and $s = (-1, 0)$ and $t = (1, 0)$ so that the above entry (k, l) is of order $s_k - t_l$.
The operator $(\kappa_i^0)^*\kappa_i^0$ is said to be elliptic of type (s, t) in the Douglis-Nirenberg sense if it is invertible for every $(x, \xi) \in T^*\Omega$, see [1, 19]. When $(\kappa_i^0)^*\kappa_i^0$ is elliptic, it has a parametrix $Q(x, \xi)$, an inverse modulo smooth terms, and applying Q to (20) yields the estimate (since QP is smooth as well)

$$\|\delta D\|_{L^2(\Omega)} + \|\delta\sigma\|_{H^1(\Omega)} \leq C\left(\|\delta\mathcal{V}_1\|_{H^2(\Omega)} + \|\delta\mathcal{V}_2\|_{H^1(\Omega)} + \|\delta D\|_{H^{s'}(\Omega)} + \|\delta\sigma\|_{H^{s'}(\Omega)}\right), \tag{28}$$

for any $s' \in \mathbb{R}$ ([19], Lemma 1.0.2', or [25]). The constant C depends on $D, \bar{D}, \sigma, \bar{\sigma}, \Omega$, on s' and the illuminations. We can now complete the proof of Theorem 1.

Proof. That boundary conditions such that $(\kappa_i^0)^*\kappa_i^0$ is elliptic of the (s, t) type exist follows from Theorem 2, and hence the estimate (28) holds. Let $s' = -1$ in (28). For $u \in L^2(\Omega) \subset$

$H^{-1}(\Omega)$, supp $u \subset B_\varepsilon$, we have

$$\begin{aligned}\|u\|_{H^{-1}(\Omega)} &= \sup_{\|v\|_{H_0^1(\Omega)}\leq 1} \left| \int_{B_\varepsilon} uv\mathrm{d}x \right| = \sup_{\|v\|_{H_0^1(B_\varepsilon)}\leq 1} \left| \int_{B_\varepsilon} uv\mathrm{d}x \right| \\ &\leq \sup_{\|v\|_{H_0^1(B_\varepsilon)}\leq 1} \|u\|_{L^2(B_\varepsilon)}\|v\|_{L^2(B_\varepsilon)} \leq \sup_{\|v\|_{H_0^1(B_\varepsilon)}\leq 1} \|u\|_{L^2(B_\varepsilon)}\tilde{C}\varepsilon\|Dv\|_{L^2(B_\varepsilon)} \\ &\leq \sup_{\|v\|_{H_0^1(B_\varepsilon)}\leq 1} \|u\|_{L^2(B_\varepsilon)}\tilde{C}\varepsilon\|v\|_{H_0^1(B_\varepsilon)} \leq \tilde{C}\varepsilon\|u\|_{L^2(B_\varepsilon)},\end{aligned}$$

where we use the Poincaré inequality to get the ε-dependence; see [16]. A priori, $u \in H_0^1(B_\varepsilon)$, and we have that

$$\|u\|_{H^{-1}(\Omega)} \leq \tilde{C}\varepsilon\|u\|_{L^2(\Omega)}, \qquad \|u\|_{H^{-1}(\Omega)} \leq \tilde{C}\varepsilon\|u\|_{H^1(\Omega)}. \tag{29}$$

Using the estimates in (27) we get that that for some constant $C_v > 0$

$$\|\delta\mathcal{V}_1\|_{H^2(\Omega)} + \|\delta\mathcal{V}_2\|_{H^1(\Omega)} \leq C_v \sum_{i=1}^{M}\sum_{j=1}^{N} \|A\delta V_{i,j}\|_{H^1(\Omega)}. \tag{30}$$

Last, the time-reversal operator A is bounded ([26], Theorem 2.4 with $\theta = 1$), and we have $\|A\delta V_{i,j}\|_{H^1(\Omega)} \leq C_A\|\delta V_{i,j}\|_{H^1([0,T]\times\partial\Omega_M)}$. Under the assumption that supp δD, supp $\delta\sigma \subset B_\varepsilon$, we make use of the inequalities (29) and rearrange (28) to get

$$(1 - C\tilde{C}\varepsilon)(\|\delta D\|_{L^2(\Omega)} + \|\delta\sigma\|_{H^1(\Omega)}) \leq C_A C_v \sum_{i=1}^{M}\sum_{j=1}^{N} \|\delta V_{i,j}\|_{H^1([0,T]\times\partial\Omega_M)}. \tag{31}$$

We can then choose ε such that $(1 - C\tilde{C}\varepsilon) > 0$. Recalling Theorem 2, we have $N = 2d$ and we get (11). □

4 Remarks and Conclusions

In a sufficiently idealized setting, where the ultrasound propagation may be modeled by a wave equation with reasonably well known sound speed (and nagging attenuation effects may be neglected [22]), quantitative PAT displays favorable mathematical properties, as a composition of two reasonably well-posed inverse problems. This holds in the setting where (sufficiently) full ultrasound measurements are available, which is not always practical [14, 27].

We show in this paper that PAT enjoys favorable stability properties as well when a similar amount of measurement is collected for optical illuminations and ultrasound detectors that are allowed to rotate during the acquisition procedure so that different parts of the ultrasound measurements are generated by different optical illuminations.

The absence of complete ultrasound measurement for each optical illumination renders the inversion of a single wave equation ill-posed. Only when measurements from all optical sources are accounted for can one expect stable reconstructions. In such settings, where only local pieces of information are available for each illumination, it is difficult to envision a direct global inversion procedure. Rather, it is the ideal setting to apply micro-local

methods, whose roles are precisely to propagate (phase-space-) local information through wave or elliptic equations.

Writing the difference of sufficiently smooth nonlinear functionals $\mathcal{F}(u) - \mathcal{F}(v)$ as a general functional $\mathcal{G}(u, v, u - v)$ that is linear in its last component, a procedure that applies in a great variety of contexts, we can write the difference of measurements as a (standard) elliptic Fourier integral operator applied to $u - v$ with a symbol that depends on (u, v). This imposes that (u, v), for us here the sound speed and the optical coefficients as well as the solutions to the diffusion equation, be sufficiently smooth. This procedure has been applied for the second step of many hybrid inverse problems in [25]. The inversion procedure then necessarily provides conditional stability, that is to say stable reconstructions up to the possibly non-trivial kernel of a compact operator. In this paper, we chose to make a smallness assumption on the support of the coefficients of interest to show that the kernel of said operator was necessarily trivial, a trick that is certainly not new [18].

These are our two main somewhat unnecessary condition: very large smoothness of coefficients and smallness of support.

Such strong smoothness assumptions are not necessary in standard PAT [11] (although no known results hold for arbitrary coefficients; for instance D and σ arbitrary measurable functions that are bounded between 1 and 2, say). They can also be avoided for more general hybrid inverse problems by writing the inverse problem as a coupled system of nonlinear partial differential equations for (D, σ, u_j), where the PDO calculus of the second step of PAT is replaced by a better behaving potential theory [7]. In that setting, one can also sometimes apply a unique continuation principle that allows one to obtain an injectivity result independently of the size of the support of the unknown coefficients.

However, in our setting of rotating measurements, it seems unclear how the micro-local pursuit of propagation of singularities can be totally avoided. Within that context, unnecessary smoothness assumptions seem necessary (realizing the inherent contradiction). As we mentioned in the introduction, all the smoothness we need is for s' in (28) to be strictly negative. This can be achieved by Taylor expansions involving finitely many terms in the definition of PDO and FIO operators as well as in their composition. How many (a sufficiently large finite number) dictates how much smoothness our coefficients need to verify.

These technical constraints notwithstanding, we expect standard quantitative PAT and QPAT in a rotating setting to display very similar resolution capabilities. As mentioned in Section 2, our method does not offer a constructive method for reconstruction, but suggests that numerical methods could be successfully adapted to the rotating setting. In the future, it would be interesting to see the development of inversion methods for the rotating setting. Many inversion methods exist for classical PAT and QPAT, e.g., [17, 21, 31, 33]; building on these methods it should be possible to achieve good results.

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Appendix B

Paper 2

Feasibility of Acousto-Electric Tomography

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Abstract

In acousto-electric tomography the goal is to reconstruct the electric conductivity in a domain from electrostatic boundary measurements of corresponding currents and voltages, while the domain is penetrated by a time-dependent acoustic wave. We explicitly model the phenomena, and we propose a complete inversion framework for acousto-electric tomography in two steps: First the interior power density is obtained from boundary measurements by solving a linear, ill-posed problem; second the interior conductivity is reconstructed from the power density by solving a non-linear, fairly well-posed problem. We perform numerical experiments on synthetic data with realistically chosen parameters. We investigate how feasibility of reconstructing the electrical conductivity from boundary measurements depends on the acousto-electric coupling constant and measurement noise. Our findings are positive, and indicate that AET is indeed feasible for interesting applications in for example medical imaging. Finally, we consider a limited angle setup and show that the conductivity is well reconstructed near the measurement boundary.

Keywords: acousto-electric tomography, electrical impedance tomography, hybrid data tomography, coupled physics imaging, inverse problems, medical imaging
MSC2000: 35R30; 65N21

1 Introduction

In a variety of applications in imaging science it is important to reliably image the electrical conductivity in some domain. In medical imaging, for instance, the electric conductivity distribution in the human body carries information about the health condition of the patient, i.e., location of tumours, lung function and brain function, and hence an image of the conductivity is very useful for medical imaging [20]. In other application domains such as Electrical Brain Stimulation it is important to have an accurate estimate of the brain's conductivity in order to compute the current density generated from the exterior. Electrical Impedance Tomography (EIT) is a fairly novel technology for medical imaging that aims at reconstructing a 3D image of a body's electrical conductivity from surface measurements of current and voltages through

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electrodes at the surface of the body. However, it is well-known that the inverse problem in EIT is notoriously ill-posed thus giving rise to low resolution images [11]. In recent years new ideas have been proposed that are often referred to as hybrid imaging. Broadly speaking, the idea of hybrid imaging is to utilize and control two separate, but coupled, physical phenomenon to obtain some extra information that makes the reconstruction problem more well-posed. One such method that has been subject to extensive research interest is acousto-electric tomography (AET) [36,37]. In AET, one perturbs the electrical conductivity of the object by acoustic waves while conducting EIT-measurements. The so-called acousto-electric effect allows one to recover first the internal electric power density and second the conductivity distribution. AET has not yet matured as a technology; it is questionable whether the measurable signals are strong enough for reconstruction purpose. The main goal of this paper is to clarify, using realistic models and parameters in a computational study, the feasibility of AET.

To formulate the problem mathematically we assume that the object of interest occupies some bounded domain $\Omega \subset \mathbb{R}^d$, $d = 2, 3$, with smooth boundary $\partial\Omega$. The object Ω has the unknown electric conductivity $\sigma \in L^\infty_+(\Omega)$ (bounded from above and below by positive constants). Through electrodes attached to the boundary, the normal current flux f is controlled, and consequently an interior electric potential u is generated. When no interior sources or sinks of charge are present inside Ω, the potential is characterized by the generalized Laplace equation

$$\begin{cases} -\nabla \cdot (\sigma(x)\nabla u(x)) = 0, & x \in \Omega, \\ \sigma(x)\partial_\nu u(x) = f(x), & x \in \partial\Omega. \end{cases} \tag{1.1}$$

Here ν denotes the outward unit normal vector to the boundary $\partial\Omega$. We require that $f \in L^2_\diamond(\partial\Omega) = \{v \in L^2(\partial\Omega) : \int_{\partial\Omega} v \, \mathrm{d}s = 0\}$. Then (1.1) has a solution, and if we ground the potential by assuming $g = u|_{\partial\Omega} \in L^2_\diamond(\partial\Omega)$, $u \in H^1(\Omega)$ is uniquely determined. In EIT the aim is to reconstruct σ from several measurements of $g = u|_{\partial\Omega}$ corresponding to a set of different input currents f.

In AET the object is, in addition to the EIT measurement setup, penetrated by an acoustic wave that is generated by some source in the exterior of the body. Let p denote the acoustic wave and S the source function. We model p by the scalar wave equation

$$\begin{cases} (\partial_t^2 - c^2(x)\Delta)p(x,t) = S(x,t), & (x,t) \in \mathbb{R}^d \times \mathbb{R}_+, \\ p(x,0) = \partial_t p(x,0) = 0, & x \in \mathbb{R}^d \end{cases} \tag{1.2}$$

equipped with reasonable decay conditions. The sound speed c and the source S is considered fully known, and in that case p is also known. When the acoustic wave travels through a material, the material is compressed and expanded. This results in a localized, time dependent change in the electrical conductivity, known as the acousto-electric effect. We denote the acoustically perturbed conductivity by $\sigma_p(x,t)$. A reasonable, yet simple model for σ_p is the following [3]:

$$\sigma_p(x,t) = \sigma(x)(1 + \eta p(x,t)). \tag{1.3}$$

Here, $\eta \geq 0$ is the acousto-electric coupling parameter. We assume throughout that η is a known constant.

Figure 1 illustrates the acousto-electric effect. The source $S(x,t)$ is situated outside the disk-shaped domain Ω and generates an acoustic wave. As the wave propagates through the domain, the conductivity is perturbed; an instantaneous image of the perturbed conductivity is seen to the right.

The perturbation of σ gives rise to a perturbation of the electric potential. We denote by $u_p(x,t)$ this potential that for fixed $t \in \mathbb{R}_+$ solves the equation

$$\begin{cases} -\nabla \cdot (\sigma_p(x,t)\nabla u_p(x,t)) = 0, & x \in \Omega, \\ \sigma_p(x,t)\partial_\nu u_p(x,t) = f(x), & x \in \partial\Omega \end{cases} \tag{1.4}$$

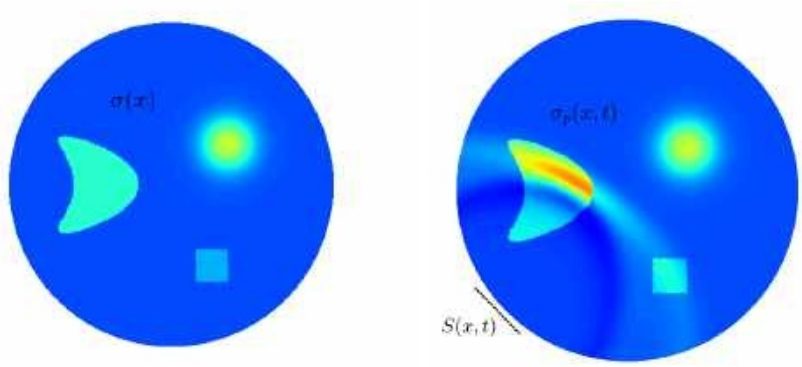


Figure 1: Right: the phantom conductivity σ. Left: The perturbed conductivity σ_p.

(again with the convention $u_p|_{\partial\Omega} \in L^2(\partial\Omega)$.) The time-dependent perturbed electric measurement is now $g_p = u_p|_{\partial\Omega}$, for $t \in [0, T]$ for some sufficiently large T.

In AET several boundary currents f and wave sources S are used, and the inverse problem is then to reconstruct the conductivity σ from measurements of the corresponding voltages $g(x), g_p(x,t),\ x \in \partial\Omega,\ t \in [0, T]$.

The typical approach to AET consists of two steps: First the interior electric power density

$$H(x) = \sigma(x)|\nabla u(x)|^2$$

is reconstructed from the boundary measurements. Second, the conductivity σ is obtained from H; this is the quantitative step. In most literature on AET the first step is overlooked, and only the second step is considered. A novelty in this paper is that we, based on a carefully designed computational approach, consider both steps. We will see that the first step can be approached through a fairly ill-posed linear integral equation, while the second step can be approached as a (regularized) non-linear optimization problem. By numerical experiments we will demonstrate that AET is a feasible method for approximating the electrical conductivity in a body, for values of the electrical coupling constant in a range of values relevant for applications.

The feasibility of AET is critically linked to the magnitude of η, or rather the ratio of η and the level of precision in the electrical measurements. This is due to the electrical noise remaining at the scale of the electrical signal, while the power-difference – used to derive the power density – will scale with η. We explore the significance of this in Sections 3 and 4, when we model the noise and perform numerical experiments.

There is a vast literature on the mathematical and computational aspects of AET. The fundamental modelling originates from [4]. In [24, 25] idealistic wave modelling was considered and an inversion scheme using the spherical Radon transform was derived. The reconstruction problem was considered in [10] using an optimal control approach. Several different linearization methods were compared in [19] giving rise to the analysis of artefacts in [6]. A numerical reconstruction method based on the Levenberg-Marquardt iteration was developed in [7] and methods using an explicit least squares optimization approach are found in [1, 2, 30]; the limited angle problem was considered in [21]. See also, e.g., [5, 29], for the case of anisotropic conductivities.

The outline of the paper is as follows: In Section 2 we discuss the modelling of the inverse problem, explain the two step inversion procedure and it's numerical implementation. In Section 3 the careful choice of physical parameters for the computational phantom is described, and

noise modelling is considered. Numerical experiments are carried out in Section 4 before the conclusion is drawn in Section 5.

2 Inversion procedure and discretization

In this section we outline the procedure for the two-step inversion. Assuming knowledge of the acoustic fields, we first propose a spectral discretization combined with a regularized least squares method that approximates $H(x)$. Next, we solve for $\sigma(x)$ using a slightly improved version of the method described in [1].

2.1 Step 1: From boundary measurements to power density

Multiplying (1.1) and (1.4) with u_p and u, respectively, we obtain by integrating by parts the time series

$$\begin{aligned} I(t) &= (f, g_p - g)_{L^2(\partial\Omega)} \\ &= \int_{\partial\Omega} f(x)\left(g_p(x,t) - g(x)\right) \mathrm{d}S = -\eta \int_{\Omega} p(x,t)\sigma(x)\nabla u(x)\cdot\nabla u_p(x,t)\,\mathrm{d}x. \end{aligned} \tag{2.1}$$

Physically, $I(t)$ measures at time t the change in the required total electric power for maintaining the current flux f at $\partial\Omega$ under the acoustic perturbation. Figure 2 illustrates how I may behave as the acoustic wave travels through an object. Under the assumption that ηp is small, the approximation $u_p \approx u$ is good (first order in η) and this yields the linear approximation

$$I(t) \approx -\eta \int_{\Omega} p(x,t)\sigma|\nabla u(x)|^2\,\mathrm{d}x = -\eta \int_{\Omega} p(x,t)H(x)\,\mathrm{d}x. \tag{2.2}$$

Allowing the source to be placed at different positions, denoted by $S = S_j,\ j = 1,\dots,N_S$, results in different corresponding acoustic fields p_j. In addition, allowing different boundary currents $f = f_i,\ i = 1,\dots,N_f$, gives rise to different boundary potentials denoted by $g = g_i$ and $g_{p_j} = g_{i,j}$, and correspondingly, different time series $I_{i,j}$. The first step in the reconstruction method in then to solve for H_i the system of integral equations

$$I_{i,j}(t) = -\eta \int_{\Omega} p_j(x,t)H_i(x)\,\mathrm{d}x, \quad 1 \le i \le N_f, 1 \le j \le N_S, t \in [0,T]. \tag{2.3}$$

The waves p_j generally possess some degree of smoothness. As a consequence (2.3) constitutes a system of Fredholm integral equations of the first kind that is in general ill-posed; the degree of ill-posedness and general solvability depends on the exact smooothness of kernel p_j of the integral operators, see for example [23], but some regularization must be employed. The above formulation emphasizes the key role of the acoustic field in the reconstruction of $H_i(x)$. Ideally, the fields $\{p_j(\cdot,t)\}_j$ should form a complete set in $L^2(\Omega)$ for H to be determined from (2.3).

We expect H to be a fairly smooth field (in particular away from singularities in σ) and therefore we propose to represent $H(x)$ by finite linear combinations of smooth, global (on Ω) basis functions, i.e. we approximate the integrals in (2.3) by a spectral quadrature method.

For a smooth domain $\Omega \subset \mathbb{R}^2$, let φ_n be solutions to the Neumann eigenvalue problem

$$-\Delta\varphi_n = \lambda_n\varphi_n \quad \text{in } \Omega, \qquad \partial_\nu\varphi_n = 0 \quad \text{on } \partial\Omega. \tag{2.4}$$

It is well-known, see e.g. [14], that the set of (normalized) eigenfunctions are smooth and constitute an orthonormal basis for $L^2(\Omega)$. For some simple geometries, analytical expressions for the eigenfunctions are known, and we will exploit that in the computational experiments

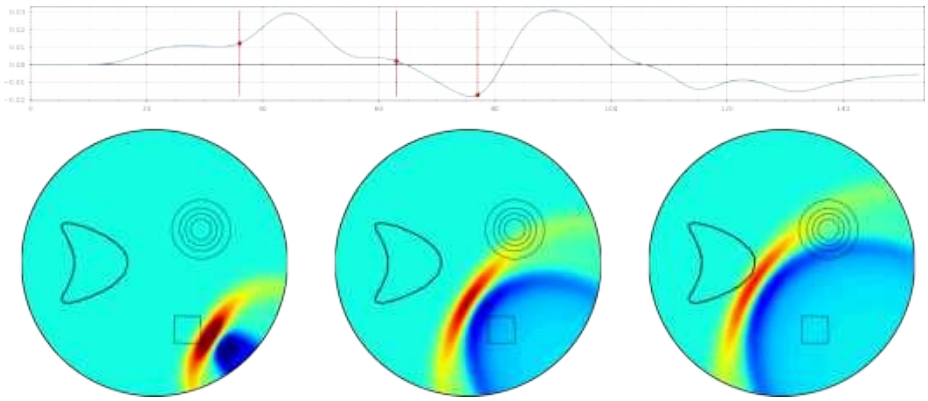


Figure 2: The top image contains the graph of the function $I(t)$ for the phantom seen in Figure 1 corresponding to a particular propagating wave and the boundary condition $f(x, y) = x$. The red vertical lines mark times corresponding to the three instant wave positions seen in the plots below.

below. In the general case there exists effective methods for computing them [31]. We represent $H_i(x)$ in the form

$$H_i(x) = \sum_{n=1}^{\infty} \hat{H}_{i,n} \varphi_n(x), \tag{2.5}$$

where $\hat{H}_{i,n} = (H_i, \varphi_n)_{L^2(\Omega)}$, and then reconstruct the coefficients $H_{i,n}$ from (2.3).

We compute $I_{i,j}$ at discrete times $t^k \equiv kT/N_t, k = 1, 2, \ldots, N_t$. Substituting (2.5) into (2.3) and truncating the series at $n = N_\varphi$ gives

$$I_{i,j}^k \approx -\eta \sum_{n=1}^{N_\varphi} \hat{H}_{i,n} \int_\Omega p_j(x, t^k) \varphi_n(x) \mathrm{d}x = \eta \sum_{n=1}^{N_\varphi} \hat{H}_{i,n} w_{j,n}^k \tag{2.6}$$

with the quadrature weights

$$w_{n,j}^k = -\int_\Omega p_j(x, t^k) \varphi_n(x) \mathrm{d}x. \tag{2.7}$$

Thus by introducing

$$\hat{H}_i = [\hat{H}_{i,1}, ..., \hat{H}_{i,N_\varphi}]^T \in \mathbb{R}^{N_\varphi}, \tag{2.8}$$

$$I_i = [I_{i,1}^1, ..., I_{i,1}^{N_t}, ..., I_{i,N_S}^1, ..., I_{N_f,N_S}^{N_t}]^T \in \mathbb{R}^{N_f N_S N_t}, \tag{2.9}$$

$$K = \begin{bmatrix} w_{1,1}^1 & w_{2,1}^1 & \cdots & w_{N_\varphi,1}^1 \\ \vdots & & \ddots & \\ w_{1,1}^{N_t} & w_{2,1}^{N_t} & \cdots & w_{N_\varphi,1}^{N_t} \\ w_{1,2}^1 & w_{2,2}^1 & \cdots & w_{N_\varphi,2}^1 \\ \vdots & & \ddots & \\ w_{1,N_S}^{N_t} & w_{2,N_S}^{N_t} & \cdots & w_{N_\varphi,N_S}^{N_t} \end{bmatrix} \in \mathbb{R}^{N_S N_t \times N_\varphi} \tag{2.10}$$

we obtain the linear system

$$\eta K \hat{H}_i = I_i, \quad \text{for } 1 \leq i \leq N_f. \tag{2.11}$$

Depending on the number of measurements available, this system might be a square, under- or overdetermined system of equations. It is well-known that a discretized Fredholm equation inherits the ill-posedness of its continuous counterpart, and results in a ill-conditioned linear system. Hence, we propose a regularized least squares approach to approximate a solution of (2.11): As a regularization term, we use $\|\nabla H_i(x)\|_{L^2(\Omega)}$, a choice that is known to impose smoothness [17]. Integration by parts yields a simple expression, where the eigenvalues appears as weights on the coefficients:

$$\|\nabla H_i(x)\|^2_{L^2(\Omega)} = -\int_\Omega \Delta H_i(x) H_i(x) \mathrm{d}x = \sum_{n=1}^{N_\varphi} \lambda_n \hat{H}^2_{i,n}.$$

The discrete, regularized least squares problem now takes the form

$$\hat{H}_i = \min_{\hat{H}\in\mathbb{R}^{N_\varphi}} \|\eta K\hat{H} - I_i\|_2^2 + \beta \sum_{n=1}^{N_\varphi} \lambda_n \hat{H}^2_{i,n} \tag{2.12}$$

where β is a regularization parameter that must be tuned to achieve a suitable solution to (2.12)

2.2 Step 2: From H to σ

The problem of reconstructing σ from the interior power density data $\hat{H}$ has been explored in a variety of scenarios, see for example [4,6,16,19,25]. We approach this problem by minimizing the functional

$$\min_\sigma \mathcal{J}(\sigma) = \sum_{i=1}^{N_f} \|H_i(\sigma) - z_i\|_{L^1(\Omega)} + \beta|\sigma|_{\mathrm{TV}}, \tag{2.13}$$

where $z_i(x) = \sum_{n=1}^{N_\varphi} \hat{H}_{i,n}\varphi_n(x)$ is the reconstructed data from step 1. The L^1 data-fidelity terms depends on the assumptions on the regularity of σ. In particular, if σ has discontinuities, standard regularity theory only guarantees L^1-regularity of H; with $L^{1+\varepsilon}$-regularity for sufficiently small discontinuities, see [1].

For reconstruction we use the algorithm described in [1]. That is, we consider the following weighted quadratic problem arising from a linearization of (2.13),

$$J_{\sigma,\kappa'}(\kappa) = \frac{1}{2}\sum_{i=1}^{N_f} \int_\Omega w_i(\sigma,\kappa')|H_i'(\sigma)[\kappa] - z_{i,\sigma}|^2 \,\mathrm{d}x + \frac{\beta}{2}\int_\Omega w_0(\sigma,\kappa')|\nabla(\sigma+\kappa)|^2 \,\mathrm{d}x, \tag{2.14}$$

where $z_{i,\sigma} = z_i - H_i(\sigma)$ and the weights are

$$w_i(\sigma,\kappa') = |H_i'(\sigma)[\kappa'] - z_{i,\sigma}|_\epsilon^{-1}, \quad \text{for } 1 \le i \le N_f, \quad w_0(\sigma,\kappa') = |\nabla(\sigma+\kappa')|_\epsilon^{-1},$$

with $|\cdot|_\epsilon \equiv \sqrt{|\cdot|^2 + \epsilon^2}$.

To solve (2.13) we initialize κ' as zero, compute the weights w_i, and successively do partial minimization of (2.14) to update κ' and thus the weights. After a few rounds of this we update $\sigma := \sigma + \kappa'$ and restart the process.

As a small modification to the algorithm presented in [1] we have a slight smoothing on the step. Inspired by [7,21] we lift the step to a slightly higher regularity using the map $E^*_{\alpha,\beta}$, where $E_{\alpha,\beta} : H^1_{\alpha,\beta}(\Omega) \to L^2(\Omega)$ is the embedding map and $H^1_{\alpha,\beta}(\Omega)$ is the Sobolev space $H^1(\Omega)$ equipped with the inner product $(u,v)_{\alpha,\beta} = \alpha(u,v)_{L^2(\Omega)} + \beta(\nabla u, \nabla v)_{L^2(\Omega)}$. The chosen values are throughout $\alpha = 1, \beta = 10^{-3}$.

3 Model parameters, phantom and noise

In this section we describe the parameters and phantom we use in our numerical experiments. We also briefly discuss our choice of noise model. Furthermore, we give a detailed description of the modeling and numerical simulation of the acoustic sources and fields, and the electrical potential and measurements.

3.1 Parameters

We assume a domain $\Omega = \{x \in \mathbb{R}^2 : |x| < R\}$ with $R = 0.1\,\mathrm{m}$.

Electrical conductivity and boundary current density

Many authors have reported results on the electrical conductivity of human tissue, and from [9, 18, 22] one infers that the electrical conductivity differs by a factor in the range 5 to 15 between healthy and cancerous breast tissue and experimental values for σ are reported to lie in the range 0.01 to 1 S/m, see, e.g., [8,12,18,22]. In our experiments, we choose a high contrast conductivity phantom taking values in the range 0.1 to 1.5 S/m and a low contrast conductivity phantom with values in the range 0.1 to 0.3 S/m. Figure 3 shows the phantoms we choose for the numerical experiments. They include both smooth and discontinuous inclusions of different sizes, and the non-convex kite shaped inclusion. They differ in that the contrast between the background and the inclusions is much lower in the second phantom.

Further, the maximum allowed input current is $f_{\max} = 1\,\mathrm{mA}$ [12]. We enforce this by requiring that

$$\int_{\partial\Omega} f^+ \mathrm{d}s \leq f_{\max}, \quad \text{where} \quad f^+(x) = \max\{f(x), 0\}.$$

Acousto-electric coupling and acoustic pressure

The value of the acousto-electric coupling constant η is of great importance, since it determines how much we can perturb the conductivity, and hence the potential by the acoustic pressure. It is known that in general, a very small change in the conductivity takes place. In [26], η in a rabbit heart is found to be approximately $0.041\,\mathrm{MPa}^{-1}$, i.e., $\eta \approx 4.1 \times 10^{-8}\,\mathrm{Pa}^{-1}$, while in [33], values η in 0.9% NaCl solution is reported to be of the magnitude $10^{-9}\,\mathrm{Pa}^{-1}$. We could not find measured values of η for breast tissue, but the mentioned values indicates a range of interest. We thus perform our numerical experiments for η in the range 10^{-7} to $10^{-9}\,\mathrm{Pa}^{-1}$.

We choose the maximal amplitude of the acoustic pressure p to be $p_{\max} = 1.5\,\mathrm{MPa}$, in accordance with clinical standards [13], and assume a constant wave speed of $c_0 = 1500\,\mathrm{m/s}$ [13]. Each of the point sources constituting the transducer excites one period of sine-wave with a frequency of approximately 20 kHz. We use 9 different transducer positions, equidistantly spaced around the boundary of the lower half of Ω, and at each position we point the wave in three different directions. In Table 1, we summarize our parameter choices.

3.2 Noise modelling

Modelling the noise in AET there are a myriad of potential sources of stochasticity including, but not limited to, the exactness of the applied boundary current, the precision of boundary potential measurements, the knowledge of the wave (could be related to precision of transducer positions, transducer timing, modelling limitations, wave speed, etc.) and exactness of timestamps for measurements etc.

It is beyond the scope of this article to take everything into account here, so we will here only consider noise in the time-dependent boundary measurements $g^k_{i,j}(x) = g_{i,j}(x, t^k)$, which transmits to become noise in the power signal $I^k_{i,j}$. We apply relative additive noise as follows:

Parameter	Symbol	Values(s)	Unit
Conductivity	σ	$0.1 - 1.5$	S/m
Boundary current amplitude	$f_{\max}$	1×10^{-3}	A
Acousto-electric coupling constant	η	$\{10^{-7}, 10^{-8}, 10^{-9}\}$	Pa^{-1}
Max. acoustic pressure	$p_{\max}$	1.5×10^{6}	Pa
Acoustic wave speed	c_0	1500	m/s
Domain radius	R	0.1	m
Noise level	δ	$\{0, 10^{-2}, 10^{-1}\}$	%

Table 1: Tabel

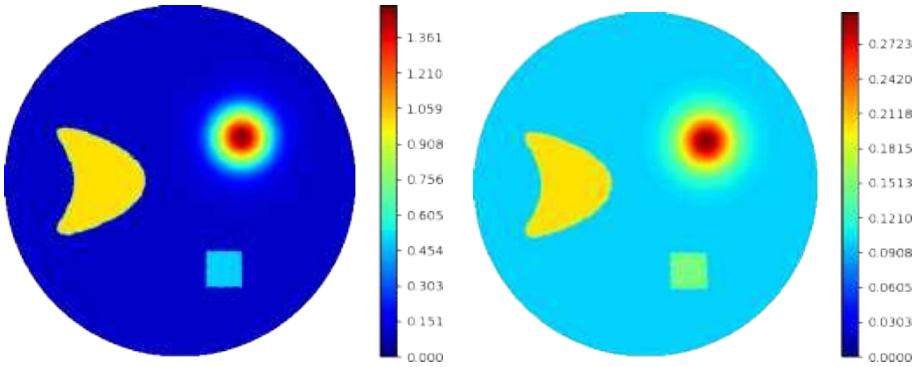


Figure 3: Left: the high contrast phantom. Right: The low contrast version of the same phantom.

We initialize a random vector $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ on the mesh boundary nodes, where $g^k_{i,j}$ lives; let $\varepsilon : \partial\Omega \to \mathbb{R}$ be the corresponding finite element function. We then compute a normalization constant $C^k_{i,j} = \|g^k_{i.j}\|_{L^2(\partial\Omega)}/\|\varepsilon\|_{L^2(\partial\Omega)}$ and denote $\widetilde{\varepsilon}^k_{i,j} = C^k_{i,j}\varepsilon$. Let δ denote the noise level, then

$$\widetilde{I}^k_{i,j} = I^k_{i,j} + \delta(f_i, \widetilde{\varepsilon}^k_{i,j})_{L^2(\partial\Omega)}, \tag{3.1}$$

where each $\widetilde{\varepsilon}^k_{i,j}$ is constructed from a unique realization of $\boldsymbol{\varepsilon}$.

We stress the point that the first term in (3.1), $I^k_{i,j}$, scales with the coupling constant, $I^k_{i,j} \sim \mathcal{O}(\eta)$, and the second term is independent of η and scales with the noise level δ. Hence the ratio η/δ, i.e., the ratio of the coupling constant and the noise level, is expected to be crucial to the solvability of the problem. If the noise level δ is fixed and η decreases by a factor 10, the effective noise level seen in the measurement $\widetilde{I}^k_{i,j}$ increases by a factor 10. In Section 4, we will see how this dependence manifests itself in numerical experiments. For noise levels we consider $\delta = 0\%$, $10^{-2}\%$ and $10^{-1}\%$. These scales of noise are based on the observed levels in the experimental studies [15, 28].

3.3 Simulation of the forward problem

Acoustic field and source terms

We simulate the acoustic field on a square finite difference grid containing Ω. k-Wave is a Matlab package that uses a $k-$space pseudospectral method to effectively and accurately solve the wave equation. To avoid the problem of having to prescribe boundary conditions on

the finite computational domain, K-WAVE uses a so-called perfectly matched layer that absorbs the wave at the domain boundary. For the source terms S_j of our acoustic field, we use a simplified transducer model. We model the action of the transducer as a linear array of N_p time dependent volume point sources. We assign to the K-WAVE source function a discrete set of N_p grid points on a line segment(see Figure 1) in the domain(a source mask), and the tone burst parameters, so that it emits one period of a sine wave at each mask point. Further, we use the steering functionality to delay the individual point source signals and orient the wave in various directions. See the K-WAVE manual [35] for further details and explanations on this, and on how K-WAVE solves the wave equation.

We simulate acoustic fields with the above parameters on a 350×350 square finite difference grid of size $31\,\text{cm} \times 31\,\text{cm}$. In each position, the transducer consists of 41 grid points on a straight line segment with a direction tangential to $\partial\Omega$, and centered at coordinates $(R_S \cos(\theta_\ell), R_S \sin(\theta_\ell))$, where $R_S = 10.6\,\text{cm}$ and $\theta_\ell = \pi + \ell\frac{\pi}{10}, 1 \leq \ell \leq 9$. In addition, we focus the acoustic wave in three different steering angles 0.35π , 0 and -0.35π, where the wave at angle 0 is directed radially inwards. The tone burst frequency is 19.2 kHz and lasts for one period. This gives $N_S = 27$ acoustic fields in total. We sample the waves at $N_t = 154$ equidistant times in the range 0 to 1.8×10^{-4} s. The acoustic fields $\{p_j(x, t_k) : 1 \leq j \leq N_S = 27, 1 \leq k \leq N_t = 154\}$ are then interpolated on a triangulated mesh with 6311 nodes.

Electric potential measurements

The electric potentials are computed by solving the PDE (1.1) using the open source computing framework FENICS [27]. A triangulated mesh Ω_h approximating Ω is produced using the program GMSH and loaded into the FENICS framework. The true phantom conductivity σ and the acoustic fields p_j are projected to σ_h and $p_{h,j}$ in a $\mathcal{P}_1$ FEM-basis on Ω_h, V_h. Picking p among $p_{h,j}$ all the perturbed conductivities $\sigma_{h,j} \equiv \sigma_{p_{h,j}}$ are computed on Ω_h according to (1.3). Using FENICS to solve the PDE (1.1) over V_h we get $u_{h,i}$ and $u_{h,i,j}$, corresponding to σ_h and $\sigma_{h,j}$, where $1 \leq i \leq N_f$. The number of boundary conditions is $N_f = 3$ with

$$f_1(x, y) = x, \quad f_2(x, y) = y, \quad \text{and} \quad f_3(x, y) = \frac{x + y}{\sqrt{2}}.$$

We then compute the measurement functions $I^k_{h,i,j}$ on the polygonal boundary $\partial\Omega_h$ according to the expression (2.1) using $g_{h,i} = u_{h,i}|_{\partial\Omega_h}$ and $g^k_{h,i,j} \equiv g_{h,i,j}(x, t^k)$, where $g_{h,i,j} = u_{h,i,j}|_{\partial\Omega_h}$, corresponding to $f = f_i$ and $p = p_{h,j}$. For the limited angle experiments we use the boundary conditions $\tilde{f}_i(\theta) = \chi_{(-\pi,0)}(\theta) \sin(2i\theta), i = 1, 2, 3$, where $0 \leq \theta < 2\pi$ is the polar coordinate angle for the point on the disc boundary.

The measurements $I^k_{i,j}$ are computed using FEniCS [27] by solving (1.1) and (1.4) and computing the inner product on the left hand side in (2.1) on the mesh boundary. Noise is then added according to (3.1) by generating normally distributed random coefficients for a $\mathcal{P}_1$ function and computing the boundary inner product.

4 Validation of the numerical method and feasibility of AET

In this section we put the above theory and numerical methods to work on the measurements generated by the phantoms in Section 3. We first demonstrate that we can reconstruct the power density, and then apply our method to simulated measurements for a range of parameters and measurement noise levels. We are especially interested in how the reconstruction quality depends on the magnitude of the acousto-electric coupling parameter and the noise level.

4.1 Reconstructing the power density

In the case of $\Omega = \{x \in \mathbb{R}^2, |x| < R\}$, analytical expressions for the eigenfunctions can be found by separation of variables [34]. The functions

$$\varphi^{\mathrm{c}}_{m,n}(r,\theta) = W_{m,n} J_m(r j'_{m,n}/R)\cos(m\theta), \qquad m \geq 0, n \geq 1, \quad 0 \leq r \leq R, 0 \leq \theta \leq 2\pi, \tag{4.1}$$

$$\varphi^{\mathrm{s}}_{m,n}(r,\theta) = W_{m,n} J_m(r j'_{m,n}/R)\sin(m\theta), \qquad m, n \geq 1, \quad 0 \leq r \leq R, 0 \leq \theta \leq 2\pi, \tag{4.2}$$

$$W_{m,n} = \begin{cases} \left(R\sqrt{\pi}|J_0(j'_{m,n})|\right)^{-1}, & \text{for } m = 0, \\ \sqrt{2}\left(R\sqrt{\pi}\left(1 - \left(\frac{m}{j'_{m,n}}\right)^2\right)|J_m(j'_{m,n})|\right)^{-1}, & \text{for } m \geq 1, \end{cases} \tag{4.3}$$

expressed here in polar coordinates, both solve (2.4) with eigenvalue $\lambda_{m,n} = (j'_{m,n}/R)^2$, for $(m,n) \in \mathbb{N}_0 \times \mathbb{N}$. The multiplicity of $\lambda_{m,n}$ is 1 for $m = 0$ and 2 for $m \geq 1$. Here J_m is the Bessel function of the first kind and order m, and $j'_{m,n}$ is the n'th zero of $J'_m(z)$. Expanding $H_i(x)$ we have

$$H_i(x) = \sum_{m=0}^{M}\sum_{n=1}^{N}\left(\hat{H}^{\mathrm{c}}_{i,m,n}\varphi^{\mathrm{c}}_{m,n} + \hat{H}^{\mathrm{s}}_{i,m,n}\varphi^{\mathrm{s}}_{m,n}\right), \tag{4.4}$$

where $\hat{H}^{\mathrm{c}}_{i,m,n}$ and $\hat{H}^{\mathrm{s}}_{i,m,n}$ are the coefficients associated with $\varphi^{\mathrm{c}}_{m,n}$ and $\varphi^{\mathrm{s}}_{m,n}$, respectively. The quadrature weights are given by

$$a^k_{j,m,n} = \int_\Omega p_j(x,t^k)\varphi^{\mathrm{c}}_{m,n}(x)\,\mathrm{d}x, \qquad b^k_{j,m,n} = \int_\Omega p_j(x,t^k)\varphi^{\mathrm{s}}_{m,n}(x)\,\mathrm{d}x, \tag{4.5}$$

and equation (2.6) takes the form

$$I^k_{i,j} \approx -\eta \sum_{m=0}^{M}\sum_{n=1}^{N}\left(\hat{H}^{\mathrm{c}}_{i,m,n}a^k_{j,m,n} + \hat{H}^{\mathrm{s}}_{i,m,n}b^k_{j,m,n}\right).$$

We set $M = 24$ and $N = 25$ in (4.4), and hence use $N_\varphi = N + 2NM = 1225$ basis functions to represent our reconstructed H. The matrix K is formed according to (2.10), and the weights in equation (2.7) is calculated using a $\mathcal{P}_1$-quadrature in FEniCS. The matrix K has $N_S N_t = 154 \times 27 = 4158$ rows and $N_\varphi = 1225$ columns. Figure 4 shows a semi-log plot of the singular values of K, showing a exponential decay. This behaviour certainly underpin the need for a regularization when when approximately solving (2.11).

Figure 5(a) shows the true H_i computed from the high contrast phantom, with boundary conditions $f_2(x,y) = y$. The regularization parameter in the least-squares problem was chosen optimally by computing the regularization-error curve and picking the minimum error choice. This was done to ensure comparability of reconstruction quality. Note the spike-like features around the edges of the inclusions. In Figure 5(b), we see the direct projection of H_i on the basis functions as in (4.4), i.e., the best possible representation. We see that the smooth basis functions capture the smoother features of H_i, but that the amplitude in the spike-like regions are diminished. Figure 5(c) shows the reconstruction of H_i from $I_{i,j}$ with no noise. Notably, it captures the main features seen in the projection very well. In Figure 5(d), 0.01% noise is added to the measurements. While the finer details of H_i are lost, most of the coarser features of H_i are still clearly present in the reconstruction. In Figure 5(e) 0.1% noise is added to the measurements and we clearly see the diminishing returns due to the signal to noise ratio. In general, our ability to reconstruct H_i strongly depends on K, and hence on the acoustic fields.

4.2 Reconstructing the electrical conductivity

As observed in Section 3.2, when η is small then it is difficult to separate the signal and the noise. To examine this correspondence systematically, we run our full inversion method on data

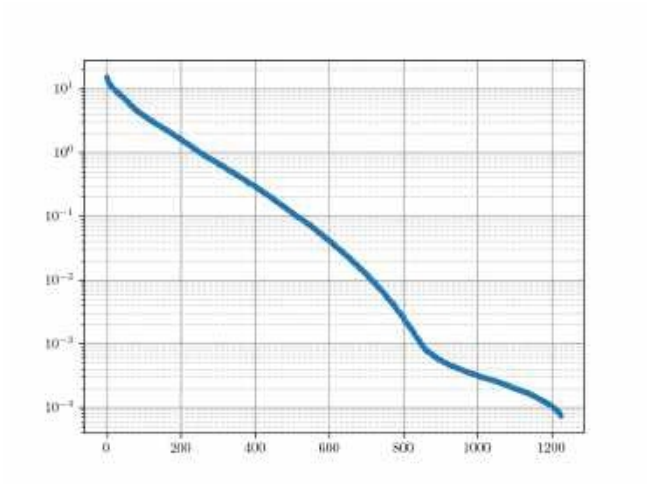
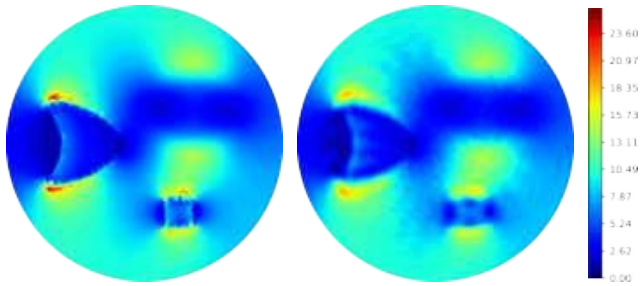
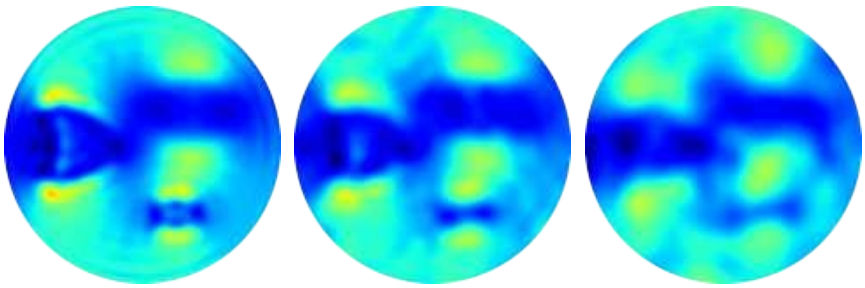


Figure 4: Plot of $(n, \log(\mu))$, where μ_n is the n'th singular value of K.



(a) Directly computed H

(b) Projected H



(c) Reconstructed H with $\delta = 0\%$, $\eta = 10^{-8}$

(d) Reconstructed H with $\delta = 0.01\%$, $\eta = 10^{-8}$

(e) Reconstructed H with $\delta = 0.1\%$, $\eta = 10^{-8}$

Figure 5: Power density data, boundary condition $f(x, y) = y$

obtained by perturbing our phantoms using three different values of η, and three different noise levels.

Figure 6 shows reconstructions of the high contrast phantom, for $\eta = 10^{-7}\,\mathrm{Pa}^{-1}$, $10^{-8}\,\mathrm{Pa}^{-1}$ and $10^{-9}\,\mathrm{Pa}^{-1}$ and noise levels 0%, 0.01% and 0.1%. The figure is organized such that η is decreasing from left to right, while the noise is increasing from top to bottom. In the first column, the reconstructions capture the features of the phantom quite well. The convex kite inclusion is clearly reconstructed, but the edges of the small square inclusion are smoothed out. The peak values of the inclusions are are slightly lower than in the phantoms. In the second column, with $\eta = 10^{-8}\,\mathrm{Pa}^{-1}$, the situation is the same, except for the reconstruction with 0.1% noise. Here, the shapes of the inclusions are severely blurred, and the small square is almost invisible. The same is true in rightmost column ($\eta = 10^{-9}\,\mathrm{Pa}^{-1}$), but with 0.1% noise the inclusions are barely visible. Last, we note that in the case of 0% noise, the small, square inclusion is better reconstructed when η decreases. This is most likely due to the fact that the error in the linearization leading to equation (2.3) decreases with η.

Figure 7 shows reconstructions of the low frequency phantom in Figure 3, again with $\eta = 10^{-7}\,\mathrm{Pa}^{-1}$, $10^{-8}\,\mathrm{Pa}^{-1}$ and $10^{-9}\,\mathrm{Pa}^{-1}$ and noise levels 0%, 0.01% and 0.1%. In the leftmost column, with $\eta = 10^{-7}\,\mathrm{Pa}^{-1}$, we see that the noisefree and 0.01% noise measurements yield good reconstructions both with respect to inclusion shapes and amplitudes. In particular, the square is quite well reconstructed. However, with 0.1% noise, the quality has already deteriorated quite a bit, and the reconstructions appears to be more blurred and contaminated by noise. The same is true in the center column; smallest inclusion is barely visible in the reconstruction with $\eta = 10^{-8}$ and 0.1% noise. With 0.1% noise for $\eta = 10^{-9}\,\mathrm{Pa}^{-1}$, we have clearly reached a limit with respect to the size of η and the noise level.

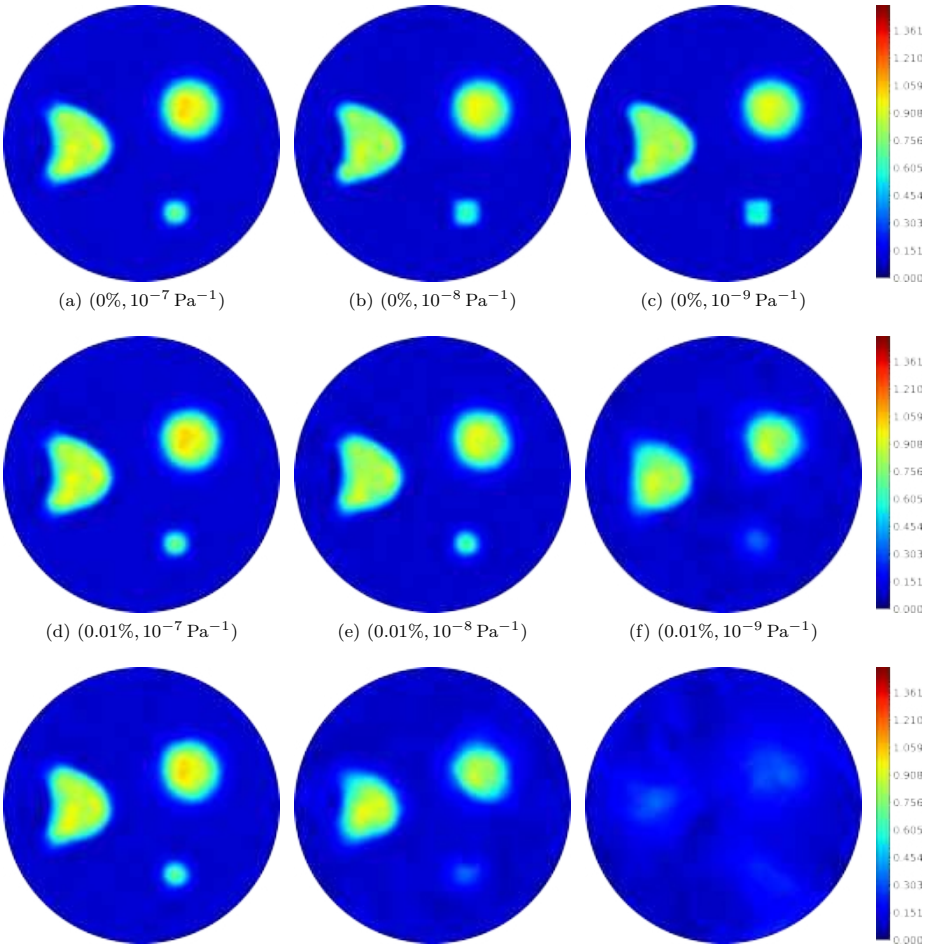
Also, note the diagonal pattern in both figures. From the consideration in Section 3.2, reconstructions, where the ratio η/δ is constant, are expected to be of similar quality. This is observed by comparing, e.g., the reconstruction with $\eta = 10^{-8}\,\mathrm{Pa}^{-1}$ and 0.1% noise and the reconstruction with $\eta = 10^{-9}\,\mathrm{Pa}^{-1}$ and 0.01% noise.

Figure 8 shows reconstruction from a limited angle situation, where the induced currents are supported on the lower half of the boundary, and measurements are taken on the same part of the boundary. Such a situation could easily occur when one does not have access to the full boundary of the object. In the experiments we use $\eta = 10^{-8}\,\mathrm{Pa}^{-1}$ and noise levels 0%, 0.01% and 0.1%. Parts of the inclusions are visible, but the amplitude is substantially diminished. Furthermore, the observation from [21], that objects closer to the measurement boundary are better reconstructed in the case of limited angle measurements, is also observed here. The small, quadratic inclusion is better reconstructed than the kite and the Gaussian.

5 Conclusions and further work

In this paper we have introduced a complete inversion framework for AET, motivated by an ambition to investigate if AET can provide a modality for stable imaging of the electrical conductivity in realistic, relevant situations. We tested our inversion method on simulated data, with satisfactory results. Further, the results from the numerical investigations in Section 4 indicates that the answer to the questions regarding feasibility of AET depends critically on the acousto-electric coupling constant and on the noise level. In producing our simulated data, we sought to use values of both η and δ in a range that can realistic in applications. It then appears (as a stroke of luck) that this indeed works. Had the noise been slightly higher or η slightly lower, we would have seen very little.

Even though we have striven to make our simulated data realistic, there are still a number of things that are idealized in our approach. Most notably, the assumption that we have accurate knowledge of the internal pressure field. It remains to do a more detailed investigation of the situation where a more realistic acoustic model is used and uncertainties in the acoustic fields is modeled. Also, the assumption that the constant acousto-electric coupling parameter η is



(a) $(0\%, 10^{-7}\,\mathrm{Pa}^{-1})$ (b) $(0\%, 10^{-8}\,\mathrm{Pa}^{-1})$ (c) $(0\%, 10^{-9}\,\mathrm{Pa}^{-1})$

(d) $(0.01\%, 10^{-7}\,\mathrm{Pa}^{-1})$ (e) $(0.01\%, 10^{-8}\,\mathrm{Pa}^{-1})$ (f) $(0.01\%, 10^{-9}\,\mathrm{Pa}^{-1})$

(g) $(0.1\%, 10^{-7}\,\mathrm{Pa}^{-1})$ (h) $(0.1\%, 10^{-8}\,\mathrm{Pa}^{-1})$ (i) $(0.1\%, 10^{-9}\,\mathrm{Pa}^{-1})$

Figure 6: Mixed high contrast phantom reconstructions. Each is marked with its corresponding (δ, η) values.

known seems to be impractical. Also, the situation of limited boundary measurements should be studied further, and a complete electrode model [32] can be introduced.

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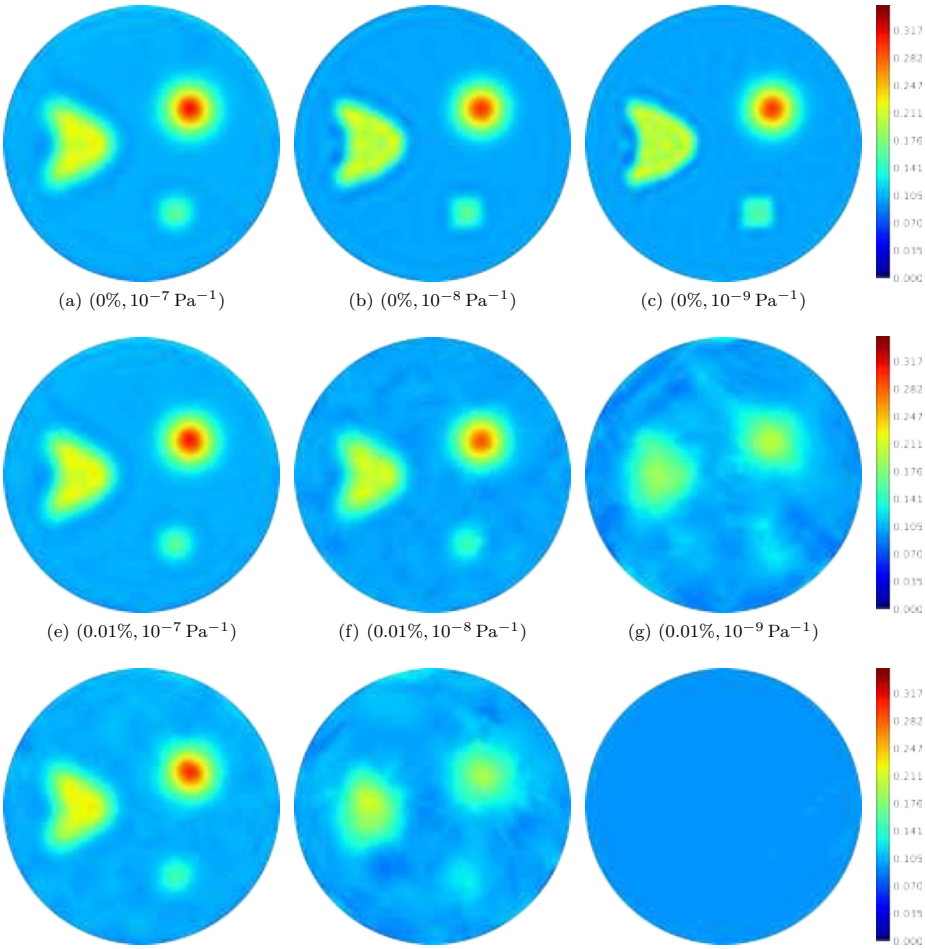

(a) $(0\%, 10^{-7}\,\mathrm{Pa}^{-1})$ (b) $(0\%, 10^{-8}\,\mathrm{Pa}^{-1})$ (c) $(0\%, 10^{-9}\,\mathrm{Pa}^{-1})$

(e) $(0.01\%, 10^{-7}\,\mathrm{Pa}^{-1})$ (f) $(0.01\%, 10^{-8}\,\mathrm{Pa}^{-1})$ (g) $(0.01\%, 10^{-9}\,\mathrm{Pa}^{-1})$

(i) $(0.1\%, 10^{-7}\,\mathrm{Pa}^{-1})$ (j) $(0.1\%, 10^{-8}\,\mathrm{Pa}^{-1})$ (k) $(0.1\%, 10^{-9}\,\mathrm{Pa}^{-1})$

Figure 7: Mixed low contrast phantom reconstructions. Each is marked with its corresponding (δ, η) values.

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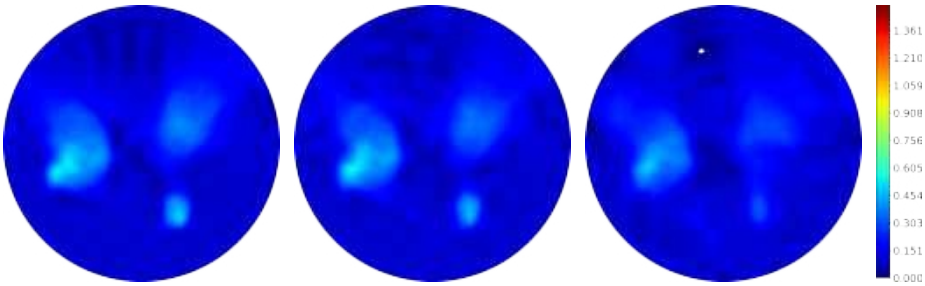

(a) $(0\%, 10^{-8}\,\mathrm{Pa}^{-1})$ (b) $(0.01\%, 10^{-8}\,\mathrm{Pa}^{-1})$ (c) $(0.1\%, 10^{-8}\,\mathrm{Pa}^{-1})$

Figure 8: Mixed high contrast phantom reconstructions with limited boundary measurements. Each is marked with its corresponding (δ, η) values.

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Appendix C

Paper 3

Stable source reconstruction from finitely many frequency observations in the Multi-frequency Inverse Source Problem

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Abstract

We consider the multi-frequency inverse source problem for the Helmholtz equation in the plane. A criterion is proved which, given a source from a certain family of finite-dimensional spaces, constructively identifies minimal finite sets of measurement frequencies sufficient for the unique reconstruction of the source. Under an additional, mild criterion on the frequencies, the reconstruction is showed to be stable, in the sense that it avoids measurement data associated with small singular values. To this end we analyze singular systems of source-to-measurement forward operators at different frequencies, and express the criterion for a minimal frequency set in terms of the positive zeros of the Bessel functions of the first kind and integer order and the source domain radius.

1 Introduction

This paper concerns the multi-frequency inverse source problem (ISP) for the scalar Helmholtz equation in the plane. For a compactly supported source, the frequency dependent radiated field is measured on a curve outside the source support, at several frequencies. The inverse problem is then to estimate the source from these measurements.

Let $D, D_0 \subset \mathbb{R}^2$ be concentric, open discs of radii R and R_0 respectively, with $0 < R_0 < R$. Let $s \in L^2(D_0)$ be the source, such that $s \subset D_0$. For a fixed, positive frequency[1] k, let u_k be the unique solution of

$$\begin{cases} (\Delta + k^2)u_k(x) = s(x), & x \in \mathbb{R}^2, \\ \lim\limits_{\|x\|\to\infty} \|x\|^{1/2}(\partial_{\|x\|} - ik)u_k = 0 & \text{uniformly for } x/\|x\| \in S^1, \end{cases} \tag{1}$$

given by

$$u_k(x) = \int_{D_0} G_k(x,y)s(y)dy \quad x \in \partial D. \tag{2}$$

[1]The proper name of the constant k in (5.2) is *wave number*, but it is closely related to the frequency, $k = \omega/c$, where c is the constant wave speed and ω is the angular frequency.

Here, $G_k(x,y) = -\frac{i}{4}H_0^{(1)}(k|x-y|)$ is the Hankel function of the first kind and of order zero. For a fixed frequency k we define the forward operator

$$\begin{aligned} F_k \colon L^2(D_0) &\to L^2(\partial D), \\ s &\mapsto U_k = u_k\big|_{\partial D}. \end{aligned}$$

F_k is a linear, compact operator (see for instance [6] for a proof of compactness). We call U_k the *measurement at frequency* k.

Given a set of frequencies $Q = \{k_j\}_{j\in I}$, where I is some index set, and a corresponding set of measurements $\{U_{k_j}\}$, the multi-frequency inverse source problem is to estimate $s \in L^2(D_0)$ such that

$$U_{k_j} = F_{k_j} s \quad \text{for all } k_j \in Q.$$

This problem has been investigated by several authors in recent years, for example in [6, 4, 5, 26, 16]. It has a wide range of applications, for example in optical tomography [3], antenna control [18] and in the acoustic reconstruction part of several hybrid imaging methods, most notably photo- and thermoacoustic tomography [19, 2], see also [6, 1].
The single-frequency ISP, i.e., when one only has a measurement at a single frequency, is not uniquely solvable due to F_k having a large kernel. In fact $\ker(F_k)$ consist of all so-called non-radiating sources (at frequency k) including functions like $s = (\Delta + k^2)u; u \in C_c^\infty(D_0)$, see also [9]. Stability estimates for the single-frequency ISP in terms of a description of the singular value spectrum are shown in [16]. For the multi-frequency ISP, uniqueness and stability of the solution are known when measurements are available over a set of frequencies with an accumulation point [6], and for a set of frequencies coinciding with eigenvalues of the Dirichlet-Laplacian [11](both situations necessarily involve infinitely many measurements). In [1] it was shown that for arbitrary $s \in L^2$, measurements at an finite number of frequencies are insufficient to obtain a unique solution to the ISP. Both iterative and direct numerical inversion methods have been proposed in the literature [7, 12, 11, 26].
In contrast to the above mentioned references we consider in this paper a more practical ISP of finite-dimensional nature. We consider measurements at a finite number of frequencies and reconstruction of sources in a certain finite-dimensional subspace spanned by low-index Fourier-Bessel functions. The main result shows that for a few, carefully chosen frequencies, we can uniquely recover such functions, and that reconstructions are robust with respect to noise in the measurements.

The rest of the paper is organized as follows: In Section 2, we introduce the Fourier-Bessel basis and singular value expansion of F_k, and show how this motivates the choice of a finite-dimensional source space. In Section 3, we construct a matrix equation relating the multi-frequency measurements and the finite-dimensional source, and show how the invertibility of this equation depends on the measurement frequencies. Building on these results, Section 4 shows how distribution of the zeros of Bessel functions allow us to substantially reduce the number of frequencies initially needed to reconstruct the finite dimensional sources. By relating our analysis with recent results in [16] we also show that, with an additional requirement on the frequency set, reconstructions will be robust towards noise in the measurements. In Section 5, we conduct a set of numerical experiments to validate our findings.

2 The finite-dimensional source space and the singular value expansion

We now introduce and relate two important bases for $L^2(D_0)$, the Fourier-Bessel basis and the singular value expansion. This will motivate the construction of the finite-dimensional source space.

First, we describe the Fourier-Bessel (FB) basis in $L^2(D_0)$. This basis appears naturally when solving PDE problems in a disc geometry using separation of variables [23]. For $f \in L^2(D_0)$ we can expand

$$f(r,\theta) = \sum_{m\in\mathbb{Z}} \sum_{n\in\mathbb{N}} \hat{f}_{m,n}\varphi_{m,n}(r,\theta), \quad 0 \le r \le R_0, 0 \le \theta \le 2\pi, \tag{3}$$

where the angular-radial basis functions are given by

$$\varphi_{m,n}(r,\theta) = \left(\sqrt{\pi J_{m+1}^2(j_{m,n})}R_0\right)^{-1} e^{im\theta} J_m(k_{m,n}r). \tag{4}$$

$J_m(x)$ is the Bessel function of the first kind and of order m, and $j_{m,n}$ denotes the n'th, real and positive zero of $J_m(x)$, i.e., $J_m(j_{m,n}) = 0 \quad \forall m \in \mathbb{N}_0, n \in \mathbb{N}$. The numbers $k_{m,n}$ are the zeros $j_{m,n}$ scaled with respect to R_0, $k_{m,n} = j_{m,n}/R_0$. The equality in (3) is is understood in the L^2-sense, and the functions $\{\varphi_{m,n}\}$ constitute an orthonormal basis for $L^2(D_0)$. The FB-coefficients $\hat{f}_{m,n}$ are computed with the standard inner product on $L^2(D_0)$, here in polar coordinates,

$$\hat{f}_{m,n} = (f, \varphi_{m,n})_{L^2(D_0)} = \int_0^{R_0}\int_0^{2\pi} f(r,\theta)\overline{\varphi_{m,n}(r,\theta)} r \mathrm{d}r \mathrm{d}\theta. \tag{5}$$

Since $J_{-m}(x) = (-1)^m J_m(x)$ for $m \in \mathbb{N}_0$, we have that[2] $\varphi_{-m,n} = (-1)^m \overline{\varphi_{m,n}}$. For real-valued $f \in L^2(D_0)$, it follows that

$$\overline{\hat{f}_{m,n}} = \overline{(f, \varphi_{m,n})}_{L^2(D_0)} = (-1)^m (f, \varphi_{-m,n})_{L^2(D_0)} = (-1)^m \hat{f}_{-m,n}. \tag{6}$$

Another consequence is that the scaled Bessel zero $k_{m,n}$ appears in both $\varphi_{m,n}$ and $\varphi_{-m,n}$.

The main idea in this paper relies on reconstructing source functions in a finite-dimensional subspace spanned by FB-functions. For $(M,N) \in \mathbb{N}_0 \times \mathbb{N}$ and D_0, we define the $(2M+1)N$-dimensional subspace

$$S_{M,N} = \mathrm{span}\{\varphi_{m,n},\ m = -M,\dots,M,\ n = 1,\dots,N\}. \tag{7}$$

The projection $P_{M,N}$ of $L^2(D_0)$ onto $S_{M,N}$ is given by

$$P_{M,N}s = \sum_{m=-M}^{M}\sum_{n=1}^{N} \hat{s}_{m,n}\varphi_{m,n}, \quad \text{where } \hat{s}_{m,n} = (s, \varphi_{m,n})_{L^2(D_0)}. \tag{8}$$

Next, we introduce the singular value expansion (SVE) of the forward operator F_k as given in [6, 16]. We write

$$F_k = \sigma_0^k (\psi_0^k, \cdot)_{L^2(D_0)}\phi_0^k + \sum_{n\in\mathbb{N}} \sigma_n^k \left((\psi_n^k, \cdot)_{L^2(D_0)}\phi_n^k + (\psi_{-n}^k, \cdot)_{L^2(D_0)}\phi_{-n}^k\right), \tag{9}$$

[2]The normalization constant is the same for indices m and $-m$ since $\int J_{-m}(k_{m,n}r)^2 r\mathrm{d}r = \int J_m(k_{m,n}r)^2 r\mathrm{d}r$

and the singular values and vectors, here given in polar coordinates, take the form

$$\sigma_n^k = \sqrt{2R}\pi R_0 |H_n^{(1)}(\kappa)| A_n(\kappa_0), \tag{10}$$

$$\psi_n^k(r,\theta) = (\sqrt{\pi} R_0 A_n(\kappa_0))^{-1} J_n(kr) e^{in\theta}, \tag{11}$$

$$\phi_n^k(\theta) = (2\pi R)^{-\frac{1}{2}} e^{i \arg H_n^{(1)}(\kappa)} e^{in\theta}, \tag{12}$$

where $n \in \mathbb{Z}$, $\kappa = kR$, $\kappa_0 = kR_0$, $H_n^{(1)}(z)$ is the Hankel function of the first kind and order n, and $A_n(\kappa) = \sqrt{J_n^2(\kappa) - J_{n-1}(\kappa) J_{n+1}(\kappa)}$. The singular vectors (11) constitute an orthonormal basis for range(F_k).

Consider now the particular frequency $k = k_{m,n}$. By using the standard recursion formula for Bessel functions [25] and the fact that $\kappa_0 = j_{m,n}$, we get that

$$A_{\pm m}(\kappa_0) = \sqrt{-J_{m+1}(\kappa_0) J_{m-1}(\kappa_0)} = \sqrt{-J_{m+1}(\kappa_0)\left(\frac{2m}{\kappa_0} J_m(\kappa_0) - J_{m+1}(\kappa_0)\right)} = \sqrt{J_{m+1}^2(j_{m,n})}, \tag{13}$$

and hence

$$\psi_{\pm m}^{k_{m,n}} = (\sqrt{\pi} R_0 A_{\pm m}(\kappa_0))^{-1} J_{\pm m}(k_{m,n} r) e^{\pm im\theta} = \left(\sqrt{\pi J_{m+1}^2(j_{m,n})} R_0\right)^{-1} J_{\pm m}(k_{m,n} r) e^{\pm im\theta} = \varphi_{\pm m,n}. \tag{14}$$

Thus, at the particular frequencies $k_{m,n}$ the singular vectors $\psi_m^{k_{m,n}}$ coincide with the FB basis functions $\varphi_{m,n}$. We now note the following: given $s = \sum_{m,n} \hat{s}_{m,n} \varphi_{m,n} \in L^2(D_0)$ and the measurement $U_{k_{m,n}} = F_{k_{m,n}} s$ at the frequency $k_{m,n}$, we can invert the SVE to find the two FB-coefficients $\hat{s}_{\pm m,n}$, since

$$\frac{(U_{k_{m,n}}, \phi_{\pm m}^{k_{m,n}})_{L^2(\partial D)}}{\sigma_{|m|}^{k_{m,n}}} = (s, \psi_{\pm m}^{k_{m,n}})_{L^2(D_0)} = (s, \varphi_{\pm m,n})_{L^2(D_0)} = \hat{s}_{\pm m,n}. \tag{15}$$

Hence, if we have measurements at all frequencies $\{k_{m,n}\}$ we could use (15) to reconstruct any $s \in L^2(D_0)$. This is similar to what was proposed in [11].

We now consider a finite-dimensional setup. Define the frequency set consisting of the scaled Bessel zeros,

$$Q_{M,N} = \{k_{m,n}\}_{m=0,n=1}^{M,N}.$$

If we have measurements at all frequencies in the set $Q_{M,N}$, we could similarly reconstruct any source $s \in S_{M,N}$, or reconstruct the projection $P_{M,N} s$ of a general source $s \in L^2$. This method would require one measurement $U_{k_{m,n}}$ for each reconstructed coefficient $\hat{s}_{m,n}$; but note that for each (m,n) only one SVE-coefficient of $U_{k_{m,n}}$ is used. This gives rise to new questions of economical nature: From a measurement at some frequency k, how many, and which, FB-coefficients can be stably reconstructed? And given a finite-dimensional source space $S_{M,N}$, can we find a possibly small set of frequencies $Q_s = \{k_j\}$ such that we can guarantee numerically stable reconstruction of any source in $S_{M,N}$ from the subset of measurements $\{U_{k_j}\}_{k_j \in Q_s}$? We will address the latter question first, and the answer to the first question will follow.

3 Reconstruction of FB-coefficients from multi-frequency measurements

In this section we obtain conditions on the set of frequencies needed to reconstruct a source in $S_{M,N}$. For multi-frequency measurements, we obtain this by constructing a change of basis matrix

K from the FB-basis to the SVE-basis. We then show that requiring that K be invertible leads to certain conditions on the frequency set. In the next section we show how knowledge about the distribution of the zeros of Bessel functions and the conditions on the frequencies allow us to substantially reduce the frequency set and still be able to stably reconstruct the source.

3.1 Change of basis: From SVE- to FB-coefficients

Let $Q_r = \{\tilde{k}_{m,n}\}_{m=0,n=1}^{M,N}$ be a set of *not necessarily distinct* frequencies. The indexing resembles that of $Q_{M,N}$; this is to emphasis that the two are closely connected, and to ease the some of the preceding arguments. Next, let $\{F_{\tilde{k}}\}_{\tilde{k}\in Q_r}$ be the family of forward operators associated with Q_r. We now define the space $V_{M,N} \subset L^2(D_0)$ to be

$$V_{M,N} = \text{span}\{\psi_{\pm m}^{\tilde{k}_{m,n}} : \tilde{k}_{m,n} \in Q_r, m = 0, 1, \ldots, M, n = 1, 2, ..., N\}. \tag{16}$$

For a source $s \in S_{M,N}$ we have the FB-coefficients $\hat{s}_{m,n}$, and we introduce the SVE-coefficients $\hat{u}_{i,j} = (s, \psi_i^{\tilde{k}_{|i|,j}})_{L^2(D_0)}$. By expanding s in the FB-basis we get the linear relation between the two sets of coefficients

$$\hat{u}_{i,j} = (s, \psi_i^{\tilde{k}_{|i|,j}})_{L^2(D_0)} = \sum_{m=-M}^{M}\sum_{n=1}^{N} \hat{s}_{m,n}(\varphi_{m,n}, \psi_i^{\tilde{k}_{|i|,j}})_{L^2(D_0)} = \sum_{n=1}^{N} \hat{s}_{i,n}(\varphi_{i,n}, \psi_i^{\tilde{k}_{|i|,j}})_{L^2(D_0)}, \tag{17}$$

where the last equality follows from the partial orthogonality relation

$$(\varphi_{m,n}, \psi_i^{\tilde{k}_{|i|,j}})_{L^2(D_0)} = \delta_{m,i} C(\tilde{k}_{|i|,j}, m, n, i)$$

where $C(\tilde{k}_{|i|,j}, m, n, i)$ is a complex number and $\delta_{m,i}$ is the Kronecker-delta. We also note that

$$(\varphi_{-m,n}, \psi_{-m}^{\tilde{k}_{m,n}})_{L^2(D_0)} = (\varphi_{m,n}, \psi_m^{\tilde{k}_{m,n}})_{L^2(D_0)}, \tag{18}$$

due to a similar argument as in (6). Equation (17) constitutes a linear relation between the coefficients in the two bases. Next, define the vectors

$$\hat{S} = (\hat{s}_{1,1}, \hat{s}_{1,2}, ..., \hat{s}_{1,N}, \hat{s}_{2,1}, ..., \hat{s}_{M,N}, \hat{s}_{0,1}, ..., \hat{s}_{0,N}, \hat{s}_{-1,1}, ..., \hat{s}_{-M,N})^T,$$
$$\hat{U} = (\hat{u}_{1,1}, \hat{u}_{1,2}, ..., \hat{u}_{1,N}, \hat{u}_{2,1}, ..., \hat{u}_{M,N}, \hat{u}_{0,1}, ..., \hat{u}_{0,N}, \hat{u}_{-1,1}, ..., \hat{u}_{-M,N})^T,$$

and the matrix $K \in \mathbb{R}^{(2M+1)N\times(2M+1)N}$ by

$$K = \begin{bmatrix} K^+ & 0 \\ 0 & K^- \end{bmatrix}, \tag{19}$$

with $K^+ \in \mathbb{R}^{(M+1)N\times(M+1)N}$ and $K^- \in \mathbb{R}^{MN\times MN}$ defined as the block-diagonal matrices

$$K^+ = \begin{bmatrix} K_1^+ & 0 & \ldots & 0 \\ 0 & K_2^+ & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & K_{M+1}^+ \end{bmatrix} \qquad K^- = \begin{bmatrix} K_1^- & 0 & \ldots & 0 \\ 0 & K_2^- & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & K_M^- \end{bmatrix} \tag{20}$$

$$K_m^+ = \left\{(\varphi_{m,n}, \psi_m^{\tilde{k}_{m,i}})_{L^2(D_0)}\right\}_{i=1,n=1}^{N,N} \quad \text{for } m = 1, 2, ...M, \tag{21}$$

$$K_{M+1}^+ = \left\{(\varphi_{0,n}, \psi_0^{\tilde{k}_{0,i}})_{L^2(D_0)}\right\}_{i=1,n=1}^{N,N}, \tag{22}$$

$$K_m^- = \left\{(\varphi_{-m,n}, \psi_{-m}^{\tilde{k}_{m,i}})_{L^2(D_0)}\right\}_{i=1,n=1}^{N,N} \quad \text{for } m = 1, 2, ...M, \tag{23}$$

where all $\tilde{k}_{m,i} \in Q_r$. The linear system (17) can now be written as $\hat{U} = K\hat{S}$. It is clear that K is a mapping from the space of FB-coefficients for $S_{M,N}$ to the space of SVE-coefficients for the space $V_{M,N}$. The SVE-coefficients are related to the measurements $\{U_{k_j}\}$ through the relation

$$\hat{u}_{i,j} = \frac{(U_{\tilde{k}_{|i|,j}}, \phi_i^{\tilde{k}_{|i|,j}})_{L^2(\partial D)}}{\sigma_{|i|}^{\tilde{k}_{|i|,j}}}. \tag{24}$$

It is evident that reconstruction of s mounts to reconstructing the vector $\hat{S}$ from $\hat{U}$, and hence depends on the invertibility of K. The properties of K depend on the choice of frequency set Q_r. From (15), it is evident that if $Q_r = Q_{M,N}$, we have that $K = I$ since

$$\hat{u}_{i,j} = (s, \psi_i^{\tilde{k}_{|i|,j}})_{L^2(D_0)} = \sum_{n=1}^{N} \hat{s}_{i,n}(\varphi_{i,n}, \psi_i^{\tilde{k}_{|i|,j}})_{L^2(D_0)} = \hat{s}_{i,j}.$$

In the case that $Q_r \neq Q_{M,N}$, then $K \neq I$, but might still be invertible, and hence all the FB-coefficients should be recoverable from the SVE-coefficients. We therefore proceed to analyze the interaction between Q_r and invertibility of K.

3.2 Invertibility of K

This section contains Theorem 1, which is our main result. It says that if the frequencies in Q_r are not too far away from those in $Q_{M,N}$, then K is invertible. Lemma 1 gives a computable bound for this distance, and is required in the proof of Theorem 1. We will need the functions

$$\mu_m = \begin{cases} \frac{R_0}{\pi} & \text{for } m \geq 1, \\ \frac{R_0}{j_{0,2} - j_{0,1}} & \text{for } m = 0, \end{cases} \quad \text{and} \quad \chi_{j,i} = \begin{cases} 1 & \text{for } i \neq j, \\ 0 & \text{for } i = j. \end{cases} \tag{25}$$

Lemma 1. *For $R_0 > 0$ and $(M, N) \in \mathbb{N}_0 \times \mathbb{N}$, let $Q_{M,N} = \{j_{m,n}/R_0\}_{m=0,n=1}^{M,N}$. Then, for any indices $0 \leq m \leq M$ and $1 \leq i \leq N$, there is a largest positive number $\Delta k_{m,i}$ such that*

$$k_{m,i} \pm \Delta k_{m,i} \in (k_{m,i-1}, k_{m,i+1})$$

and that the inequality

$$\begin{aligned} \left|\frac{k_{m,i}}{(k_{m,i} \pm \delta)^2 - k_{m,i}^2}\right| \geq \chi_{1,i} &\left(\frac{\mu_m}{2} \log\left(\frac{(k_{m,i} \pm \delta)^2 - k_{m,1}^2}{(k_{m,i} \pm \delta)^2 - k_{m,i-1}^2}\right) + \frac{k_{m,i-1}}{(k_{m,i} \pm \delta)^2 - k_{m,i-1}^2}\right) \\ &+ \chi_{N,i}\left(\frac{\mu_m}{2} \log\left(\frac{k_{m,N}^2 - (k_{m,i} \pm \delta)^2}{k_{m,i+1}^2 - (k_{m,i} \pm \delta)^2}\right) + \frac{k_{m,i+1}}{k_{m,i+1}^2 - (k_{m,i} \pm \delta)^2}\right) \end{aligned} \tag{26}$$

holds for all $\delta \in (0, \Delta k_{m,i})$.
Furthermore, let $\Delta k = \min_{m,i} \Delta k_{m,i}$. *Then the inequality*

$$\left| \frac{k_{m,i}}{(k_{m,i} \pm \delta)^2 - k_{m,i}^2} \right| > \sum_{n=1, n \neq i}^{N} \left| \frac{k_{m,n}}{(k_{m,i} \pm \delta)^2 - k_{m,n}^2} \right| \tag{27}$$

holds for all $0 < \delta < \Delta k$ *and all indices* $0 \leq m \leq M$ *and* $1 \leq i \leq N$.

Proof. For $R_0 > 0$ and $(M, N) \in \mathbb{N}_0 \times \mathbb{N}$, let $Q_{M,N} = \{k_{m,n}\}_{m=0,n=1}^{M,N}$, where $k_{m,n} = j_{m,n}/R_0$. For fixed $0 \leq m \leq M$, the sequence $\{k_{m,n}\}_{n=1}^{N}$ is strictly increasing and positive (see e.g. Proposition 1 in Section 4.1). We fix m and i, and look at the inequality

$$\left| \frac{k_{m,i}}{x^2 - k_{m,i}^2} \right| > \sum_{n=1, n \neq i}^{N} \left| \frac{k_{m,n}}{x^2 - k_{m,n}^2} \right| \tag{28}$$

with the assumption $k_{m,i-1} < x < k_{m,i+1}$. We estimate the sum om the right hand side. Define the functions

$$S^-(x,k) = \frac{k}{x^2 - k^2} \quad \text{for } 0 < k < x, \quad \text{and} \quad S^+(x,k) = \frac{k}{k^2 - x^2} \quad \text{for } x < k,$$

and write

$$\sum_{n=1, n \neq i}^{N} \left| \frac{k_{m,n}}{x^2 - k_{m,n}^2} \right| = \sum_{n=1}^{i-1} S^-(x, k_{m,n}) + \sum_{n=i+1}^{N} S^+(x, k_{m,n}).$$

Since $S^-(x, k_{m,n}) \leq S^-(x,s)$ for $k_{m,n} \leq s < x$, and $S^+(x, k_{m,n}) \leq S^+(x,s)$ for $x \leq s < k_{m,n}$, we obtain the inequalities

$$\begin{aligned}
S^-(x, k_{m,n}) &< \mu_m \int_{k_{m,n}}^{k_{m,n+1}} \frac{s}{x^2 - s^2} \mathrm{d}s \\
&= \frac{\mu_m}{2} (\log(x^2 - k_{m,n}^2) - \log(x^2 - k_{m,n+1}^2)), \quad \text{for } x < k_{m,i-1}, \\
S^+(x, k_{m,n}) &< \mu_m \int_{k_{m,n-1}}^{k_{m,n}} \frac{s}{s^2 - x^2} \mathrm{d}s \\
&= \frac{\mu_m}{2} (\log(k_{m,n}^2 - x^2) - \log(k_{m,n-1}^2 - x^2)), \quad \text{for } x > k_{i+1}.
\end{aligned}$$

The multiplication by μ_m, defined in (25), corrects for the integration being done over an interval of length $k_{m,i+1} - k_{m,i}$, since $k_{m,i+1} - k_{m,i} \geq \frac{\pi}{R_0}$ for $m, i \geq 1$, and $k_{0,i+1} - k_{0,i} \geq \frac{j_{0,2} - j_{0,1}}{R_0}$ for $i \geq 1$. These properties are described in Section 4, where we get more in to detail about the properties of the zeros of Bessel functions. The estimates are quite sharp away from the diagonal, except for

$S^-(x, k_{m,n-1})$ and $S^+(x, k_{m,n+1})$, so we include these terms directly in the sums and get

$$\begin{aligned}\sum_{n=1}^{i-1} S^-(x, k_{m,n}) &< \frac{\mu_m}{2}(\log(x^2 - k_{m,1}^2) - \log(x^2 - k_{m,i-1}^2)) + S^-(x, k_{m,i-1}) \\ &= \frac{\mu_m}{2}\log\left(\frac{x^2 - k_{m,1}^2}{x^2 - k_{m,i-1}^2}\right) + \frac{k_{m,i-1}}{x^2 - k_{m,i-1}^2} \\ \sum_{n=i+1}^{N} S^+(x, k_{m,n}) &< \frac{\mu_m}{2}(\log(k_{m,N}^2 - x^2) - \log(k_{m,i+1}^2 - x^2)) + S^+(x, k_{m,i+1}) \\ &= \frac{\mu_m}{2}\log\left(\frac{k_{m,N}^2 - x^2}{k_{m,i+1}^2 - x^2}\right) + \frac{k_{m,i+1}}{k_{m,i+1}^2 - x^2}.\end{aligned}$$

To have the estimate valid for the cases $i = 1$ and $i = N$, we need the function $\chi_{i,j}$ defined in (25). We then have that

$$\begin{aligned}\sum_{n=1, n\neq i}^{N} \left|\frac{k_{m,n}}{x^2 - k_{m,n}^2}\right| &< \chi_{1,i}\left(\frac{\mu_m}{2}\log\left(\frac{x^2 - k_{m,1}^2}{x^2 - k_{m,i-1}^2}\right) + \frac{k_{m,i-1}}{x^2 - k_{m,i-1}^2}\right) \\ &\quad + \chi_{N,i}\left(\frac{\mu_m}{2}\log\left(\frac{k_{m,N}^2 - x^2}{k_{m,i+1}^2 - x^2}\right) + \frac{k_{m,i+1}}{k_{m,i+1}^2 - x^2}\right). \end{aligned} \tag{29}$$

We rewrite x as a perturbation of $k_{m,i} \in Q_{M,N}$ on the form $x = k_{m,i} \pm \delta$, with $\delta > 0$ and such that $x \in (k_{m,i-1}, k_{m,i+1})$. Let $\Delta k_{m,i}$ be the largest number such that

$$\begin{aligned}\left|\frac{k_{m,i}}{(k_{m,i} \pm \delta)^2 - k_{m,i}^2}\right| &\geq \chi_{1,i}\left(\frac{\mu_m}{2}\log\left(\frac{(k_{m,i} \pm \delta)^2 - k_{m,1}^2}{(k_{m,i} \pm \delta)^2 - k_{m,i-1}^2}\right) + \frac{k_{m,i-1}}{(k_{m,i} \pm \delta)^2 - k_{m,i-1}^2}\right) \\ &\quad + \chi_{N,i}\left(\frac{\mu_m}{2}\log\left(\frac{k_{m,N}^2 - (k_{m,i} \pm \delta)^2}{k_{m,i+1}^2 - (k_{m,i} \pm \delta)^2}\right) + \frac{k_{m,i+1}}{k_{m,i+1}^2 - (k_{m,i} \pm \delta)^2}\right), \end{aligned} \tag{30}$$

holds for all $\delta \in (0, \Delta\kappa_{m,i})$. Such a $\Delta\kappa_{m,i}$ exists, since for fixed m and i and a closed interval $[k_{m,i} - \delta, k_{m,i} + \delta] \subset (k_{m,i-1}, k_{m,i+1})$ the estimate on the right hand side of (29) is continuous and bounded, and hence attains a maximum M. On the other hand, the function $\left|k_{m,i}/((k_{m,i} \pm \delta)^2 - k_{m,i}^2)\right|$ is unbounded from both sides as $\delta \to 0$, and we can find a δ such that $\min\left\{\left|k_{m,i}/((k_{m,i} \pm \delta)^2 - k_{m,i}^2)\right|\right\} \geq M$.
It follows from (29) that

$$\left|\frac{k_{m,i}}{(k_{m,i} \pm \delta)^2 - k_{m,i}^2}\right| > \sum_{n=1, n\neq i}^{N} \left|\frac{k_{m,n}}{(k_{m,i} \pm \delta)^2 - k_{m,n}^2}\right| \tag{31}$$

holds for all $\delta \in (0, \Delta k_{m,i})$, and that with $\Delta k = \min_{m,i} \Delta k_{m,i}$, it holds for all indices $0 \leq m \leq M$ and $1 \leq i \leq N$. □

We are now in the position to prove our main result.

Theorem 1. *Consider an ISP geometry specified by R and R_0, and a finite-dimensional source space $S_{M,N}$, and let $\Delta k > 0$ be as specified in Lemma 1. If then, for each $\tilde{k}_{m,i} \in Q_r$ and corresponding $k_{m,i} \in Q_{M,N}$, we have that $|\tilde{k}_{m,i} - k_{m,i}| < \Delta k$, the matrix K is invertible, and any $s \in S_{M,N}$ is uniquely recoverable from measurements at the frequencies $\tilde{k} \in Q_r$.*

Proof. (Thorem 1) We will show that under the assumption in Theorem 1, K is strictly diagonally dominant. Recall that a matrix $A \in \mathbb{R}^{m\times m}$ is called strictly diagonally dominant if

$$|a_{i,i}| > \sum_{j=1, j\neq i}^{m} |a_{i,j}| \quad \text{for } i = 1, 2, ...m. \tag{32}$$

It is well-known, see e.g. [24], that strictly diagonally dominant matrices are invertible.

We start by computing the entries of the matrix K, as given in (21)-(23). For the matrix-block K_m^+, the entries of the i'th row are given as

$$(\varphi_{m,n}, \psi_m^{\tilde{k}_{m,i}})_{L^2(D_0)}, \quad \text{for } n = 1, 2, ..., N.$$

Using (4) and (9), we compute

$$\begin{aligned}
&(\varphi_{m,n}, \psi_m^{\tilde{k}_{m,i}})_{L^2(D_0)} = \int_0^{R_0}\int_0^{2\pi} \psi_m^{\tilde{k}_{m,i}} \overline{\varphi_{m,n}} r \mathrm{d}\theta \mathrm{d}r \\
&= \frac{1}{\pi R_0^2 \sqrt{J_{m+1}^2(j_{m,n})} A_m(R_0 \tilde{k}_{m,i})} \int_0^{2\pi} e^{im\theta} e^{-im\theta} \int_0^{R_0} J_m(\tilde{k}_{m,i} r) J_m(k_{m,n} r) r \mathrm{d}\theta \mathrm{d}r \\
&= \frac{2}{R_0^2 \sqrt{J_{m+1}^2(j_{m,n})} A_m(R_0 \tilde{k}_{m,i})} \int_0^{R_0} J_m(\tilde{k}_{m,i} r) J_m(k_{m,n} r) r \mathrm{d}r.
\end{aligned}$$

The partial orthogonality appearing in (17) is apparent from from the second line. The evaluation of the last integral can be found in, e.g., [25], and with some further manipulation we get that

$$(\varphi_{m,n}, \psi_m^{\tilde{k}_{m,i}})_{L^2(D_0)} = -\mathrm{sign}(J_{m+1}(j_{m,n})) \frac{2J_m(\tilde{k}_{m,i} R_0)}{R_0 A_m(\tilde{k}_{m,i} R_0)} \frac{k_{m,n}}{(\tilde{k}_{m,i}^2 - k_{m,n}^2)}. \tag{33}$$

Due to the block diagonal structure of K, and since (18) implies that $K_m^+ = K_m^-$, it is sufficient to check that each block K_m^+ is diagonally dominant. On the i'th row of K_m^+, the inequality in (32) takes the form

$$|(\varphi_{m,i}, \psi_m^{\tilde{k}_{m,i}})_{L^2(D_0)}| > \sum_{n=1, n\neq i}^{N} |(\varphi_{m,n}, \psi_m^{\tilde{k}_{m,i}})_{L^2(D_0)}|, \tag{34}$$

or, in terms of (33),

$$\left| \frac{2J_m(\tilde{k}_{m,i} R_0)}{R_0 A_m(\tilde{k}_{m,i} R_0)} \frac{k_{m,i}}{(\tilde{k}_{m,i}^2 - k_{m,i}^2)} \right| > \sum_{n=1, n\neq i}^{N} \left| \frac{2J_m(\tilde{k}_{m,i} R_0)}{R_0 A_m(\tilde{k}_{m,i} R_0)} \frac{k_{m,n}}{(\tilde{k}_{m,i}^2 - k_{m,n}^2)} \right|. \tag{35}$$

We note that due to the relation in (14), we have that when $\tilde{k}_{m,i} \to k_{m,i}$, the left hand side of (35) equals 1, while the right hand side equals 0. Hence (35) holds. When $\tilde{k}_{m,i} \neq k_{m,i}$, the factor $(2J_m(\tilde{k}_{m,i} R_0))/(R_0 A_m(\tilde{k}_{m,i} R_0))$ enters as a constant on both sides, and can therefore be neglected. Hence we need only look at the inequality

$$\left| \frac{k_{m,i}}{\tilde{k}_{m,i}^2 - k_{m,i}^2} \right| > \sum_{n=1, n\neq i}^{N} \left| \frac{k_{m,n}}{\tilde{k}_{m,i}^2 - k_{m,n}^2} \right|. \tag{36}$$

Recognizing the above expression as equation (27) in Lemma 1, we see that if $|\tilde{k}_{m,i} - k_{m,i}| < \Delta k$, with Δk as specified in Lemma 1, then (36) must hold. It follows that if $|\tilde{k}_{m,i} - k_{m,i}| < \Delta k$ holds for all $0 \leq m \leq M$ and $1 \leq i \leq N$, then (36) holds for each row in each $K_m^{\pm}$, and that K is strictly diagonally dominant and hence invertible. □

In Figure 1 we plot both sides of (26) and the right hand side of (27), for different choices of m and i and fixed N. We see that the interval where the inequality holds is quite insensitive to changes in indices. Also, numerical investigation indicates that when $M, N \leq 150$, it is sufficient to check the inequality only for $m = 0$ and $n = 1$, i.e., the first row in K, since the smallest admissible Δk occurs there (We did not try with $M, N > 150$). We suspect this is related to the convexity/concavity of the zeros, and a proof of this would definitevely be interesting.

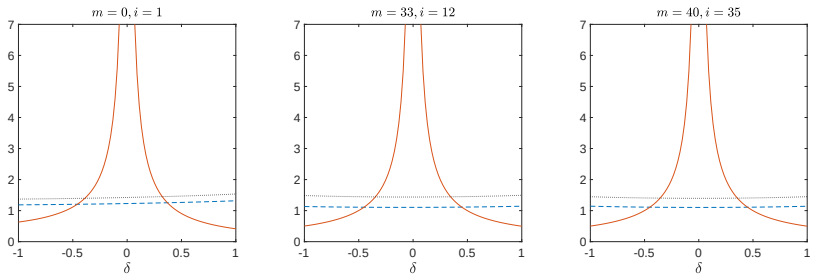

Figure 1: The red and the (dotted) black lines are the left and right hand sides of (30), respectively. The dashed blue line is the estimated off-diagonal sum $\sum_{n=1,n\neq i}^{N} \left| k_{m,n}/(\tilde{k}_{m,i}^2 - k_{m,n}^2) \right|$. We use $\delta \in [-1, 1]$, and set $R_0 = 1$ and $N = 50$.

Lemma 1 gives a bound Δk on admissible perturbations of frequencies in $Q_{M,N}$, sufficient to guarantee invertibility of K. With the set $Q_{M,N}$ available it is straight forward to compute Δk. In Section 4 we analyze the distribution of the zeros of Bessel functions, and show how one can choose a set of frequencies Q_s that are admissible perturbations of the frequencies $Q_{M,N}$, but such that Q_s is substantially smaller than $Q_{M,N}$. The set Q_r in Theorem 1 then constructed using the frequencies in Q_s.

3.3 Reconstruction of sources not in $S_{M,N}$

When we take the source to be any function $s \in L^2(D_0)$, we introduce an error in the reconstruction of the projection s on $S_{M,N}$. Let $U_k = F_k s$. The error occurs in the SVE-coefficients; Let $\tilde{u}_{i,j}$ be a coefficient computed from a measurement of source in L^2. From (17) we then have

$$\tilde{u}_{i,j} = \frac{(U_k, \phi_m^k)_{L^2(\partial D_0)}}{\sigma_{|i|}^k} = (s, \psi_i^k)_{L^2(D_0)} = \sum_{n=1}^{\infty} \hat{s}_{i,n} (\varphi_{i,n}, \psi_i^k)_{L^2(D_0)}. \tag{37}$$

Splitting the last sum, we get

$$\tilde{u}_{i,j} = \sum_{n=1}^{N} \hat{s}_{i,n} (\varphi_{i,n}, \psi_i^k)_{L^2(D_0)} + \sum_{n=N+1}^{\infty} \hat{s}_{i,n} (\varphi_{i,n}, \psi_i^k)_{L^2(D_0)} = \hat{u}_{i,j} + E(i, N). \tag{38}$$

Hence, we get an additional error term $E(i, N)$ in our coefficients, and the reconstruction of the projection of s on $S_{M,N}$ will not be exact. If the source is assumed to have certain regularity properties, one can estimates on the size of E, but in general, one cannot. However, in numerical simulations, the effect of $E(i, N)$ on reconstructions appear to be small, even for discontinuous sources.

4 Invertibility and stability with a reduced number of frequencies

In this section we investigate the consequences of the findings in Section 3 for a reduction of the measurement frequencies. By considering the distribution of the zeros of Bessel functions, we will see that a lot of the frequencies in $Q_{M,N}$ are redundant. We also analyze the stability and regularizing effect that comes with considering finite-dimensional reconstructions.

4.1 Density of the zeros of Bessel functions in $\mathbb{R}^+$

As a consequence of the findings in section 3, we are interested in the distribution of the positive zeros of Bessel functions on the real line. We want to investigate frequencies that are in the vicinity such zeros, particularly frequencies that are in the vicinity of multiple zeros. Given positive α and Δj, and the interval $I_\alpha(\Delta j) = (\alpha - \Delta j, \alpha + \Delta j)$, we want to know how many zeros $j_{m,n}$ are contained within $I_\alpha(\Delta j)$, and how this depends on α and Δj.
The literature on Bessel functions is vast, and we gather here a few properties of their zeros that proved useful for our purpose. They are found in [8], [10],[22] and [20], respectively.

Proposition 1.

1) $\lim_{n\to\infty}(j_{m,n+1} - j_{m,n}) = \pi \quad$ for $m \in \mathbb{R}$.

2) $j_{m,n} > m + n\pi - \frac{\pi}{2} + \frac{1}{2}, \quad m \geq -1/2$.

3) If $|m| \neq \frac{1}{2}$, then $(j_{m,n+1} - j_{m,n} - \pi)(|m| - \frac{1}{2}) > 0$.

4) Interlacing property: $j_{m,1} < j_{m+1,1} < j_{m,2} < j_{m+1,2} < j_{m,3} < \cdots$

From 2) it is evident that the zeros are strictly increasing functions of both index and order, and that they are shifted upwards on the positive real axis with increasing order. Also, 1) and 3) shows that the distribution of the zeros for a fixed order becomes, eventually, quite regular, as the difference between consecutive zeros tend to π. The interlacing property 4) shows how zeros are spaced between each other as a function of order and index. This suggests that a zero $j_{m,n}$ may be close to some zero $j_{i,j}$, or $j_{m,n} \approx j_{i,j}$, when $m > i$ and $j > n$, or vice versa. In Figure 2, we have plotted the 7 first zeros of J_m, $m = 0, 1, 2, ..., 7$. The horizontal lines indicate an interval $I_\alpha(\Delta j)$. Observe that different zeros of J_m's of different order lie within the strip spanned by $I_\alpha(\Delta k)$ in the $\mathbb{R}^+ \times \mathbb{N}_0$-plane. Figure 3 shows the number of $j_{n,l}$ such that $|\alpha - j_{n,l}| < 0.5$ for a increasing α. The numerical results indicate that the density of zeros increases as one moves up the real line. In fact, one can produce an approximate formula for the number of close zeros to a given point from the line argument. The reasoning follows from considering a horizontal interval centered at a point α in the $\mathbb{R}^+ \times \mathbb{N}_0$-plane, as in Figure 2.

A horizontal line passing through a point $\alpha > 0$ will intersect the intervals $(j_{m,1}, \infty)$ for all m such

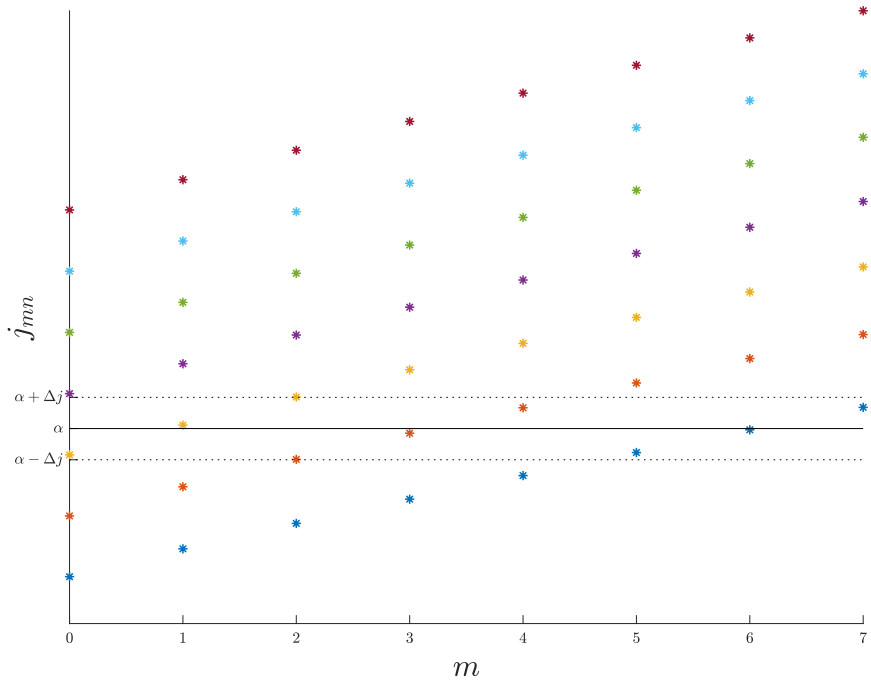

Figure 2: A plot of the 7 first zeros of J_m, $m = 0, 1, 2, ..., 7$. The dashed horizontal lines bound an interval $(\alpha - \Delta j, \alpha + \Delta j)$, with $\alpha = 10$ and $\Delta j = 1.6$.

that $\alpha > j_{m,1}$. Since the distance between consecutive zeros tends to π, we can approximate the expected number of zeros of $J_m(x)$ in an interval $I_\alpha(\Delta j), \alpha \geq j_{m,1}$, to be $\frac{2\Delta j}{\pi}$, and note that this approximation will get better as α increases. We now estimate the number of intervals $(j_{m,1}, \infty)$ the horizontal line passing through α intersects. Since $j_{m,1}$ is an increasing function of m, this is equivalent to estimating the largest $M \in \mathbb{N}$ such that $\alpha \geq j_{M,1}$ holds. To this end we use the following upper bound on $j_{M,1}$ [25]:

$$j_{M,1} < \left(\frac{4}{3}(M+1)(M+5)\right)^{\frac{1}{2}}, M > 0.$$

Requiring $j_{M,1} < \alpha$, we now want the largest M such that the following inequality holds:

$$\left(\frac{4}{3}(M+1)(M+5)\right)^{\frac{1}{2}} \leq \alpha. \tag{39}$$

Hence we get an estimate for M,

$$M = \left\lfloor \left(\frac{1}{2}\left(3\alpha^2 + 16\right)^{\frac{1}{2}} - 3\right) \right\rfloor.$$

Combining M and the approximation of the expected number of zeros, we have that the total number of zeros number of zeros within the interval $[\alpha - \Delta j, \alpha + \Delta j]$, $N(\alpha, \Delta j)$, is approximately

$$N(\alpha, \Delta j) \approx M \frac{2\Delta j}{\pi} = \left\lfloor \left(\frac{1}{2} \left(3\alpha^2 + 16\right)^{\frac{1}{2}} - 3 \right) \right\rfloor \frac{2\Delta j}{\pi} \tag{40}$$

This approximation is also plotted in Figure 3. We see that $N(\alpha, \Delta j) = O(\alpha \Delta j)$. The formula implies that for a fixed Δj, the number of zeros contained in an interval $I_\alpha(\Delta j)$ increases with increasing α. This is in agreement with the numerical findings.

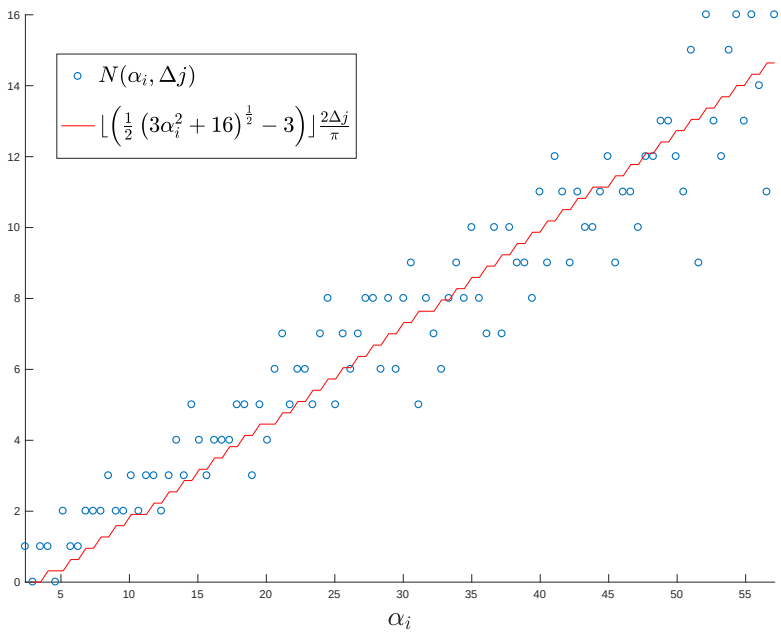

Figure 3: The plot shows $N(\alpha_i, 0.5)$ for $\alpha_i = j_{0,1} + j_{50,1}/(N+1)i = 2.4048 + 0.5769i, i = 0, 1, 2, ..., 99$. The red line is a plot of the approximative formula for $N(\alpha_i, (\Delta j)$ in (40).

.

To further substantiate the assertion that the zeros of the Bessel functions $J_m(x), m = 0, 1, 2...$ become more densely distributed as their numerical value increases, we include the following result: In [15, 10] it is shown that if we consider the set $\cup_{j=1}^{\infty}\{x_j\} = \cup_{m,n=1}^{\infty}\{j_{m,n}\}$, where $x_1 < x_2 < x_3 \cdots$, i.e., the set of all zeros sorted increasingly, then

$$\lim_{j \to \infty} x_{j+1} - x_j = 0.$$

Also, in [21], the authors did a numerical investigation of the distribution of the zeros of $J_m(x), m = 0, 1, 2, ...2000$ for $x \in [0, 3000]$, and computed the distances d_i between the (sorted) zeros. They

found that the maximum distance d_i between two zeros was max $d_i = 1.426$, and that the mean and standard deviation where $\mu(d_i) = 0.0028$ and $\sigma(d_i) = 0.0062$, respectively.

Together, the above results imply that the answer to our initial question about the existence of frequencies that can are suitable perturbations of Bessel zeros, is positive. Furthermore, it shows that the number of zeros being in the vicinity of a given frequency increases as the frequency increases. Theorem 1 and the results in this section implies that we can reduce the number of different frequencies needed for K to be invertible.

4.2 Reducing the frequency set

Once we have decided on a finite dimensional source space, $S_{M,N}$ we can use Theorem 1 to compute a perturbation level Δk such that the SVE- to FB-coefficient mapping K is invertible under perturbation of frequencies $Q_{M,N}$. Given $Q_{M,N}$, set $k^- = \min k_{m,n}$ and $k^+ = \max k_{m,n}$. We define $Q_s(\Delta k)$ to be *the smallest member of the family of sets*

$$\Big\{k \in [k^-, k^+] : \text{ for every } k_{m,n} \in Q_{M,N} \quad \exists k \text{ such that } |k_{m,n} - k| < \Delta k\Big\}.$$

The set Q_s is sometimes called a minimal subcover. Table 4.2 contains some numbers on the size of $Q_s(\Delta k)$ compared to $Q_{M,N}$ for different values of Δk and M and N. The increasing density of the Bessel zeros is evident, as the relative size of the reduced frequency set Q_s is much smaller for the full frequency set $Q_{M,N}$ for larger M and N.

Δk		0	0.25	0.5	0.75	1	1.25
$M = N = 7$	$\frac{\lvert Q_s\rvert}{\lvert Q_{M,M}\rvert} \cdot 100\%$	100%	62.5%	41.1%	30.4%	23.2%	19.6%
$M = N = 7$	$\lvert Q_s\rvert$	56	35	23	17	13	11
$M = N = 50$	$\frac{\lvert Q_s\rvert}{\lvert Q_{M,M}\rvert} \cdot 100\%$	100%	15%	8.2%	5.6%	4.2%	3.5%
$M = N = 50$	$\lvert Q_s\rvert$	2550	382	210	143	108	88

Table 1: The numerical value in the first row gives the reduction percentage for the reduced frequency set, and the second row contains the number of frequencies in the different Q_s's. For dimensions $m = n = 7$ and $M = N = 50$ and $R_0 = 1$, one has $|Q_{m,n}| = 56$ and $|Q_{M,N}| = 2550$.

Once the reduced frequency set is computed, one can construct K using the smaller number of perturbed frequencies in Q_s instead of those in $Q_{M,N}$. We assemble the frequency set Q_r of *not necessarily distinct frequencies* in Theorem 1 by substituting those $k_{m,n} \in Q_{M,N}$ by their closest $k_j \in Q_s$. Hence Q_r will have several repeated entries.

4.3 Stability and regularization through the reduced frequency set

Next, we analyze the stability of the equation $K\hat{S} = \hat{U}$ with respect to additive noise in the measurement data. Note that the invertibility of K is unaffected by noisy data. We assume a noisy measurement at at frequency k_j is of the form $U^{\varepsilon}_{k_j} = U_{k_j} + \varepsilon$, where $\varepsilon \in L^2(\partial D)$ is the noise

and U_{k_j} is the true measurement. Also, we assume the relative noise level is bounded by a $\delta > 0$, $\|\varepsilon\|_{L^2}/\|U_{k_j}\|_{L^2} \leq \delta$. To see how a noisy measurement influences our solution, we define the sets

$$\begin{aligned} I_{k_j} &= \{(m,n) : k_{m,n} \in Q_{M,N} \text{ and } |k_{m,n} - k_j| < \Delta k\}, \\ M_{k_j} &= \{m : k_{m,n} \in Q_{M,N} \text{ and } |k_{m,n} - k_j| < \Delta k\}. \end{aligned}$$

I_{k_j} the set of indices of those $k_{m,n} \in Q_{M,N}$ that are perturbations of $k_j \in Q_s$, and M_{k_j} contains only the order index m. Recalling equation (24) from the construction of the linear system $K\hat{S} = \hat{U}$, we see that I_{k_j} contains the indices of the singular vectors $\phi_i^{k_j}$ and singular values $\sigma_i^{k_j}$ used to compute the SVE-coefficients $\hat{u}^{\varepsilon}_{m,n}$ from the measurement $U^{\varepsilon}_{k_j}$:

$$\hat{u}^{\varepsilon}_{\pm m,n} = \frac{(U^{\varepsilon}_{k_j}, \phi^{k_j}_{\pm m})_{L^2(\partial D)}}{\sigma^{k_j}_m}, \quad (m,n) \in I_{k_j}. \tag{41}$$

It is well-known that the stability of inverting a compact operator depends on the behaviour of the singular value spectrum, and that small singular values amplify the (high-frequency) noise. Such is the case here:

$$|\hat{u}_{m,n}| = \left| \frac{(U^{\varepsilon}_{k_j}, \phi^{k_j}_m)_{L^2}}{\sigma^{k_j}_m} \right| \leq \left| \frac{(U_{k_j}, \phi^{k_j}_m)_{L^2}}{\sigma^{k_j}_m} \right| + \left| \frac{(\varepsilon, \phi^{k_j}_m)_{L^2}}{\sigma^{k_j}_m} \right| \leq |\hat{u}_{m,n}| + \frac{\|\varepsilon\|_{L^2}}{\sigma^{k_j}_m}.$$

It is evident that when $\sigma_m << 1$, we get an amplification of the noise. Due to the fact [17] that $\lim_{m\to\infty} \sigma^{k_j}_m = 0$, it becomes important to study the pre-asymptotic behaviour of the singular values. In [16], a characterization of the bandwidth of the operator F_{k_j} is presented. The bandwitdh β_{k_j} is defined as the singular value index whereafter the singular values of F_{k_j} become strictly decreasing:

$$\beta_{k_j} = \text{argmin}_{m\in\mathbb{N}_0}\{\sigma^{k_j}_{m+n} > \sigma^{k_j}_{m+n+1} \text{ for all } n \in \mathbb{N}_0\}.$$

Knowing the value of β_{k_j} we can say whether noise in the SVE-coefficient $\hat{u}_{m,n}$ will be amplified. In [16], a tight lower for the bandwidth is proved, and a tight upper bound is conjectured. Both bounds are numerically validated. We state the result on the lower bound M_-:

The bandwidth β_{k_j} of the forward operator $F_{k_j} : s \mapsto U_{k_j}$ associated with (1) satisfies $M_- \leq \beta_{k_j}$, with M_- given by

$$M_- = \text{argmin}_{m\in\mathbb{N}_0}\{j_{m,1} \geq k_j R_0\}.$$

Hence, to avoid the amplification of noise from a measurement $U^{\varepsilon}_{k_j}$, we only want to extract the following SVE-coefficients:

$$\hat{u}^{\varepsilon}_{\pm m,n} = \frac{(U^{\varepsilon}_{k_j}, \phi^{k_j}_{\pm m})_{L^2(\partial D)}}{\sigma^{k_j}_m}, \quad m \in M_{k_j}, (m,n) \in I_{k_j} \text{ and } m \leq \beta_{k_j}.$$

We now ask if we can avoid $m \in M_{k_j}$ such that $m > \beta_{k_j}$. The answer is given by the following proposition:

Theorem 2. *If $\Delta k \leq 1/R_0$ then all indices m in M_{k_j} are within the bandwidth of the forward operator F_{k_j}, that is,* $\max_{M_{k_j}} m \leq \beta_{k_j}$.

Proof. We first need to show that $j_{m+1,1} - j_{m,1} > 1$ for $m \in \mathbb{N}_0$. From [14], we have that $\frac{\mathrm{d}j_{m,n}}{\mathrm{d}m} > 1$ whenever $m > -1$ and $n = 1, 2, 3, \ldots$. Hence it follows by the fundamental theorem of calculus that

$$j_{m+1,1} - j_{m,1} = \int_m^{m+1} \frac{\mathrm{d}j_{m,n}}{\mathrm{d}m} \mathrm{d}m > \int_m^{m+1} \mathrm{d}m = 1. \tag{42}$$

We set $m = \operatorname{argmin}_{\mu \in N_0} \{j_{\mu,1} \geq k_j R_0\}$, i.e., $m = M_-$. If we have that $j_{m,1} \in R_0(k_j - \Delta k, k_j + \Delta k) \subset (k_j R_0 - 1, k_j R_0 + 1)$, it follows from (42) that there can be no $j_{\tilde{m},1}$ with $\tilde{m} > m$ within the same interval. This implies that all the zeros $k_{\tilde{m},n} \in (k_j - \Delta k, k_j + \Delta k)$ must have order $\tilde{m} \leq m = M_-$ and hence that $\max_{m \in M_{k_j}} m \leq M_- \leq \beta_{k_j}$. □

The result shows that with a proper choice of Δk, we regularize the multi-frequency ISP by making a selective SVE [13] for each frequency, and it becomes clear that, under the additional requirement $\Delta k \leq 1/R_0$, the FB-coefficients of the source are stably recoverable with respect to noise in the measurements.

To recapitulate the developments of Theorem 1 and Theorem 2 in terms of reconstruction of sources, consider an ISP geometry specified by R and R_0, as well as a finite-dimensional source space $S_{M,N}$. Assume Δk is the largest number such that $\Delta k \leq 1/R_0$ and furthermore such that the inequality (26) holds for all indices $0 \leq m \leq M$ and $1 \leq i \leq N$. If, for each $k_{m,i} \in Q_{M,N}$, there is some $k_j \in Q_r$ satisfying $|k_j - k_{m,i}| < \Delta k$, then any source $s \in S_{M,N}$ is uniquely recoverable from measurements at the frequencies $k_j \in Q_r$. Furthermore, the reconstruction is stable in the sense that it does not incorporate data associated with singular values with indices above the bandwidth β_{k_j} for any of the measurements $U_{k_j}, k_j \in Q_r$.

We end this section commenting that our analysis offers an explanation to a phenomenon observed by many authors working on the ISP; namely that the well-posedness of the ISP increases when the frequency increases. The results in Section 4.1 show that the density of the zeros of Bessel functions J_m increases as one moves up the real line. Together with the results in Section 4.3, this implies that one can expect to stably approximate more SVE-coefficients at from measurements at higher frequencies. Hence, in general, we are able to stably recover more information about the source from higher frequency measurements.

5 Numerical method and experiments

5.1 Numerical method

The numerical method consists of computing a reduced frequency set, implementing the K-matrix mapping FB-coefficients to SVE-coefficients, and computing the SVE-coefficients. We summarize the procedure below, and briefly elaborate on some of the steps.

1) Given a domain D, a source domain D_0, and wave speed c, one decides on a preferable resolution expressed in terms of the finite dimensional space $S_{M,N}$.

2) Compute the reduced frequency set Q_s and assemble Q_r.

3) Use the frequencies from Q_r to assemble the matrix K, according to equations (23)-(19).

4) Record the measurements $\{U_{k_j}\}_{k_j \in Q_s}$, where $k_j = \frac{\omega_j}{c}$ and $U_{k_j} = F_{k_j} s$. Use a quadrature method to compute the SVE-coefficients according to equation (41).

5) Solve $\hat{U} = K\hat{S}$ for $\hat{S}$, and express s as $s = \sum_{i=-M}^{M} \sum_{j=1}^{N} \hat{s}_{i,j} \varphi_{i,j}$.

The problem specification in 1) dictates a choice of the measurement frequency set Q_s in 2). From (33), we have available the formula for the entries of the matrix K, and we construct it using the entries in Q_r according to (23)-(19).
In step 4) we sample each measurement U_k on ∂D; the spatial sampling rate has to be small enough to avoid aliasing. Last, in 5), we express the solution as a finite linear combination of FB-vectors on a suitable mesh.

5.2 Numerical experiments

We test the method on three sources with different characteristics. The first source is a simple linear combination in $S_{3,3}$. Next, we take a smooth, slowly varying source, and last, a discontinuous source with some smaller, sharper features. In all cases we compute the reconstruction for different choices of M, N and Δk, and also let Δk violate the invertibility condition. The largest admissible Δk from Lemma 1 is written in bold. The measurements are computed by solving equation with the built in 2D-quadrature in MATLAB, and sampled at 200 uniformly distributed points on ∂D. We add complex Gaussian white noise to the measurements with relative noise level 0.2 (20%). The inner product in (41) is approximated using Simpson's rule. We compute a number of interesting L^2 error-norms for the reconstructions; they are listed in Tables 2-6. In the following, we denote the source by s, the reconstructed source by s_r, the source reconstructed from noisy measurements by s_r^ε, and the projection of s on $S_{M,N}$ by s_p. In all reconstructions we let D_0 and D have radii 1 and 1.5, respectively, and assume a wave speed $c = 1$.

5.2.1 Source in $S_{3,3}$

First, we consider the following linear combination of FB-basis functions:

$$s = 2\varphi_{0,1} + \pi\varphi_{3,3}. \tag{43}$$

We expect small errors, due only to the sampling and numerical approximations. This agrees with the observations in Table 2. The robustness towards noise, described in Section 4.3 is also evident, as the influence of the noise is seen to be very small.

5.2.2 Smooth source

Inspired by the the papers [7, 26], we consider the source

$$\begin{aligned} s(x,y) = 0.3(1-3x)^2 \exp(-(3x)^2 - (3y+1)^2) - (0.2(3x) - (3x)^3 \\ -(3y)^5) \exp(-(3x)^2 - (3y)^2) - 0.03 \exp(-(3x+1)^2 - (3y)^2). \end{aligned}$$

We do reconstructions in $S_{3,3}$ and $S_{7,7}$, both with a number of reduced frequency sets corresponding to different choices of Δk. Results are presented in Tables 3 and 4, and plots of the source, the reconstruction, and the pointwise error are found in Figure 4. We observe that the we get a quite accurate approximation of s already by choosing $S_{3,3}$, and that we obtain fine reconstructions with as few as 5 frequencies. Also, all reconstructions are stable with respect to noise in the

$\lvert Q_s \rvert$	9	8	6	**5**	4
Δk	0.25	0.5	0.75	**0.91**	1.5
$\frac{\lVert s-s_r \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	1.3%	1.3%	1.4%	1.40%	1.3%
$\frac{\lVert s-s_r^\varepsilon \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	2.7%	2.9%	2.4%	2.2%	%4.6

Table 2: Relative errors in percentage, with and without noise, for a source is $S_{3,3}$, measured in the L^2-norms.

measurements. By comparing $\|s - s_r\|_{L^2}/\|s\|_{L^2}$ and $\|s - s_p\|_{L^2}/\|s\|_{L^2}$, we see that the effect of $s \notin S_{M,N}$ is small. Here, and later, we notice that K is invertible also even for larger Δk, emphasizing that the invertibility condition in Theorem 1 is only a sufficient condition. For reconstructions in $S_{7,7}$, the projection becomes a more accurate approximation than the reconstruction, and we suspect this is due to the projection not suffering from the the approximation errors described in Section 3.3.

$\lvert Q_s \rvert$	9	8	6	**5**	4
Δk	0.25	0.5	0.75	**0.91**	1.5
$\frac{\lVert s-s_r \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	15.3%	17.5%	20.8%	16.6%	16.3%
$\frac{\lVert s-s_p \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	15.1%	15.1%	15.1%	15.1%	15.1%
$\frac{\lVert s_p-s_r \rVert_{L^2}}{\lVert s_p \rVert_{L^2}} 100\%$	2.5%	8.8%	14.4%	7.1%	6.2%
$\frac{\lVert s-s_r^\varepsilon \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	15.5%	18.1%	21.5%	17.4%	16.4%
$\frac{\lVert s_p-s_r^\varepsilon \rVert_{L^2}}{\lVert s_p \rVert_{L^2}} 100\%$	3.3%	10.0%	15.4%	8.6%	6.4%

Table 3: Number of frequencies and relative errors in percentage for the smooth source, measured in the L^2-norms, with $M = N = 3$.

5.2.3 Discontinuous source

Next, we consider the source

$$s_d(x,y) = 0.1 + \begin{cases} 1 & \text{for } (x,y) \in \hat{B}((-0.4,-0.08),0.05), \\ 0.5 & \text{for } (x,y) \in \{(x,y) : |x-0.2| \le 0.15, |y+0.4| \le 0.15\}, \\ 2 & \text{for } (x,y) \in \{(x,y) : |x+0.2| \le 0.2, |y-0.4| \le 0.3\}. \end{cases} \tag{44}$$

We consider reconstructions in $S_{5,5}$ and $S_{15,15}$, also with different choices of reduced frequency set. Table 5 and 6 contains error norms. A plot of the source, the reconstruction and the point-wise

$\lvert Q_s \rvert$	35	**21**	17	13	10
Δk	0.25	**0.61**	0.75	1	1.5
$\frac{\lVert s-s_r \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	6.5%	7.8%	19.2%	19.0%	20.7%
$\frac{\lVert s-s_p \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	1.6%	1.6%	1.6%	1.6%	1.6%
$\frac{\lVert s_p-s_r \rVert_{L^2}}{\lVert s_p \rVert_{L^2}} 100\%$	6.3%	7.6%	19.2%	18.9%	20.6%
$\frac{\lVert s-s_r^\varepsilon \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	7.6%	8.6%	19.8%	18.8%	19.5%
$\frac{\lVert s_p-s_r^\varepsilon \rVert_{L^2}}{\lVert s_p \rVert_{L^2}} 100\%$	7.4%	8.5%	19.7%	18.7%	19.4%

Table 4: Number of frequencies and relative errors in percentage for the smooth source, measured in the L^2-norms, with $M = N = 7$.

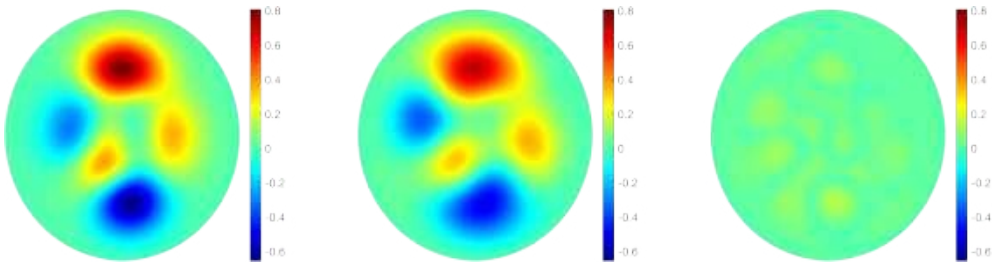

Figure 4: Smooth source. Left: Source. Middle: Reconstruction with $\Delta k = 0.91$, using 5 frequencies, and measurements with 20% additive Gaussian noise. Right: Pointwise absolute error.

error is shown in Figure 5 and 6. It is evident from the plots that reconstructions in $S_{5,5}$ does not reproduce the smaller features, but that reconstructions in $S_{15,15}$ does quite well. We also notice the error around the edges, a common phenomenon when approximating with generalized Fourier-series. It is clear from the tables that the main error comes from the projection, as expected. Also, for reconstructions in $S_{15,15}$, the method shows significantly robustness to noise in the measurements.

$\lvert Q_s \rvert$	20	**14**	11	9	7
Δk	0.25	**0.7**	0.75	1	1.5
$\frac{\lVert s-s_r \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	42.4%	43.2%	42.9%	43.4%	43.8%
$\frac{\lVert s-s_p \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	42.2%	42.2%	42.2%	42.2%	42.2%
$\frac{\lVert s_p-s_r \rVert_{L^2}}{\lVert s_p \rVert_{L^2}} 100\%$	3.2%	9.9%	8.6%	11.3%	12.6%
$\frac{\lVert s-s_r^\varepsilon \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	42.5%	43.3%	43.0%	43.4%	44.1%
$\frac{\lVert s_p-s_r^\varepsilon \rVert_{L^2}}{\lVert s_p \rVert_{L^2}} 100\%$	4.2%	10.2%	9.1%	11.2%	14.1%

Table 5: Relative errors in percentage for the discontinuous source, measured in the L^2-norms, with $M = N = 5$.

$\lvert Q_s \rvert$	91	**56**	40	30	22
Δk	0.25	**0.5**	0.75	1	1.5
$\frac{\lVert s-s_r \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	25.9%	26.2%	27.3%	27.3%	30.7%
$\frac{\lVert s-s_p \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	25.3%	25.3%	25.3%	25.3%	25.3%
$\frac{\lVert s_p-s_r \rVert_{L^2}}{\lVert s_p \rVert_{L^2}} 100\%$	5.0%	6.9%	10.2%	10.4%	17.4%
$\frac{\lVert s-s_r^\varepsilon \rVert_{L^2}}{\lVert s \rVert_{L^2}} 100\%$	26.3%	26.6%	27.5%	27.9%	30.1%
$\frac{\lVert s_p-s_r^\varepsilon \rVert_{L^2}}{\lVert s_p \rVert_{L^2}} 100\%$	7.1%	8.3%	10.7%	12.0%	17.8%

Table 6: Relative errors in percentage for the discontinuous source, measured in the L^2-norms, with $M = N = 15$.

6 Conclusion

We have analyzed the multi-frequency inverse source problem with measurements at a finite number of frequencies. Our main result are the criteria which allows for a reduction of the frequency set while at the same time guaranteeing stable reconstructions of sources in finite-dimensional subspaces $S_{M,N} \subset L^2$. Further, we have deviced a direct reconstruction scheme to reconstruct such sources. The method is stable with respect to noisy measurements, and regularization is implicit in the choice of the finite-dimensional source space and the corresponding measurement frequencies. Results from numerical experiments support our findings.

In the future, it would be interesting to see how our analysis can be generalized to more arbitrary geometries and higher dimensions, and to the Helmholtz equation with variable wave speed.

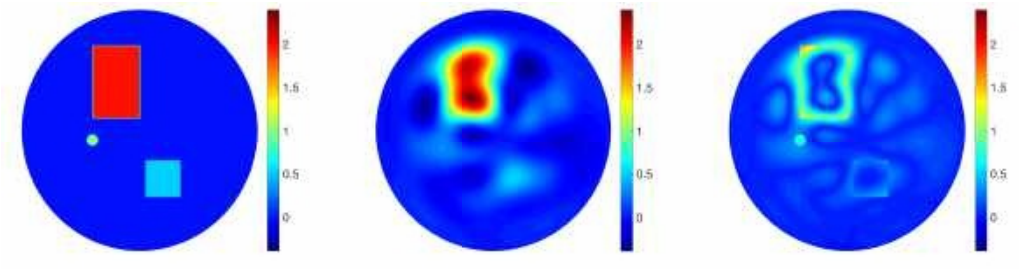

Figure 5: Discontinuous source. Left: Source. Middle: Reconstruction in $S_{5,5}$ with $\Delta k = 0.7$, using 14 frequencies, and measurements with 20% additive Gaussian noise. Right: Pointwise absolute error.

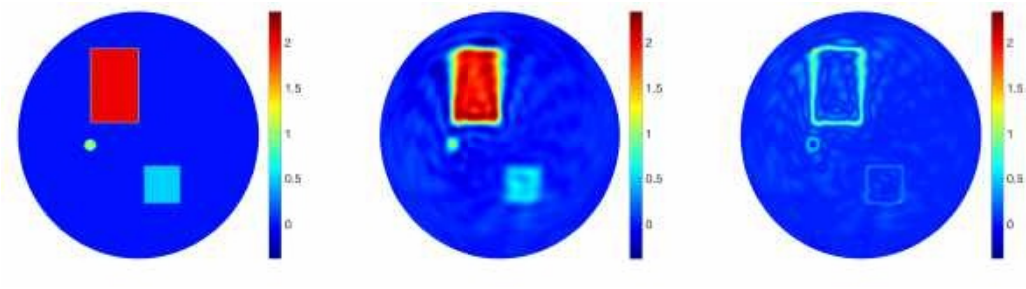

Figure 6: Discontinuous source. Left: Source. Middle: Reconstruction in $S_{15,15}$ with $\Delta k = 0.5$, using 56 frequencies, and data with 20% additive Gaussian noise. Right: Pointwise absolute error.

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Appendix D

Paper 4

Stability of the inverse source problem for the Helmholtz equation in $\mathbf{R}^3$

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Abstract

We consider the reconstruction of a compactly supported source term in the constant-coefficient Helmholtz equation in $\mathbf{R}^3$, from the measurement of the outgoing solution at a source-enclosing sphere. The measurement is taken at a finite number of frequencies. We explicitly characterize certain finite-dimensional spaces of sources that can be stably reconstructed from such measurements. The characterization involves only the measurement frequencies and the problem geometry parameters. We derive a singular value decomposition of the measurement operator, and prove a lower bound for the spectral bandwidth of this operator. By relating the singular value decomposition and the eigenvalue problem for the Dirichlet-Laplacian on the source support, we devise a fast and stable numerical method for the source reconstruction. We do numerical experiments to validate the stability and efficiency of the numerical method.

1 Introduction

Assume $R > R_0 > 0$ and let $B_0, B \subset \mathbf{R}^3$ be concentric open balls centered at the origin and of radius R_0 and R, respectively. Let $s \in L^2(B_0)$, and fix a positive frequency (wavenumber) k. Let u_k be the unique outgoing solution of the Helmholtz equation in $\mathbf{R}^3$ with the inhomogeneous term s,

$$\begin{cases} (\Delta + k^2)u_k &= s \quad \text{in } \mathbf{R}^3, \\ \lim\limits_{|x|\to\infty} |x|(\partial_{|x|} - ik)u_k &= 0 \quad \text{uniformly for } x/|x| \in S^2. \end{cases} \tag{1}$$

We have

$$u_k(x) = \int_{B_0} G_k(x-y)s(y)dy, \quad x \in \mathbf{R}^3, \tag{2}$$

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where G_k is the outgoing fundamental solution of the Helmholtz equation in $\mathbf{R}^3$,

$$G_k(x) = \frac{\exp(ik|x|)}{4\pi|x|}, \quad x \in \mathbf{R}^3 \setminus \{0\}. \tag{3}$$

The forward operator, or measurement operator, F_k associated with the frequency k maps the source to the measurement, $F_k s = u|_{\partial B} = U_k$. The mapping $F_k \colon L^2(B_0) \to L^2(\partial B)$ is linear and compact [6] . We call the trace U_k the 'measurement at frequency k.' The inverse source problem (ISP) associated with (1) is:

> Given the measurements $\{U_{k_j} = u_{k_j}|_{\partial B}\}$ over a set of frequencies $Q = \{k_j\}$, estimate the source term s in (1).

The multi-frequency ISP arises when the set Q has more than one element.

The ISP for the Helmholtz equation in $\mathbf{R}^3$ appears, e.g., in the analysis and solution of inverse acoustic scattering problems and linearized acoustic inverse obstacle and inverse medium problems, as well as photo- and thermoacoustic tomography [1, 4, 8, 10, 12]. Uniqueness, stability and inversion methods for the (multi-frequency) ISP was treated in, for instance, in [2, 3, 5–7, 11, 13–17, 20, 23]. The purpose of this work is to investigate the stability of the single-frequency and multi-frequency inverse source problem for the Helmholtz equation in $\mathbf{R}^3$, as well as to present a stable and effective reconstruction method for the multi-frequency problem. In section 2 we derive a singular value system of the forward operator F_k, and in section 3 we use this to give a non-asymptotic spectral characterization of F_k. The main result there is Theorem 2. The analysis in sections 2 and 3 follows our earlier work [13] in $\mathbf{R}^2$ rather closely. Section 4 concerns the multi-frequency ISP, and here we identify certain finite-dimensional spaces of sources that can be stably reconstructed from the boundary measurements. The main result there is Theorem 3, where we give a sufficient criterion for reconstruction of sources belonging to a class of finite-dimensional subspaces of $L^2(B_0)$. In section 5 we conduct numerical experiments to verify our analysis and the reconstruction method, and we finally offer our conclusions and suggestions for further work in section 6.

Remark

Our analysis is applicable to more general measurement boundaries than the sphere, since in certain cases one may extend a measurement continuously to the sphere. More precisely, let Γ be the boundary of some C^2 domain Ω that includes the source support B_0, and take B to be the smallest ball that includes Ω. Let $g = u_k|_\Gamma$ be the measurement of the field u_k on Γ. Then g has a unique extension to a measurement U_k on ∂B that depends continuously on g (cf. Theorem 3.11 in [8]).

2 A singular system of F_k

Our results on the stability of solution of the multi-frequency inverse source problem rely on detailed spectral analysis of the measurement operator F_k. Thus, we first describe a singular value decomposition of F_k in terms of spherical harmonics and spherical Bessel functions. Let $h_m^{(1)}$ be the spherical Hankel function of the first kind and order m, $j_m(z)$ the spherical

Bessel function of order m, and Y_m^n, $m \in \mathbf{N}_0$, $n = -m, ..., m$, the spherical harmonics. Write $\kappa_0 = kR_0$ and $\kappa = kR$, and let

$$\begin{aligned} a_m(\kappa_0) &= \frac{2\sqrt{k/\pi}}{R_0}\left(\int_0^{R_0} j_m(kr)^2 r^2\,dr\right)^{1/2} = \frac{2\sqrt{k/\pi}}{R_0}\left(\int_0^{R_0} r^2\left(\sqrt{\frac{\pi}{2kr}}J_{m+1/2}(kr)\right)^2 \mathrm{d}r\right)^{1/2} \\ &= \frac{\sqrt{2}}{R_0}\left(\int_0^{R_0} rJ_{m+1/2}(kr)^2\mathrm{d}r\right)^{1/2} \\ &= \left(J_{m+1/2}(\kappa_0)^2 + J_{m-1/2}(\kappa_0)^2 - \frac{2m+1}{\kappa_0}J_{m+1/2}(\kappa_0)J_{m-1/2}(\kappa_0)\right)^{1/2} \\ &= \left(J_{m+1/2}(\kappa_0)^2 + J_{m+3/2}(\kappa_0)^2 - \frac{2m+1}{\kappa_0}J_{m+1/2}(\kappa_0)J_{m+3/2}(\kappa_0)\right)^{1/2}, \end{aligned} \tag{4}$$

where we use the well-known relation $j_m(z) = \sqrt{\pi/2z}J_{m+1/2}(z)$. The last equality in (4) follows after using the recurrence relation

$$J_{m-1/2}(\kappa_0) = \frac{2(m+1/2)}{\kappa_0}J_{m+1/2}(\kappa_0) - J_{m+3/2}(\kappa_0).$$

For positive arguments, the function j_m is real-valued and has zeros over a subset of the real axis with measure zero. Therefore, we have for any $m \in \mathbf{N}_0$ and any positive κ_0 that

$$a_m(\kappa_0) = 2\sqrt{k/\pi}R_0^{-1}\left(\int_0^{R_0} j_m(kr)^2 r^2\,dr\right)^{1/2} > 0.$$

Theorem 1. *The measurement operator F_k admits the singular value decomposition*

$$F_k = \sum_{m\in\mathbf{N}_0}\sum_{n=-m}^{m} \sigma_m^k(\cdot, u_{m,n}^k)_{L^2(B_0)} v_{m,n}^k \tag{5}$$

with

$$\text{the singular values} \quad \sigma_m^k = \frac{RR_0\sqrt{k\pi}}{2}|h_m^{(1)}(\kappa)|a_m(\kappa_0), \tag{6}$$

$$\text{the left singular vectors} \quad u_{m,n}^k(x) = \frac{2\sqrt{k/\pi}}{R_0 a_m(\kappa_0)}j_m(k|x|)Y_m^n(x/|x|),\ x \in B_0, \quad \text{and} \tag{7}$$

$$\text{the right singular vectors} \quad v_{m,n}^k(x) = (i/R)\mathrm{e}^{i\arg h_m^{(1)}(\kappa)}Y_m^n(x/|x|),\ x \in \partial B. \tag{8}$$

The m'th singular value σ_m^k has the multiplicity $2m+1$.

Proof. The fundamental solution (3) of the Helmholtz equation in $\mathbf{R}^3$ has an expansion in terms of the spherical harmonics [8, Theorem 2.11, page 31],

$$G_k(x-y) = ik\sum_{m=0}^{\infty}\sum_{n=-m}^{m} h_m^{(1)}(k|x|)Y_m^n(x/|x|)j_m(k|y|)\overline{Y_m^n(y/|y|)}, \quad |x| > |y|.$$

The L^2-adjoint $F_k^* : L^2(\partial B) \to L^2(B_0)$ of F_k is given by

$$F_k^* U(x) = \int_{\partial B} \overline{G_k(x-y)}U(y)\mathrm{d}S(y), \quad U \in L^2(\partial B),\ x \in B_0. \tag{9}$$

Since $\overline{B_0} \subset B$, we have $|z| > |x|$ whenever $z \in \partial B$ and $x \in B_0$. Also, $\{Y_m^n\}$ is an orthonormal basis of $L^2(S^2)$, and $G_k(x-y) = G_k(y-x)$, so for every $s \in L^2(B_0)$ and $x \in B_0$ it holds that

$$\begin{aligned}
F_k^* F_k s(x) &= \int_{\partial B} \overline{G_k(z-x)} \int_{B_0} G_k(z-y) s(y) \mathrm{d}y \mathrm{d}S(z) \\
&= \int_{B_0} s(y) \int_{\partial B} \overline{G_k(z-x)} G_k(z-y) \mathrm{d}S(z) \mathrm{d}y \\
&= -\int_{B_0} s(y) \int_{\partial B} ik \sum_{m=0}^{\infty} \sum_{n=-m}^{m} \overline{h_m^{(1)}(k|z|) Y_m^n(z/|z|) j_m(k|x|)} Y_m^n(x/|x|) \\
&\times ik \sum_{\mu=0}^{\infty} \sum_{\nu=-\mu}^{\mu} h_\mu^{(1)}(k|z|) Y_\mu^\nu(z/|z|) j_\mu(k|y|) \overline{Y_\mu^\nu(y/|y|)} \mathrm{d}S(z) \mathrm{d}y \\
&= \kappa^2 \sum_{m=0}^{\infty} \sum_{n=-m}^{m} |h_m^{(1)}(\kappa)|^2 j_m(k|x|) Y_m^n(x/|x|) \int_{B_0} s(y) j_m(k|y|) \overline{Y_m^n(y/|y|)} \mathrm{d}y.
\end{aligned}$$

This is, in fact, an eigendecomposition of the operator $F_k^* F_k$,

$$F_k^* F_k = \sum_{m \in \mathbf{N}_0} \sum_{n=-m}^{n} (\sigma_{m,n}^k)^2 (\cdot, u_{m,n}^k)_{L^2(B_0)} u_{m,n}^k, \tag{10}$$

with (normalized) eigenvectors

$$u_{m,n}^k(x) = \frac{j_m(k|x|) Y_m^n(x/|x|)}{\|j_m(k|x|) Y_m^n(x/|x|)\|_{L^2(B_0)}}, \quad x \in B_0,\ m \in \mathbf{N}_0,\ n = -m, \ldots, m,$$

and eigenvalues

$$(\sigma_{m,n}^k)^2 = \kappa^2 |h_m^{(1)}(\kappa)|^2 \|j_m(k|x|) Y_m^n(x/|x|)|\|_{L^2(B_0)}^2, \quad m \in \mathbf{N}_0,\ n = -m, \ldots, m.$$

Now each spherical harmonic Y_m^n has unit norm in $L^2(S^2)$, so

$$\begin{aligned}
\|j_m(k|x|) Y_m^n(x/|x|)|\|_{L^2(B_0)}^2 &= \int_{B_0} |j_m(k|x|) Y_m^n(x/|x|)|^2 \, \mathrm{d}x \\
&= \int_0^{R_0} \int_0^{2\pi} \int_0^{\pi} j_m(kr)^2 |Y_m^n(\theta, \varphi)|^2 r^2 \sin\theta \mathrm{d}\theta \mathrm{d}\varphi \mathrm{d}r \\
&= \int_0^{R_0} r^2 j_m(kr)^2 \mathrm{d}r = \frac{R_0^2 \pi}{4k} a_m(\kappa_0)^2.
\end{aligned}$$

Hence

$$u_{m,n}^k(x) = \frac{2\sqrt{k/\pi}}{R_0 a_m(\kappa_0)} j_m(k|x|) Y_m^n(x/|x|), \quad x \in B_0,\ m \in \mathbf{N}_0,\ n = -m, \ldots, m,$$

and

$$(\sigma_{m,n}^k)^2 = \frac{k R^2 R_0^2 \pi}{4} |h_m^{(1)}(\kappa)|^2 a_m(\kappa_0)^2, \quad m \in \mathbf{N}_0,\ n = -m, \ldots, m.$$

In particular, the eigenvalues $(\sigma_{m,n}^k)^2$ of $F_k^* F_k$ are degenerate with the degree of degeneracy $2m+1$. In the following, we write $(\sigma_m^k)^2$ for these eigenvalues. By definition, the operator F_k admits the singular values σ_m^k, as in (6). By Theorem 4.7 and its proof in [8, pp. 100–101] we

have the associated left and right singular vectors of F_k given respectively by $u^k_{m,n}$, as in (7), and by

$$\begin{aligned}
v^k_{m,n}(x) &= \frac{1}{\sigma^k_{m,n}} F_k u^k_{m,n}(x) = \frac{4}{\pi R_0^2 R |h^{(1)}_m(\kappa)| a_m(\kappa_0)^2} \int_{B_0} G_k(x-y) j_m(k|y|) Y^n_m(y/|y|) \mathrm{d}y \\
&= \frac{4\mathrm{i}k}{\pi R_0^2 R |h^{(1)}_m(\kappa)| a_m(\kappa_0)^2} \sum_{\mu=0}^{\infty} \sum_{\nu=-\mu}^{\mu} h^{(1)}_\mu(\kappa) Y^\nu_\mu(x/|x|) \int_{B_0} j_m(k|y|) j_\mu(k|y|) Y^n_m(y/|y|) \overline{Y^\nu_\mu(y/|y|)} \mathrm{d}y \\
&= \frac{4\mathrm{i}k \exp(\mathrm{i} \arg h^{(1)}_m(\kappa))}{\pi R_0^2 R a_m(\kappa_0)^2} Y^n_m(x/|x|) \int_{r=0}^{R_0} r^2 j_m(kr)^2 \mathrm{d}r \\
&= \frac{\mathrm{i} \exp(\mathrm{i} \arg h^{(1)}_m(\kappa))}{R} Y^n_m(x/|x|), \quad x \in \partial B,
\end{aligned}$$

as in (8). □

Remark 1. *In [13, Lemma 1, p. 4] we followed Bao, Lin and Triki [6] and found the following expression for singular values of the measurement operator for the Helmholtz equation in the plane, at frequency k:*

$$\sigma^k_{2\mathrm{D},\,m} = \sqrt{2R}\pi R_0 |H^{(1)}_m(\kappa)| A_m(\kappa_0), \quad m \in \mathbf{N}_0.$$

Here $H^{(1)}_m = J_m + \mathrm{i}Y_m$ is the Hankel function of the first kind and order m, Y_m is the Bessel function of the second kind (also called the Neumann function) of order m, and

$$A_m(\kappa_0) = \left(J_m(\kappa_0)^2 + J_{m+1}(\kappa_0)^2 - \frac{2m}{\kappa_0} J_m J_{m+1}(\kappa_0) \right)^{1/2}, \quad m \in \mathbf{N}_0.$$

Since the Bessel functions are defined for all real orders, we may extend the expression for $\sigma^k_{2\mathrm{D},\,m}$ to $m > 0$ and use the relation $h^{(1)}_m(z) = \sqrt{\pi/2z} H^{(1)}_{m+1/2}(z)$ to express the singular values (6) *by*

$$\sigma^k_m = \frac{1}{4} \sigma^k_{2\mathrm{D},\,m+1/2}, \quad m \in \mathbf{N}_0. \tag{11}$$

3 Spectral bandwidth of F_k

As we did in [13], we here define the *bandwidth* $\mathscr{B}$ of the operator $F_k : L^2(B_0) \to L^2(\partial B)$ to be

$$\mathscr{B} = \mathrm{argmin}_{m \in \mathbf{N}_0} \{ \sigma^k_{m+n} > \sigma^k_{m+n+1} \text{ for all } n \in \mathbf{N}_0 \},$$

that is, the smallest singular value index after which the singular values, ordered as in Theorem 1, are strictly decaying. In [13] we proved and validated numerically an explicit tight lower bound for the bandwidth of F_k in $\mathbf{R}^2$, as well as argued analytically and numerically for a conjectured tight upper bound. In [14] we used the lower bound from [13] to ensure stability of solution of the multi-frequency problem in $\mathbf{R}^2$. We now obtain a lower bound for the bandwidth $\mathscr{B}$ of F_k in $\mathbf{R}^3$ analogous to that in the two-dimensional case. First we need a technical result; let $m \in \mathbf{N}_0$.

Lemma 1. *If $j_\xi(\kappa_0) = 0$ for some $\xi \in [m, m+1]$ then $a_m(\kappa_0) \le a_{m+1}(\kappa_0)$.*

Proof. Since $j_m(z) = \sqrt{\pi/2z}J_{m+1/2}(z)$, we have from (4) that

$$\begin{aligned} a_m(\kappa_0) &= \sqrt{J_{m+1/2}(\kappa_0)^2 + J_{m-1/2}(\kappa_0)^2 - \frac{2m+1}{\kappa_0}J_{m+1/2}(\kappa_0)J_{m-1/2}(\kappa_0)} \\ &= \sqrt{\frac{2\kappa_0}{\pi}}\sqrt{j_m(\kappa_0)^2 + j_{m-1}(\kappa_0)^2 - \frac{2m+1}{\kappa_0}j_m(\kappa_0)j_{m-1}(\kappa_0)}. \end{aligned}$$

Omitting, for simplicity, the arguments (κ_0), and using the well-known recurrence relation

$$j_{m-1} + j_{m+1} = \sqrt{\frac{\pi}{2\kappa_0}}\left(J_{m-1/2} + J_{m+3/2}\right) = \sqrt{\frac{\pi}{2\kappa_0}}\frac{2(m+1/2)}{\kappa_0}J_{m+1/2} = \frac{2m+1}{\kappa_0}j_m,$$

we find

$$\begin{aligned} \frac{\pi}{2\kappa_0}(a_m^2 - a_{m+1}^2) &= j_{m-1}^2 - j_{m+1}^2 - \frac{2m+1}{\kappa_0}j_m j_{m-1} + \frac{2m+3}{\kappa_0}j_m j_{m+1} \\ &= (j_{m-1} - j_{m+1})(j_{m-1} + j_{m+1}) - \frac{2m+1}{\kappa_0}j_m(j_{m-1} - j_{m+1}) + \frac{2}{\kappa_0}j_m j_{m+1} \\ &= \frac{2m+1}{\kappa_0}j_m(j_{m-1} - j_{m+1}) - \frac{2m+1}{\kappa_0}j_m(j_{m-1} - j_{m+1}) + \frac{2}{\kappa_0}j_m j_{m+1} \\ &= \frac{2}{\kappa_0}j_m j_{m+1}. \end{aligned}$$

It was shown in Lemma 2 in [13] that if $\mu_2 > \mu_1 > 0$ and $J_{\mu_1}(\kappa_0) = J_{\mu_2}(\kappa_0) = 0$ then $\mu_2 - \mu_1 > 1$. This result implies that if $\mu_2 > \mu_1 > 0$ and $j_{\mu_1}(\kappa_0) = j_{\mu_2}(\kappa_0) = 0$ then $\mu_2 - \mu_1 > 1$. Thus, if $j_\xi(\kappa_0) = 0$ for some $\xi \in [m, m+1]$ then the function $\mathbf{R} \ni \mu \mapsto j_\mu(\kappa_0)$ equals zero at m, or it equals zero at $m+1$, or it changes sign exactly once in the interval $[m, m+1]$. In either case, we have

$$a_m^2 - a_{m+1}^2 = \frac{4}{\pi}j_m j_{m+1} \leq 0. \tag{12}$$

and it remains to recall that $a_m(\kappa_0) > 0$ for all $m \in \mathbf{N}_0$ and $\kappa_0 > 0$. □

Lemma 2. *The function $m \mapsto |h_m^{(1)}(\kappa)|$ is strictly increasing for $m > -1/2$ and $\kappa > 0$.*

Proof. The argument is completely analogous to that in Remark 1 in [13]. By Nicholson's integral theorem [21, pp. 441–444], we have

$$\begin{aligned} \left|h_m^{(1)}(\kappa)\right|^2 &= |j_m(\kappa) + iy_m(\kappa)|^2 = j_m(\kappa)^2 + y_m(\kappa)^2 \\ &= \frac{\pi}{2\kappa}\left(J_{m+1/2}(\kappa)^2 + Y_{m+1/2}(\kappa)^2\right) = \frac{\pi}{2\kappa}\left|H_{m+1/2}^{(1)}(\kappa)\right|^2 \\ &= \frac{4}{\pi\kappa}\int_0^\infty \cosh\left(2\left(m+\frac{1}{2}\right)t\right)K_0(2\kappa\sinh t)\,dt \\ &= \frac{4}{\pi\kappa}\int_0^\infty \cosh\left(2\left(m+\frac{1}{2}\right)t\right)K_0(2\kappa\sinh t)\,dt, \end{aligned}$$

where K_0 is the modified Bessel function of the second kind. Differentiating w.r.t. order, we get

$$\partial_\nu \left| h_\nu^{(1)}(\kappa) \right|^2 = \frac{8}{\pi\kappa} \left(\nu + \frac{1}{2} \right) \int_0^\infty \sinh\left(2 \left(\nu + \frac{1}{2} \right) t \right) K_0 \left(2\kappa \sinh t \right) dt,$$

and since K_0 and the hyperbolic sine are positive for positive arguments, it holds that

$$\partial_m \left| h_m^{(1)}(\kappa) \right|^2 > 0 \qquad \text{for all } m > -1/2.$$

This implies that $|h_m^{(1)}(\kappa)|$ is strictly increasing for $m > -1/2$ and $\kappa > 0$. □

For $m \geq 0$ and $n \in \mathbf{N}$, let $j_{m+1/2,\,n}$ be the n'th positive zero of the Bessel function $J_{m+1/2}$, that is, the n'th positive zero of the spherical Bessel function j_m.

Theorem 2. *A lower bound $\mathscr{B}_-$ for $\mathscr{B}$ is given by $\mathscr{B}_- = \operatorname{argmin}_{m \in \mathbf{N}_0} \left\{ j_{m+1/2,\,1} \geq \kappa_0 \right\}$.*

Proof. Let $m \in \mathbf{N}_0$. If $j_{m+1/2,\,1} < \kappa_0$ then, due to the well-known continuity and strictly increasing behavior of the function $\mu \mapsto j_{\mu,1}$, the value κ_0 is the first positive zero of some Bessel function $J_{\xi+1/2}$ with $\xi > m$, hence it is the first positive zero of some spherical Bessel function j_ξ with $\xi > m$. There are thus $n \in \mathbf{N}_0$ and $\xi \in [m+n, m+n+1]$ satisfying $j_\xi(\kappa_0) = 0$, and, by lemma 1, we have $a_{m+n}(\kappa_0) \leq a_{m+n+1}(\kappa_0)$. Also, by lemma 2, the function $\mathbf{N}_0 \ni \mu \mapsto |h_\mu^{(1)}(\kappa)|$ is strictly increasing. Since the singular value σ_μ from (6) is proportional to $|h_\mu^{(1)}(\kappa)| a_\mu(\kappa_0)$ for $\mu \in \mathbf{N}_0$, we have $\sigma_{m+n} \leq \sigma_{m+n+1}$, hence $m < \mathscr{B}$. In conclusion,

$$\mathscr{B} \geq \operatorname{argmax}_{m \in \mathbf{N}_0} \{ j_{m+1/2,\,1} < \kappa_0 \} + 1 = \operatorname{argmin}_{m \in \mathbf{N}_0} \{ j_{m+1/2,\,1} \geq \kappa_0 \}.$$

□

In [13] we found a lower bound $\mathscr{B}_{2D,-}$ for the two-dimensional case to be

$$\mathscr{B}_{2D,-} = \operatorname{argmin}_{m \in \mathbf{N}_0} \left\{ j_{m,\,1} \geq \kappa_0 \right\}.$$

In light of the strictly increasing behavior of the function $\mu \mapsto j_{\mu,1}$ for nonnegative orders μ, we interpret the additional '+1/2' in the order of the zero in the three-dimensional case as an indication of greater stability of the inverse source problem in $\mathbf{R}^2$ compared to the three-dimensional case. However, other aspects of the change of setting from $\mathbf{R}^2$ to $\mathbf{R}^3$ may offset this. We finally note that it might be possible to exploit the simple connection (11) between the singular values of the measurement operator in $\mathbf{R}^2$ and in $\mathbf{R}^3$ in an analysis along the lines of [13, pp. 9–10] and potentially obtain estimates of tight upper bounds on the bandwidths of both near-field and far-field measurement operators in $\mathbf{R}^3$. We defer this to future work, in particular since it suffices to use the lower bound $\mathscr{B}_-$ in our stability analysis of the multi-frequency inverse source problem below.

With the singular value decomposition and bandwidth results for the measurement operator in place, we begin the analysis of the multi-frequency inverse source problem.

4 Stability of the multi-frequency inverse source problem in $\mathbf{R}^3$

It is well-known [2] that the multi-frequency inverse source problem, as described in our introduction and in [2, Problem 3.1], is uniquely solvable only if measurements are available over a set of frequencies with an accumulation point, possibly at infinity. In practical settings, it is impossible to perform infinitely many measurements, so we turn to reconstructing sources in finite-dimensional subspaces of $L^2(B_0)$. The analysis in this section is a natural extension of the methods in [14] from $\mathbf{R}^2$ to $\mathbf{R}^3$, formulated in a simpler and more flexible way.

4.1 Finite-dimensional source spaces

Consider first the eigenvalue problem for the Dirichlet-Laplacian in B_0,

$$\begin{aligned} -\Delta\varphi &= \lambda\varphi \quad \text{in } B_0, \\ \varphi &= 0 \quad \text{on } \partial B_0. \end{aligned} \tag{13}$$

One can check [18] that the eigenfunctions and eigenvalues are given uniquely up to a scaling by

$$\varphi_{\alpha,\beta,\gamma} = \sqrt{\frac{2}{R_0^3}}\frac{j_\alpha(\mu_{\alpha,\beta}|x|/R_0)}{|j_{\alpha+1}(\mu_{\alpha,\beta})|}Y_\alpha^\gamma(x/|x|), \qquad \alpha \in \mathbf{N}_0, \beta \in \mathbf{N}, \gamma \in \{-\alpha, ..., \alpha\}, \tag{14}$$

$$\lambda_{\alpha,\beta} = \left(\frac{\mu_{\alpha,\beta}}{R_0}\right)^2, \qquad \alpha \in \mathbf{N}_0, \beta \in \mathbf{N}, \tag{15}$$

where $\mu_{\alpha,\beta}$ is the β'th positive zero of the spherical Bessel function of order α, i.e., $j_\alpha(\mu_{\alpha,\beta}) = 0$ for $\beta \in \mathbf{N}$ [1]. Each eigenvalue $\lambda_{\alpha,\beta}$ has multiplicity $2\alpha + 1$.
It is well-known that the set $\{\varphi_{\alpha,\beta,\gamma}\}$ of normalized eigenfunctions is an orthonormal basis of $L^2(B_0)$. Any source $s \in L^2(B_0)$ thus has an eigenfunction expansion

$$s(x) = \sum_{\alpha,\beta,\gamma} \hat{s}_{\alpha,\beta,\gamma}\varphi_{\alpha,\beta,\gamma}(x), \qquad \hat{s}_{\alpha,\beta,\gamma} = \int_{B_0} s\overline{\varphi}_{\alpha,\beta,\gamma}\mathrm{d}x. \tag{16}$$

For $M \in \mathbf{N}$, let $\Lambda_M = \{\lambda_{\alpha,\beta}\}$ be a set of any M eigenvalues in (15). We associate with Λ_M the finite-dimensional source space

$$S_{\Lambda_M}(B_0) = \mathrm{span}\{\varphi_{\alpha,\beta,\gamma} : \lambda_{\alpha,\beta} \in \Lambda_M, -\alpha \le \gamma \le \alpha\} \subset L^2(B_0). \tag{17}$$

Due to the multiplicity of the eigenvalues, it follows that $M \le \dim S_{\Lambda_M}(B_0) \le M(1 + 2A)$, with $A = \max\{\alpha,\ \lambda_{\alpha,\beta} \in \Lambda_M\}$. A natural example of such finite-dimensional source space is given by the sources of the form

$$s = \sum_{\alpha=0}^{N_1}\sum_{\beta=1}^{N_2}\sum_{\gamma=-\alpha}^{\alpha} \hat{s}_{\alpha,\beta,\gamma}\varphi_{\alpha,\beta,\gamma}. \tag{18}$$

Then $\Lambda_M = \{\lambda_{\alpha,\beta}\}_{\substack{\alpha=0,...,N_1\\ \beta=1,...,N_2}}$, written $\{\lambda_{\alpha,\beta}\}_{\alpha=0,\,\beta=1}^{N_1,\,N_2}$ in the following, and hence $M = N_2(N_1 + 1)$ and $\dim\ S_{\Lambda_M}(B_0) = N_2(N_1 + 1)^2$.

[1] For clarity we here use the notation $\mu_{\alpha,\beta} = j_{\alpha+1/2,\beta}$.

As we will see, the spaces $S_{\Lambda_M}(B_0)$ allow us to relate in a constructive manner the source information contained in the measurements at different frequencies to a common basis. We now address the following questions.

1. Given a finite-dimensional source space $S_{\Lambda_M}(B_0)$, can we identify a set of measurement frequencies $Q = \{k_j\}$ such that $|Q| \ll \dim S_{\Lambda_M}(B_0)$ and such that any $s \in S_{\Lambda_M}(B_0)$ can be reconstructed exactly from measurements at these frequencies?

2. Vice versa: Given a finite set of measurement frequencies Q, can we identify the finite-dimensional source spaces $S_{\Lambda_{M_j}}(B_0)$ in which the sources can be reconstructed exactly from measurements at these frequencies?

3. Will such reconstructions be stable with respect to imperfect or noisy measurements? By 'stable' we mean that noise in the measurements is not amplified in the reconstructed source.

The main results, and answers to the above questions, are summarized as follows.

Theorem 3. *Let $s \in S_{\Lambda_M}(B_0)$, for some finite set Λ_M of eigenvalues. There is a $\Delta k \in (0, 1/R_0]$ such that, if for each $\lambda_{\alpha,\beta} \in \Lambda_M$ there is a measurement frequency $k_j \in Q$ satisfying $|k_j - \lambda_{\alpha,\beta}^{1/2}| < \Delta k$, then the source s can be stably reconstructed from the set of measurements $\{U_{k_j}\}_{k_j \in Q}$.*

Corollary 1. *Let Q be a finite set of measurement frequencies. There is a $\Delta k \in (0, 1/R_0]$ such that for any set of eigenvalues Λ_{M_j} satisfying $\min_{k_j \in Q} |\lambda_{\alpha,\beta} - k_j| < \Delta k$ for all $\lambda_{\alpha,\beta} \in \Lambda_{M_j}$, any source $s \in S_{\Lambda_{M_j}}(B_0)$ can be stably reconstructed from the measurements $\{U_{k_j}\}_{k_j \in Q}$.*

Remarks

- Theorem 3 says that if we have measurements at a set of frequencies Q that is close enough to the set Λ_M of eigenvalues, then any source $s \in S_{\Lambda_M}(B_0)$ can be stably reconstructed from these measurements. Conversely, given a set Q of measurement frequencies, we can find sets of eigenvalues satisfying the hypothesis in Theorem 3. These sets then give rise to finite-dimensional source spaces, the elements of which are stably reconstructible. We will later also show that, due to the distribution of the eigenvalues of the Dirichlet Laplacian on B_0, a single frequency k can be close to multiple eigenvalues. Consequently, sources from high-dimensional $S_{\Lambda_M}(B_0)$ are reconstructible even from quite few measurement frequencies.

- If $s \notin S_{\Lambda_M}(B_0)$, we reconstruct an approximation $\tilde{s} = P_{S_{\Lambda_M}(B_0)} s$, the projection of s onto $S_{\Lambda_M}(B_0)$. The quality of the approximation depends on $\|\tilde{s} - s\|_{L^2(B_0)}$, and hence on $S_{\Lambda_M}(B_0)$.

- Given $S_{\Lambda_M}(B_0)$, we can extract a (not necessarily unique) smallest set of measurement frequencies Q_m sufficient for any $s \in S_{\Lambda_M}(B_0)$ to be stably reconstructed from measurements $\{U_{k_j}\}_{k_j \in Q_m}$. However, our analysis does not provide a necessary criterion for

such a set Q_m, and hence it might be possible to reconstruct sources from even smaller frequency sets.

The above results are achieved by utilizing the special relationship between the singular vectors of the measurement operator F_k and the eigenfunctions of the Dirichlet-Laplacian, summarized in the following lemma.

Lemma 3. *For any positive measurement frequency k we have*

$$(\varphi_{\alpha,\beta,\gamma}, u^k_{m,n})_{L^2(B_0)} = \delta_{\alpha,m}\delta_{\gamma,n}\,\mathrm{sign}(j_{m+1}(\mu_{m,\beta}))\frac{2j_m(\kappa_0)}{R_0 a_m(\kappa_0)}\frac{\lambda^{1/2}_{m,\beta}}{\lambda_{m,\beta}-k^2}, \tag{19}$$

where $\delta_{m,n}$ is the Kronecker delta. In addition, if $k = \lambda^{1/2}_{m,i}$ for some m, i, then the left singular vectors of the measurement operator F_k are the eigenvectors of the Dirichlet Laplacian: $u^k_{m,n} = \varphi_{m,i,n}$, for each $n \in [-m, ..., m]$.

Proof. From the orthonormality of the spherical harmonics, we immediately have that

$$(\varphi_{\alpha,\beta,\gamma}, u^k_{m,n})_{L^2(B_0)} = \delta_{\alpha,m}\delta_{\gamma,n}\frac{2}{R_0^3}\left(j^2_{\alpha+1}(\mu_{\alpha,\beta})a^2_m(\kappa_0)\right)^{-1/2}\int_0^{R_0} j_\alpha(\mu_{\alpha,\beta}/R_0 r)j_m(kr)r^2\mathrm{d}r.$$

Setting $\alpha = m$, evaluating the above integral and using the recurrence relations for Bessel functions, as well as that $\lambda_{\alpha,\beta} = (\mu_{\alpha,\beta}/R_0)^2$, we get

$$\int_0^{R_0} j_m(\mu_{m,\beta}/R_0 r)j_m(kr)r^2\mathrm{d}r = \frac{R_0^2\lambda^{1/2}_{m,\beta}}{\lambda_{m,\beta}-k^2}j_m(kR_0)j_{m+1}(\mu_{m,\beta}).$$

This shows the first statement. For the second part, we rewrite $a_m(z)$ as

$$\begin{aligned} a_m(z) &= \left(j_m(z)^2 - j_{m+1}(z)\left(\frac{2m}{z}j_m(z) - j_{m+1}(z)\right)\right)^{1/2} \\ &= \left(j_{m+1}(z)^2 - j_{m+1}(z)\frac{2m}{z}j_m(z) + j^2_m(z)\right)^{1/2}. \end{aligned}$$

Hence $a_m(\mu_{m,i}) = |j_{m+1}(\mu_{m,i})|$. Recall that $\lambda^{1/2}_{m,i} = \mu_{m,i}/R_0$ and $\kappa_0 = kR_0$. Taking $k = \lambda^{1/2}_{m,i}$, we have that

$$\begin{aligned} u^k_{m,n}(x) &= \frac{\sqrt{2}}{R_0^{3/2}a_m(\kappa_0)}j_m(k|x|)Y^n_m(x/|x|) \\ &= \frac{\sqrt{2}}{R_0^{3/2}\left|j^2_{m+1}(\mu_{m,i})\right|}j_m(k|x|)Y^n_m(x/|x|) \\ &= \varphi_{m,i,n}(x), \qquad m \in \mathbf{N}_0,\ n \in \{-m, ..., m\}. \end{aligned}$$

□

To prove Theorem 3 we also need the following result.

Lemma 4. *For any numbers $k_1, \dots, k_N$ satisfying $0 < k_1 < \cdots < k_N$ there exists a largest $\Delta k > 0$ such that for any $1 \le i \le N$ and any real x, $|k_i - x| < \Delta k$ implies*

$$\left|\frac{k_i}{k_i^2 - x^2}\right| > \sum_{j=1, j\neq i}^{N} \left|\frac{k_j}{k_j^2 - x^2}\right|.$$

Proof. For any index i, assume $x \in [k_i - \Delta k, k_i + \Delta k] \subset (k_{i-1}, k_{i+1})$, with the convention that $k_0 = 0$ and $k_{N+1} > k_N$. The function

$$S(x) = \sum_{j=1, j\neq i}^{N} \left|\frac{k_j}{k_j^2 - x^2}\right|, \quad x \in [k_i - \Delta k, k_i + \Delta k],$$

is convex and has a maximum M at one of the endpoints $k_i \pm \Delta k$. Next, the function

$$D(x) = \left|\frac{k_i}{k_i^2 - x^2}\right|, \quad x \in [k_i - \Delta k, k_i + \Delta k],$$

is unbounded at k_i, and it is even. We can therefore find a largest Δk such that for all $x \in (k_i - \Delta k, k_i + \Delta k)$, we have $D(x) > M \ge S(x)$. □

We now construct the mapping between the multi-frequency measurements and the finite-dimensional source space. Let Λ_M be a set of M eigenvalues and Q a finite set of measurement frequencies, with $|Q| \le M$. For the sake of bookkeeping, we introduce some new notation.

1. The auxiliary frequency set Q_M: For each $\lambda_{\alpha,\beta} \in \Lambda_M$, take the corresponding $k_{\alpha,\beta} \in Q_M$ to be $k_{\alpha,\beta} = \operatorname{argmin}_{k_j \in Q} |\lambda_{\alpha,\beta}^{1/2} - k_j|$, i.e., the measurement frequency closest to $\lambda_{\alpha,\beta}^{1/2}$. Then Q_M is a set of M *not necessarily distinct* numbers from Q.

2. We label a measurement at frequency $k_{\alpha,\beta} \in Q_M \subset Q$ as $U_{\alpha,\beta}$. That is, if $k_{\alpha,\beta} = k_j \in Q$, then $U_{\alpha,\beta} = U_{k_j}$.

3. Let I_{Λ_M} be the increasingly ordered set of indices α such that $\lambda_{\alpha,\beta} \in \Lambda_M$. For each $\alpha \in I_{\Lambda_M}$, we define the set $I_\alpha = \{\beta' \in \mathbf{N} : \lambda_{\alpha,\beta'} \in \Lambda_M\}$, also increasingly ordered.
   If for example $\Lambda_4 = \{\lambda_{1,2}, \lambda_{3,1}, \lambda_{3,3}, \lambda_{7,3}\}$, we have that $I_{\Lambda_4} = \{1, 3, 7\}$, and that $I_1 = \{2\}$, $I_3 = \{1, 3\}$ and $I_7 = \{3\}$.

4. We associate to each measurement $U_{\alpha,\beta}$ a set of coefficients:

$$\hat{u}_{\alpha,\beta,\gamma} = \frac{(U_{\alpha,\beta}, v_{\alpha,\gamma}^{k_{\alpha,\beta}})_{L^2(\partial B)}}{\sigma_\alpha^{k_{\alpha,\beta}}}, \quad k_{\alpha,\beta} \in Q_M,\ \gamma \in \{-\alpha, ..., \alpha\}. \tag{20}$$

   Here $v_{\alpha,\beta}^{k_{\alpha,\beta}}$ and $\sigma_\alpha^{k_{\alpha,\beta}}$ are the singular vectors and singluar values of $F_{k_{\alpha,\beta}}$, respectively.

We now construct a family of matrix operators mapping the eigenfunction expansion coefficients $\{\hat{s}_{\alpha,\beta,\gamma}\}$ of a source $s \in S_{\Lambda_M}(B_0)$ to the measurement coefficients $\{\hat{u}_{\alpha,\beta,\gamma}\}$.

Expressing $s \in S_{\Lambda_M}(B_0)$ in the above notation we have

$$s = \sum_{\alpha \in I_{\Lambda_M}} \sum_{\beta \in I_\alpha} \sum_{\gamma=-\alpha}^{\alpha} \hat{s}_{\alpha,\beta,\gamma} \varphi_{\alpha,\beta,\gamma}.$$

Fixing $m \in I_{\Lambda_M}$ and letting $U_{m,n}$ be the measurement associated with the source s yields

$$\begin{aligned}\hat{u}_{m,n,i} =& \frac{(U_{m,n}, v_{m,i}^{k_{m,n}})_{L^2(\partial B)}}{\sigma_m^{k_{m,n}}} \\ =& (s, u_{m,i}^{k_{m,n}})_{L^2(B_0)} \\ =& \sum_{\beta \in I_m} \hat{s}_{m,\beta,i} (\varphi_{m,\beta,i}, u_{m,i}^{k_{m,n}})_{L^2(B_0)}, \quad n \in I_m,\, i \in \{-m, ..., m\}.\end{aligned} \tag{21}$$

This is a linear system relating the source coefficients with the measurement coefficients. Setting $N = |I_m|$, we define

$$\bar{U}_m^i = (\hat{u}_{m,j,i})_{j \in I_m} \in \mathbf{C}^N, \tag{22}$$

$$\bar{S}_m^i = (\hat{s}_{m,j,i})_{j \in I_m} \in \mathbf{C}^N, \tag{23}$$

$$K_m = \Big((\varphi_{m,j,i}, u_{m,i}^{k_{m,n}})_{L^2(B_0)} \Big)_{n,j \in I_m} \in \mathbf{C}^{N \times N}. \tag{24}$$

Notice that the matrix K_m does not depend on the index i. Equation (21) now takes the form

$$\bar{U}_m^i = K_m \bar{S}_m^i, \quad i \in \{-m, ..., m\}. \tag{25}$$

To reconstruct all the coefficients of $s \in S_{\Lambda_M}(B_0)$, we thus have to solve the system of equations

$$\bar{U}_m^i = K_m \bar{S}_m^i, \quad \forall m \in I_{\Lambda_M}, i \in \{-m, ..., m\}, \tag{26}$$

and so we have to show that the matrices K_m are invertible.

To this end, we show that there exists a $\Delta k > 0$ such that if for each $\lambda_{\alpha,\beta} \in \Lambda_M$ there is a corresponding measurement frequency $k_{\alpha,\beta} \in Q_M \subset Q$ close to $\lambda_{\alpha,\beta}$ in the sense that $|\lambda_{\alpha,\beta}^{1/2} - k_{\alpha,\beta}| < \Delta k$, then each matrix K_m is invertible. In addition, we show that if $\Delta k \leq 1/R_0$, then the inversion is stable with respect to noisy measurements.

Invertibility of the matrices K_m is shown by proving that Δk can be chosen such that each matrix K_m is strictly diagonally dominant. Recall that an $M \times M$ matrix $A = (a_{i,j})$ is strictly diagonally dominant if $|a_{i,i}| > \sum_{j \neq i} |a_{i,j}|$ for each $1 \leq i \leq M$, and that such matrices are invertible [19].

Proof of Theorem 3, (reconstruction): From equation (24) we get that for K_m, $m \in I_{\Lambda_M}$ to be strictly diagonally dominant, the inequality

$$|(\varphi_{m,n,i}, u_{m,i}^{k_{m,n}})_{L^2(B_0)}| > \sum_{\beta \in I_m, \beta \neq n} |(\varphi_{m,\beta,i}, u_{m,n}^{k_{m,n}})_{L^2(B_0)}| \tag{27}$$

has to hold for each row. Recall (21) and that the index $n \in I_m$ is row-dependent. With the help of Lemma 3 we write (27) as

$$\left| \frac{2 j_m(\kappa_0)}{R_0 a_m(\kappa_0)} \frac{\lambda_{m,n}^{1/2}}{\lambda_{m,n} - k_{m,n}^2} \right| > \sum_{\beta \in I_m, \beta \neq n} \left| \frac{2 j_m(\kappa_0)}{R_0 a_m(\kappa_0)} \frac{\lambda_{m,\beta}^{1/2}}{\lambda_{m,\beta} - k_{m,n}^2} \right|. \tag{28}$$

The factor $2j_m(\kappa_0)/R_0 a_m(\kappa_0)$ is constant in each row and can be neglected. Hence the simplification

$$\left|\frac{\lambda_{m,n}^{1/2}}{\lambda_{m,n}-k_{m,n}^2}\right| > \sum_{\beta\in I_m,\beta\neq n}\left|\frac{\lambda_{m,\beta}^{1/2}}{\lambda_{m,\beta}-k_{m,n}^2}\right|. \tag{29}$$

For fixed m and increasing $\beta \in I_m$, the eigenvalues $\{\lambda_{m,\beta}\}_{\beta\in I_m}$ form a strictly increasing sequence. Assuming that $k_{m,n} \in (\lambda_{m,n-1}^{1/2}, \lambda_{m,n+1}^{1/2})$, Lemma 4 shows that there exists a largest $\Delta k_{m,n} > 0$ such that if $|\lambda_{m,n}^{1/2} - k_{m,n}| < \Delta k_{m,n}$ then the inequality (29) holds. We take Δk_m to be $\Delta k_m = \min_{n\in I_m} \Delta k_{m,n}$, i.e., the smallest of the $\Delta k_{m,n}$'s such that (29) holds for each row in K_m. Next, we set $\Delta k = \min_{m\in I_{\Lambda_M}}$. Assume now that for each $\lambda_{m,n} \in \Lambda_M$ there is a measurement frequency $k_j \in Q$ such that $|\lambda_{m,n}^{1/2} - k_j| < \Delta k$. Then, for $k_{m,n} = k_j \in Q_M$, we immediately have that

$$|\lambda_{m,n}^{1/2} - k_{m,n}| < \Delta k \leq \Delta k_{m,n},$$

and hence the inequality (29) holds for each row in each K_m. Hence each K_m is strictly diagonally dominant and invertible. □

Remark

The above proof finds a single Δk that serves as a "global" constraint on how close it suffices for the measurement frequencies to be to the eigenvalues in Λ_M. This is done to make the statement of Theorem 3 simpler. In a more refined analysis, one could aim to find a Λ_M-dependent Δk. Moreover, the above is merely an existence proof for Δk. Clearly, we want Δk as large as possible (but, as we will see, smaller than or equal to $1/R_0$ for stability reasons). This can of course be checked numerically, by computing the rank or condition number of the matrices K_m as a function of Δk, and doing this reveals that one can choose $\Delta k = 1/R_0$. However, we have not yet been able to obtain a proof of this.

We now analyze the stability of the above inversion procedure with respect to noise. We will see that under the simple constraint that $\Delta k \leq 1/R_0$, where R_0 is the radius of the source domain, we can guarantee stable reconstructions in the sense that noise is not amplified in the calculation of the measurement coefficients $\hat{u}_{m,n,i}$.

For $\varepsilon \in L^2(\partial B)$, we consider an additive noise model where the noisy measurements are of the form $U_{m,n}^{\varepsilon} = U_{m,n} + \varepsilon$. We assume the noise ε to be bounded in the sense that $\|\varepsilon\|_{L^2(\partial B)} \leq \delta \|U_{m,n,i}\|_{L^2(\partial B)}$ for some noise level $0 < \delta < 1$. Recalling equation (20) for the measurement coefficients,

$$\hat{u}_{m,n,i} = \frac{(U_{m,n}^{\varepsilon}, v_{m,i}^{k_{m,n}})_{L^2(\partial B)}}{\sigma_m^{k_{m,n}}} = \hat{u}_{m,n,i} + \frac{(\varepsilon, v_{m,i}^{k_{m,n}})_{L^2(\partial B)}}{\sigma_m^{k_{m,n}}}, \tag{30}$$

it is evident that the noise will be amplified if $\sigma_m^{k_{m,n}} \ll 1$. Hence we want to avoid the coefficients associated with small singular values, i.e., coefficients $\hat{u}_{m,n,i}$ where $m > \mathscr{B}$, where $\mathscr{B}$ is the bandwidth of the operator $F_{k_{m,n}}$. We now show that if $\Delta k \leq 1/R_0$, then $m \leq \mathscr{B}_-$.

Proof of Theorem 3 (stability): The result follows from Theorem 2 in [14]. We adapt the

proof to the current setting, including all the relevant details for completeness.

To each measurement frequency $k_j \in Q$ we associate the index set

$$I_{k_j} = \{\alpha : \lambda_{\alpha,\beta} \in \Lambda_M \text{ and } |\lambda_{\alpha,\beta}^{1/2} - k_j| < \Delta k\}.$$

For each measurement U_{k_j}, we defined I_{k_j} to be the set of indices m of the measurement coefficients that are computed from U_{k_j}. Consequently, by (30), the elements of I_{k_j} are also the indices of the singular values involved in the computation. We show that if $\Delta k \leq 1/R_0$, we have $\max_m I_{k_j} \leq \mathscr{B}_-$. Recalling Theorem 2, we have

$$\mathscr{B}_- = \operatorname{argmin}_{m\in\mathbf{N}_0}\{j_{m+1/2,\,1} = \mu_{m,1} \geq \kappa_0 = k_j R_0\}.$$

As shown in [14], for $m > -1$ and $n \in \mathbf{N}$ it holds that $\mu_{m+1,n} - \mu_{m,n} > 1$. Recalling that $\lambda_{m,n} = (\mu_{m,n}/R_0)^2$, we have that

$$|\lambda_{m,n}^{1/2} - k_j| < \Delta k \Longleftrightarrow \mu_{m,n} \in (k_j R_0 - 1, k_j R_0 + 1).$$

Assume first that the largest index $\widetilde{m} \in I_{k_j}$ is associated with some $\mu_{\widetilde{m},1}$ satisfying

$$\mu_{\widetilde{m},1} \in (k_j R_0 - 1, k_j R_0 + 1).$$

Then either $\mu_{\widetilde{m},1} < k_j R_0$ and so $\widetilde{m} < \mathscr{B}_-$, or $\mu_{\widetilde{m},1} \in [k_j R_0, k_j R_0 + 1)$ and so $\widetilde{m} = \mathscr{B}_-$, since no two $\mu_{m,1}$ can occur within an interval of length 1, and $\mu_{\mathscr{B}_-,1}$ is the smallest $\mu_{m,1} \geq k_j R_0$. Last, assume the largest index $\widetilde{m}$ comes from $\mu_{\widetilde{m},n}$, $n > 1$, and that $\widetilde{m} > \mathscr{B}_-$. Then $\mu_{\widetilde{m},n} > \mu_{\widetilde{m},1} > \mu_{\mathscr{B}_-,1}$, which is a contradiction since then $\mu_{\widetilde{m},n} \notin (k_j R_0 - 1, k_j R_0 + 1)$. □

4.1.1 Density of eigenvalues and subspace dimension

A key observation in [14] was that the eigenvalues become more densely spaced as one moves up the real line. An estimate for the expected number $N_\lambda(k, \Delta k)$ of $\lambda_{\alpha,\beta}^{1/2}$ that lay in an interval of length $2\Delta k$ centered at k was in [14] given by the formula

$$N_\lambda(k, \Delta k) \approx \left\lfloor \left(\frac{1}{2}(3k^2 + 16)^{1/2} - 3\right) \right\rfloor \frac{2\Delta k}{R_0 \pi}. \tag{31}$$

Roughly, this implies that for a set of N linearly spaced frequencies $Q = \{2\Delta k \cdot n\}_{n=1}^N$, we have

$$|\Lambda| \approx \frac{(\Delta k)^2}{R_0 \pi} 3N(N+1),$$

and hence that the cardinality of Λ_M increases approximately quadratically with N. Equation (31) also has the interpretation that high-frequency measurements carry more information in terms of the dimension of $S_{\Lambda_M}(B_0)$.

5 Numerical verification of the multi-frequency inversion method

The proof of Theorem 3 immediately suggests a reconstruction method:

1. For a given finite-dimensional source space S_Λ, find a set of measurement frequencies $Q = \{k_j\}$ satisfying $|k_j - \lambda_{\alpha,\beta}^{1/2}| < \Delta k$, as in Theorem 3.

2. Compute the measurement coefficients from the measurements $\{U_{k_j}\}_{k_j \in Q}$ as in equation (21).

3. Solve the linear systems in (26) to obtain the source coefficients and a reconstruction of the source.

Step 1 above can also be done in the reversed direction, i.e., given a set of measurement frequencies Q, find a space S_{Λ_M} satisfying Corollary 1 and in which a stable reconstruction can be achieved.

For numerical experiments and validation of the above reconstruction method, we introduce two finite-dimensional source spaces. Let $\Lambda_L = \{\lambda_{\alpha,\beta}\}_{\alpha=0,\beta=1}^{5,5}$ and $\Lambda_H = \{\lambda_{\alpha,\beta}\}_{\alpha=0,\beta=1}^{15,15}$, and let $S_{\Lambda_L}$and S_{Λ_H} be the corresponding source spaces. Further, let the source support B_0 have radius 1, and assume measurements are taken on ∂B, where B has radius 2. By numerical testing, we find that we can choose $\Delta k = 0.7$ in Theorem 3 and we find the smallest (non-unique) sets of measurement frequencies Q_L and Q_H sufficient for stable reconstruction of sources in $S_{\Lambda_L}$and S_{Λ_H}, respectively. We get

$$|Q_L| = 12, \quad |\Lambda_L| = 30, \quad \dim S_{\Lambda_L} = 180, \tag{32}$$

$$|Q_H| = 42, \quad |\Lambda_H| = 240, \quad \dim S_{\Lambda_H} = 3840. \tag{33}$$

$$\tag{34}$$

Due to the density of the eigenvalues, we see that we can reconstruct 3840 source coefficients from measurements at only 42 different frequencies.
Let $\chi_{B_{r,x_0}}(x)$ be the characteristic function for a ball of radius r centered at x_0 in $\mathbf{R}^3$. We consider the two sources

$$s_E(x) = 1\varphi_{2,1,1} + 2\varphi_{3,3,3} + 4\varphi_{4,3,2} + 0.5\varphi_{12,6,7}, \tag{35}$$

$$s_B(x) = 1\chi_{B_{r_1,x_1}}(x) + 2\chi_{B_{r_2,x_2}}(x) + 3/2\chi_{B_{r_3,x_3}}(x), \tag{36}$$

where the radii and centers of the characteristic functions are

$$r_1 = 0.4,\ x_1 = (0, 0.1, 0.3),\ r_2 = 0.1,\ x_2 = (0, -0.4, -0.3),\ r_3 = 0.3,\ x_3 = (0.2, 0.5, -0.4).$$

Clearly, $s_E \in S_{\Lambda_H}$ but $s_E \notin S_{\Lambda_L}$. On the other hand, s_B does not belong to neither S_{Λ_L} or S_{Λ_H} and we can only reconstruct the projection of the source onto the source spaces.
To compute the measurements $\{U_{k_j}\}_{k_j \in Q}$, we do as follows:
For a source $\chi_{B_{r,x_0}}(x)$ supported in B_0 we have

$$\begin{aligned} F_k \chi_{B_{r,x_0}}(x) &= \int_{B_0} G_k(x,y)\, \chi_{B_{r,x_0}}(y) \mathrm{d}y \\ &= \int_{B_{r,x_0}} \frac{e^{ik|x-y|}}{4\pi |x-y|} \mathrm{d}y = \frac{1}{4\pi} \sum_{n=0}^{\infty} \frac{(ik)^n}{n!} \int_{B_{r,x_0}} |x-y|^{n-1} \mathrm{d}y. \end{aligned} \tag{37}$$

The integrals in the sum can be evaluated by noting that the function $|x-y|^{n-1}$ is constant for $y \in \partial B_{z,x}$. Let $d = |x-x_0|$. For $d-r \leq z \leq d+r$ the sphere $\partial B_{z,x}$ intersects B_{r,x_0}, and the part

of $\partial B_{z,x}$ enclosed in B_{r,x_0} is a spherical cap with area $A(z) = 2\pi z\left(z - (d^2 - r^2 + z^2)/2d\right)$, see [22]. We thus have that

$$\int_{B_{r,x_0}} |x-y|^{n-1}\mathrm{d}y = \int_{d-r}^{d+r} z^{n-1} 2\pi z \left(z - \frac{d^2 - r^2 + z^2}{2d}\right)\mathrm{d}z. \tag{38}$$

This integral can be evaluated by standard methods and we get

$$\int_{B_{r,x_0}} |x-y|^{n-1}\mathrm{d}y = \frac{2\pi\left((d + r(2+n))(d-r)^{n+2} + ((n+2)r - d)(d+r)^{n+2}\right)}{d(n+1)(n+2)(n+3)}. \tag{39}$$

A lengthy calculation shows that

$$\begin{aligned}&\sum_{n=0}^{\infty} \frac{(ik)^n}{4\pi} \frac{2\pi\left((d + r(2+n))(d-r)^{n+2} + ((n+2)r - d)(d+r)^{n+2}\right)}{dn!(n+1)(n+2)(n+3)}\\ &= \frac{(i-kr)\mathrm{e}^{ik(d-r)} - (i+kr)\mathrm{e}^{ik(d+r)}}{2dk^3},\end{aligned}$$

and hence the measurement is given by

$$F_k\chi_{B_{r,x_0}}(x) = \frac{(i-kr)\mathrm{e}^{ik(|x-x_0|-r)} - (i+kr)\mathrm{e}^{ik(|x-x_0|+r)}}{2|x-x_0|k^3}, \quad x_0 \in B_0, x \in \partial B. \tag{40}$$

The measurement corresponding to an eigenvector source $\varphi_{\alpha,\beta,\gamma}$ is given by (19) in Lemma 3, and we have that

$$F_k\varphi_{\alpha,\beta,\gamma}(x) = \sigma_\alpha^k v_{\alpha,\gamma}^k(x)\,\mathrm{sign}(j_{\alpha+1}(\mu_{\alpha,\beta}))\frac{2j_\alpha(\kappa_0)}{R_0 a_\alpha(\kappa_0)}\frac{\lambda_{\alpha,\beta}^{1/2}}{\lambda_{\alpha,\beta} - k^2}.$$

We avoid the high computational cost of simulating high-frequency solutions to the Helmholtz equation by generating the measurements of the sources using the above closed-form solutions. We sample each of the measurements $\{U_{k_j}\}_{k_j\in Q}$ at equidistant points $\{x_i\}_{i=1}^N$ on ∂B. For a noise level δ, we add a noise vector $\varepsilon \in \mathbf{C}^N$ of normally distributed random numbers[2] to the measurement vector $\left(U_{k_j}(x_i)\right) \in \mathbf{C}^N$, with the convention that $|\varepsilon| \le \delta|U_{k_j}|$. We denote the noisy measurement by $U_{k_j}^\varepsilon = U_{k_j} + \varepsilon$.
To compute the measurement coefficients, we approximate the inner products appearing in (21) by a simple quadrature method

$$\left(U_{k_j}^\varepsilon, v_{\alpha,\gamma}^{k_j}\right)_{L^2(\partial B)} \approx 4\pi R^2 \frac{1}{N}\sum_{i=1}^N U_{k_j}^\varepsilon(x_i)\,\overline{v_{\alpha,\gamma}^{k_j}(x_i)}. \tag{41}$$

All entries in the matrices K_m in (21) are explicitly given, and are implemented in Matlab. We compute the source coefficients by solving

$$\bar{S}_m^i = K_m^{-1}\bar{U}_m^i, \quad \forall m \in I_\Lambda, i \in [-m, ..., m], \tag{42}$$

where $\bar{U}_m^i$ are the vectors of measurement coefficients calculated from the noisy measurements in (41). Ultimately, the source is expanded in terms of the eigenvectors in S_Λ. The number of points necessary for a good estimate varies with the nature of the measurement and the dimension of the source space.

[2]For each component ϵ_i the real and imaginary part is generated separately by the Matlab-function randn().

5.1 Reconstructions

The measurements are generated as outlined above, and we add noise with a noise level $\delta = 0.1$, i.e., 10% noise to all measurements before we perform the reconstruction. We sample the measurements at $N \approx 10000$ points. We plot the $(0, y, z)$-sections of the sources, the reconstructions and their (log) pointwise absolute difference. Figures 1-2 show the results of the numerical simulations. Figure 1 shows that the source s_E agrees remarkably well with the reconstruction from high-frequency measurements $\{U_{k_j}\}_{k_j \in Q_H}$. The outstanding agreement in the presence of noisy measurements might be due to the small number of measurement coefficients being non-zero because of the very simple structure of s_E. From the low-frequency measurements $\{U_{k_j}\}_{k_j \in Q_L}$ we can only reconstruct the projection $P_{S_{\Lambda_L}} s_B$, and hence we miss the component $0.5\varphi_{12,6,7}$ of s_E. In Figure 2, we see that the reconstruction of s_B from the low-frequency measurements captures the characteristic functions with larger supports quite well, but that the smallest ball is almost invisible. This is as expected, since S_{Λ_L} does not contain the high-frequency vectors needed to resolve smaller features. On the contrary, the reconstruction of s_B from the high-frequency measurements captures all characteristic functions quite well and is a satisfactory approximation to s_B. As we can see in the difference plots, both the low and high-frequency reconstructions suffer from larger pointwise errors around the discontinuities of s_B, and we notice the well-known Gibbs phenomenon.

The results verify that the proposed reconstruction method offers stable reconstruction of sources in $S_\Lambda(B_0)$ and of projections $P_{S_{\Lambda_L}} s$ for $s \in L^2(B_0)$, and that the quality of such reconstructions depends on the number of measurement frequencies. Moreover, the method has a low computational cost, requiring only addition and inversion of low-dimensional matrices.

5.2 Approximation properties of S_{Λ_M}

We now study numerically the approximation properties of S_{Λ_M}. The approximation property we are interested in the distance $\|P_{S_{\Lambda_M}} s - s\|_{L^2(B_0)}$. We want to compare approximations of $s \in L^2(B_0)$ in S_{Λ_M} and in E_Q, where E_Q is a large set of left singular vectors of the measurement operators F_{k_j}, $k_j \in Q$.

Let $Q = \{k_j\}_{j=1}^N$ be the smallest frequency set associated with Λ_M and $s \in L^2(B_0)$. We define the truncated measurement operator at frequency k_j by

$$F_{k_j}^T = \sum_{m=1}^{\beta_-^{k_j}} \sum_{n=-m}^{m} \sigma_m^{k_j} v_{m,n}^{k_j} (\cdot, u_{m,n}^{k_j})_{L^2(\Omega)}, \quad k_j \in Q, \quad \beta_-^{k_j} = \operatorname{argmin}_{m \in \mathbf{N}_0} \{\mu_{m,1} \geq k_j R_0\}.$$

Next, we define the multi-frequency and truncated multi-frequency measurement operators by, respectively,

$$F_Q = \begin{bmatrix} F_{k_1} \\ F_{k_2} \\ \vdots \\ F_{k_N} \end{bmatrix}, \qquad F_Q^T = \begin{bmatrix} F_{k_1}^T \\ F_{k_2}^T \\ \vdots \\ F_{k_N}^T \end{bmatrix}. \tag{43}$$

We have that $F_Q, F_Q^T : L^2(B_0) \to \big(L^2(\partial B)\big)^N$. Since F_Q^T is finite-rank, it has a finite singular system $\{\sigma_m, \phi_m, \psi_m\}_{m=1}^M$, and the inverse of F_Q^T is

$$(F_Q^T)^{-1} = \sum_{m=1}^{M} \frac{(\cdot, \psi_m)_{(L^2(\partial B))^N}}{\sigma_m} \phi_m.$$

We say that $(F_Q^T)^{-1}$ is the truncated singular value decompoistion (TSVD) inverse of F_Q, and we have that $\mathrm{Im}(F_Q^T)^{-1} = \mathrm{span}\{\phi_m\}$. Since the inversion method using S_{Λ_M} only extracts parts of the recoverable information from the measurements, it is of interest to compare it to approximations in $E_Q = \mathrm{span}\{\phi_m\}$, which in a sense is the largest space of stably recoverable information we can reconstruct from the measurements $\{U_{k_j}\}_{k_j \in Q}$.

To this end, we proceed as follows. For $k_j \in Q$, let E_{k_j} be the set of right singular vectors included in $F_{k_j}^T$, i.e.,

$$E_{k_j} = \{u_{m,n}^{k_j} : 0 \le m \le \mathscr{B}_-^{k_j}, -m \le n \le m\}. \tag{44}$$

For our investigation, we need the following lemma.

Lemma 5. $\mathrm{Im}(F_Q^T)^{-1} = \mathrm{span}\big\{\bigcup_{k_j \in Q} E_{k_j}\big\}$.

Proof. First, note that $\mathrm{Ker} F_Q^T = \bigcap_{k_j \in Q} \mathrm{Ker} F_{k_j}^T$, and that $\mathrm{Ker}\, F_{k_j} = L^2(\Omega) \ominus \mathrm{span}\, E_{k_j}$. Hence

$$\mathrm{Ker}\, F_Q^T = \bigcap_{k_j \in Q} \big(L^2(\Omega) \ominus \mathrm{span}\, E_{k_j}\big) = L^2(\Omega) \ominus \mathrm{span}\big\{\bigcup_{k_j \in Q} E_{k_j}\big\}.$$

The result now follows by noting that

$$\mathrm{Im}(F_Q^T)^{-1} = (\mathrm{Ker}\, F_Q^T)^\perp = \left(L^2 \ominus \mathrm{span}\Big\{\bigcup_{k_j \in Q} E_{k_j}\Big\}\right)^\perp = \mathrm{span}\big\{\bigcup_{k_j \in Q} E_{k_j}\big\}.$$

□

Hence we have $E_Q = \mathrm{span}\big\{\bigcup_{k_j \in Q} E_{k_j}\big\}$. To compare approximations, we compute the distances $d(s, S_{\Lambda_M})$ and $d(s, E_Q)$ for different $s \in L^2(\Omega)$ and different Λ_M and Q.
Since S_{Λ_M} is the span of orthonormal vectors,

$$d(s, S_{\Lambda_M})^2 = \|s\|^2 - \sum_{\varphi_{\alpha,\beta,\gamma} \in S_{\Lambda_M}} |(s, \varphi_{\alpha,\beta,\gamma})|^2. \tag{45}$$

From Chapter 7.6 in [9], we have that the distance from a finite-dimensional subspace $M = \mathrm{span}\{\phi_m\}_{m=1}^M$ of a infinite-dimensional Hilbert space X to any $s \in X$ is given by the formula

$$d(s, M)^2 = \|s\|_X^2 - \beta^T G^{-1} \beta. \tag{46}$$

Here $\beta = [(s, \phi_1), \cdots, (s, \phi_M)]$ and G^{-1} is the inverse of the Gram matrix $G = \big\{(\phi_i, \phi_j)_X\big\}_{i,j=1}^M$. Since $E_Q = \mathrm{span}\big\{\bigcup_{k_j \in Q} E_{k_j}\big\}$, the entries of G are given by $(u_{m,n}^{k_i}, u_{\alpha,\beta}^{k_j})_{L^2(\partial B)}$. By an analogous calculation to that in the proof of Lemma 3, we find that

$$(u_{m,n}^{k_i}, u_{\alpha,\beta}^{k_j})_{L^2(B_0)} = \\ \delta_{m,\alpha}\delta_{n,\beta} \frac{2}{R_0 a_m(k_i R_0) a_m(k_j R_0)} \frac{1}{k_i^2 - k_j^2} \big(k_j j_m(k_i R_0) j_{m-1}(k_j R_0) - k_i j_{m-1}(k_i R_0) j_m(k_j R_0)\big).$$

For $n = 3, 4, ..., 8$, let $\Lambda_{M_n} = \{\lambda_{\alpha,\beta}\}_{\alpha=0,\beta=1}^{n,n}$, and take Q_{M_n} to be the smallest frequency set such that the hypothesis of Theorem 3 holds. We set $R_0 = 1$ and compute the distances from the spaces $S_{\Lambda_{M_n}}$ and E_{Q_n} to the functions

$$s_1(x) = \exp\left(-|x|^2/0.1\right)\sin(2\tan^{-1}(x_2/x_1))\cos^{-1}(x_3/|x|),$$
$$s_2(x) = \chi_{r,x_0}, \quad r = 0.2, x_0 = (0.2, 0.3, 0.1).$$

The inner products in equations (45) and (46) are approximated with the Matlab quadrature integral3(). Figure 3 shows the (relative) distances to the sources for increasing n and the dimensions of $E_{Q_n} = \mathrm{Im}(F_{Q_n}^T)^{-1}$ and $S_{\Lambda_{M_n}}$. For both sources the approximation in E_{Q_n} is slightly better than in $S_{\Lambda_{M_n}}$, but the dimension, and hence the computational cost associated with the reconstruction/approximation increases dramatically with n. Hence the results indicate that there is a large number of redundant vectors in E_{Q_n}, and that the lower-dimensional space S_{Λ_M} does capture the essential information about the sources present in the multi-frequency measurements.

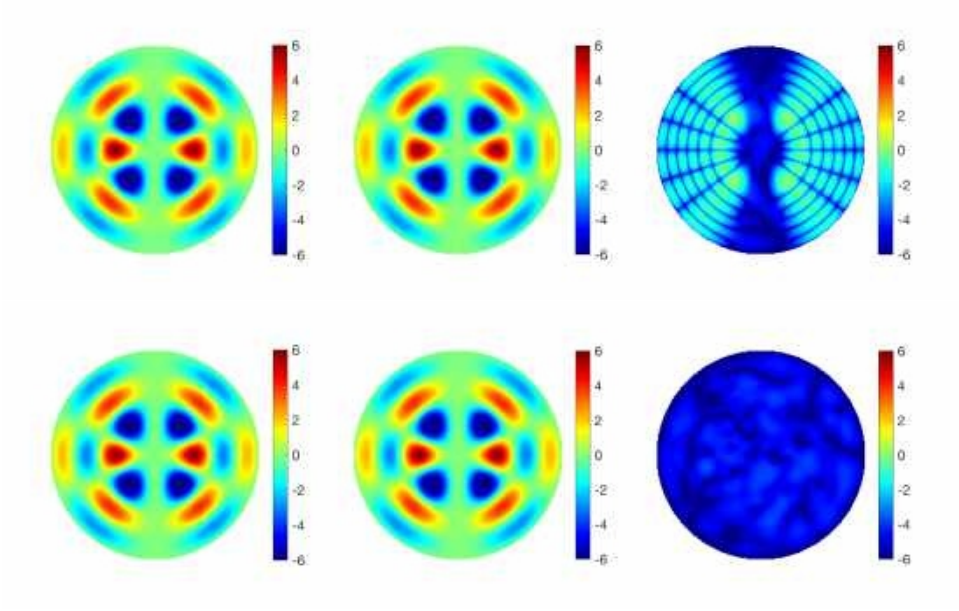

Figure 1: All plots are in the $(0, y, z)$-plane of B_0. Top-left: Source $s_E(x)$. Top-middle: Reconstruction $\hat{s}_E^L(x)$ of $s_E(x)$ from low-frequency measurements $\{U_{k_j}\}_{k_j \in Q_L}$. Top-Right: Logarithm of pointwise absolute error, $\log|\hat{s}_E^L(x) - s_E(x)|$. Bottom-left: Source $s_E(x)$. Bottom-middle: Reconstruction $\hat{s}_E^H(x)$ of $s_E(x)$ from high-frequency measurements $\{U_{k_j}\}_{k_j \in Q_H}$. Bottom-Right: Logarithm of pointwise absolute error, $\log|\hat{s}_E^H(x) - s_E(x)|$.

6 Conclusion

We have analyzed the feasibility and the stability of solution of the multi-frequency inverse source problem for the Helmholtz equation in $\mathbf{R}^3$. Given any finite set of measurement

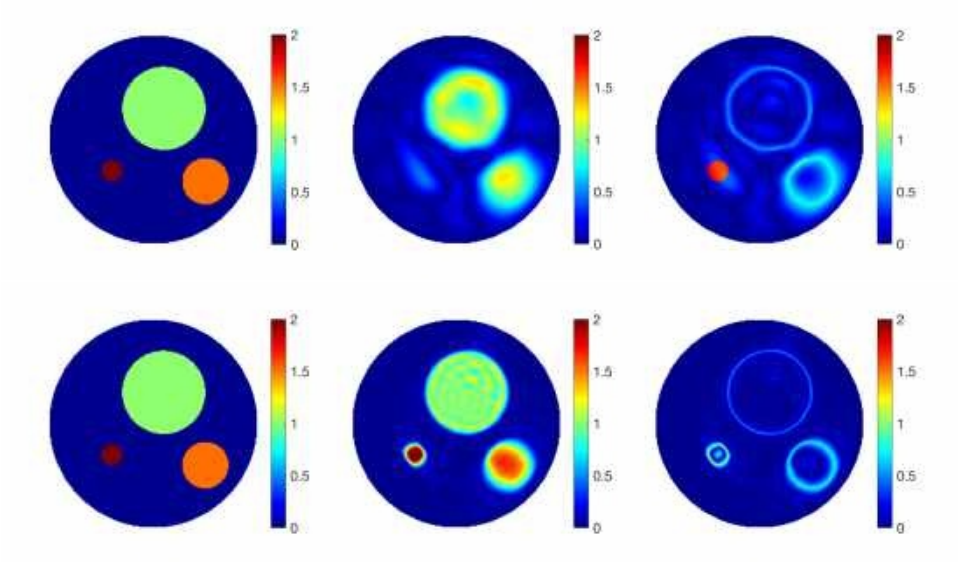

Figure 2: All plots are in the $(0, y, z)$-plane of B_0. Top-left: Source $s_B(x)$. Top-middle: Reconstruction $\hat{s}_B^L(x)$ of $s_B(x)$ from low-frequency measurements $\{U_{k_j}\}_{k_j \in Q_L}$. Top-Right: Pointwise absolute error $|\hat{s}_B^L(x) - s_B(x)|$. Bottom-left: Source $s_B(x)$. Bottom-middle: Reconstruction $\hat{s}_B^H(x)$ of $s_B(x)$ from high-frequency measurements $\{U_{k_j}\}_{k_j \in Q_H}$. Bottom-Right: Pointwise absolute error $|\hat{s}_B^H(x) - s_B(x)|$.

frequencies, we have characterized certain finite-dimensional source spaces in which exact source reconstruction is guaranteed if the measurements are noise-free. Conversely, for each of the finite-dimensional source spaces we have found the smallest set of measurement frequencies that is sufficient for exact reconstruction in the absence of noise. Regarding stability, we have extended our results from the two-dimensional case and proved a lower bound on the 'bandwidth' of the measurement operator, at any measurement frequency. We showed that our inversion method for multi-frequency measurements is stable and highly efficient; since the method does not rely on discretization of the measurement operator, one avoids the computational cost associated with wave phenomena in higher dimensions. It would be interesting to see if the method can be generalized to measurement boundaries other than spheres. Finally, it remains to investigate whether also in the non-constant coefficient case a similar relationship exists between the eigenvalues and eigenvectors of an elliptic PDE and the frequency dependence of the SVD of the measurement operator.

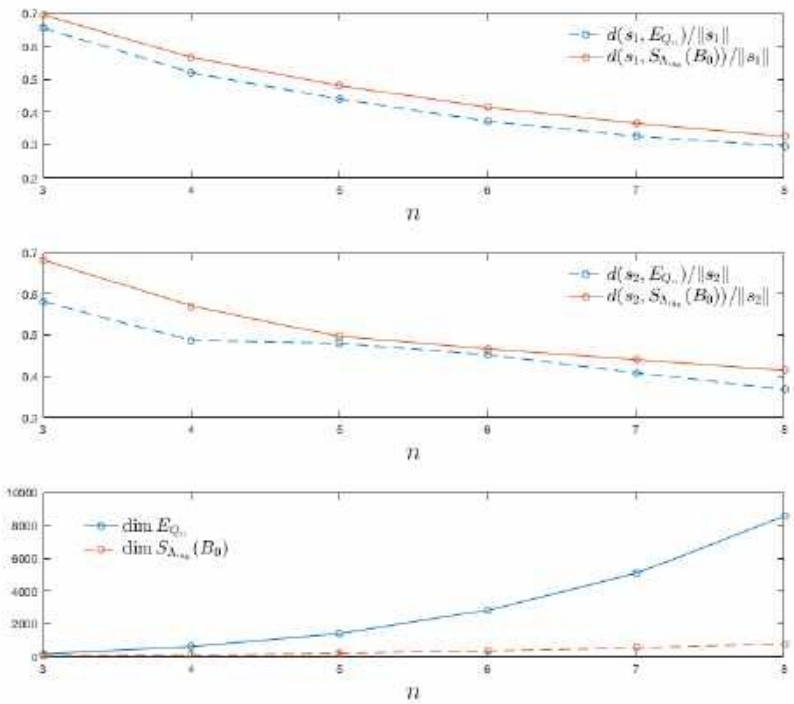

Figure 3: The top and middle plot shows the relative approximation errors in $S_{\Lambda_{M_n}}$ and E_{Q_n}. The bottom plot shows the dimension of the two spaces.

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